



Routledge Studies in the Biblical World

THE DAWN OF AGRICULTURE AND THE EARLIEST STATES IN GENESIS 1–11

Natan Levy



The Dawn of Agriculture and the Earliest States in Genesis 1–11

This book invites a close textual encounter with the first 11 chapters of Genesis as an intimate drama of marginalised peoples wrestling with the rise of the world's first grain states in the Mesopotamian alluvium.

The initial 11 chapters of Genesis are often considered discordant and fragmentary, despite being a story of beginnings within the context of the Bible. Readers discover how these formative chapters cohere as a cross-generational account of peoples grappling with the hegemonic spread of domesticated grain production and the concomitant rise of the pristine states of Mesopotamia. The book reveals how key episodes from the Genesis narrative reflect major societal revolutions of the Neolithic period in Mesopotamia through a three-fold hermeneutical method: literary analysis of the Bible and contemporary cuneiform texts; modern scholarship from archaeological, anthropological, ecological, and historical sources; and relevant exegesis from the Second Temple and rabbinical era. These three strands entwine to recount a generally sequential story of the earliest archaic states as narrated by non-elites at the margins of these emerging state spaces.

The Dawn of Agriculture and the Earliest States in Genesis 1–11 provides a fascinating reading of the first 11 chapters of Genesis, appealing to students and scholars of the Hebrew Bible and the Near East, as well as those working on ecological injustice from a religious vantage point.

Dr. Natan Levy is a Lecturer at Leo Baeck College in the United Kingdom and Head of Operations at Faiths Forum for London. He received his rabbinical ordination from Rabbi Brovender and Rabbi Riskin, his doctorate in Environmental Theology from Bristol University, and an MA in Jewish Studies from King's College, London. Rabbi Levy was the environmental liaison to the former Chief Rabbi of the Commonwealth, Lord Sacks. Rabbi Levy is the co-author of *Sharing Eden: Green Teachings from Jews, Christians and Muslims*.

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**To my children, Temima, Sa'adya, Salomé, Emuna, Ezra,
and Zohar, for whom angels guarding entrances aren't threats
but promises**



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1 Subversive Scripture

I'd like to begin with a question. In the book of Genesis, specifically Genesis 3:17–19, God curses both the human (אָדָם) and the ground (הָאָדָמָה) with grain farming. “And to Adam (אָדָם) He said: ‘Because you have listened to the voice of your wife and have eaten of the tree of which I commanded you saying: Do not eat from it! Cursed is the ground (הָאָדָמָה) because of you; in suffering you shall eat of it all the days of your life. Thorns and thistles it shall bring forth to you, and you shall eat of the grain of the field. In the sweat of your face you shall eat bread until you return to the ground.’”¹ Why would an agrarian culture, such as the hill-terraced dryland farming communities of ancient Israel, incarnate a curse on their own foundational subsistence activity? Why would farmers curse their farming?

To compound the question, we could note that no cuneiform text yet unearthed disparages the core food crop as Genesis does, let alone reifies a grain curse through divine malediction. These approximately contemporaneous narratives incised with reeds on hardening clay surely run the spectrum from outright praise for grain farming to grudging acknowledgement of its singular importance. Of the former, an Old Babylonian cosmology, *The Disputation Between Ewe and Wheat*, describes the domestication of wheat and sheep as a wonderful advance for human society. “The people of those distant days, they knew not bread to eat, they knew not cloth to wear; they went about in the Land with naked limbs eating grass with their mouths like sheep, and drinking water from the ditches. At that time, at the birthplace of the Gods, in their home, the Holy Hill, they (the Gods) fashioned Ewe and Wheat ... They gave them to Mankind as sustenance.”² Of the latter, the Sumerian aetiological tale of *Enki of Ninmah* is upfront about the horrific sweat equity of irrigated grain production; yet this is necessary work to put bread upon the divine table. “In those days, in the days when heaven and earth were [created], ... the gods who baked their daily bread (and) set therewith their tables—the senior gods did oversee the work, while the minor gods were bearing the toil/suffering.”³ And thus, when the minor gods rebel and smash their tools in union-style boycott, Enki and Ninmah create humans, a more pliant replacement to farm the gods’ bread.⁴ In the cuneiform, farming is not a curse, at best it is a divine gift, at worst, a task we were all

2 *Subversive Scripture*

created for. Scholars of ancient Near Eastern textual traditions, from Hermann Gunkel at the start of the 20th century to Nahum Sarna⁵ at the end, have noted the singular incongruity of the Bible's farming curse: "It opposes the opinion, surely common in its time, that farming was a gift of the deity," writes Gunkel, "with the contention that the field is cursed."⁶

Why does Genesis forge its foundational narrative with this cut against the grain? Modern biblical scholarship often downplays this question, and I will explore why as this chapter progresses. When the question is addressed directly, the most prevalent current academic response is that this curse on bread cultivation and consumption is a literary stowaway from a nomadic tradition that somehow or other burrowed its way into the guts of Genesis and retained its singular hold there, even after the legend passed into the mouths and quills of an agrarian community.⁷ According to this subset of source criticism, we have simply uncovered within Genesis an oral nomadic diatribe against the sedentary farming lifestyle.⁸ A curse on the dull food procurement strategy of farmers trapped on their fields, uttered by nimble passersby, akin to the Berber's disdainful comment on farming: "raiding is our agriculture."⁹

Dissecting a text into separate sources based on incongruities in genre, philology, and context would seem the perfect method to relieve the friction of farmers sacralising a curse on their main food procurement strategy. In the early 20th century, Otto Eissfeldt, building on the work of Rudolf Smend, argued that the Yahwist narrative (J) of Genesis, which contains our curse, actually comprised two distinct strands. A nomadic storyteller Eissfeldt entitled the Lay (L) source, which was rooted in a primitive desert pastoral tradition, juxtaposed against a sedentary, agricultural source of J proper. Eissfeldt writes, "It appears that the L strand in the primeval history pictured men as nomads, whereas J and P clearly think of them as husbandmen ... It is quite clear that in Israel which became an agricultural people from being a nomadic people, an outline of their history which places nomads at the beginning must be older than one which pictures the first men as husbandmen ... L is no doubt giving expression in this to his enthusiasm for the nomadic ideal."¹⁰ Later scholarship by George Fohrer¹¹ and Peter Ellis¹² would further bifurcate Hebrew scripture into desert and sown typologies. The biblical curse on agriculture, like the disparagement of Cain, the farmer, compared to Abel, the pastoralist, is simply the fossilised remnants of this pre-farming tradition.¹³

Reading the aetiological curse on farming as the outcry of the nomadic pastoralist runs into two difficulties. First, even if there really is a desert L thread running through our narrative, and our problematic curse is now endowed with pristine nomadic provenance, it still retains its troublesome contra-distinctive presence. Why would farmers retain an anti-farming slogan adapted from their desert forebears? And second, Eissfeldt and similar 19th- and early 20th-century scholars¹⁴ relied upon the then prevailing theory that societies follow a set pattern of aggregation from bands of hunter-gatherers to tribes of pastoralists to villages of farmers. The curse was simply a vestige

from that earlier evolutionary stage of pastoralism. Yet, current research in archaeology and anthropology suggests that social “evolution” is rarely an arrow of progression.¹⁵ What we do find within both fields is that hunter-gatherer societies will adapt some partial elements of domesticating plants and animals, perhaps during mild climate conditions, or for a particular part of the year, without committing in any degree to a pastoral or agricultural track.¹⁶ The social ecologist, Murray Bookchin, termed this diverse and flexible food procurement strategy an “ecology of freedom,” meaning that societies who first began to cultivate crops didn’t automatically evolve into full-time farmers, but moved freely across a broad food web, penning animals or hunting wild fauna when the crops failed, to return in other seasons or years for another planting.¹⁷ Thus, we can’t really say that Israel ever existed as a pristine nomadic society,¹⁸ and we certainly cannot argue for an early nomadic source for the farming curse, when these societies were constantly shifting between cultivation, pastoralism, and wild procurement.¹⁹ We could as well expect a curse on pastoralism during intermittent farming episodes as a curse on farming when society turned again to pastoralism.²⁰

Yet, perhaps we should not dismiss the underlying insight of the multi-stranded compositional approach to these early chapters of Genesis. Genesis really does present a wandering storyline in its first 11 chapters. We begin in the densely forested hill country with Adam and Eve, only to find their grandchildren ensconced in an urban milieu in the alluvial plain, then we escape frantically with Noah into the high mountains of Ararat, before descending again into the valley of Shinar to build a city and tower in Babel. It’s no wonder that biblical scholars, influenced by Hermann Gunkel’s insight that texts are formed by their cultural, social, and natural life-setting, their *Sitz-im-Leben*,²¹ would read into this diversity of backdrops a crowd of variant sources, plundered in piece-meal from cosmopolitan storytellers, dessert bards, and the farming hearth.²² No wonder then that Claus Westermann’s analysis of Genesis 1–11, considered a key representation of the compositional methodology,²³ can chisel the text into no less than five different strata, teasing apart—for example—verse 8 and 9 of Genesis chapter 2, depending on whether Eden was a fitting zone for human habitat or a landscape of mythic trees.²⁴

The problem with this decompositional reading of Genesis is that these early chapters have traces of underlying integrity-in narrative flow, in repetitive wordplay, and recurring motifs—that appear like gossamer threads artfully woven between these variant sources.²⁵ Or as George Coats concluded in his 1983 analysis of Genesis: “The structure of the unit does not support arguments that the paradise-gained story is independent of the paradise-lost story. Rather, the two elements stand together as one unit, the one a reflex to the other.”²⁶ Thus, the curse against farming grain should be viewed as a response to Adam and Eve eating the forbidden fruit from the tree of knowledge of good and evil. This sounds like a rather obvious case to make. After all, it’s precisely what God intones: “And to Adam (אָדָם) He said: ‘Because you have listened to the

4 *Subversive Scripture*

voice of your wife and have eaten of the tree of which I commanded you saying: Do not eat from it! Cursed is the ground (אדמה) because of you; in suffering you shall eat of it all the days of your life.”²⁷ Yet, you can quickly begin to feel the attraction of splitting these stories into separate traditions. There really isn’t any sort of obvious link between eating a forbidden fruit and becoming a farmer.

This is what makes these first 11 chapters so challenging. Source critics are correct to note that our contextual terrain is a patchwork of ecologies. We are moving at a dizzying pace from a story in the high mountains somewhere near the headwaters (ראשי) ²⁸ of the Tigris and Euphrates, to a cityscape built—as every early Mesopotamian city was—in the alluvial floodplains below, then retreating back into the mountains to escape a flood, and like some out of control rollercoasting ending up again in the flat valley floor for a terse vignette on the rise and fall of Babel. It’s a dizzying topographical journey. Yet, it is not a haphazard one.²⁹ How are we to read a consistent plotline into a story of such marked movement?

Perhaps, the fluid ecological jumps of Genesis would have been less remarkable for the subsistence farmers in the highlands of Canaan. These were communities who had migrated en masse to the sparsely populated mountains from elsewhere, sometime in the 13th and 12th century BCE. A narration that moved readily between mountains and valleys might simply reflect a populace much less sedentary than our own. Perhaps, it is the movement of these early stories across terrains that contributes to their very sanctity. Though it’s still a matter of ongoing debate amongst Near Eastern archaeologists, the prevailing model holds that the majority of the populace who would latterly gain the ethnic rubric of Israel were a motley collection of dissidents and refugees escaping the Late Bronze-age collapse of Canaanite cities and farms in the coastal lowlands to colonise the mountain frontier. The archaeologist William Dever, revitalising earlier theories espoused by biblical scholars Norman Gottwald, George Mendenhall, and Marvin Chaney, labelled this migration pattern a “withdrawal” from the Egyptian vassal city-states towards a hinterland in the mountains of Canaan.³⁰ Not only did these migrants abandon the cities of the Canaanite plains, but in their mountain redoubts they actively pursued strategy after strategy that evaded its tax collectors, levelled the military advantages of the feared imperial chariot with inaccessible terrain, and generally eschewed its hegemonic impositions. Consider the marked increase in hewn cisterns lined with waterproof lime-plaster to collect rainwater run-off. These cisterns would ensure a continuous and independent water supply.³¹ In a similar manner, the grain pit silos found in most early Israelite towns and cities arguably represented a decentralised storage system in stark contradistinction to the central grain temple storehouses located in the main Canaanite cities of the plains.³² Lawrence Stager likewise points to the rock terraces instituted in the hills during this period, but notes that the Israelite inhabitants began to cultivate them with ill-suited crops of wheat and barley. “These [terraces] were best suited to vine, olive, and nut cultivation. But the earliest terrace farmers at Ai

and Khirbet Raddana grew cereals. This inefficient use of terraces suggests an attempt by the highlanders to maintain a subsistence cereal agriculture free from the Canaanite and Philistine spheres, where the primary 'bread baskets' were located."³³ These interlocking strategies created a radical anomaly. "The particularity of earliest Israel," asserts Norman Gottwald, "is that it enjoyed the longest stretch of tribute-free communal life known to us from any ancient Near Eastern sources."³⁴

I have outlined this rough and ready sketch of the farmers who are telling the stories that will eventually be canonised within Genesis in order to tighten our initial question.³⁵ Why would these highland subsistence farmers, whose forebears had evaded the collapsing Canaanite city-states in the coastal plains, sanctify a curse on grain farming? Or stated a bit differently: Did the very anomaly of this highland population relatively free from extractive regimes bear any significance for the anomalous farming curse that emerged from amongst its ranks?

This deprecation on grain production is not the only curse contained within the first 11 chapters of Genesis. The Hebrew root word for curse (אָרַר) repeats at five critical junctures within these early tales. To put this density of cursing into context, the same root word appears only five further times throughout the remaining 39 chapters of Genesis. Map this cluster of five cursed moments against the fluid altitude shifts in our concatenated Genesis stories and an intriguing pattern emerges. Namely, curses portend moments of descent. I will, very superficially, describe here each curse and its aftermath to offer a bird's eye view of this pattern, recognising that a more intensive grappling with each curse lies waiting in the chapters ahead.

The first curse of the Bible is God's laceration of the snake for enticing Eve to eat the forbidden fruit. "The Lord God said to the snake, because you did this, cursed are you from among all the animals, and from all the beasts of the field, you will go on your belly, and you will eat dust all the days of your life."³⁶ This formative curse curls the link between cursing and descent into the body of the snake. The accursed snake falls into the dust. In this sense, it represents the DNA of an emerging schematic.

The second curse is our farming diatribe. Oddly, this divine curse on growing grain is the outlier in this series of cursing. *Prima facie*, it doesn't lead to any direct descent. Adam and Eve are surely banished from the Garden of Eden to work the newly accursed ground—"cursed is the ground because of you."³⁷ And surely the evocative imagery of bending to tend the field is another embodied descent when contrasted to the moments of reaching upward to harvest the trees of Eden; yet there is no terminology of falling within the text, nor does the curse lead directly to a narrative at a lower elevation point. Indeed this is odd, because this curse is ground zero of the colloquial notion of the Fall. A concept so definitive that it requires capitalisation. Though the Fall of Adam is nominally defined as an entirely spiritual descent, it's a particular turn of phrase. Why not refer to this moment of transgression as the Sin, or the Rebellion of Adam? Might the taken-for-granted nomenclature of falling carry

a mimetic trace of a real migration from a highland Eden into lower terrain? After all, it is difficult to cultivate a field of grain on the mountain slopes. So when “the Lord God ushered him out of the Garden of Eden to work the ground that he was taken from,”³⁸ perhaps we readers insert an intuited declination within the sparse text.

The third curse of Genesis is the divine punishment meted out to Cain after he murders his brother. “Now, cursed are you from the ground that opened its mouth to take your brother’s blood from your hand. When you work the ground it will no longer give its strength to you, so you will be restless and itinerant on the earth.”³⁹ Cain’s curse is so freighted with the lexicon of Adam’s curse that it can be read as a malignant growth from one generation to the next. Whereas in Adam’s rebuke, the ground was cursed because of a human act, here the human is cursed because the ground acted in swallowing Abel’s blood. As though the soil’s infection has spread from the field into the farmer.⁴⁰ Whereas Adam was forced to abandon the fecund forest to cultivate nutrient poor ground of thorns and thistles, Cain’s ground is even further reduced to a state of total exhaustion where nothing will sprout.

Leaving behind his barren fields, Cain erects the Bible’s first city. Though the text does not state so explicitly, every urban complex in Mesopotamia began in the alluvial valley floor of the Tigris and Euphrates watershed.⁴¹ We will shortly explore just why the proto-state took root in this lowland terrain; here I will simply foreground the cursed trajectory of these intergenerational stories. Adam, forced from the headwaters of the Tigris and Euphrates to perspire over his fields of grain, raises another cursed farmer in Cain, who eschews the nutrient-depleted soil and heads down to the alluvium. At every node of decline we find curses. The snake that must slither and strike at the heel,⁴² the man who must leave the forest to dig in the earth and die in the dust,⁴³ and his son, who buries his brother deep in the ground,⁴⁴ before descending into the valley below, curses foreshadow each downward turn.

As Genesis slips into the valley floor, Adam’s curse is recalled again in Gen 5:28–29. “Lamech lived 182 years, and he had a son. He called his name Noah, meaning this one shall relieve us from our activities and the misery of our hands, from the ground cursed by God.” This reiteration intensifies our very first question. Why do farmers keep telling tales of a damned ground and miserable work, when this toil on this ground provides their very sustenance? Likewise, it repeats the pattern of cursing and falling on two fronts. First, following directly from Noah’s naming, the Bible segues immediately into a vignette that introduces the *Nefilim*, literally “the fallen,” the first usage of falling in the biblical text.⁴⁵ And second, as soon as the narrative of the *Nefilim* is concluded, the tale of the flood commences. How can a story descend even further than the valley floor at sea level? If that same tale sinks its drama below the waters.

The flood narrative is the topographic nadir of these pre-Abrahamic chapters of Genesis. From this point, the story elevates again with Noah’s ark, riding the waters to the mountains of Ararat. But this high altitude

sojourn is fleeting, and for the fifth time a curse will prefigure a fall into the valley floor. “He [Noah] said: Cursed is Canaan, he will become a slave of slaves to his brothers!”⁴⁶ Though Lamech referenced the divine curse on Adam, Noah is the first human to articulate an original curse. His bitter indictment of his own grandchild is the biblical aetiology of slavery. And though the text will first describe an intervening population dispersal post-deluge, the eventual concentration of peoples in the lowland of Shinar⁴⁷ and the subsequent building of a tower looming over the flat terrain are motifs of a society caged and watched. Here in the valley floor, Noah’s curse of captured labour becomes the blueprint for Babel’s infrastructure.

When we graph the first 11 chapters of Genesis against assumed elevation, the link between curses and topographic descent emerges like this (Figure 1.1):

It’s apparent that these five curses have gravity. They bend the story out of the mountains towards riverine valleys. Which begs the obvious corollary, what’s so accursed in the Mesopotamian lowlands?

If I have convincingly argued that curses in the early chapters of Genesis are always preludes to a story slipping downhill, then we can expand slightly

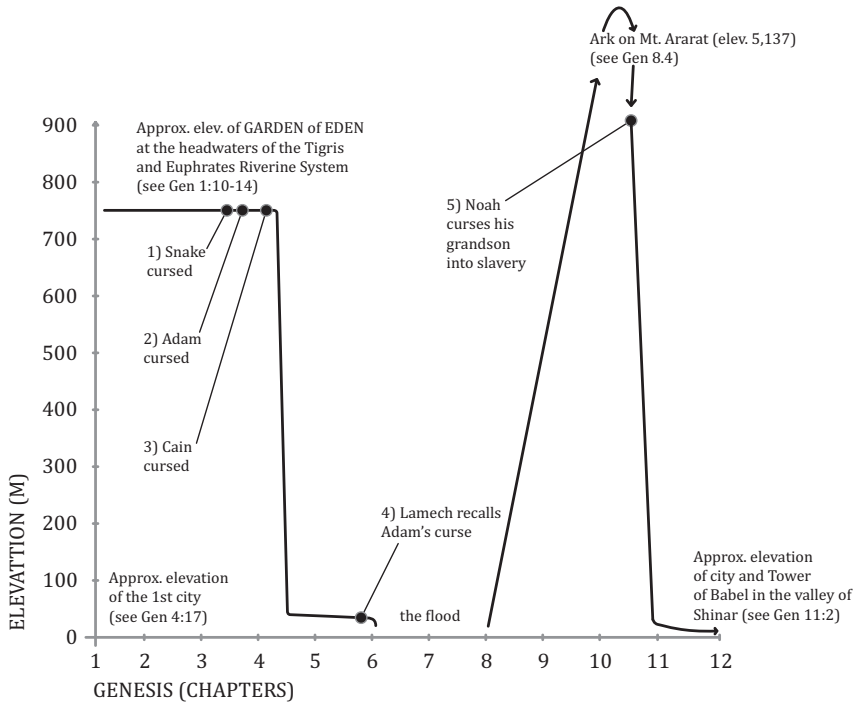


Figure 1.1 A graph of the narrative topography of Genesis 1–11. (Graphic design: Strange Diva).

further upon our original question. Why would the highland agriculturalists of Israel tell stories of accursed farming that catalyse narratives of descent? The key may be in the type of crop cursed by divine fiat, namely grain.

The curse on Adam is not a curse on farming *per se*, but a particular diatribe against the domesticated grasses of wheat and barley. “And you shall eat the grain of the field, in the sweat of your face you shall eat bread.”⁴⁸ The 20th-century biblical scholar Umberto Cassuto attends to the shift between the tree-fruit of Eden and this cereal crop: “Even when you succeed in eating of the fruit of your land, it will no longer be the desirable fruit of the garden of Eden but the *grain of the field*, that is the bread that you will produce from grasses like wheat and barley and other kinds of cereal.”⁴⁹ Leaving Eden demands a radical new diet. God initially stations humans in an edenic ecosystem of harvestable fruit, “And the Lord God made to grow out of the ground, every tree that is pleasant to the sight, and good for food.”⁵⁰ Grain agriculture, this farming curse, is an *ex post facto* capitulation to human intervention in the garden ecology. But why does God focus solely on wheat and barley production? Why not include the pulses of lentils, peas, broad-beans, and chickpeas that were domesticated at approximately the same time and the same place as these cereals and formed part of the agriculture assemblage found across the ancient Levant?⁵¹ What stands accursed, it would seem, is the transition from a highland forest diet to a mono-cropped field of grain. This is an important point not least because it alters the common perception that Adam’s curse was a punishment from leisure to work, rather than a shift in cultivation methods. The anthropologist David Graeber bases our current Western attitude on work, in part, to this literary moment: “In both the story of the Garden of Eden and in the myth of Prometheus, the fact that humans have to work is seen as their punishment for having defied a divine Creator.”⁵² But Graeber, or rather the theological worldview Graeber is paraphrasing, has overlooked Genesis 2:15! “The Lord God took the man and placed him in the Garden of Eden to work it and to watch it.” Eden was never intended as a primaeval leisure resort, but a cultivated highland forest that required upkeep and care.⁵³ And the punishing curse does not encompass work itself, but only a narrowly defined subset of work, namely the production of bread.

I don’t think we should easily dismiss this salient detail. Genesis is an aetiological touchstone, seeking to explain just how ideologies and institutions became normative via a narrative of a couple, their children, and a few offspring. Yet, this familial story maps out across the exact landscape where grain was first domesticated and where the pristine states were formed. That our most famous tale of beginnings happens to occur in the very region of the globe where the accoutrements of civilisation—agriculture, urbanism, writing, state administration—all got a first head-start might explain why these simple tales have staying power. But by specifying grain production as the cursed overture of history, Genesis may be doing more than simply recording events. Perhaps, these opening chapters of Genesis are the accounts of marginalised peoples cursing the state.

In order to suggest that a curse on grain farming is a curse on state-making, we will need to explore the intrinsic link between wheat and states. Like our prior engagement with the pattern of curses and descent, I will attempt to describe here the contours of this interplay between the early state and cereal crops, leaving the details for successive chapters.

These first ploughed fields of domesticated wheat and barley were not simply another subsistence technique on display, but the beginning of the state and the root of its power over people.⁵⁴ Non-state cultures, and any strata of society that felt themselves under the thumb of taxes, conscription, or outright bondage, would curse the cereal crops that were the source of the state's legitimacy and their own disempowerment. In his description of the first Neolithic agro-complexes in the Mesopotamian alluvium, James Scott writes, "Thus it was possible and not uncommon at the time to have sedentary farming populations on alluvial soils practising irrigation without any state. But *there was no such thing* as a state that did not rest on an alluvial, grain-farming population."⁵⁵ Grain produced *en masse* in nutrient rich alluvial (i.e. flood produced) soils on flat valley floors to aid transportability was the essential method to create the appropriable surplus required to form the state retinue of non-producers: clerks, artisans, soldier, priests, and aristocrats.⁵⁶ And "grains make states," as Scott intones, because cereal grains grow above ground (unlike tubers),⁵⁷ so that they are visible and thus appropriable by tax collectors⁵⁸; grains ripen nearly simultaneously (unlike legumes), so that they are easily assessable at a single visit by an administrator, and are light and transportable (unlike vegetables), to feed an elite population at the state core or soldiers at the frontiers.⁵⁹ Without the domestication of wheat and barley and the super-abundance it generated to feed a non-farming populace, the pristine states of Mesopotamia could not have risen.⁶⁰

Yet, as political theorist Francis Fukuyama darkly reminds us: "The transition from tribe to state involves huge losses in freedom and equality The weaknesses and gaps in all of the explanations [of pristine state formation] that are primarily economic in focus point to violence as an obvious source of state formation."⁶¹ The violence and oppression inflicted by these early states on their populace were assuredly accursed by all the non-elites forced within its bounds.⁶²

When I refer to the earliest states, I have in mind Scott's definition as "late-Neolithic multispecies resettlement camps"⁶³ replete with all the pejorative implications this terminology conjures. These first grain states were places of dense human, animal, and resource bundling,⁶⁴ or "social caging,"⁶⁵ in Michael Mann's evocative description. The state required a cage because as long as arable land was plentiful, populations preferred to abandon the drudgery of the field in favour of shifting cultivation and hunting and gathering.⁶⁶ In short, when given the option of remaining sedentary citizens of emergent states, monotonously farming a fixed-field of grain in order to reap an extracted surplus, or slipping away to subsist on a wide spectrum of

foodstuff mainly derived from hunting, foraging, forest- and bush-fallow cultivation across a diverse range of ecological zones, these peoples chose the later option.⁶⁷ Not least because these early sedentary societies created an oversimplified ecosystem, reliant on just a few fragile species of plants and animals and deeply vulnerable to the slightest variations in weather, disease, or pest. Densely packed cores of animals, people, and plants were likewise a breeding ground for uninvited commensals from rats and insects down to viruses.⁶⁸

Despite its many drawbacks, fixed-field agriculture had one overriding feature in its favour. It allowed a population to live in a densely packed area. The problem with hunting and foraging or shifting cultivation is that these food procurement strategies require far more land to meet the caloric needs of a family than a dense and concentrated wheat field.⁶⁹ The further the geographic spread of a population, the harder it becomes for the state to retain central authority, and the more costly and difficult it is to collect and redistribute surplus produce from its subjects. These early proto-statelets could not easily collect the harvest for taxes, nor mobilise the populace for warfare or building works, nor effectively assess their subjects and their productivity, without packing people into concentrated and permanent zones of control.⁷⁰ These state cores required flat ground with navigable waterways for transport and rich topsoil for intensive cropping; in Mesopotamia that meant the Tigris and Euphrates lowland alluvium. The encircling mountains had neither the soil nor terrain conducive to the prerequisites of state formation, and thus a fluid dichotomy emerged between lowland state-space and highland non-state space. A malleable boundary not marked on any map, but shrinking and expanding as populations were incorporated by or evaded state reach.⁷¹

A Chinese manual on governance from the first century, when arable land and natural resources were plentiful, and thus population dispersion was an ever-present threat to state power, cautions thusly,

If the multitudes scatter and cannot be retained, the city-state will become a mound of ruins ... Even if there are wooded hills nearby, and the plants and trees there are in flourishing condition ... Even if the pools and wetlands spread far and wide, and the fish and the turtles are plentiful, there must be regulations on the use of nets ... This is not the state taking plants and trees to be its private possessions, nor is it grudging the people their fish and turtles. [Rather] How can the people be allowed to stop producing grain?⁷²

This ancient report on statecraft is implicitly concerned with maximising the ideal polity of the earliest city-states: densely packed cultivators of permanent grain fields, who produced a considerable annual surplus.⁷³ Yet, these sophisticated legal restrictions against foraging and hunting wild turtles and fish beyond the bounds of state appropriation and control would only be required if large numbers of people did indeed flee the grain core on a regular

basis. Scott describes this process of human seepage out of state territory as a reaction against the fundamental attitude of statecraft in compelling its subjects to continuously produce a surplus for elite appropriation.⁷⁴ The more the state coerced its peasantry, the more alluring non-state space became. “Do what it might to discourage and punish flight—and the earliest legal codes are filled with such injunctions—the archaic state lacked the means to prevent a certain degree of leakage under normal circumstances. In hard times occasioned by, say, crop failure, unusually heavy taxes, or war, this leakage might quickly become a haemorrhage.”⁷⁵

Let’s return, again, to our first question, enfolding this short overview of the early grain states. Why do highland agrarians sacralise stories that both curse the monocropping of wheat and barley and set in motion an accursed migration from non-state mountains into a state held valley? These farmers aren’t cursing their own mountain terrace farming; they are cursing the foundational mechanisms of state-making. This curse against the proto-states, or rather this dialectic of multiple curses, remains alive and traceable within the first 11 chapters of Genesis. The project of this book is to follow that trace.

Before we take up our story of just how Genesis wrestles with the formative states of Mesopotamia, it’s worthwhile to interrogate an assumption at the root of the question that got us here, namely why would farmers curse farming. The tension between a living agricultural community and a covenantal scripture that envisions agriculture as accursed only grates if the curse is essentially stripped clean of metaphor. Only if Genesis is about real-life events, which is to say it is intrinsically a mimetic tale, only then does God inveigh with a hefty curse against the same sort of seeding, weeding, and ploughing activities that shaped the lives of the agrarian settlements of Israel.

Therefore, when the Bible describes the sweat of a cursed farmer engrossed with his harvest, or a woman screaming in labour, whilst these characters may also have symbolic resonance, this is mainly a drama concerned with actual harvesting and birthing. While the pain of labour can be interpreted, for example, as a metaphorical punishment for original sin, I assume throughout this book that the meta-symbolism of birth pangs should never occlude the pathos and particularity of Eve giving birth at the cusp of leaving a hilltop tree filled locale called Eden. If the garden and the curse and the banishment and the subsequent lives of Adam, Eve, and their progeny were entirely subsumed by the mythic or metaphorical tradition, our question would lose its resonance and its perplexity, for we could never hold up the text as a useful foil to the real world. Eden would simply represent some imaginary concept, some moral system, or fantastical exploration, and Adam and Eve’s banishment from such a realm would be of symbolic reckonings, rather than a political and geographical event.⁷⁶

The farmers of Israel would not be bothered *qua* farmers by a metaphorical curse that would transform the tangible attack on wheat, barley, and cereal “bread” farming into a reframed moral lesson. Perhaps, this moralisation of

Eden's collapse is the precise reason that the question of why the Bible curses farming to an audience of farmers is so little explored by modern biblical exegetes. For example, the ground cursed because of the "hubris that characterised Adam,"⁷⁷ as Norman Wirzba asserts, is not a problematisation of farming, just ego. Nor would these agrarians feel much perturbed, one imagines, if the curse to farm a weedy field of thorns and briars was recast as the "best index"⁷⁸ of Adam's disrupting of the "biotic unity"⁷⁹ between humanity, God, and nature, as Ellen Davis describes the aftermath of the curse. For Davis, who distinguishes between a wise and harmonious (and presumably briar-free) agrarian lifestyle practised by Israel versus the way of "thorns and briars"⁸⁰ that indicated a wicked farming ethic, such curses are simply allegorical engines to generate larger messages. One does not question why an allegory reads counter to lived reality, as long as it drives home its point, especially if that point is to praise the simple Israeli farming tradition over and against Mesopotamian land-use techniques that sought to control their natural systems.⁸¹

At issue with these metaphorical thematics is that they so often feel forced. The biblical narrative is complex and fluid; yet the symbolic worlds they engender are usually binary and rather static. When the text is read purely within its symbolic frame, one gets the impression that elements of the narrative are simply ignored or discarded to keep the allegory intact. Because his actions are so ambiguous and his life so peripatetic, Noah is potentially the best foil in Genesis on whether a writer has let their particular ideology subsume a story about a man and his family escaping a valley flood to dwell in the highlands. Davis describes the story of Noah as "a story that exemplifies the line of succession that can be secured by land care."⁸² And certainly the biblical label for Noah as a "man of the fertile soil"⁸³ could be read, as Davis does, as a "new heroic epithet,"⁸⁴ if not for the fact that what Noah plants in that fertile soil is a vineyard to manufacture enough wine to pass-out naked, wake-up in a rage, and curse his grandchild into perpetual slavery, before the whole story slinks back into the tragedy of Babel in the valley once more. Davis, however, never addresses this dark finale. A tormented, multitudinous Noah planting so unwisely undermines Davis' flat reading of Noah as the righteous agrarian.

Likewise, Davis does not ask why farmers would sacralise a text that cursed the farming act despite the fact that Davis is adamant that it was farmers, rather than nomads, who engender these Genesis accounts. "This look to the beginning is recognisably that of an Israelite, whose social world was dominated and indeed made possible by a mixed agricultural economy of rain-fed crops and small animal husbandry."⁸⁵ For Davis, the divine curse on agriculture isn't a real curse on real farming, only a moral warning to develop an "agrarian mindfulness" for the land.⁸⁶

Theodore Hiebert should be troubled by the God who curses the farmer. For he asserts throughout his work, *The Yahwist's Landscape*, that the stories of Genesis unfold within a community of "sedentary agriculture."⁸⁷ Yet, his

thesis is so thoroughly wedded to a “leitmotif” in which the Genesis narrative is primarily about the interrelationship between “human morality and the soil’s productivity,”⁸⁸ that the curse of farming is substantially reduced to a reward and punishment account with an agricultural stick. Adam disobeys God, and God pays him back with thorns and thistles.⁸⁹ This reliance on a simple biblical equation between good people and good farms leads Hiebert to some uncomfortable conclusions. Hiebert, like Davis, casts Noah as a “righteous hero”⁹⁰ of the post-diluvium, because of his successful vineyard planting. “Nothing in the story,” Hiebert contends, “compromises this fact.”⁹¹ And, like Davis, Hiebert conveniently fails to mention Genesis 9:25–27, when Noah wakes up from a bender to curse his grandson into bondage.⁹² The vines don’t seem to care how bad Noah acts; they produce abundant grapes just the same.

I think that Davis, Hiebert, and Wirzba⁹³ are reading the first 11 chapters as a timeless moral play, rather than a series of events in the same temporal plane as that of the listening audience, because nearly every modern engagement with the historicity or social milieu of the Bible begins with Abraham in chapter 12.⁹⁴ Everything prior is simply described as a legend, a myth, or a piece of propaganda created to undermine some existent and threatening myth of a neighbouring tribe.⁹⁵ The actions within these first 11 chapters of Genesis would then take place within a self-enclosed “absolute past” that keeps equal distance to a moving present.⁹⁶ What Mircea Eliade described as the “in illo tempore” of mythic time.⁹⁷ The modern biblical scholar Herbert Schneidau suggests with broad—and pejorative—strokes that the locked-in-time aspect of mythic storytelling turns them into engines of totalitarianism because they are “impelled by the principle of closure.”⁹⁸ This principle of absolute closure could explain why Davis, Hiebert, and Wirzba each choose to ignore the ugly epilogue to the Noah narrative. Each envisions these first chapters of Genesis as rigid moral folktales hung neatly on a stable plotline. Unfortunately, Noah’s drunken escapade turns our tale into an ambiguous mess, with an indeterminate dynamism mirroring the ambiguities of real life. Davis, Hiebert, and Wirzba close their case just before this destabilising outburst, with Noah’s idyllic viticulture redeeming Adam’s farming curse in a pleasantly sealed loop. But the Bible obstinately undermines such mythic style closure and subverts any fixed messaging, with a character as lurching as the storyline. Thus, when biblical historian Hershel Shanks intones that the “the first 11 chapters of the Bible, in the judgement of modern critical scholars, are mythic, not historic,”⁹⁹ we would do well to interrogate this claim. Shanks is restating a cleavage point between myth and sacred history staked out by biblical scholarship since the 19th century. Yet, the text itself offers no indication at the end of chapter 11 that a genre shift is afoot simply because Abraham plans to move to Canaan.¹⁰⁰ I would next like to present a philologist cum refugee who did not approach these early chapters differently than the rest of the book. For Erich Auerbach, the Bible is speaking about real events in the actual material world straight through.

Mimesis and Subversion

If Genesis is mimetic, then the divine curse on agriculture is a curse on the way real farmers make their actual living. So our question only grows from within a biblical hermeneutic cultivated by Auerbach's seminal work in biblical mimesis. Yet, Auerbach makes a more radical claim than merely noting that Genesis mirrors real-life events. Because the Bible engages so thoroughly with the everyday and the mundane, it is receptive to social life in all its facets, embracing and giving a voice to the marginalised, the lower classes, and the enslaved. For Auerbach, a mimetic text is intrinsically a subversive text as well.

Before describing how Auerbach lays the foundational hermeneutic for this book, a slight detour is required. My hypothesis is that chapters 1 through 11 of Genesis retells key developments of the late Neolithic from the perspective of non-elites.¹⁰¹ However, this should not imply that these biblical tales originated whole-cloth precisely at that point in early history, nor that mimesis is a perfect correlation with historical events. The Bible is an unreliable witness to the dawn of farming approximately 10,000 to 8,000 years ago. Or a case of "Chinese whispers" would be more apt, for the most archaic sections of biblical composition are potentially no more than 3,000 years old.¹⁰² Even if we follow David Damrosch's method in *The Narrative Covenant* that certain biblical stories have a cuneiform provenance that pushes their origin a thousand years earlier, these are still modern constructs held against the ancient Neolithic timeline.¹⁰³ A 4,000 year old tale is closer to our own present moment, in fact, than as a witness to the onset of agriculture.¹⁰⁴ So the Hebrew Bible¹⁰⁵ can never be an utterly authentic historical recording of events that predated writing itself. This does not necessarily imply that these stories are fabricated. They could well be regarded as oral traditions told and retold about much, much earlier events in the real world, with all the lacunae and transformations that such retelling would impart.¹⁰⁶

Though 4,000 years is a chasm of time, the Neolithic revolution—or rather the non-linear, halting and backsliding few millennia of patchwork Neolithic revolutions—transformed the Levant¹⁰⁷ people and landscape so fundamentally that it would be rather odd if an origin document did not address its seismic impact.¹⁰⁸ Perhaps, we should simply consider these early Genesis chapters as a literary response by a particular ethnic group to this radical new tact in human–nature relations. In fact theologians,¹⁰⁹ philosophers,¹¹⁰ political scientists,¹¹¹ archaeologists,¹¹² and anthropologists¹¹³ have all crossed disciplinary boundaries to remark on how Genesis provides a unique mirror into the key historical leverage points that follow on from crop domestication. It's not terribly complicated to envision the expulsion from Eden to take up farming, cityscaping, human bondage, and mega-construction as a microcosm of the cascading real-world events catalysed by the Neolithic revolution. Usually, these reflections are passing allusions. Scientists and historians, in particular, seem keen to anchor down their complex hypotheses with familiar

biblical tropes. James Scott, for example, harnesses the farming curse to his larger project: to overturn our presumed doctrinal theory on the positive outcomes of grain domestication, remarking that “while the planting of crops has seemed, in the standard narrative, a crucial step toward a utopian present, it cannot have looked that way to those who first experienced it: a fact some scholars see reflected in the biblical story of Adam and Eve’s expulsion from the Garden of Eden.”¹¹⁴ Archaeologist Ian Morris heads his sub-chapters on early Levantine farming sites with evocative phrases from Genesis such as “the Garden of Eden,” “Paradise Lost,” and “Going Forth and Multiplying,” and historian Yuval Noah Harari rides the narrative flow of Genesis in his chapter headings from “The Tree of Knowledge” to “A Day in the Life of Adam and Eve” to “The Flood,” even though he is grappling with prehistory and the agricultural revolution, rather than the Bible.¹¹⁵

But arguing that a story mirrors historical situations is not sufficient to label it a mimetic text, at least not according to Auerbach. In his influential 1946 critical analysis *Mimesis: The Representation of Reality in Western Literature*, Auerbach constructs a dialectical model for recognising the mimetic quality of any literary work. Tales that portray actual events drawn from lived reality are recognised by their “pendulation.” These stories are replete with descriptions of characters in transitional arcs through time—they live, as we do, with joys and sorrows, rises and falls, akin to the swing of a pendulum in space. The antithesis of a mimetic tale is a schema of flat figures interacting within a timeless present. For Auerbach, the foundational example in Western literature of a narrative undergoing pendulation is the Hebrew Bible. At the opposite end, Auerbach offers the Greek works of antiquity, specifically the Homeric epic. The profound insight of *Mimesis*’ first two chapters resides in recognising that this narrative fluidity, when compared against the static set-pieces offered up by the Bible’s Greek and Roman contemporaries, is a radical literary phenomenon. Auerbach here describes the difference between the Genesis characters and Odysseus:

There is hardly one of them who does not, like Adam, undergo the deepest humiliation—and hardly one of them who is not deemed worthy of God’s personal intervention and personal inspiration ... The poor beggar Odysseus is only masquerading, but Adam is really cast down, Jacob really a refugee, Joseph really in the pit and then a slave to be bought and sold ... The reader clearly feels how the extent of the pendulum’s swing is connected with the intensity of the personal history ... And very often, indeed generally, this element of development gives the Old Testament stories a historical character, even when the subject is purely legendary and traditional.¹¹⁶

Auerbach’s observation on embodied pendulation enlivens the dynamic pattern identified early in this chapter. Genesis really is a narrative on the move, both of the topographic variety from mountain to valley and within the churning interiority of Adam. But it’s not simply these transformations

that fascinate Auerbach, but the underlying question on why the Bible, nearly alone amongst the classical texts, represents life in motion when the Greek and Roman tales do not.

Auerbach begins with his evocative description of two biblical stories, Abraham's near infanticide of Isaac in the Hebrew Bible and Peter's repudiation of Jesus on Golgotha in the Gospels, as narratives "fraught with background."¹¹⁷ The sparse dialogue between Isaac and Abraham as they make their inevitable way up the mountain with the wood on Isaac's back, the fire and knife in Abraham's hands, but lacking any animal to kill, reads as mimesis, because that is always how we encounter real-life events, with insufficient details foregrounded and accessible, and much left in nebulous shadow.¹¹⁸

The Bible's fraught background signifies a representation of history as opposed to legend. Legend, argues Auerbach, "runs smoothly ... All cross-currents, all friction, all that is casual, secondary to the main event and themes, everything unresolved, truncated, and uncertain, which confuses the clear progression of the action and the simple orientation of the actors, has disappeared."¹¹⁹ Historical accounts are, conversely, not smooth reading. Their telling is a jagged and truncated affair, full of half-realised moments and unresolved happenstances. "The historical event which we witness, or learn from the testimony of those who witnessed it, runs more variously, contradictorily and confusedly."¹²⁰ Think for a moment of Abraham coming down from the mountain directly after the near-sacrifice of his son in Genesis 22:9. "Abraham returned to his attendants and they departed together for Be'er Sheva; and Abraham stayed in Be'er Sheva." And then ask yourself, whatever has happened to Isaac, who explicitly did not return with his father to Be'er Sheva? A confused and contradictory conclusion, and therefore a mimetic one. Auerbach is explicit, however, that the biblical narrative is not verifiably true because its genre is historical. Only that its author¹²¹ was attempting to represent a sense of reality, rather than propaganda.¹²²

This vacillating, churning, and messy text makes the Bible a dangerous text as well. Auerbach sets scriptural dynamism against the "fixed aprioristic model concepts"¹²³ of Greek and Latin writings of antiquity. Homer, Petronius, and Tacitus, he argues, ignored societal "growth processes in the depths"¹²⁴ because of an "aristocratic reluctance"¹²⁵ to consider historical change, either intellectually or materially. For these elite authors and their audience, the status quo, and their own power which flowed from its continued existence, was tantamount.¹²⁶

The works of Greece and Rome were written by and for the social elite, whereas the Bible is concerned with the life of the lower classes. Therefore, scripture is free of the "aristocratic reluctance" to authentically portray social change, and can openly record those historical forces cutting against the grain of the status quo. This is most obvious in the story of exodus, but every intrigue in the field, every massive tower brought tumbling down, even the very audacity of informing the reader that farming is an aetiological curse to

a readership of farmers, shakes the foundation of the accepted societal order, turning the *Es Muss Sein*¹²⁷ into an open question. Auerbach repeatedly emphasises this point:

Thus we become conscious of the fact that in the Homeric poems life is enacted only among the ruling class ... As a social picture, this world is completely stable; wars take place only between different groups of the ruling class; nothing ever pushes up from below. In the early stories of the Old Testament the patriarchal condition is dominant too, but since the people involved are individual nomadic or half nomadic tribal leaders, the social picture gives a much less stable impression; class distinctions are not felt.¹²⁸

The Bible is the first text in the Western oeuvre that speaks of the “commonly realistic”¹²⁹ and tells the story of the underclass as a pushing up from below. And because the poor do not care for class stability, “the deep subsurface layers, which were static for the observers of classical antiquity, began to move.”¹³⁰ The reason that the Bible is so *fraught* with background is because the subsurface of the social fabric is always in motion. This is a subversive literary gesture indeed.

Auerbach is not alone in recognising the Bible’s narrative capacity to shake the status quo of societal order. Robert Alter alerts us to the Bible’s incessant “probing”¹³¹ of the established ideological assumptions of the age. Alter, ever steeped in the background of a literary theorist, attributes this subversive writing style to an intrinsic quality of the narrative art. The biblical writer is foremost a literary artist, and literature, argues Alter, runs against the grain of received values and traditional forms:

The literary imagination qualifies or challenges prevalent ideological assumptions not necessarily—in fact, not usually—because it has an ideological program of its own but, on the contrary, because its inventive, associative, and even formal engagement with its own verbal, narrative, and referential materials leads it to peer over the side, or at the underside, of things as they are ordinarily seen.¹³²

Like Auerbach, Alter foregrounds the uniquely radical design of biblical storytelling through a comparative gesture. Specifically, Alter invites a comparison of creation accounts between the Babylonian epic, *Enuma Elish*, and chapter 2 of Genesis. Introducing key verses from the *Enuma Elish*, such as this one: “Verily, savage-man I will create. He shall be charged with the service of the gods,” Alter argues that the Akkadian verses portray humanity as essentially “locked into a set of fixed hierarchies.” How “radically different” is the creation of Adam and Eve, endowed with “morally problematic interiority” and authored as complex active agents rather than passive functionaries.¹³³

For the legal scholar Robert Cover, the “thick”¹³⁴ narrative stories in Genesis are so potently subversive that they can even undermine the internal codes of the Bible itself. In particular, he points to the discontinuity between the law of succession in Deuteronomy 21:15–17, which favours the first-born with a double-portion of inheritance, upended by the multiple stories of the Bible where the younger son is favoured by God, father, or nation. From the moment where God favours the animal sacrifice of Abel, the younger son, over the produce of Cain, the elder, to the political succession of Moses, David, and Solomon over their elder siblings, the biblical narrative appears intent on prying apart its own internal coding of primogeniture. Cover, in fact, argues that it’s precisely this intrinsic narrative subversion which marked the text as holy. “Divine destiny is likely to manifest itself precisely in overturning this specific rule.”¹³⁵

It may be more than artistic imagination that makes the Bible such a subversive tale. Herbert Schneidau argues at length in his work *Sacred Discontent* that the Hebrew Bible is essentially an anti-epic arrayed against earlier and contemporary Near Eastern texts. Still other biblical scholars attribute this concern with the lower classes to a rather unique interstice in ancient history. According to this view, the biblical authors developed an “ideology of egalitarianism”¹³⁶ during the pre-monarchic period of Israel. The great empires of the Late Bronze Age had just collapsed and no new major superpowers had formed to fill the vacuum. Thus, incipient Israel—which in biblical terms is described as the time of Judges—was nominally free to create independent social and political structures at the town and regional level. As inter-regional trade and hegemonic political control diminished, these small communities became self-supporting and free of tax demands from distant kings engaged in war; this in turn caused social hierarchies to diminish and local food distribution to spread more equitably.¹³⁷ This brief span of relative egalitarianism, coupled with a deep unease with the strong hierarchical structure of the despised Canaanite society, imbued the writings of Israel with a nearly classless dimension.¹³⁸

Still other scholars of antiquity claim the rate of literacy was higher in Israel than in the surrounding societies of Mesopotamia and Egypt.¹³⁹ These authors generally hold that literacy ran higher in Palestine because the Israelites used the alphabet, which has tens of characters to learn, rather than the thousands of ideograms required to master cuneiform or hieroglyphics.¹⁴⁰ If these scholars are correct that literacy permeated across the socio-economic strata of ancient Israel—though there is research that argues persuasively against this view¹⁴¹—then one could expect that scripture would reflect and cater to its entire readership, flattening class divides and elitist conventions. Perhaps, then, it was the alphabet that engendered a subversive biblical text.

What emerges from these variant models for the textual and social uniqueness of early Israel is a narrative frame where marginal and disenfranchised voices take centre stage. Perhaps, all of the separate reasons I’ve offered on just why the Bible should strike so many scholars as this uniquely

subversive work of literature can all be subsumed under the rubric of feminist standpoint theory.¹⁴² Which is to say that in any society built on systematic inequality, those on the bottom of the heap expend a great deal of imaginative energy interpreting the smallest whims and wishes of those on top, whilst those on top remain largely oblivious to the perspectives of those below them and the huge input of imaginative labour that is required to maintain their elaborate zones of fantasy. According to David Graeber, this phenomenon generates the social creation of lopsided structures of the imagination where the least powerful have the most at stake in anticipating the wants of others: “Whether one is dealing with masters and servants, men and women, employers and employees, rich and poor, structural inequality—what I’ve been calling structural violence—invariably creates highly lopsided structures of the imagination. Since I think [Adam] Smith was right to observe that imagination tends to bring with it sympathy, the result is that victims of structural violence tend to care about its beneficiaries far more than those beneficiaries care about them.”¹⁴³ It would make good sense then that well-to-do Sumerians, Akkadians, Babylonians, Greeks, and Romans would tend to write epic works mainly revolving around their politics, their palaces, and their erotic and martial fantasies without considering what all those servants, slaves, women, retainers, and so on might be doing or thinking whilst they scrubbed those palaces. Of course, if the Bible, as Auerbach and these other thinkers argue, really is the output—at least in part—of those scrubbers and cooks, it would look very much as we have it. Namely, a fascinating portrayal of bottom of the heap stories of cursed farmers, painful childbirth, and slave revolts, mingled with detailed insider information on when and where the Pharaoh of Egypt would take his bathroom break.¹⁴⁴

I believe that the first 11 chapters of Genesis intrinsically adhere to this framing. This book is an invitation to take another look at these stories as though they were molotov cocktails thrown against the walls of formative Mesopotamian civilisation. What if we whispered to each other that farming does not produce a blessed bread for the gods, but a cursed crop; or that the first urban planner wasn’t a Gilgamesh-esque hero, just a marked murderer on the run; or that valley civilisation was better flooded into oblivion, or that it’s highest tower would be better collapsed into ruin as an inverted monument to the limits of hubris and a celebration of linguistic diversity. What if these formative etiologies were both mimetic in their depiction of reality, and subversive in their critique of statecraft?

The story of the destruction of Eden offers a salient example of the Bible’s willingness to report on a destabilised eco-phenomenon over and against a static and mythic realm. For the biblical composer, the distant past contained a verdant, fecund Mediterranean forest at the headwaters of the Tigris and Euphrates. Yet, even in the early monarchical period, these headwaters were denuded into a scrubland of exposed limestone and a few hearty bushes and grasses. Because Eden existed *in situ* in real hills, near the source of actual rivers, its material disappearance begs the question: How did this happen?

How did the lush garden described in the text disappear so completely? This is not the type of question that the elite would normally consider. It invites further investigations from outside the forcefield of societal stability. These are questions that imply revolution. They ponder how things came to be this way, which implies that such things need not, in essence and perpetuity, remain this way.¹⁴⁵

A fleeting survey of building works described in cuneiform epics offers a glimpse at the biblical antithesis. In the cuneiform accounts, walls are nearly eternal, pre-ordained fixtures upon the landscape. Consider the first chapter of the Epic of Gilgamesh. “[Gilgamesh] built the wall of Uruk-Haven, the walls of the sacred Eanna Temple, the holy sanctuary. Look at its walls which gleam like copper, inspect its inner wall, the likes of which no one can equal! Take hold of the threshold stone—it dates from ancient times!”¹⁴⁶ Or, again, this *primaeva* Sumerian account, “When the royal sceptre was coming down from heaven ... [the king] laid the bricks of those cities in pure spots.”¹⁴⁷ Or, this snippet from the Sumerian collection of temple hymns, which begins with a hymn to Enki’s temple in the city of Eridu, called E-unir, and hallows both the monumental structure and the wheat-bread offerings which took place there. “E-unir which has grown high, (uniting) heaven and earth ... House, holy mound, where pure food is eaten ... your great [...] wall is kept in good repair ... Where the oven brings bread to eat on your Ziggurat, the lofty shrine, stretching towards the sky, where the oven rivals the Holy of Holies.”¹⁴⁸ These texts represent the normative attitude for a royal scribal description of such muscular urban structures. In comparison, the tower and city of the biblical Babel is both short-lived and denigrated. “So the Lord scattered them abroad from there, over the face of the whole earth, and they ceased to build the city. Therefore its name was called Babel [confusion], because there the Lord confused the language of the whole earth.”¹⁴⁹ Whilst the Epic of Gilgamesh was a state medium, one senses the ironic tones of the storyteller-as-outsider in the biblical account. The city and its tower are neither ancient, nor imposing, but fragile little objects are brushed aside by God, just as the Tower of Babel narrative itself is brushed aside in only 11 terse verses in the biblical text.

The Hebrew Bible is concerned with the life of the common, and therefore it can ask radical questions on how walls and buildings rose and Eden disappeared. And its answer, indeed the answer that creates the central motif of the first 11 chapters of Genesis, is that state-imposed farming is a cursed act. Grain farming reaps forbidden fruit that afflicts the ground (“... cursed is the ground because of you”¹⁵⁰), poisons relationships between siblings (“Now, cursed are you from the ground that opened its mouth to swallow your brother’s blood.”¹⁵¹), problematises transmissions between fathers and sons (“He called his name Noah saying this one will relieve us from our work and our hurting hands upon the God-cursed ground.”), and enslaves grandchildren (“... he will become a slave of slaves to his brothers”¹⁵²). So whilst we can never say for certain that Genesis speaks authentically about the Neolithic revolution, amongst contemporary classical works, it stands as

perhaps the singular text that would dare to look past the status quo to gaze upon farming's discontents.¹⁵³ Only the poorest farmers would decry farming as accursed and only a very radical text would record such dispositions. These first chapters of Genesis are the aetiological background for that curse, and the set stage for all the drama of Israel that will unfold over and against that curse.

Methodological Cross-Currents

How should we interrogate the very ancient past? We are accustomed to examining archaeological heaps and the ruins of concentrated settlement cores and poring over the ancient texts of cuneiform clay tablets and hieroglyphics etched in stone. Yet, these are artefacts of the state. Clay, stone, and brick are the heavy accoutrements of sedentary living, whilst the reeds, bamboo, animal components, and wood spread thinly across the landscape by the "barbarians"¹⁵⁴ sliding just beyond state control are lost to us. And yet these early states were "a mere smudge on the map of the ancient world," Scott warns us. "Not much more than a rounding error in a total global population estimated at roughly twenty-five million in the year 2,000 BCE."¹⁵⁵ So, I have brought the foundational chapters of the Bible to cast attention on the stories of peripheral state folk at the formative moment when grain agriculture and, concomitantly, state structures entered into the sphere of human existence. This method is based on two basic assumptions that inform the chapters ahead. First, that these early chapters are mimetic descriptions of ancient Mesopotamian engagements with pre- and formative statehood. And second, that the Bible is wary of these early states. This second assumption is essentially anchored on the primary question of this book. Simply stated, state-centric actors don't curse their foundational food source, nor will the elites at the state core portray the adaptation of surplus bread cultivation as a fall from paradise, nor will temple and palace representatives belittle a monumental urban tower as inconsequential and fragile. Genesis, I argue, offers a unique glimpse of early Mesopotamian society through the lens of the have-nots, the disenfranchised, and the peripheral peoples. Unique, not because ancient Near Eastern artefacts are scarce, but rather because these artefacts are almost always biased towards statehood. I argue that the Bible, when offset against these cuneiform tablets and ancient bricks and stone, subverts that bias again and again with an outsider's perspective. And yet, this methodology of using biblical texts to narrate a decentralised ancient past is fraught with three decisive risks.

First, the social theorists, political historians, anthropologists, archaeologists, and ecologists I engage with most frequently are often on a mission to de-privilege the primacy of the state's role in the historical record. Scholars such as Scott, Jered Diamond, Clive Ponting, Ester Boserup, Pierre Clastres, Owen Lattimore, and the majority of others you will encounter in the chapters ahead tend to downplay the good deeds of past state regimes, whilst

simultaneously extolling the virtues of life beyond its clutches. As this often dovetails with my own interpretation of these early biblical chapters, I give valence to those thinkers that support my basic claim that grain farming state subjects were a cursed lot, whilst marginalising, or outright ignoring, theories about pristine state regimes, in which its subjects may have freely chosen the sedentary farming life. One does not need an angel with a flaming sword to police the way back into the mountains, unless the way forward into alluvial statehood is particularly unpleasant.¹⁵⁶ For Adam, Eve, and their progeny, the Mesopotamian alluvial state is where the cursed go. Thus, I have chosen interlocutors with interpretative methods that skew towards critical examinations of early statehood.¹⁵⁷

Consider, for example, the clay bevelled rim bowls found in huge numbers in all excavated Uruk culture sites (Figure 1.2).

Following the research of Scott, Harvey Weiss, Susan Pollack, and Hans Nissen, I conclude that these uniformly measured bowls represented the meagre grain rations of the unfree in an early state regimen already calcified into bureaucratic and monolithic hierarchy. Thus, these bowls stand as testament to certain ideas of empirical homogeneity. And this flat and rigid rationing with bevelled rim bowls textures the hermeneutic frame I build around the Tower of Babel story with its stark singularity of language and



Figure 1.2 Bevelled Rim Bowls (BRB) from the late Uruk period (3300–3100 BCE) excavated at Nippur (southeastern Iraq). Courtesy of the Metropolitan Museum of Art, New York.

cityscape. Which might all be well, if not for Assyriologist Petr Charvát's countering interpretation. Charvát argues that Uruk society was essentially egalitarian. That, for the most part, everyone in the Uruk state had equal access to resources, that its writing system and other technological advances were noble efforts to ensure greater efficiency and transparency in the fair distribution of goods, and that those bevelled rim bowls all had equal shapes because they were moulds for bread or sweetmeat eaten throughout the empire as symbols of equality.¹⁵⁸ Sadly, I have reduced Charvát's findings to a single footnote, because his claims are essentially disruptive to my argument. And not only Charvát! I am particularly enamoured of the idea that the name Cain is bound to the root for ownership, which fits well with his position as a capital hungry farmer and urbanist. Yet, there is a good body of scholarship that links the etymology of his name to the biblical Kenite tribe,¹⁵⁹ and still another line of enquiry that associates Cain with the terminology of tool sharpening.¹⁶⁰ Again, I have buried these variant readings in footnotes, and biased my interpretation of Cain's name towards a decidedly anti-capitalist hermeneutic linking ownership to greed, and ultimately to murder. To be sure, the glare of state self-representation,¹⁶¹ to use Scott's phrase, is often blinding. It requires a concerted and palpable shift in focus to turn to the scant historical and textual ripples at the margins. Yet, the tendency to recalibrate too far against state hegemony may lock our thinking into caricatures and grotesques: evil states of valley coercion in tension with an edenic hunter-gathering mountain primitivism.¹⁶² It is worthwhile to recall throughout this book that whilst the Bible curses agriculture, neither does it advocate a full return to Eden; the biblical tale is gnarled with such complexities.

This is not to say that these early narratives urge a passive acceptance of state-making. I will argue throughout this work that these first 11 chapters construct an oscillating movement into and out of state space that runs from narrative to narrative, and mountain to valley. In order to catch that rhythmic trope, we must take another methodological risk, namely, crossing disciplines to posit that long-run historical processes are occurring in concentrated vignette style within terse bits of biblical storytelling. This method freights the storyline with the weight of social theory, which in turn creates two major stress points. First, we cannot focus overlong on any particular story if we hope to catch these processes at work. Thus, the pacing of this book limits each narrative to a single chapter and touches upon some elements with only a very superficial brush. I never dwell on the fascinating exchange between the snake and Eve in her allurements into grain cultivation, nor the full parameters of God's dialogue with Noah after the flood. Much detail and commentary has been left on the cutting room floor in order to create the impression of continuity within these stories.

So too, key eco-biblical scholars who would otherwise stand as the organic dialogical partners with this book are side-lined because they never fully consider the capacity of peoples to travel from one livelihood to another, and from state to non-state zones in quick succession. Theologians such as Theodor

Hiebert, Ellen Davis, and Walter Brueggemann, who are committed to providing the fullest *in situ* ecological context to the Genesis narrative, are here referenced only as peripheral foils. Each of these scholars settles on a singular hermeneutic for Genesis 1–11, and then each bends the storyline around their particular fixed frame. These theories feel so hermetically sealed because they struggle to accommodate the narrative fluctuations from non-state to state space, or characters eluding easy description. Brueggemann simply bisects the aetiological drama into landed people “living towards expulsion” from Genesis 1–11, followed hard by dislocated folk anticipating land, from chapters 12–50.¹⁶³ Brueggemann’s theory cannot fully attend to the complex panoplies at play in these early chapters. Cain’s pendulation, for example, from a hill farmer to an itinerant to an urbanite has no place in Brueggemann’s scheme. Likewise, Davis and Hiebert both envision the Bible as a “drama of the soil,” to evoke Hiebert’s lovely phrase.¹⁶⁴ But their drama follows rigid taxonomies. For Hiebert, well-cultivated crops are the Bible’s cue to moral uprightness.¹⁶⁵ For Davis, agrarian land farmers are almost always ethical and wise.¹⁶⁶ As we described earlier, Noah’s viticulture achievements are viewed similarly by both scholars as “heroic”¹⁶⁷ and “restorative.”¹⁶⁸ Neither Davis nor Hiebert dwell on the bitter conclusion that it was this very crop that led Noah to drink and then curse-out his grandson with eternal slavery. Though I will note a few of my disagreements with these scholars, this book is not intended as a critique of their transformative theories. It is rather a means to advance the biblical conversation towards elasticity, pendulation, and mimetic subversion. Chapters 1–11 are narratives on the move, and I remain wary of non-eclectic frames of reference as we follow the drama from the mountains to the alluvial state, back to the mountains, and then back to the valley below.

Likewise, I conflate various historical periods and geographic nodes in order to make certain claims about agriculture and the state in relation to these seminal biblical events. Whilst the bulk of the presented evidence stems from Mesopotamia, and runs from the early domestication of wheat in approximately 8000 BCE to the collapse of the Ur III dynasty in 2004 BCE, where this evidence is thin or disputed, I will briefly turn to other early state projects at other times and corners of the globe. Scott describes his similar forays between the grain states of Mesopotamia and the padi empires of Southeast Asia as a type of triangulation in search of patterns.¹⁶⁹ Comparisons between slavery in classical Greece, Imperial Rome, or under the Qin Dynasty in China may not tell us precisely what bondage looked like in the Mesopotamian kingdom of Ur III, but it may provide an educated guess on why early states such as Ur III were so keen to entrap and hold human beings.

The problem with triangulation is that it subsumes the nuance and granularity of the particular for the sake of the larger picture. However, these first 11 chapters of Genesis are similarly problematised. These are stories of temporally and spatially confined individuals, a wheat grower on a hill, an urban planner in the valley below, a refugee from a flood; yet each story is

also located here, within Genesis, as a harbinger of how the world was remade with grains and states. Thus, each story is here because it transcends its particularities in order to explain just how the world we inhabit came into existence.

Thus, we touch upon the third fundamental methodological risk within this book. In order to read the mimetic arc of events from paradise straight through to Babel, I find myself eschewing many of the critical tools nominally applied to modern biblical scholarship. Interpretive biblical methods that disentangle the textual weave are less coherent than the act of stretching the entire narrative tapestry tight around its social milieu and natural ecology. Thus, I prioritise exegeses of these stories that seek both internal consistency and implied functionality within the wider narrative sweep of Genesis. Many respected biblical thinkers claim that the odd snippets of incongruous storylines within these chapters, such as Lamech's strange poetic intrusion or the children of god episode that cuts through Noah's early life, are only the borrowed shards of fuller and now forgotten myths. Likewise, the entirely missing dialogue of Cain and Abel may simply be a scribal error, and the first and second chapters of Genesis may be the work of two separate authors and an inattentive redactor. Yet, I am inclined otherwise in each of these moments. Are these oddities perhaps less odd when read together as integrated elements towards a more complex, realistic, and socially engaged storyline leading from Eden to Abraham? Daniel Boyarin calls this engagement with the fullness of the text a powerful nonreductive way of reading the Bible. A way of implying that God is the author of the Bible, not as a theological or dogmatic claim, but simply a semiotic one. Writes Boyarin: "If God is the implied author of the Bible, then the gaps, repetitions, contradictions, and heterogeneity of the biblical text must be *read*, as a central part of the system of meaning production of that text. In midrash the rabbis respond to this invitation and challenge."¹⁷⁰ Boyarin's final point on midrashic intent as the multiple and indeterminate gap-filling that allows "the Bible to read itself"¹⁷¹ is an important argument for this book. Throughout this work, I will turn to various midrashic interpretations to expand the fraught background of the biblical text in order to construct a deeper coherency.¹⁷² I too follow in the wake of this midrashic project, reading into the heterogeneity of a divine text.

Thumbnail Itinerary of the Chapters Ahead

The following chapters of this book are primarily concerned with building plausible links between the first 11 chapters of Genesis and the societal adaptations caused by the domestication of plants and animals, followed by increased sedentism and social hierarchy, and the eventual formation of, and reaction to, the pristine states of alluvial Mesopotamia. As such, each chapter will focus on a key biblical narrative that can generally be assigned to a particular node in both space and time. In this overview, I will attempt to

briefly elucidate these chapters according to four dovetailed components. First, the biblical verses under discussion. Second, the very approximate time span when the narrative events may have taken place. Third, the major societal innovations that were then occurring in the Levant at that time. And fourth, the geographical area where the biblical tale may have unfolded, including a general description of terrain and topography.

At the onset, I would like to draw the reader's attention to the topography indicator, as it is the unlikely lynchpin around which some of the other demarcative components are settled. This is because the Bible is never overly interested in the precise location of narrative happenings, but rather the positional relationships between peoples, notably in these early chapters, the relationships between state and non-state actors. And such dialectics often unfold in the transition between higher hilly elevations and lower alluvial valleys. Throughout these first 11 chapters, the characters are consistently migrating in and out of state space.¹⁷³ Adam is banished from a mountainous Eden, Cain wanders into the land of Nod to build a city, Noah is flung up upon the mountains of Ararat, and his offspring return to build a tower in the valley of Shinar. Topography plays a key role here, because the first pristine grain states of Mesopotamia are all valley enterprises.¹⁷⁴ So, the city of Cain and the Tower of Babel are clear markers of events on the valley floor, whereas Eden and Ararat are the stories that play out in the hills above. And such hilly zones are the enclaves for self-governing, non-state peoples. In the rugged highlands, the reach of state power is diminished, if not altogether absent, due to the "friction of terrain" model championed by James Scott.

Though Scott's research is focused primarily on Southeast Asian pre-modern state-making, and the wet-rice cultivation which undergirds its formation, he freely acknowledges that all early grain states share similar geographic constraints, as a direct result of the "friction of terrain."¹⁷⁵ In Scott's reading, the ideal "state-space" is consistently equated with the ideal "space of appropriation" of foodstuffs, *corvée* labour, soldiers, tribute, and tradable goods. In order to appropriate these goods most effectively, and bring them into the core of control efficiently, the state sought to concentrate its fields and manpower as close to the centre as possible. "Such concentration," writes Scott, "is all the more imperative in pre-modern settings where the economics of oxcart or horse-cart travel set sharp limits to the distance over which it makes sense to ship grain. A team of oxen, for example, will have eaten the equivalent of the cartload of grain they are pulling before they have travelled 250 kilometres over flat terrain ... The non-grain producing elites, artisans, and specialists at the state core must then be fed by cultivators who are relatively near."¹⁷⁶ Hilly terrain, marshes, deserts, forests, and swamps represent considerable barriers to the state's appropriative reach. Not only do these zones represent areas of increasing friction of transport versus a flat alluvial plane, they are also more difficult to cultivate intensively, which means the carrying capacity per area of land is considerably less than a well-irrigated valley terrain. This, Scott argues, creates a clear demarcation between a flat and open

geography more amenable to state control and appropriation, and a mountainous or swampy geography intrinsically resistant to state control.¹⁷⁷

Scott creates a wonderful thought model to illustrate his point. Imagine holding a map on which altitudes were represented by the physical relief of the map itself. Now, envision the grain-growing alluvial cores as a reservoir of red paint. If one would tilt this map in each direction, the red paint would begin to flow out. It will spill first along level ground and along the lowland water courses, but as you increase the angle of tilt, the red paint would flow slowly or abruptly, depending on the steepness of the terrain to higher and higher elevations. "The angle at which you had to tilt the map," Scott concludes, "to reach particular areas would represent, very roughly, the degree of difficulty the state would face in trying to extend its control that far."¹⁷⁸ Some areas of higher elevation may reveal just a tinge of red, whereas even higher terrain would remain entirely white, indicating that the inhabitants of these zones were rarely, if ever, under state rule.

By extrapolating the approximate elevation of each biblical narrative, we can build a working model that describes the level of state subjugation acting upon the characters at any given point. I'd like to revisit our earlier diagram from Figure 1.1, this time however, let's over-layer a grossly simplified schematic of Scott's red-paint friction of distance model as a blurry boundary where the Mesopotamian state-influence on the valley floor wanes into non-state hill country as elevation increases (Figure 1.3).

Framed this way, the initial Genesis storyline looks like a series of failed attempts to elude the gravity of Mesopotamian state hegemony. Edenic hilltop refugees wandering into the urban floodplain, Ararat mountain flood survivors now baking bricks in the valley of Shinar, these chapters tell the drama of a family straining against the bit of the cursed state.

Chapter 2 entitled "Paradise Lost" references Genesis 2:4 until the end of Chapter 3. It is concerned with the formative, but fleeting, spell of human existence in the Garden of Eden, the transgressive act of eating the forbidden fruit, the curses against Eve and Adam, and the banishment from the garden. As these events are riven with the divine curse to adopt agriculture, let us assume that it reflects humanity's first, fraught transition towards the domestication of plants and livestock from approximately 9000 BCE to sometime after 6000 BCE.¹⁷⁹ Again, using the key textual fulcrum of the farming curse, let us further postulate that this narrative can be located at a proposed site of first crop domestication somewhere in the Hilly Flanks, the arc of rolling hills curving around the Tigris, Euphrates, and Jordan valleys in Southwest Asia, and the proposed cradle of the Neolithic revolution.¹⁸⁰ To further illustrate the transition between the "before" and "after" of agriculture, including its deleterious effects upon the flora and fauna of these hills, I will propose the particular environs of Göbekli Tepe, 760 m above sea level. This location and time puts us firmly at both the inception of habitual farming, and close to the headwaters of the Tigris and Euphrates riverine system, which is to say at the jumping-off point for Genesis.

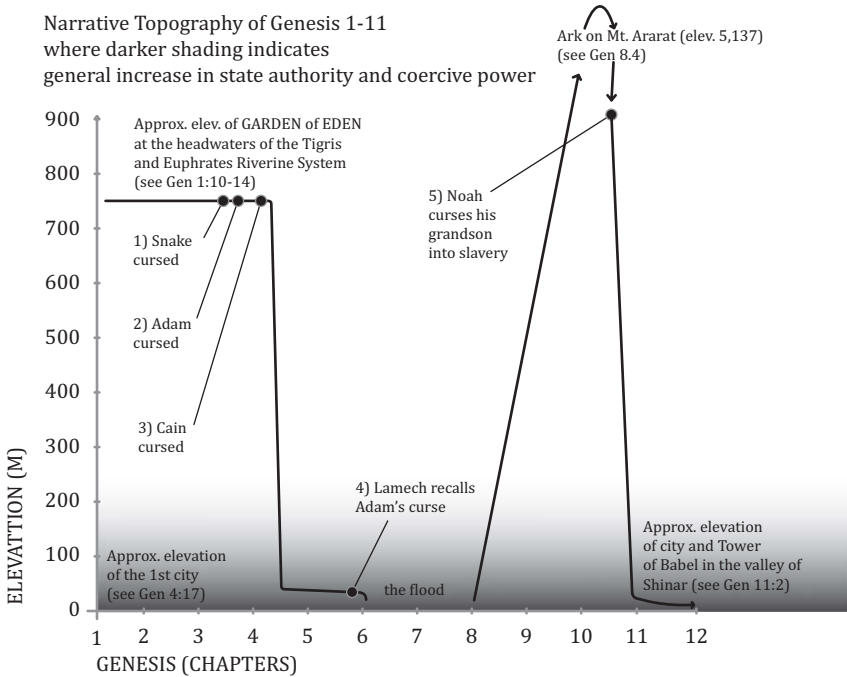


Figure 1.3 A graph of the narrative topography of Genesis 1–11 with state control decreasing as elevation increases. (Graphic design: Strange Diva).

Chapter 3, “Acquiring Man,” examines the fraternal conflict from Genesis 4:1 until 4:17. Though our clues here are scarce, the given domestication of both crops and animals would place this narrative in one of the early agrarian villages of the hilly flanks, circa 5000 BCE. For the sake of focus on a thoroughly excavated longitudinal site, we will map this tale on Çayönü Tepesi, a Neolithic settlement located at 830 m above sea level in the Taurus foothills.

For our Chapter 4, “A Tale of Two Wives and a City,” we encounter the Bible’s first city in Genesis 4:17–23.¹⁸¹ If we assume that the biblical urban initiation corresponds to any of the earliest city-states that dotted the Mesopotamian flood plain, then we are temporally situated in approximately 3200 BCE. From the current archaeological and historical evidence, this demarcates when the city of Uruk definitively erected the basic foundations of state apparatus, arguably the first such occurrence, with walls, tax collection, and officials.¹⁸² Not only is Uruk potentially the first city-state, but it is also the most well documented amongst the early centres of Mesopotamia.¹⁸³ Thus, both in this and later chapters, we will often draw on the findings unearthed during the excavations at Uruk, as well as

satellite sites in Syria, southern Turkey, and western Turkey that all display Uruk period artefacts and buildings.¹⁸⁴ Uruk was situated on an ancient channel of the Euphrates, at 16 m above current sea level. With the advent of cityscaping, Genesis has migrated into the flat alluvium, the ideal location for early state forming.¹⁸⁵

Chapter 5, “Noah the Ploughman,” continues to dwell in the Mesopotamian plain. This is mainly because there are no distinctive terrain shifts mentioned in the primary text, Genesis 5:28–6:8. And there is one prominent midrashic source that confers on Noah the invention of the plough, which was primarily a tool for loose soil and flat tracts of farmland, indicative of alluvial grain farming. Let us assume that this section unfolds somewhere in the Early Dynastic era of city-states, from approximately 3000 BCE to 2500 BCE. Let us say, for the sake of location, the narrative takes place in Shuruppak, in southern Mesopotamia, the traditional home of the flood hero, Utnapishtim.¹⁸⁶

By the end of Chapter 6, “The Flooded State,” we are narratively swept up out of the state core and upon the mountains of Ararat at a dizzying height of 5,137 m above sea level. This section covers the flood account from Genesis 6:9 to 9:17. I am loath to attribute any historical epoch defined in relation to ancient state complexes such as Akkadian or Ur III to such an interregna drama in which the main phenomenon is annihilation of the state. Let us simply note, in looking forward to the Tower of Babel account, that perhaps we are located at the cusp of empire at approximately 2500 BCE.

Chapter 7, “A Slave to Your Brothers,” continues to recount the aftermath of the deluge in Genesis 9:17–9:25. Though there is no shift of landscape, the text speaks of slavery for the very first time. Often the capture and forcible transfer of peoples from non-state into state zones was the main method for building state population density. One ideogram for “slave” in third-millennium Mesopotamia was the combination of the sign for “mountain” with the sign for “woman,” signifying that women were taken as captives from the hills to become subjects to the state regime and the scribal record both.¹⁸⁷ Thus, this chapter paves the way for a descent into the alluvial plains and state space.

Chapter 8, entitled “King of the Four Corners of the Universe,” ensnares the narrative arc back into the Mesopotamian alluvium. Whether this fragile city and its tower refers specifically to the Akkadian, Ur III, or later empiric dynasty is unclear. Perhaps, this brief episode from Genesis 11:1 to 11:9 simply amalgamates the ill-fated and short-lived span of these multiple and fleeting early empires into a single tale. Yet, in order to focus on specific archaeological discoveries which point to an abrupt climactic collapse to the Akkadian regime, let us propose that this narrative occurs in approximately 2100 BCE at the conclusion of Akkadian rule and at an elevation of approximately 20 m on the Tigris and Euphrates plain.

A map of these sites is here in Figure 1.4.

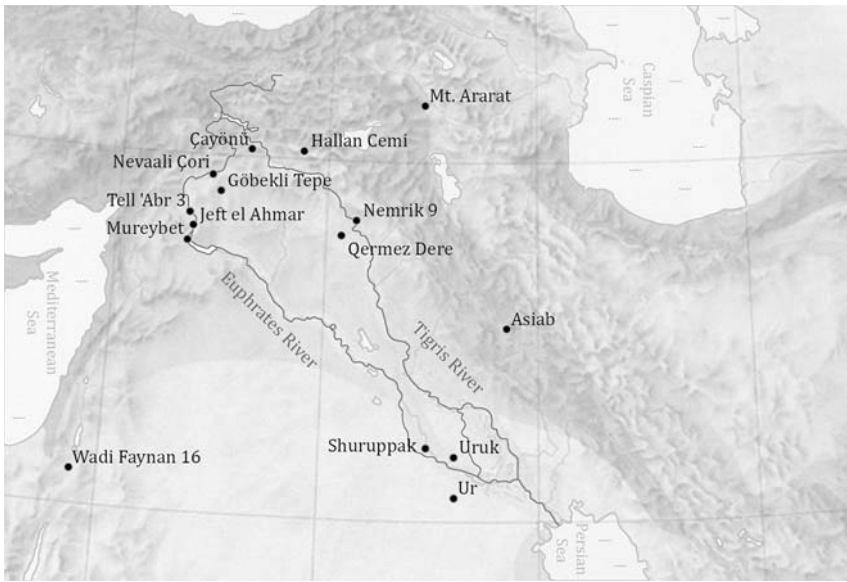


Figure 1.4 Mesopotamia alluvium and surrounding highlands with selected archaeological sites (Graphic design: Strange Diva).

Excluding Lists

Whilst stating upfront that the book you hold is a historical-philological interpretation¹⁸⁸ of the first 11 chapters of Genesis, the careful reader will note that I skip over the majority of the text in question, specifically the long genealogical sections of Genesis 5 (the Sethite Genealogy or *Sethitenstammbaum*), Genesis 10 (Table of Nations or *Volkertafel*¹⁸⁹), and Genesis 11:10–32. There are three reasons for the omission of these sections. The first is prosaic and in keeping with the insight of Robert Alter. The more a text is imbued with literary imagination, the more capable its capacity to “peer over the side”¹⁹⁰ of the status quo. And, as I hope to argue, peering at the underside of the state is precisely where we are headed.¹⁹¹ I simply have to admit that the genealogical lists appear rather flat reading against the complex narratives that surround them. They may not argue against the framing of Genesis as a subversive story, but they do not appear to buttress the argument either.

Second, these lists bear certain hallmarks of written, rather than oral compositions. Specifically, the intricate, non-repetitive lifespans of the characters of Genesis 5 and 11:10–32 indicate a written record. Cassuto points this out in his commentary to Genesis 5, in which the term “book” (ספר) is expressly used to introduce the ethnographical list. “Stories such as those narrated in the previous preceding chapters can be recorded in writing in a book, or they may be recited by heart; but the present section which includes so many numbers, is conceivable only in writing. Hence it is called a book.”¹⁹² Lists of people and

ethnicities in a very general sense represented the nomenclature of Levantine state power through standardisation, abstraction, and legibility.¹⁹³ In effect, notes James Scott, written records were the “information software” of early statecraft that could transform a variegated mass of peoples into a legible polity and thus facilitate appropriation of their goods, land, and labour.¹⁹⁴ Therefore, I read these genealogical lists as essentially later scribal records imposed upon existent oral traditions of proto-Israelite communities.¹⁹⁵ Because this thesis is mainly engaged with events that either preceded or took place outside the rubric of scribal authority, these genealogical lists appear cut from a state cloth and thus of a different purpose.

Just what that purpose may be is open to speculation. Cassuto observes an overall narrative pattern in the genealogical lists that occur throughout the book of Genesis.¹⁹⁶ A summary genealogy of a biblical character indicates a type of dismissal of a now extraneous role, before the narrative can hone in on the more central figure. Thus, the Bible presents a brief genealogy of the offspring of Cain in Genesis 4:17–24, and then Cain’s children all perish in the flood. Subsequently, in Genesis chapter 5, the pedigree of Seth is detailed in great length. And then the story turns exclusively to Noah and his family. Noah’s descendants are elucidated in Genesis 10 and 11 before these nations are largely ignored, and the text turns specifically to Abraham. This pattern continues with the brief listing of Keturah’s and then Ishmael’s offspring in Genesis 25, before the story of Isaac and his children commences, and the other branches of Abraham’s family slip into the narrative background. Again, Esav’s children are succinctly listed and dismissed in Genesis 36, before the story of Jacob and his children is recounted at length. These lists, ironically, indicate a stripping of individuality and identity. Once labelled, these characters no longer play a unique role in the story, and are rarely referred to again.

If Cassuto’s pattern is correct, and genealogical lists indicate a biblical curtain call, then perhaps these lists are telling us something about the underlying reasons for list-making in the first place. Who gains from the classification of ethnicities and for what purpose? Scott notes that the tribal designations imposed on the Southeast Asian hill peoples were mainly exonyms, “applied by outsiders and not used at all by the people so designated.”¹⁹⁷ The amorphous and fluid identities of the Southeastern hill peoples are paradigmatic of fracture zones at the edge of every early state. These zones absorbed wave after wave of people in retreat from the state core, resulting in a constant drift and flux of dialects, customs, and identity. States and empires, notes Scott, “create tribes precisely to cut through the flux and formlessness that characterize vernacular social relations ... At some level it simply did not matter how arbitrary the invented tribes were; the point was to put an administrative end to the flux by instituting units of governance and negotiation.”¹⁹⁸

Perhaps, we can biblically observe the plasticity of tribal identity by tracing the duplicated names of Havila and Sheba in the ethnogenesis account of

Genesis, chapter 10. In Genesis 10:7, Havila and Sheba are labelled as the son and grandson of Kush; yet in Genesis 10:28–29, they appear again as the sons of Joktan. Whilst some biblical scholars see this duplication as a sign that the list in our hands is really an amalgamation of two source traditions, Cassuto offers a variant reading. “Although one man can have only one father, a tribe may be comprised of different elements, and in such instances the Bible apparently intends to indicate that these tribes [Havila and Sheba] comprised some ethnic elements pertaining to Cush and others belonging to Joktan.” In other words, the “tribes” of Havila and Sheba were actually comprised of two separate ancestral identities, joined together. Or perhaps the tribes of Havila and Sheba were so “labile and porous”¹⁹⁹ that they could as easily be classified as the children of Joktan as of Kush. Such was the case with the Lua/Lawa: a swiddening, Mon-Khmer speaking, animist people of Southeast Asia, who were so intimately familiar with the Thai language, padi-cultivation, and Buddhism, that they could be categorised as Lua on one day, and as Tai the next.²⁰⁰ Robert Montagne notes a similar fluidity amongst the Berber society in Morocco, who could oscillate between relatively democratic assembly rule to “ephemeral tribal tyrannies” dependent mainly on accommodation to the adjacent state.²⁰¹ Michel Kordosky’s study of the Kalmyk nomads also points to the “jellyfish” social forms that Kalmyk clans would adapt, determined mainly on their tactical alignment vis-à-vis the Russian state.²⁰²

Yet, the particularly messy case of Havila and Sheba is an anomaly against a precise demarcation of peoples, their language, and their territories set forth in Genesis 10 and 11. These are ethnographical lists that congeal the jelly-like nature of adaptable, mobile groups into a defined language, lineage, and place. For each of Noah’s three children, their genealogies conclude with a rigid taxonomy. “By these are the isles of the nations divided in their lands: everyone after his language, after their families, in their nations.”²⁰³ Again, in Genesis 10:20, we read: “These are the sons of Ham, after their families, after their languages, in their lands, in their nations.”²⁰⁴ And nearly identical wording is found in Genesis 10:31 when the offspring of Seth are defined.

James Scott and Hjørleifur Jonsson both point out that the documentation of peoples into “tribes” was primarily a project of control. “The demarcation of tribes in Africa,” Scott notes, “was ... an official imperial project ... A small army of specialists was busy drawing ethnic boundaries, codifying customs, assigning territories and appointing chiefs to create manageable units of imperial rule, often over stateless people. Some grid of classification had to be imposed on a bewildering cultural variety so as to yield named units of tributes, taxes and administration.”²⁰⁵ The Han state referred to such documentation of previously uncivilised hill people as “cooking barbarians.”²⁰⁶ In summing up an extensive review of research of tribes across the globe, Scott concludes, “Given what we now know, it would be more accurate to say that states preceded tribes and, in fact, largely invented them as an instrument of rule.”²⁰⁷

“Cooking” previously unincorporated non-state peoples into well-defined tribal groups, which creates a pathway towards eventual absorption and assimilation into the alluvial grain state. Edmund Leach described the ethnic succession by which neighbouring hill peoples became valley subjects of the Shan as a gradual transformation from egalitarian (gumloa) social groups, to stratified (gumsa) tribes, to full Shan.²⁰⁸ This closely mirrors the Han civilisation series from “raw barbarians” beyond state documentation, to “cooked barbarians” who had recognisable social hierarchies and some alliance with the Han state core, to full subjects who had “entered the map” into sedentary grain production, taxes, and conscription.²⁰⁹

Here, we can bring in the third consideration for not lingering over-long on the genealogical lists of Genesis. Once people are listed, they become nearly irrelevant to the narrative, just as much as they enter under the gaze of the state. These lists are indicative of a kind of dead-end to certain narrative branches; at the same moment, they often signify the dead-end of assimilation for many non-state peoples. Documenting and defining ethnicity and tribal land is the first step towards assimilation and disappearance into the state core, just as remaining undocumented and unlisted is a strong technique of state evasion. Because the narrative force of the Bible is centrifugal, whereas the state’s pull is centripetal, a demarcated social group pulled into orbit around, and eventually within the state, becomes absorbed into the anonymity of state culture, and nearly always lost to the biblical narrative.²¹⁰ Thus, as Cassuto noted above, once a character or nation in Genesis has been defined by a genealogical listing, they immediately become irrelevant for the remainder of the biblical account. To be registered in a genealogical table is to “enter the map” of civilisation. Ironically, to enter the map of civilisation is the precise point, and the very signifier, of a parting of ways with the biblical storyline. Certainly, key figures such as Noah and Abraham are included in their respective genealogies; yet because they also embark on a rich and complex narrative life, moving centrifugally beyond state grasp, they escape from the fixed structure imposed by these ensnaring lists.

Notes

- 1 Gen 3:17–19a. Hebrew translations not otherwise identified are the author’s own, with reference to *Tanakh: The Holy Scriptures: The New JPS Translation According to the Traditional Hebrew Text* (Philadelphia: Jewish Publication Society, 2003). At different points within this chapter, I will render the same Hebrew word or phrase from the same text into variable English translations in order to articulate a particular nuance of the word or phrase in the context of the specific discussion. For example, I will translate the Hebrew word אדמה as: dirt, soil, (arable) land, humus, earth, or ground; and אדם as human, man, Adam, earthly, humanity, or person.
- 2 H.L.J. Vanstiphout, “The Disputation Between Ewe and Wheat,” in *The Context of Scripture: Canonical Compositions from the Biblical World*, Vol. 1, ed. William W. Hallo (Leiden: Brill, 1997), 575.

- 3 Jacob Klein, "Enki and Ninmah," in *The Context of Scripture: Canonical Compositions from the Biblical World*, Vol. 1, ed. William W. Hallo (Leiden: Brill, 1997), 516, and footnote 6, for the dual reading of DU-LUM as both "toil" and "suffering."
- 4 Klein, "Enki," 516–517.
- 5 Nahum M. Sarna, *The JPS Torah Commentary: Genesis* (Philadelphia: The Jewish Publication Society, 1989), 18–19.
- 6 Hermann Gunkel, *Genesis: Translated and Interpreted*, trans. Mark E. Biddle (Georgia: Mercer University Press, 1997), 22.
- 7 See Theodor Hiebert's multifaceted argument based on both textual and artifactual evidence that farming, rather than pastoral nomadism, was the main food procurement strategy of Israel's "ancestors." "The diet of Israel's ancestors illustrates this reliance on cereal production. In the meals of Israel's ancestors, of Abraham ([Gen.]:5–6) and Lot (19:3), of Isaac (27:17) and his sons Jacob and Esau (25:34), and of Jacob's sons (37:25), bread is the staple. In fact, bread is so basic to the ancestral diet that, in J's diction, bread becomes almost synonymous with 'food' (e.g. 47:12, 15) and 'eating bread' synonymous with partaking of a meal (e.g. 37:25, 43:25, 32)." Theodore Hiebert, *The Yahwist's Landscape: Nature and Religion in Early Israel* (New York: Oxford University Press, 1984), 90–91.
- 8 For a colloquial framing of the nomadic narrative hidden within Genesis, see Daniel Quinn, *Ishmael* (New York: Bantam, 1992), 177–178: "One of the clearest indications that these two stories [of the forbidden tree and the farming curse] were not authored by your cultural ancestors is the fact that agriculture is not portrayed as a desirable choice, freely made, but rather as a curse. It was literally inconceivable to the authors of these stories that anyone would *prefer* to live by the sweat of his brow. So the question they asked themselves was not, 'Why did these people adopt this toilsome life-style?' It was, 'What terrible misdeed did these people commit to deserve such a punishment? What have they done to make the gods withhold the bounty that enables the rest of us to live a carefree life?' ... In our own cultural history the adoption of agriculture was a prelude to ascent. In these stories, agriculture is the lot of the fallen."
- 9 As recorded in James C. Scott, *Against the Grain: A Deep History of the Earliest State* (New Haven: Yale University Press, 2017), 34.
- 10 Otto Eissfeldt, *The Old Testament: An Introduction*, trans. P.R. Ackroyd (New York: Harper & Row, 1965), 195. As cited in Theodore Hiebert, *The Yahwist's Landscape: Nature and Religion in Early Israel* (Oxford: Oxford University Press, 1996), 26. Rudolf Smend, *Die Erzählung des Hexateuch auf ihre Quellen untersucht* (Berlin: G. Reimer, 1912), as cited in Hiebert, *Yahwist's Landscape*, 178, ft. 75.
- 11 George Fohrer, *Introduction to the Old Testament*, trans. David Green (Nashville: Abingdon, 1968), 165. Fohrer introduced his own source called N, to single out the perspective of the Nomadic Source Stratum.
- 12 Peter Ellis, *The Yahwist, The Bible's First Theologian* (Collegeville, MN: The Liturgical Press, 1968), 127.
- 13 See Gen 4:3–4.
- 14 Notably Albrecht Alt, *Essays on Old Testament History and Religion* (Oxford: Oxford University Press, 1966); and Martin Noth, *The Deuteronomistic History* (Sheffield: JSOT Press, 1981).
- 15 Marvin Chaney, *Peasants, Prophets and Political Economy: The Hebrew Bible and Social Analysis* (Eugene, OR: Cascade Books, 2017), 5: "In a sharp break with nineteenth century concepts which still dominate OT scholarship, modern pre-historians and anthropologists no longer regard pastoral nomadism as an evolutionary interval between hunting and gathering and plant cultivation. Instead it is viewed as a marginal specialisation from the animal husbandry that came to be

associated with horticulture and agriculture. Thus, although some pastoral nomads might later sedenterize, the evolutionary flow was from the cultivated areas of the middle east toward the steppe and desert, not out of the desert to the sown.“ See also William G. Dever, *Who Were the Early Israelites and Where Did They Come From?* (Michigan: Erdmans, 2003), 51–52. “The ‘peaceful infiltration’ model is badly flawed by its dependence on typical 19th-century European misconceptions about Bedouin life. At that time, most investigators of Middle East pastoral nomadic societies and lifestyles knew little Arabic. They observed local tribespeople only superficially—rarely, for example, accompanying them year-round throughout their entire annual cycle of migrations as to experience firsthand all aspects of the lives of tribal people on the move.”

- 16 One such example of fluid and seasonal farming is “flood retreat” or *decrue* farming, which takes place on the margins of seasonally flooding lakes or rivers, and was widely utilised in the more arid, lowland sectors of the Fertile Crescent, and particularly the Levantine Corridor. Because the waters ebb and flow did much of the labour of tillage, soil preparation, and irrigation, these “farming communities” spent much of their time hunting, foraging, fishing, trading, and so on. See David Graeber and David Wengrow, *The Dawn of Everything* (Milton Keynes: Allen Lane, 2021), 235.
- 17 Murray Bookchin, *The Ecology of Freedom* (Palo Alto: Cheshire Books, 1982). As elucidated in David Graeber and David Wengrow, *The Dawn of Everything*, 260: “The ecology of freedom describes the proclivity of human societies to move (freely) in and out of farming.” See also Melinda Zeder, “The Origins of Agriculture in the Near East,” *Current Anthropology* 52, no. S4 (2011): S230–231. “Stable and highly sustainable subsistence economies based on a mix of free-living, managed, and fully domesticated resources seem to have persisted for 4,000 years or more before the crystallization of agricultural economics based primarily on domestic crops and livestock in the Middle East.” See also Scott, *Against the Grain*, 61. “A major exponent of a geo-graphical perspective on history rejected any categorical distinction among hunter-gatherers, pastoralists, and agriculturalists, emphasizing that for safety’s sake, most people have preferred to straddle at least two of these subsistence niches.”
- 18 A. M. Khazanov, *Nomads and the Outside World* (Cambridge: Cambridge University, 1984), 85–86. “In this connection I must point out that the Bible gives chronological priority not to pastoralism, and certainly not to pastoral nomadism, but to agriculture.”
- 19 Hiebert, *Yahwist’s Landscape*, 75–76. “In sum, J does not describe beginnings from the perspective of a nomadic pastoralist. His primaeval society reflects the diversified rural economy typical of the Mediterranean hill country ... in which grain-based agriculture is combined with sheep and goat herding to meet subsistence needs and reduce the risks inherent in this environment.” See also Oded Borowski, *Agriculture in Iron Age Israel: The Evidence from Archaeology and the Bible* (Winona Lake, Indiana: Eisenbrauns, 1987), 10. “The impact agriculture had on ancient Israel could be seen in almost every facet of daily life, economic, social, and cultural. From the beginning, Israelite settlement depended heavily on successful adaptation to natural conditions and available resources.”
- 20 Genesis 26:12 recounts how Isaac freely shifts from pastoralism to farming in response to a regional drought in Canaan. “Isaac sowed in that [coastal] land, and reaped in that year one-hundred-fold, for God blessed him.” Clearly, within such fluid food procurement strategies, farming, when utilised, was not a toil-some curse, but an opportunity to survive.
- 21 Richard N. Soulen and R. Kendall Soulen, *Handbook of Biblical Criticism* (Louisville: Westminster John Knox Press, 2001), 173: “*Sitz-im-Leben* (Ger: setting

in life, or life situation) has become a technical term in form criticism to refer to that sociological setting within the life of Israel ... in which particular rhetorical forms first took shape."

- 22 Hermann Gunkel, *Genesis*, xlix: "These legends, themselves, bear the particular testimony to the fact that they have come together from a wide variety of sources. They are extraordinarily multicoloured ... a few of them envision the fathers and patriarchs as farmers (Paradise legend), others as shepherds ... The influence of a higher [cosmopolitan Babylonian] culture also explains the fact that, whereas the patriarchs lived from animal husbandry according to the oldest patriarchal legends, the primal legends envision farmers (cf. 8:22, 'sowing and harvesting') and speak of viticulture (9:20), indeed even of great structures, cities, towers, and ships."
- 23 T. Stordalen, *Echoes of Eden: Genesis 2–3 and Symbolism of the Eden Garden in Biblical Hebrew Literature* (Leuven: Peeters, 2000), 192. "Even if no one would agree with Westermann on all points, this work is still representative in method and in many exegetical positions."
- 24 Claus Westermann, *Genesis 1–11: A Continental Commentary*, trans. John J. Scullion (Minneapolis: Fortress, 1994), 194: "A certain perceptiveness is needed here: the garden that God planted for his creature in v.8 has nothing to do with paradise. It serves to provide nourishment to the creature and nothing more. The two sentences, vv.9 and 15, however portrayed a particular garden, behind which stands the very ancient and widespread description of a paradise."
- 25 For a sensitive rebuttal of the "composite" authorial reading of Genesis (specifically chapters 2 and 3), see T. Stordalen, *Echoes of Eden*, 197–200. "The point is that in the case of Genesis 2–3 traditional preoccupation with the presumed authorial process has not produced a reasonable approach to the given text." Recent critique of the decompositional reading of Genesis specifically, and the Bible generally, has emerged from scholars of literature. For example, Gabriel Josipovici, *The Book of God: A Response to the Bible* (New Haven: Yale University Press, 1988), 14–15. "It is not that the documentary hypothesis is necessarily wrong in substance; Genesis is clearly made up of a number of traditions, which have been combined at different stages. But is it not the task of the critic to try to come to grips with the final form as we have it, and to give the final editor or redactor the benefit of the doubt, rather than to delve beyond his work to what was there before? The inventors of the documentary hypothesis believed that by trying to distinguish the various strands they were getting closer to the truth, which in good nineteenth century fashion, they assumed to be connected with origins. But in practice the contrary seems to have taken place. For their methodology was necessarily self-fulfilling: deciding in advance what the Jahwist or the Deuteronomist should have written, they then called whatever did not fit this view an interpolation. But this leads, as all good readers know, to the death of reading; for a book will never draw me out of myself if I only accept as belonging to it what I have already decreed should be there." See also Robert Alter, *The Art of Biblical Narrative* (New York: Basic Books, 1981), 19–20. "It may actually improve our understanding of the Torah to remember that it is quoting documents, that there is, in other words, a purposeful documentary montage that must be perceived as a unity." See also J.L. McKenzie, "The Literary Characteristics of Genesis 2–3," *Theological Studies*, 15(4), (1954): 541–572.
- 26 George Coats, *Genesis: With an Introduction to Narrative Literature* (Michigan: Eerdmans Publishing Co., 1983), 51.
- 27 Gen 3:17.
- 28 Gen 2:10. The verse in full reads, "A river emerged from Eden to water the garden, and from there it would part and become four headwaters."

- 29 Coats, *Genesis*, 57. "Despite complexities in structure that point to a tradition history, the story ends where the plot intends it to end."
- 30 William Dever, *Who Were the Early Israelites and Where Did They Come From?* (Grand Rapids, MI: Eerdmans, 2003), 178–179: "Why would peasants, any more than pastoral nomads, want to uproot themselves, and in this case migrate to the inhospitable hill country frontier? The answer may be rather simple, if one accepts my picture above of the miserable conditions in the heartland of Canaan toward the end of the 13th century B.C. Canaan was on the verge of total collapse." See also George Mendenhall, "The Hebrew Conquest of Palestine," in *The Biblical Archeologist Reader*, vol. 3, eds., E.F. Campbell and D.F. Freedman (Garden City: Doubleday, 1970), 107–109. "In other words, there was no statistically important invasion of Palestine at the beginning of the twelve tribe system of Israel. There was no radical displacement of population, there was no genocide, there was no large scale driving out of population, only of royal administrators (of necessity!). In summary, there was no real conquest of Palestine in the sense that has usually been understood; what happened instead may be termed, from the point of view of the secular historian interested only in socio-political processes, a peasant's revolt against the networking of interlocking Canaanite city-states." And, Marvin Chaney, *Peasants, Prophets and Political Economy: The Hebrew Bible and Social Analysis* (Eugene, OR: Cascade Books, 2017), 32.
- 31 Albright, *The Archeology of Palestine*, 113. "Thanks to the rapid spread of the art, then recent, of constructing cisterns and lining them with waterproof lime plaster instead of previously used limy marl or raw-lime plaster, the Israelites were able to settle in any site where there was rain, whereas their earlier Canaanite precursors had been forced to restrict their occupation in general to sites near springs or perennial springs." See also Dever, *Who Were the Early Israelites*, 116–7.
- 32 J. Callaway, "Excavating Ai (et-Tell): 1964–72," *Biblical Archeology* 39 (1976): 29–30. And Dever, *Who Were the Early Israelites*, where he estimates the capacity of the 110 grain silos excavated at the proto-Israelite village of Izbet Sartah could have stored 54 tons of wheat and 21 tons of barley annually, twice the amount needed to feed the estimated population of 100.
- 33 Lawrence E. Stager, "Agriculture," in *The Interpreter's Dictionary of the Bible Supplementary Volume*, ed. Keith Crim (Nashville: Abingdon, 1976), 13.
- 34 Norman Gottwald, *Ideology, Class, and the Hebrew Bible* (Eugene, OR.: Cascade Books, 2018), 7. "In pre-state Israel we meet the anomaly of a period of about two centuries when the grip of the Canaanite city-state tributary control over the mountainous hinterland was broken and the previously dominated agrarian and pastoral populace was largely free of surplus extraction."
- 35 Dever champions the term "proto-Israelites" to refer to this emergent population cluster in the Canaanite highlands beginning in Iron I, though they cannot yet be reliably classed under the ethnic label of Israel proper, see *Who Were the Early Israelites*, 194: "Thus the people of the highlands were not yet citizens of a State of Israel with fixed boundaries, a unified sense of ethnic identity, and passports saying 'Israelite.' But I will argue that these were the ancestors—the authentic and direct progenitures—of those who later became the biblical Israelites." However, Dever's call for a distinct pre-monarchic ethnic identity for these highland Canaanite farmers is debatable (see Diana Edelman, "Ethnicity and the Bible," in *Ethnicity and the Bible*, ed. Mark G. Brett (Boston: Brill, 2002), 25–55.
- 36 Gen 3:14.
- 37 Gen 3:17.
- 38 Gen 3:23.

- 39 Gen 4:11–12.
- 40 I attribute this concept of Cain's curse as a cancerous continuation of Adam's curse to the 11th century exegete Rabbi Solomon ben Isaac (Rashi). Rashi describes the ground as the agent of intergenerational transmission. See his commentary to Gen 4:11, in particular: "Now (says God to Cain) I will add a further curse (on the ground) on your account, 'it will no longer give its strength to you.'"
- 41 See Ester Boserup, *Population and Technological Change* (Chicago: University of Chicago Press, 1981), 64–74, for a cogent overview of why larger river valleys in general, and Mesopotamia in particular, were the locus of early urbanisation.
- 42 See Gen 3:15: "I will put enmity between you and the woman, and between your seed and her seed; he will bruise you on the head, and you will bruise him on the heel."
- 43 See Gen 3:19.
- 44 See Gen 4:10: "[God] said: 'What did you do! Your brother's blood is crying out to me from the ground.'"
- 45 Francis Brown, et al., *The Brown-Driver-Briggs Hebrew and English Lexicon: with an Appendix Containing the Biblical Aramaic: Coded with the Numbering System from Strong's Exhaustive Concordance of the Bible* (Peabody, MA: Hendrickson Publishers, 2012), 658. The common translation as "giants" follows the Greek Septuagint, Vulgate, and Syriac manuscripts, probably derived from the description of the *Nefilim* as בני ענק of Nu. 13:33. See also Samson Raphael Hirsch, *The Pentateuch: Genesis*, trans. Isaac Levy (New York: The Judaica Press, 1971), 129. Hirsch suggests that the connection between falling and gigantic proportions could be due to the biblical lexical association between the notion of falling and the overpowering of a weaker force by a more powerful one.
- 46 Gen 8:25.
- 47 From Gen 11:2.
- 48 Gen 18b–19.
- 49 U. Cassuto, *Commentary on Genesis, Part I, From Adam to Noah, Genesis I–VI 8*, trans. Israel Abrahams (Jerusalem: Magnes Press, 1961), 169. Likewise, Hiebert's translation of *eseb hassadeh* links it directly with the staple grain crops of ancient Mediterranean agriculture, wheat, and barley. "Set apart for human consumption and placed in a parallel relationship to bread in these lines, *eseb* is undoubtedly the grain from which bread is made. Thus *eseb hassadeh* can be translated literally as 'grain of the field' or more idiomatically as 'field crops' ... directly identified with wheat and barley (Exodus 9:22, 25, 31, 32: 10:12, 15). Grain is in fact the field crop that, for biblical Israel, was the basis of its highland Mediterranean agriculture, as archeological evidence and biblical references both attest." *Yahwist's Landscape*, 37. See also E.A. Speiser, *Genesis AB* (Garden City, NY: Doubleday, 1964), 14. What distinguishes these field crops (עשב השדה) from the simple grasses (עשב) included in Adam's divinely proposed diet in Gen 1:29 is the additional conjunctive noun (שדה) that locates these grasses specifically within a tilled field.
- 50 Gen 2:9. See Sarna, *JPS Genesis*, 18. "The idea is that man's food is ever ready at hand. The attractive, nutritious, delectable qualities of the fruit are stressed with the next episode in mind. The human couple will not be able to plead deprivation as the excuse for eating the forbidden fruit."
- 51 Clive Ponting, *A New Green History of the World* (London: Vintage, 2007), 42–44.
- 52 David Graeber, *Bullshit Jobs: The Rise of Pointless Work and What We Can Do About It* (London: Penguin Book, 2018), 221.
- 53 There are numerous translations of the key dyadic phrase contained in Gen 2:15 "to work and to watch" (לעבדה ולשמרה), many referenced by Jeremy Benstein in

- The Way into Judaism and the Environment* (Woodstock, VT: Jewish Lights Publishing, 2006), 47–49. Benstein contributes his own preferred translation: “sustainable development.”
- 54 James Scott, *The Art of Not Being Governed: An Anarchist History of Upland Southeast Asia* (New Haven: Yale University Press, 2009), 9. “Fixed-field grain agriculture has been promoted by the state and has been, historically, the foundation of its power.”
 - 55 Scott, *Against the Grain*, 117. The italics are mine.
 - 56 Scott, *Grain*, 124.
 - 57 Tubers, such as potatoes or cassava, can be left in the ground for an additional year or even two after ripening. Frederick the Great of Prussia understood that by ordering his subjects to plant potatoes as opposed to cereal grains, they could not be so easily dispersed by opposing armies. Cited by Scott, *Grain*, 130.
 - 58 Scott, *Grain*, 130–131: “It helps to place yourself in the sandals of an ancient tax-collection official interested, above all, in the ease and efficiency of appropriation ... The above-ground simultaneous ripening of cereal grains has the inestimable advantage of being legible and assessable by the state tax collectors.”
 - 59 Scott, *Grain*, 131–2.
 - 60 See Francis Fukuyama, *The Origins of Political Order* (London: Profile Books, 2011), 89. “We seem to be getting closer to a fuller explanation for pristine state formation. We need the confluence of several factors. First, there needs to be a sufficient abundance of resources to permit the creation of surpluses above what is necessary for subsistence ... Second, the absolute scale of society has to be sufficiently large to permit the emergence of a rudimentary division of labour and a ruling elite. Third, that population needs to be physically constrained so that it increases in density when technological opportunities present themselves, and in order to make sure that subjects cannot run away when coerced.”
 - 61 Francis Fukuyama, *Origins*, 85.
 - 62 Though Fukuyama, James C. Scott, Ester Boserup, and Pierre Clastres, amongst other archeologists, anthropologists, assyriologists, and political historians, envision the earliest pristine states as coercive and rigidly hierarchical entities, their view has not gone entirely unchallenged. The Czech Assyriologist Petr Charvát argues, for example, that the early city-state of Uruk (c. 4000–3200 BCE) was a primaevial “welfare state” and essentially egalitarian. See his *Ancient Mesopotamia: Humankind’s Long Journey into Civilization* (Prague: Oriental Institute, 1993). Thus, whereas most archeologists interpret the ubiquitous bevelled rim bowls of Uruk cultural sights as containers given to labourers and slaves to hold a sparse quota of grain, Charvát postulates that these bowls were uniform moulds for bread and sweetmeat, eaten by everyone in every location as a symbolic gesture of equality. See Gwendolyn Leick, *Mesopotamia: The Invention of the City* (London: Penguin Books, 2002), 46–7.
 - 63 Scott, *Against the Grain*, 73.
 - 64 A translation of *emboitement*, a term coined by George Condominas to describe the statecraft of the Tai *muang* in *From Lawa to Mon, from Saa’aa to Thai: Historical and Anthropological Aspects of Southeast Asian Social Space*. As quoted by James Scott, *The Art of Not Being Governed* (New Haven, Yale University, 2009), 65.
 - 65 A term to describe the similar work of the earliest states to circumscribe a population, coined by Michael Mann in *The Sources of Social Power*, Vol. I: *A History of Power from the Beginning to A.D. 1760* (Cambridge: Cambridge University, 1986), 54–58, as quoted by Scott, *Art*, 351, ft. 4.
 - 66 As well as the intrinsic fragility of the grain state, another factor that may contribute to the impetus for flight could include the general rule that labour input

for foraging, or long-fallow, is much less than the input for fixed-field cereal agriculture, to obtain similar nutritional levels in output. See E. Boserup, *The Conditions of Agricultural Growth: The Economics of Agrarian Change Under Population Pressure* (Abingdon: Earthscan, 1965), 45. "The impression left by such studies is that members of sparse tribes of primitive agriculturalists usually work much shorter and less regular hours than members of densely settled peasant communities."

- 67 Boserup argues that this choice is mainly determined by a comparative weighing up of time and labour investment. "I am convinced that the former one [forest fallow] usually requires less hard human labour than the latter [ploughed cultivation] ... There are many examples of Government offers to supply ploughs and draught animals being turned down because the cultivators found it easier to continue with the system of forest fallow." *Agricultural Growth*, 33 and ft. no.1. David Graeber and David Wengrow argue from first-hand historical accounts of Europeans captured by indigenous foraging populations who would often make the choice to live with their former captors rather than return to "civilization" because of the relative equality and social bonds that ensured support for the destitute and infirm amongst these non-state communities. *Dawn of Everything*, 20.
- 68 Scott, *Grain*, 112–113. "An illness—of crops, livestock, or people—a drought, excessive rains, a plague of locusts, rats, or birds, could bring the whole edifice down in the blink of an eye."
- 69 Shifting cultivation imposes an upper limit of population density of about 20–30 inhabitants per square kilometre. Anthony Reid, *Southeast Asia in the Age of Commerce, 1450–1680*, Vol. 1: *The Lands Below the Winds* (New Haven, Yale University, 1988), 20; as cited by Scott, *Art*, 65.
- 70 Boserup, *Population and Technological Change*, 65. "Fertility of land and agricultural technology were secondary factors; the prime condition for early urbanization was a sufficiently large and dense population."
- 71 Boserup, *Population*, 83. "The urbanized, more densely populated areas increased their advantage in both population density and technological levels by skimming off the population surplus in surrounding areas by means of forced or voluntary migration to the urban centers and their immediate neighbourhood ... [yet] many urbanized societies had temporary or lasting periods of regression."
- 72 Quoted in Mark Elvin, *The Retreat of the Elephants: An Environmental History of China* (New Haven: Yale University, 2004), 104–105. See also Scott, *Art*, 66.
- 73 See Scott, *Art*, 65.
- 74 Scott, *Grain*, 152.
- 75 Scott, *Grain*, 153.
- 76 Perhaps, this strict barrier I construct between mythmaking and mimesis is rather porous, and collective folk memories can shape mythical accounts. See Ofer Bar-Yosef, "Forward," in *The Neolithic Revolution in the Near East: Transforming the Human Landscape*, ed. Alan Simmons (Arizona: University of Arizona Press, 2007), ix. "The socioeconomic transition from foraging to farming left its imprint on the collective human memory, expressed in various parts of the world in mythological stories. Judging by the myths, the shift from the seemingly leisurely lifeways of hunting and gathering to toiling on the land, sowing, planting, harvesting, and storing surplus grain, was a major shift in the role of both work and gender in society. In the biblical story it is retold as the expulsion of Adam and Eve from the Garden of Eden. Thus, the transition in lifeways caused by the Neolithic Revolution resulted in deep-seated cultural shock retained in the oral history of the populations of the Near East."
- 77 Norman Wirzba, *The Paradise of God: Renewing Religion in an Ecological Age* (New York: Oxford University Press, 2003), 34.

- 78 Ellen F. Davis, *Scripture, Culture and Agriculture: An Agrarian Reading of the Bible* (New York: Cambridge University Press, 2009), 8. "Overall, from a biblical perspective, the sustained fertility and habitability of the earth, or more particularly of the land of Israel, is the best index of the health of the covenant relationship."
- 79 Davis, *Scripture*, 12.
- 80 Davis, *Scripture*, 8. "When humanity ... is disobedient, thorns and briars abound."
- 81 Davis, *Scripture*, 31.
- 82 Davis, *Scripture*, 32.
- 83 From Gen 9:20, the translation is from Davis, *Scripture*, 32. Her translation of the full verse reads: "And Noah got started [as] a man of the fertile soil, and he planted a vineyard." Every translation is commentary and any writer intent on imparting a large-scale pattern to Scripture, myself included, should be regarded with suspicion when presenting their own translation as definitive. It is entirely reasonable to translate the same verse as: "Noah, the lord of dirt, degraded himself in planting a vineyard." This translation follows the commentary of Rashi ad loc. s.v. ויחל.
- 84 Davis, *ibid.*
- 85 Davis, *Scripture*, 31. See also Theodore Hiebert, *The Yahwist's Landscape: Nature and Religion in Early Israel* (New York: Oxford University Press, 1984), 80.
- 86 Davis, *Scripture*, 40.
- 87 Hiebert, *Yahwist's Landscape*, 87. "The Yahwist's portrait of Israel's ancestors contains numerous details that associate them directly with sedentary agriculture and point to this setting as the perspective from which their stories are told."
- 88 Hiebert, *Yahwist's Landscape*, 69. "This curse introduces a theme which is to become the leitmotif of the primeval narrative, the relationship between human morality and the soil's productivity."
- 89 Hiebert, *Yahwist's Landscape*, 69. "The primary consequence of immorality will be the soil's sterility and the disruption of the relationship between farmer and land."
- 90 Hiebert, *Yahwist's Landscape*, 48. Later on that page, Hiebert praises Noah as "humanity's first righteous ancestor."
- 91 Hiebert, *Yahwist's Landscape*, 48.
- 92 Hiebert, *Yahwist's Landscape*, 48. Hiebert translates Noah's drunken state as "celebrative," despite the fact that he ends up naked, and wakes up cursing.
- 93 Like Davis and Hiebert, Norman Wirzba also ignores the tragic conclusion of the Noah narrative, to preserve, one assumes, his rigid taxonomy of hubristic versus humble biblical types. "There is an integrity to creation that depends on humans seeing themselves as properly placed within a network of creation and God. The drama shows us that neither God nor the creation itself can tolerate violence, manipulation or shame. Instead of the hubris that characterized Adam and his descendents, Noah stands out as a beacon of the humbled *adam* who is faithful to the needs of *adamah*." *Paradise*, 34.
- 94 Gerhard von Raad, *Old Testament Theology*, trans. D.M.G. Stalker (Louisville: Westminster John Knox Press, 2001), 153: "The transition from primeval to sacred history occurs abruptly and surprisingly in verses 1–3 [of chapter 12]." See Josipovici's critique of this division in *The Book of God*, 15: "Why does he [von Raad] find the transition surprising? First of all because he relegates the last two verses of chapter 11 to the Priestly writer and the start of chapter 12 to the Jahwist; second because he holds—this is another dogma of nineteenth century scholarship—that Genesis is made up of a 'primeval' and a 'sacred' history, which meet precisely here in chapter 12; and finally because he is so busy trying to

- assign each verse to its own strand that he seems never to have attended to the rhythm of the whole.”
- 95 As propaganda, see Umberto Cassuto, *A Commentary on the Book of Genesis: Part I, from Adam to Noah: Genesis I-VI* 8, trans. Israel Abrahams (Jerusalem: The Magnes Press, Hebrew University, 1961), 12. “But the Torah, which is not written in verse but in prose, and employs as a rule simple, not figurative, language, and weighs every word scrupulously, was careful not to introduce ingredients that were not completely in accord with its doctrines. Nay more, whenever necessary it voiced, in its own subtle way, its objection in principle to concepts suggestive of an alien spirit as, for instance, the myth of the revolt of the sea against its Creator.”
 - 96 Jan Assmann, “Myth as *Historia Divina* and *Historia Sacra*,” in *Scriptural Exegesis: The Shapes of Culture and the Religious Imagination*, ed. Deborah A. Green and Laura S. Lieber (Oxford: Oxford University Press, 2009), 15. “Mythical time belongs to another temporality than that to which historical time belongs.”
 - 97 Mircea Eliade, *Le Mythe de l'éternel retour* (Paris: Gallimard, 1949), as cited by Assmann, “Myth,” 15.
 - 98 Herbert Schneidau, *Sacred Discontent* (Baton Rouge: Louisiana State University Press, 1977), 292: “A cosmology of hierarchical continuities, as in mythological thought, exhibits strong metaphorical tendencies ... In this world, movement tends to round itself into totalization, impelled by the principle of closure.”
 - 99 Hershel Shanks, *Ancient Israel: From Abraham to the Roman Destruction of the Temple* (Washington D.C.: Biblical Archeology Society, 1999), xix. See also Herman Gunkel, *The Legends of Genesis*, trans. W.H. Carruth (London: Open Court Publishing, 1901), 13–14. Also, Hiebert, *Yahwist's Landscape*, 78–79, where he cites Gunkel as the seminal advocate of this separate status for Gen 1–11. “The primeval narrative is considered distinctively mythical, to be distinguished in this regard from the following material—the traditions about the ancestors, Egypt, and the desert. Herman Gunkel, who set out the parameters for twentieth-century discussions of myth in the Hebrew Bible, separated the stories of the primeval era from the legends of the patriarchs because he saw in them a ‘more decidedly mythical character.’”
 - 100 Josipovici in *The Book of God* contends that the literary coherence between chapters 11 and 12 is artfully manifest in the tension built into the narrative between Sarah's barrenness as introduced in verses 29 and 30 of Genesis 11 and the divine promise of Genesis 12 that Abraham will become a great nation. A tension only resolved with the birth of Isaac in chapter 21. “Thus to split the last two verses off the end of chapter 11, as von Raad does, and then to start a whole new section of commentary at 12:10, calling it ‘The Biblical Patriarchal History,’ is to deny the reader the right to experience this as a growing narrative.”
 - 101 While most scholars concur that the Neolithic revolution was “an economic transformation that involved the domestication of wild food resources and the establishment of permanent settlements,” dating this Neolithic revolution is challenging. First, because it attempts to demarcate a flexible range of societal behaviours that shifted gradually and rarely linearly over thousands of years. And second, because the Neolithic revolution was not a singular innovation, but includes a cascading spectrum of independent events that occurred at different locations and different times around the world. Its maximal approximate dates are 12000 BCE to 6000 BCE. See Alan H. Simmons, *The Neolithic Revolution in the Near East: Transforming the Human Landscape* (Arizona: University of Arizona Press, 2007), 4.
 - 102 Robert Alter, *The Art of Biblical Narrative*, 49. “And even that reach backwards is only in the shards of poetry embedded into the larger biblical prose, and the song of Deborah.”

- Dever offers a pithy overview of the general timeline of biblical composition in *Who Were the Early Israelites*, 7: "Traditionally it was thought that J, dated as early as the 10th century B.C., and E, perhaps composed in the 9th century, were edited together in the 8th century or so. Then a final 'Priestly school' (known as P) edited both together into the work that we now have, adding much priestly legislation, sometime during the exilic or post-exilic period (6th century B.C)."
- 103 David Damrosch, *The Narrative Covenant: Transformations of Genre in the Growth of Biblical Literature* (New York: Harper and Row, 1987), 91 ft 4.
 - 104 However, cf. E.A. Speiser goes so far as to suggest that the "canonical tradition among the people of the Book, must be older than the age of the Pentateuchal writers, older indeed than the time of Moses himself." "The Biblical Idea of History in Its Common Near Eastern Setting," *Israel Exploration Journal* 7, no. 4 (1957): 208. <http://www.jstor.org/stable/27924711>.
 - 105 See Alter, *Art*, 48, in utilising this terminology to describe the work known in Hebrew as the Tanakh.
 - 106 Indeed, oral legends can have tremendous staying power. In the Australian state of Victoria, the indigenous people, the Gunditjmara, tell stories about a local volcano called Budj Bim spitting "liquid fire from between his teeth." Budj Bim last erupted 37,000 year ago. Erin L. Matchan, David Phillips, Fred Jourdan, Korien Oosting, "Early human occupation of southeastern Australia: New insights from ⁴⁰Ar/³⁹Ar dating of young volcanoes," *Geology* (2020): <https://doi-org.bris.idm.oclc.org/10.1130/G47166.1>
 - 107 I define the Levant broadly following the lead of Alan Simmons, who himself relies generally on the definitions of Malcolm Colledge and Ofer Bar-Yosef, the latter of whom calls it the "Mediterranean Levant." Simmons writes: "It encompasses the area east of the Mediterranean Sea and is bounded in the north by the Taurus and Zagros mountains. In the northeast, the boundary is the Euphrates River Valley, and in the south and southeast, the Negev, Sinai, and Syro-Arabian deserts. The Levant has a north-south length running some 1,100 km and an east-west width of 250–350 km; it can be subdivided into three general zones. These are the northern Levant, which includes parts of southeastern Turkey and the Euphrates Valley; the central Levant, consisting primarily of Syria and Lebanon; and the southern Levant, which comprises Israel and Jordan. Within this region, there exists a multitude of diverse environments. While these three subdivisions are somewhat arbitrary, they conform to many major developments that occurred during the Neolithic." Alan H. Simmons, *The Neolithic Revolution in the Near East: Transforming the Human Landscape* (Arizona: University of Arizona Press, 2007), 30.
 - 108 Roger Matthews, Wendy Matthews, Amy Richardson and Kamal Rasheed Raheem, "The Neolithic Transition in the Eastern Fertile Crescent: Project Themes, Aims and Objectives." In *The Early Neolithic of the Eastern Fertile Crescent: Excavations at Bestansur and Shimshars, Iraqi Kurdistan*, ed. Roger Matthews, Wendy Matthews, Amy Richardson and Kamal Rasheed Raheem (Oxford: Oxbow Books, 2020), 1–2: "Many researchers see the Neolithic transition as representing a first stage in the formulation of a new relationship between humans and their environments whereby new modes of intensified human engagement with plants, animals and entire landscapes could be characterised as 'the single most dramatic (and ultimately the most catastrophic) set of changes that human society has experienced since the mastery of fire.'" Quoting C. Hillman and M Davies, "Domestication Rates in Wild Wheats and Barley under Primitive Cultivation." In *Prehistory of Agriculture*, ed. P.C. Anderson (Los Angeles: Institute of Archaeology, University of California, 1999), 70.

- 109 Michael S. Northcott, *A Political Theology of Climate Change* (London: Society for the Promotion of Christian Knowledge Press, 2014), 29: "In Eden, Adam and Eve are forest-dwelling gatherers of fruits and herbs, whereas the exile from Eden requires them to raise their food through agriculture. In this ancient myth of origin, agriculture is the source of alienation between human beings that leads to the original murder, of Abel by Cain, using an agricultural implement, and the rise of hierarchical civilisations in which men and women visit violence on one another and reduce one another to slavery. It is also the beginning of a mark of humanity on the earth which changes the destiny of all creatures and the earth, and not only the destiny of humanity." This book essentially follows Northcott's thesis, excepting the detail of Cain's murder weapon as an "agricultural implement" which is never mentioned in the Bible, nor alluded to in any classical or mediaeval exegesis. See also Hiebert, *Yahwist's Landscape*, 48–49: "Human life starts in a garden which could represent a hunting and gathering stage, proceeds at the expulsion from Eden to the domestication and cultivation of grains, and then to the mixed economy of grain farming and the domestication and herding of sheep in the generation of Cain and Abel. These developments are followed by urbanization and emerging specializations such as specialized pastoralism and metalworking. Finally, the vine—and possibly other fruit species—is domesticated by Noah and grown for its produce ... All in all this is not a bad chronological survey of the development of food production in antiquity, which moved from hunting and gathering to domestication of grains (c. 7000 BCE), then animals (c. 5000), then fruit trees and vines (c. 3000)." Hiebert, however, ultimately rejects this approach because Adam "cultivated" the Garden of Eden, and therefore, should be viewed from the onset as a fully-fledged farmer and not a gatherer (citing Genesis 2:5, 7, and 15).
- 110 Max Oelschlaeger, *The Idea of Wilderness: From Prehistory to the Age of Ecology* (New Haven: Yale University Press, 1991), 31: "If the thesis that agriculture underlies humankind's turn upon the environment, even if out of climatological exigency, is cogent, then the ancient Mediterranean theatre is where the 'fall from Paradise' was staged, from here began extensive humanization of the natural landscape." Also, Evan Eisenberg, *The Ecology of Eden: Humans, Nature and Human Nature* (London: Picador, 1998), 95–98.
- 111 James C. Scott, *Against the Grain: A Deep History of the Earliest State* (New Haven: Yale University Press, 2017).
- 112 Ian Morris, *Why the West Rules for Now: The Patterns of History and What They Reveal About the Future* (London: Profile Books, 2011). In particular, Part 1, Chapters 1–3. Also, Desmond Collins, *The Human Revolution: From Ape to Artist* (Oxford: Phaidon Press, 1976), 196. "Adam and Eve fell because they ate the forbidden fruit and were therefore subject to all the moral and social problems of modern man. Cromagnon man also ate a 'forbidden' fruit. He cultivated plants, notably wheat and barley, and within a few thousand years he was dependent on them and passed a point of no return ... Within a few hundred years, descendants of the classless hunters found themselves peasants, tied to the land and near the base of a pyramidal hierarchy with rulers, nobles, warriors, priests, and scribes standing above them. A fall indeed!"
- 113 Peter J. Wilson, *The Domestication of the Human Species* (New Haven: Yale University Press, 1988), 24.
- 114 Scott, *Grain*, 10.
- 115 Yuval Noah Harari, *Sapiens: A Brief History of Humankind* (London: Vintage, 2011).
- 116 Auerbach, *Mimesis*, 18.
- 117 Erich Auerbach, *Mimesis: The Representation of Reality in Western Literature*, trans. Willard R. Trask, Fifteenth anniversary printing (Princeton: Princeton University Press, 2003), 12.

- 118 Biblical scholars, such as Robert Alter, have criticised Auerbach's theory as over simplistic and methodology as limited to just two tales. See Alter, *Art*, 17: "[Auerbach's] key notion of biblical narrative as a purposely spare text 'fraught with background' is at once resoundingly right and too sweepingly general. Distinctions have to be made for narratives by different authors, of different periods, and written to fulfil different generic or thematic requirements."
- 119 Auerbach, *Mimesis*, 19.
- 120 Auerbach, *Mimesis*, 19.
- 121 My use of the general, and admittedly vague, term "author" to describe the biblical creative mind closely follows the wide parameters of this term endorsed by Kerel Van Der Toorn, in his *Scribal Culture and the Making of the Hebrew Bible* (Cambridge, Mass: Harvard University Press, 2007), 47. For the ancient Near East society, the idea of the author could include a collective group or an individual who either created or edited the text. However, unlike the authors of modernity, writers of antiquity did not revel in their individuality but rather in their communal scribal skillset. "If the author," writes Van DerToorn, "is a representative of the scribal craft, anonymity is a fitting phenomenon ... It confirms the fact that the authors of the time did not write as individuals but functioned as constituent parts of a social organism." This notion of constituent or amorphous authorship dovetails with a particular approach to reading the Bible, championed by Umberto Cassuto. "The sources ... are not therefore documents of P or J or the like, from which fragments were extracted mechanically. Without doubt the material was derived from different sources, but the whole variegated content was smelted down into a single mass by the fire of literary creation, as silver is smelted in a crucible, and it is no longer possible to discern in the work before us each of the various elements that become amalgamated in it, just as in the waters of the Jordan that pour into the Dead Sea one cannot distinguish between the waters that come from the Yarmuk or the Jabbok or the other tributaries." Cassuto, *Genesis II*, 185. In a similar manner, Robert Alter describes the "esemplastic" power of the Biblical authors/redactors who exhibited a definitive "faculty for molding disparate elements into an expressly unified whole." Following the phraseology of Sir Edmund Leach, Alter reads the atomising activity of the historical biblical critics as mainly conjectural attempts to "unscramble the omelette." Alter, *World*, 24. In this sense, the term "author" is synonymous with the biblical redactor of the scriptural text we now possess. See also Eduard Nielson, *Oral Tradition* (London: SCM Press, 1961), 102. "Thereby we have seen that a *mechanical* division of the text into two independent sources does not do justice to the present text. Such a division disregards the fact that our present text is a work of art, composed of different traditions, it is true, but in such a way that a unified work has been the result."
- 122 Auerbach, *Mimesis*, 20. "Even in the legendary passages of the Old Testament, historical structure is frequently discernible—of course, not in the sense that the traditions are examined as to their credibility, according to the methods of scientific criticism; but simply to the extent that the tendency to a smoothing down and a harmonizing of events, to a simplification of motives, to a static definition of characters which avoids conflict, vacillation, and development, such as are natural to the legendary structure, does not predominate in the Old Testament world of legend." Auerbach's insistence on the 'historical' quality of scripture is a not so veiled critique against the prevalent scholarly view that the biblical narrative was primarily redacted as a plastic device of state or religious propaganda. To highlight only a few of the better known examples of this utilitarian view of the text, see, e.g., J. Alberto Soggin, *Introduction to the Old Testament*, trans. John Bowden (Philadelphia: Westminster, 1976), 102. "We may therefore

- consider the J collection to be something like an apologetic writing intended to justify and legitimate the monarchy in Israel, a form of government which was certainly new and the subject of lively debate, if we consider the sources; this legitimation was achieved by means of an extremely shrewd choice from ancient material enriched by early prophecies which were probably *ex eventu*." Also, R.E. Clements, *Abraham and David: Genesis XV and Its Meaning for Israelite Tradition* (London, SCM Press: 1967); and Delbert R. Hillers, *Covenant: The History of a Biblical Idea* (Baltimore, MD: John Hopkins University Press: 1969).
- 123 Auerbach, *Mimesis*, 39.
 - 124 Auerbach, *Mimesis*, 38.
 - 125 Auerbach, *Mimesis*, 38.
 - 126 Assman makes a similar claim regarding ancient Egypt. He first cites Herodotus. "Within this time-span [11,340 years of Egyptian kingly rule], the sun is reported to have changed its course several times. Twice it has risen where it now sets and has set where it now rises. But this did not cause any changes in the Egyptian world, neither in the vegetation nor in the activity of the river, nor with regard to diseases and death in the human world' (2.143)." Then Assman concludes that, "This seemingly absurd remark nonetheless offers a deep insight into the non-narrative structure of the Egyptian concept of the past. Narrative is about time as change. Time without change can only be counted or measured, not narrated." Assman, "Myth," 16. Likewise, in a more general rubric, Claude Levi-Strauss distinguishes between cold and hot societies. Cold societies freeze change and thus "strive, by means of the institutions they are giving themselves, to erase quasi-automatically the effect which historical factors could have on their equilibrium and their continuity." Whereas, hot societies "are characterised by a desire for change" (Cited in Assman, "Myth," 16). In this sense, the Hebrew Bible produces cultural "heat."
 - 127 Ludwig van Beethoven, *String Quartet No. 16 in F major*, Op. 135, October 1826.
 - 128 Auerbach, *Mimesis*, 21.
 - 129 Auerbach, *Mimesis*, 31.
 - 130 Auerbach, *Mimesis*, 45.
 - 131 Alter, *World*, 44.
 - 132 Alter, *World*, 45.
 - 133 Alter, *Art*, 29.
 - 134 Robert M. Cover. "The Supreme Court, 1982 Term—Foreword: Nomos and Narrative," *Faculty Scholarship Series Paper* 2705 (1983): Specifically 19–25, in the section entitled "The Thickness of Legal Meaning." Cover borrows the term "thickness" from the anthropologist Clifford Geertz's landmark essay "Thick Description: Toward an Interpretive Theory of Culture." For Cover, a thick reading of the Bible requires that contradictions between proscriptions and narratives are not neatly smoothed over, but interpreted with their frictions intact.
 - 135 Cover, *Nomos*, 22.
 - 136 Nathan MacDonald, *What Did the Ancient Israelites Eat? Diet in Biblical Times* (Michigan: Eardmans Publishing, 2008), 77.
 - 137 See Aharon Sasson, "The Pastoral Component in the Economy of Hill Country Sites in the Intermediate Bronze and Iron Ages: Archaeo-Ethnographic Case Studies," *Tel Aviv* 25:1 (1998): 50. See also Shimon Dar, *Landscape and Pattern: an Archaeological Survey of Samaria 800 B.C.E.–636 C.E.* (Oxford: B.A.R., 1986), 160–161.
 - 138 On whether this biblical egalitarianism was nominally a utopian fantasy or a lived reality, see Dever, *Who Were the Early Israelites and Where Did They Come From*, 187: "Some would dismiss the biblical tradition as fiction: late, unhistorical, and romantic—the produce of nostalgia for a past that never was, hope for a future that never would be. But insofar as this tradition is both ancient and unique among the

- other literature of the time, some explanation is required. I suggest simply that the biblical notion of what we would now call 'agrarian land reform' is deeply rooted in the historical memory, specifically in a long oral tradition that reaches back to the social and economic realities of Canaan of the end of the Bronze Age."
- 139 See Joseph Naveh, *Early History of the Alphabet: An Introduction to West Semitic Epigraphy and Paleography* (Jerusalem: Magnes, 1982), 75–76; Alan R. Millard, "An Assessment of the Evidence of Writing in Ancient Israel," in *Biblical Archaeology Today: Proceedings of the International Congress, Jerusalem, April 1984*, ed. Janet Amitai (Jerusalem: Israel Exploration Society, 1985), 301–312; Aaron Demsky, "Writing in Ancient Israel: The Biblical Period," in *Mikra: Text, Translation, Reading and Interpretation of the Hebrew Bible in Ancient Judaism and Early Christianity*, ed. Martin-Jan Mulder (Philadelphia: Fortress, 1988), 2–20.
 - 140 For example, Rykle Borger lists 907 Akkadian signs in his cuneiform list; *Mesopotamisches Zeichenlexikon* (Münster: Ugarit Verlag, 2004).
 - 141 See Meir Bar-Ilan, "Illiteracy in the Land of Israel in the First Centuries C.E.," in *Essays in Social Scientific Study of Judaism and Jewish Society*, Vol. 2, eds. Simcha Fishbane et al. (New York: Ktav, 1992), 46–51; Ian M. Young, "Israelite Literacy: Interpreting the Evidence," *VT* 48 (1998): 239–253.
 - 142 See, for example, *The Feminist Standpoint Theory Reader: Intellectual and Political Controversies* (London: Routledge, 2004).
 - 143 David Graeber, *The Utopia of Rules: On Technology, Stupidity, and the Secret Joys of Bureaucracy* (Brooklyn, NY: Melville House Publishing, 2016), 72.
 - 144 Ex. 7:15. "Go to Pharaoh in the morning, behold he goes to the water, and you should stand facing him on the bank of the Nile ..." See Rashi ad loc. "He goes to the water to relieve himself because he had fashioned himself as a god, and proclaimed that he never needed the toilet, so he had to get up early and go to the Nile to take care of his needs."
 - 145 In this sense, mimetic narrative runs close to Scott's understanding of historical analysis. See Scott, *Against the Grain*, 5. "History at its best, in my view, is the most subversive discipline, inasmuch as it can tell us how things that we are likely to take for granted came to be."
 - 146 Maureen Gallery Kovacs, trans., *The Epic of Gilgamesh* (Stanford, CA: Stanford University Press, 1989), 3.
 - 147 As cited by Hiebert, *The Yahwist's Landscape*, 42. "When compared with origin narratives from the great urban cultures of antiquity, the distinctive character of the Yahwist's treatment of the city becomes clear. In Mesopotamian traditions about the preflood era, cities dominate the landscape ... By comparison, the Yahwist's presentation of the primeval age is decidedly nonurban."
 - 148 As quoted in Gwendolyn Leick, *Mesopotamia: The Invention of the City* (London: Penguin Books, 2002) 26. These hymns are presented on tablets from the Old Babylonian period (c. 1800 BCE), but according to the colophones they were originally edited in the 23rd century BCE.
 - 149 Genesis 11: 8–9.
 - 150 Gen 3:17.
 - 151 Gen 4:11.
 - 152 Gen 9:25.
 - 153 Consider, for example, the contradistinction between the Genesis narrative which curses the abandonment of the highland forest to take up farming and city-living, with the religio-scribal texts at Ugarit which castigate the seditious citizens who leave the city, and it's divine protector, Baal Haddu, for the highlands. "Baal's enemies take to the woods, Haddu's foes to the sides of the mountain crag." *The Cuneiform Alphabetic Texts from Ugarit, Ras Ibn Hani and Other Places*, 2nd ed., eds. Manfred Dietrich et al. (Munster: Ugarit, 1995), 1.4:7.35–37. As quoted by

- Chaney, *Peasants*, 13. Chaney contends that the divine Baal Haddu “functions in many regards as a city-state king writ large.”
- 154 See Scott, *Against the Grain*, 32. “The term ‘barbarian’ we know, was originally applied by the Greeks to all non-Greek speakers ... ‘Ba-ba’ was meant to be a parody of the sound of non-Greek speech. In one form or another the term was reinvented by all early states to distinguish themselves from those outside the state.”
- 155 Scott, *Against the Grain*, 14.
- 156 An allusion to Gen 3:24.
- 157 To offer just one indicative phrase (and perhaps allude to an element of post-Marxist realism), see Scott’s epigraph in *The Art of Not Being Governed*, which is a quote’s from Pierre Clastres. “It is said that the history of peoples who have a history is the history of class struggles. It might be said with at least as much truthfulness, that the history of peoples without history is a history of their struggle against the state.” My own methodology takes this approach one step further, “if the history of peoples without a history, should be narrated as destiny, the result is a scripture against the state.”
- 158 Petr Charvát, *Ancient Mesopotamia: Humankind’s Long Journey into Civilization* (Prague: Oriental Institute, 1993).
- 159 See Cassuto, *Genesis I*, 179. Also Hiebert, *The Yahwist’s Landscape*, 181, ft. 23.
- 160 See Cassuto, *Genesis I*, 180.
- 161 Scott, *Against the Grain*, 17.
- 162 To give but one example of the lopsided privileging I bestow on anti-state scholars, consider the concept of “corvée” in ancient Mesopotamian labour practice. Throughout this work, I will assume that the obligatory practice of seasonal labour exacted from free Mesopotamian subjects, termed “corvée,” was generally coerced from above, and deserves to be associated with pejorative practices of subjugation such as servitude and conscription. In this, I am deeply influenced by James Scott, *Grain*, e.g. 157–158: “The case for the importance of slavery in the Mesopotamian polities is, I hope to show, convincing. When other forms of unfree labor, such as debt bondage, forced resettlement, and corvée labor are taken into account, the importance of coerced labor for the maintenance and expansion of the grain-labor module at the core of the state is hard to deny.” Yet, other scholars put much greater valence on sources that paint a far rosier version of corvée, c.f. Graeber and Wengrow, *Dawn*, 299–300: “Royal [Mesopotamian] hymns describe the ‘happy faces’ and ‘joyous hearts’ of corvée workers. No doubt there’s an element of propaganda here, but it’s clear that, even in periods of monarchy and empire, these seasonal projects were undertaken in a festive spirit, labourers receiving copious rewards of bread, beer, dates, cheese, and meat. There was also something of the carnival about them. They were occasions when the moral order of the city spun on its axis, and distinctions between citizens dissolved.”
- 163 Brueggemann, *Land*, 15. “These two histories set the parameters of land theology in the Bible: presuming upon the land and being expelled from it; trusting toward a land not yet possessed, but empowered by anticipation of it.”
- 164 Hiebert, *Yahwist’s Landscape*, 68. “The Yahwist’s primeval narrative might be described, without exaggeration, as a drama of the soil, a narrative designed to explain and define the relationship between arable land and its farmers.” Davis uses this phrase to similar affect in *Scripture, Culture and Agriculture*, 50.
- 165 See, for example, Hiebert, *Yahwist’s Landscape*, 69. “This curse [from Gen 3:17] introduces a theme that is to become the leitmotif of the primeval narrative, the relationship between human morality and the soil’s productivity.”
- 166 See, for example, Davis, *Scripture, Culture, and Agriculture*, 38. “The Israelites, in contrast to their materially more fortunate neighbours, never had enough water

- or arable land to waste. They managed to establish themselves in the steep, rocky hill country because it was the only part of the land of Canaan that nobody else wanted. They survived as farmers by becoming intimate with the land, by learning to meet its expectations and its needs ... Probably that is why they developed agrarian insight."
- 167 Davis, *Scripture, Culture and Agriculture*, 32. "At his birth, Noah's father Lamekh declares: 'This one will bring us relief' from the long-accursed condition of the soil (5:29; cf. 3:17). That destiny is fulfilled, but only after the flood has washed away the human-initiated 'ruination' of the earth (6:11–13) ... At that point, Noah the righteous man receives a new 'heroic epithet,' complementary to the first: 'And Noah got started [as] a man of the fertile soil, and he planted a vineyard' (9:20)."
- 168 Hiebert, *The Yahwist's Landscape*, 50. "Thus for the Yahwist, Noah's work represents a restoration rather than a revolution of agricultural practices." For an analogously glorified treatment of Noah's farming (and silence on Noah's cursing), see Norman Wirzba, *The Paradise of God*, 34. "Instead of the hubris that characterized Adam and his descendants, Noah stands out as a beacon of the humbled Adam who is faithful to the needs of *adamah*."
- 169 Scott, *Against the Grain*, xiv.
- 170 Daniel Boyarin, *Intertextuality and the Reading of Midrash* (Bloomington, Indiana University Press, 1990), 40.
- 171 Boyarin, *Intertextuality*, 41.
- 172 A similar methodology to the one that guides Cassuto's commentary on Genesis. See, Cassuto, *Genesis I*, 2. "In order to determine the content of the traditions that were current among Israelites, and their origin and development, I sought to gather and examine scattered reference in the other books of the Bible to the subjects under discussion; and since even the Apocrypha and Rabbinic Literature, though written at a later date, incorporate ancient elements capable of shedding light on the Israelite traditions that were extant in the earliest period, I made use of them too, in my researches." Where Cassuto cites midrashic sources to clarify the biblical context, Rober Alter will often introduce midrashic elements to elucidate an otherwise obscure artistic intention of the biblical text. See, Alter, *Art*, 30. "What is especially interesting about this miniature dramatization in the Midrash is where it might have come from in the text, for the literary insights of the midrashic exegetes generally derive from their sensitive response to verbal cues." For my sake, I believe that midrash can function simultaneously to both provide a contextual background for the Bible and transform its indeterminacy into art.
- 173 One of the first exegetes to note the migratory dynamic of populations within these first 11 chapters of Genesis was Nachmanides. In his first comment to the Bible (Gen 1:1, s.v. בְּרֵאשִׁית בְּרָא אֱלֹהִים), he lists each of the exilic peregrinations of these chapters, from Adam leaving Eden, to Cain forced to wander, to the generation of the flood exiled from the entire earth, and the builders scattered from Babel, and concludes that this pattern represents a universal moral red-line to landownership. "If they [Israel] should sin, then He [God] will expunge them from the land [of Israel], as He once expunged the nations that had dwelled there beforehand." *Exegesis of the Ramban on the Torah: Vol. 1: Genesis, with Explanation of the Tov Yerushalayim* [Heb], ed. Pinchas Yehuda Leiberman (Jerusalem: Tuviah, 1997), 23.
- 174 Scott, *Against the Grain*, 122. "Not just in Mesopotamia but virtually everywhere, it seems, early state battens itself onto this new source of sustenance. The dense concentration of grain and manpower on the only soils capable of sustaining them in such numbers—alluvial or loess soils—maximized the possibilities of appropriation, stratification, and inequality ... It is, of course, only in the context of rich

- soils and available water that the ecological capacity for the further intensification of agriculture and population growth was possible, and thus it was only in such settings that the first bureaucratic states were likely to arise.”
- 175 Scott, *The Art of Not Being Governed*, 43.
- 176 Scott, *The Art of Not Being Governed*, 41.
- 177 Scott, *Art*, 48.
- 178 Scott, *Art*, 57.
- 179 For these dates, see Scott, *Against the Grain*, 5 & 26. Also, Gwendolyn Leick, *Mesopotamia: The Invention of the City* (London: Penguin, 2001), xxi-xxii, and Otto T. Solbrig, and Dorothy J. Solbrig, *So Shall You Reap: Farming and Crops in Human Affairs* (Washington D.C.: Island Press, 1994), 38–40, with specific reference to early wheat domestication, specifically einkorn wheat (*Triticum monococcum*) and emmer wheat (*T. dicoccoides*).
- 180 Ian Morris, *Why the West Rules for Now: The Patterns of History and What They Reveal About the Future* (London: Profile Books, 2011), 97.
- 181 That the biblical city of Hanoch fits the generally accepted definition of the early city-state is conjectural. The etymology for the Hebrew word for city, עִיר, is entomologically linked to the Sabeian word for “fortified height,” *BDB* (Peabody, MA: Hendrickson Publishers, 2012), 746; and “to be awake,” as cited in Samson Raphael Hirsch, *The Pentateuch: Genesis*, trans. Isaac Levy (New York: The Judaica Press, 1971), 109; these lexical links to fortifications and alert activity may indicate that it was the presence of military walls or ramparts that defined a cityscape. Other modern biblical scholars have noted the link between agriculture (Cain, the farmer), concomitant sedentism, and urbanism. See Claus Westerman, *Genesis 1–11: A Continental Commentary*, trans. John J. Scullion (Minneapolis: Fortress Press, 1984), 327. “The founding of the city is reported in the genealogy as the first achievement of civilization. When this is ascribed to the son of Cain, it can only mean in the context that the founding of the city is considered part of sedentary civilization, the basis of which was agriculture (Cain, the farmer). The passage shows that Israel preserved the memory that the building of cities was something that happened prior to and outside its own history.” See also Nahum Sarna, *The JPS Torah Commentary: Genesis* (Philadelphia: The Jewish Publication Society, 1989), 36. “Furthermore, the notion that the farmer originated the city is consistent with the fact that the rise of urban centers historically followed in the wake of the development of agriculture.”
- 182 Scott, *Against*, 118. Also, Hans J. Nissen, *The Early History of the Ancient Near East, 9000–2000 BC* (Chicago: University of Chicago Press, 1993), 127.
- 183 Uruk is thought to be the biblical city Erech in Gen 10:10, one of the four Mesopotamian plain cities where Nimrod’s reign began. See Sarna, *JPS Torah: Genesis*, 74. These four cities were all located in “valley of Shinar,” of which the general scholarly consensus is that Shinar refers to Southern Mesopotamia, perhaps a dialectic form of ‘Sumer.’ See Yigal Levin, “Nimrod the Mighty, King of Kish, King of Sumer and Akkad,” *Vetus Testamentum* 52.3 (2002): 352.
- 184 Gwendolyn Leick, *Mesopotamia: The Invention of the City* (London: Penguin Press, 2001), 35.
- 185 Scott, *Grain*, 124. “The late Neolithic resettlement camp located on rich alluvial soil was the already existing nucleus of people and grain from which a state could be elaborated ... only the richest soils were productive enough per hectare to sustain a large population in a compact area and to produce a taxable surplus ... Alluvia, the historic gift of the annual floods of the Tigris and Euphrates and their tributaries were the sites of state making in Mesopotamia: no alluvium, no state.”
- 186 Leick, *Mesopotamia*, 61.

- 187 Scott, *Grain*, 158. See also Robert K. Englund, "The Smell of the Cage." *Cuneiform Digital Library Journal* 4, (2009). <https://cdli.mpiwg-berlin.mpg.de/articles/cdli/2009-4>: "The sign *geme₂* ('female slave') appears in ED IIIa texts (Fara period, ca. 2600 BC) ... This component KUR of the compound sign has in all discussion of *geme₂* been considered a geographical qualifier, thus literally 'mountain-woman,' where, with ample textual justification, the chattel slaves of early Babylonia were believed to have been purchased, or taken, by force or threat of force, from the mountains, or more generally foreign lands, to the east or north of the Mesopotamian alluvium." And also I.J. Gelb, "Prisoners of War in Early Mesopotamia." *Journal of Near Eastern Studies* 32, no. 12 (1973), 77–78. Gelb cites a set of administrative texts from the reign of Bur-Sin, the third king of the Third Dynasty of Ur (2052–2043 BCE) in which the overwhelming majority of the 197 prisoners of war have names that indicate that they are not indigenous to Mesopotamia, but rather reflect the "ethno-linguistic situation of the mountain regions east and north of Mesopotamia."
- 188 The method is derived from Cassuto, *Genesis I*, 1. "The aim of this commentary is to explain, with the help of an historico-philological method of interpretation, the simple meaning of the Biblical text, and to arrive, as nearly as possible, at the sense that the words of the Torah were intended to have for the reader at the time when they were written."
- 189 See Cassuto, *Genesis II*, 174. Gunkel describes Gen 10 as "a first step in Ethnography," as cited by Cassuto, *Genesis II*, 174.
- 190 Alter, *World*, 45. "The literary imagination qualifies or challenges prevalent ideological assumptions not necessarily—in fact, not usually—because it has an ideological program of its own but, on the contrary, because its inventive, associative, and even formal engagement with its own verbal, narrative, and referential materials leads it to peer over the side, or at the underside, of things as they are ordinarily seen."
- 191 Here, and throughout this book, I am using a fluid definition of "stateness" defined by James Scott as: "an institutional continuum, less an either/or proposition than judgement of more or less. A polity with a king, specialised administrative staff, social hierarchy, a monumental center, city walls, and tax collection and distribution is certainly a 'state' in the strong sense of the term." As Scott notes, states of this sort came into existence in the last centuries of the fourth millennium BCE, and are well attested at the latest by the strong Ur III territorial polity of 2100 BCE. James Scott, *Against the Grain*, 23–24.
- 192 Cassuto, *Genesis I*, 273. For the relatively late compositional character of the "Table of Nations" list, see, B. Oded, "The Table of Nations (Genesis 10)—a Socio-Cultural Approach," *Zeitschrift Für Die Alttestamentliche Wissenschaft* 98 (1986): 16. "It is widely recognized that Genesis 10 is a late composition, probably not earlier than the 8th century BCE. The Table is an artificial composition in a genealogical pattern reflecting no reality in any historical period."
- 193 Scott, *Against the Grain*, 142–144. "The earliest administrative tablets from Uruk (Level IV), circa 3,300–3,100 BCE, are lists, lists and lists—mostly of grain, manpower and taxes ... A preoccupation at Uruk IV and later in other centers is the population roll ... The entire exercise in early state formation is one of standardization and abstraction required to deal with units of labour, grain, land, and rations. Essential to that standardization is the very invention of a nomenclature, through writing, of all the essential categories—receipts, work orders, labor dues, and so on."
- 194 Scott, *Against the Grain*, 147.
- 195 On the prevalence of such pre-biblical oral traditions, with specific reference to their capacity to intrude even within the genealogical lists of Genesis, see

Cassuto, *Genesis II*, 185–6. “Another kind of material emanates from epic poetry, the ancient heroic poesy. Just as in the Sumerian King List there are often interpolated, among the official data concerning the names of kings, the cities of their kingdoms and the length of their reigns, notices about the valorous deeds of the most famous monarchs, drawn from the epic poetry of Mesopotamia, so there are introduced here [Gen 10] references to the mighty acts of Nimrod from an epos that spoke of him and his deeds at length.”

196 Cassuto, *Genesis I*, 251.

197 Scott, *Art of Not Being Governed*, 238.

198 Scott, *Art of Not Being Governed*, 257.

199 Scott, *Art*, 254.

200 Scott, *Art*, 254.

201 As cited in Ernest Gellner, *Saints of the Atlas* (London: Weidenfeld and Nicolson, 1969), 26.

202 Michael Khodarkasky, *Where Two Worlds Met: The Russian State and the Kalmyk Nomads, 1600–1771* (Ithaca, Cornell University Press, 1992), 47.

203 Gen 10:5.

204 Gen 10:20.

205 Scott, *Art*, 263.

206 Scott, *Art*, 257. See also Hjørleifur Jonsson, *Mein Relations: Mountain People and State Control in Thailand* (Ithaca: Cornell University Press, 2005), 19–34.

207 Scott, *Against the Grain*, 236.

208 Edward Leach, *The Political Systems of Highland Burma: A Study of Kachin Social Structure* (Cambridge: Harvard University Press, 1954), 40–41.

209 Scott, *Art*, 273.

210 Here, I am tweaking a social theory developed by Hjørleifur Jonsson, who compares the centripetal valley systems, which flatten culture, in contrast to the centrifugal hill systems, which appear to diversify identity. Thus, the valley state creates a “rather uniform” closed ranked system, whilst the highlands exhibit a “bewildering variety” of open-ranked tendencies. Hjørleifur Jonsson, *Shifting Social Landscapes*, 218. As cited by Scott, *Art*, 251. One way to visualise the first 11 chapters of Genesis as mainly centrifugal is to consider the fate of city-state builders against the fate of non-sedentary actors. Cain, the murderer, builds a city, and his offspring who inhabit it are lost in the flood. This marks the conclusion of the Bible’s first nod towards centripetal state-making. Noah, contrastingly, is pushed centrifugally out of the flooded valley and into the hills of Ararat. Then, the city of Babel is built by a centripetally ingathered workforce, only to be immediately destroyed. This allows the Genesis account to turn to Abraham who wanders, centrifugally away from the valley core of Ur, to reside in the Palestinian highlands.

2 Paradise Lost

In Chapter 1, I suggested that the curse of farming within the Genesis narrative was a critique against state-based systems of grain domestication. This chapter seeks to refine that claim on two fronts. First, I suggest that the Eden tale from Genesis 2:4 to 3:24, which spans an arc from the formation of Eden to its ultimate collapse, can be read as the story of the rise and fall of humanity's first bout with farming. Though one must remain wary of binding the Genesis trajectory too tightly to historical dates and specific locations, I will argue that the robust link between Eden and the first domesticated grain crops might infer that Eden is a tale about the Anatolian highlands, set approximately 6,000 to 10,000 years ago, in the period known as the late Mesolithic or early Neolithic.¹ This place and this time mark ground zero of the earliest extensive use of semi-permanent agriculture. Though it would take another 4,000 years before these first grain cultivators would abandon nomadic hunting and gathering and become the earliest agro-pastoral societies, this liminal four-millennia gap where farming begins to take root is precisely where Eden is uprooted.² The uprooting of Eden is the second discursive touchstone of the chapter. It takes in Adam and Eve's devourous eating from the tree of knowledge, the divine curses of labour pains and farming, and the banishment from paradise that follows. These interlocking events are all linked with the domestication of wheat and the devastating environmental impacts of agriculture. In this chapter, I will argue that Eden was located in the approximate place and time where farming first emerged on the world stage, that Eve and Adam generally represent the world's first farmers, the forbidden fruit was the world's first domesticated wheat, and the fall from paradise, the world's first ecocide.

Göbekli Tepi and Humanity's First Ecocide

In the low-lying Taurus Mountain range of the Anatolian highlands in eastern Turkey squats Göbekli Tepe, Turkish for the "pot-bellied hill." It is a nondescript and barren scrubland, gnawed clean of all but the most hardy grasses by millennia of overuse. But underneath the yellowing thorn bushes and hard-packed earth lies a ruin of such staggering ancientness and complex

design, that it leaves the experts agog. “Göbekli Tepe is too extraordinary for my mind to understand,” admits archaeologist Steve Mithen.³ “Göbekli Tepe changes everything,” adds Ian Hodder.⁴

What delineates Göbekli Tepe so uniquely is its age. Carbon dating reveals that this complex of large standing stones is at least 12,000 years old.⁵ That places it approximately 7,000 years prior to more popular constructs of early civilisations such as Stonehenge or the Pyramids in Giza. The carved stones of Göbekli Tepe are simply so ancient that at 10,000 BCE, the term “civilisation” seems a misnomer. The human population then lived as nomadic hunter-gatherers, which archaeologists had previously assumed made massive cooperative ventures impossible. “It is pre-pottery, pre-writing, pre-everything,” wrote journalist Tom Knox after a visit to the site in 2006. “Göbekli hails from a part of human history that is unimaginably distant, right back in our hunter-gatherer past.”⁶

But it is not just age that sets this site apart. There is a passionate creative outpouring carved into the slender T-shaped stone pillars, some over 10 feet in height. The artists depicted an ecology utterly foreign to the barren and thorny landscape of today’s site. The stones are alive. Cranes and ducks of granite are depicted peeking shyly from the long marsh grasses. A boar runs down the side of another pillar, snout still wrinkled in an eternal grimace. Sinuous serpents, crayfish, and lions are all carved with an attention to detail that intimates a strong familiarity with these creatures. An abundance of animal and fowl bones of various species found at the site, dovetailed with soil samples and pollen analysis, indicates a complex and healthy ecological milieu⁷ (Figure 2.1).

Sitting at the headwaters of the Tigris and Euphrates River system, Göbekli Tepe was once “a paradisiacal place” according to the lead archaeologist at the site, Klaus Schmidt.⁸ Herds of game grazed in the meadows, ringed by woodlands of oak and wild pistachio. According to Schmidt, it was the very fecundity of the place that caused the fall of Göbekli. As these Neolithic people gathered together at the grounds of a prime hunting and gathering site, their increased numbers meant increased mouths to feed. “To build such a place as this, the hunters must have joined together in large numbers ... But they found they could not feed so many people with regular hunting and gathering. So I think they began cultivating the wild grasses on the hills.”⁹

In order to discover the true origin of our fundamental crops, such as domesticated wheat or chickpeas, archaeologists will employ a number of methods. One method, according to biologist and environmental historian Jared Diamond, is to “examine a map of the geographic distribution of the crop’s wild ancestors and to reason that domestication took place in the area where the wild ancestor occurred.”¹⁰ For example, wild einkorn wheat with the closest genetic match to the domesticated variety is located in the Karacadag mountain range of southeastern Turkey.¹¹ These regional genetic maps can then be collaborated, refined, and dated with support from archaeological indicators that trace the domesticated debut of that crop in the field. The earliest

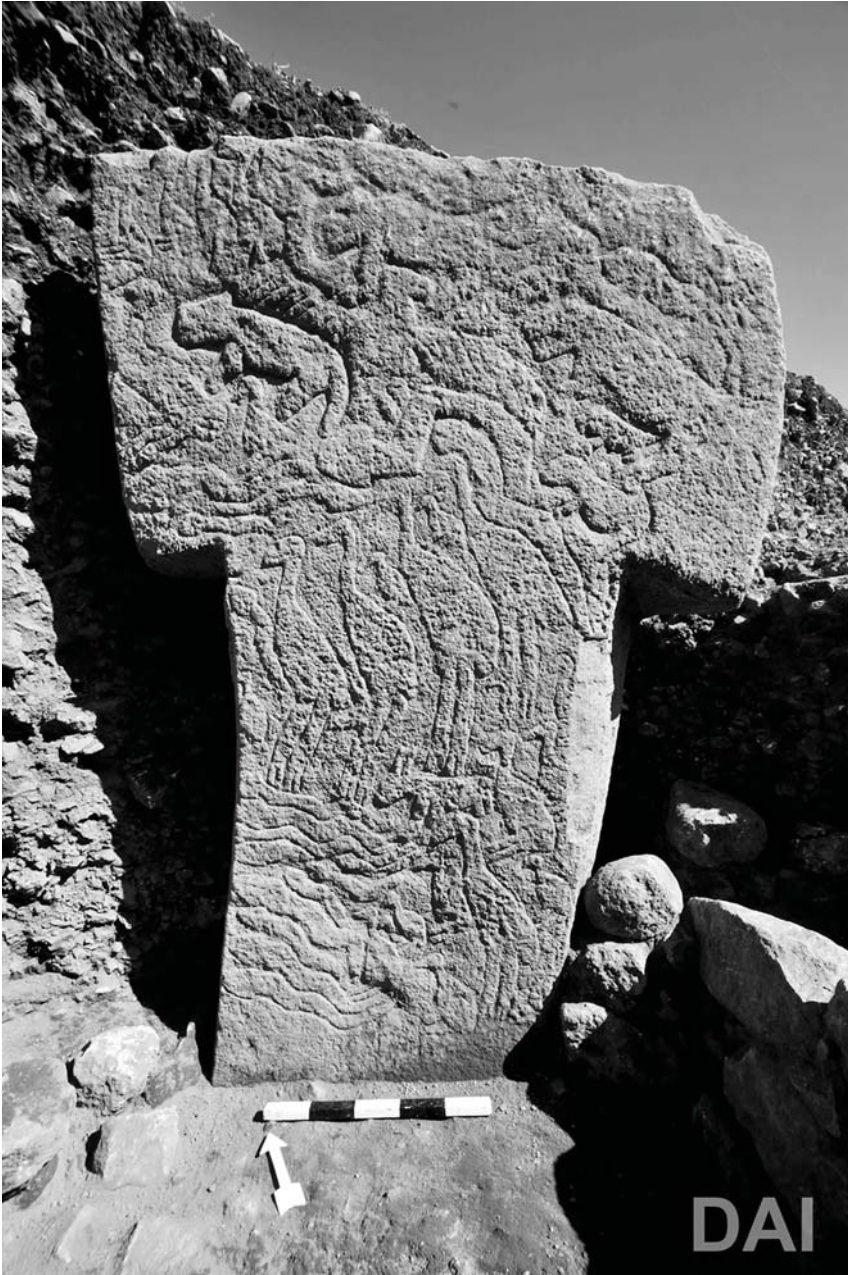


Figure 2.1 One of Göbekli Tepe's carved pillars (Pillar 56 within enclosure H). Photo courtesy of the German Archaeological Institute, Nico Becker.

examples of proto-domesticated einkorn wheat were discovered amongst the charred plant remains from Neolithic hearth sites along the Euphrates River tributaries in the foothills of the Karacadag, from approximately 8000 BCE, and within 100 miles of Göbekli.¹²

Much more than a particular strand of wheat emerges in its earliest domesticated form in the Anatolian hills and valleys. Chickpeas, olives, rye, barley, emmer wheat, and oats, as well as the sheep and the goat, are all posited for first domestication within the small geographic radius of the Levant. Though there is currently understood to be between five to nine regions dotted around the world in which independent food production emerged *sui generis*, the oldest by far is Southwest Asia.¹³ By 8000 BCE, this curve of the Fertile Crescent hit first upon the agrarian trick that would in time encompass the globe. Thus, there is no better place in the world to measure the very long-term effect of concentrated human intervention upon a landscape than the cradle of domestication itself.

The ecological results of this first foray into agriculture are dismal.¹⁴ The increasing population encouraged the inhabitants to adopt ever more intensive forms of agriculture, infringing upon wider and wider swaths of virgin forests.¹⁵ Without the trees and foliage to root the ground in place, the continuous tilling of the land left the fertile topsoil susceptible to erosion down the mountain slopes, leaving behind an underlying layer of nutrient deficient rock and clay substrata.¹⁶ Gone forever was the “paradisiacal place” that was Göbekli Tepe, replaced with a landscape of stones, thorny bushes, and a lone mulberry tree which the local Kurdish shepherds consider holy. To use a term coined by Jared Diamond in his book *Collapse*, it was quite possibly humanity’s very first ecocide¹⁷ (Figure 2.2).

When I argue that Southwest Asia marked the site of civilisation’s first evidential fall into unintended ecological suicide, this is not to imply that this Anatolian society was morally worse off than earlier cultures or neighbouring ones. Human beings have a long penchant for destroying flora and fauna for their own gains. Though controversial, and often plagued with political associations, current and prevailing paleontological theories suggest that pre-farming hunters and gatherers either caused or exacerbated the mass extinctions of large land animals in America, Australia, and throughout the Pacific Rim islands.¹⁸ What is unique in the Taurus mountain range is the whole-scale collapse of the ecological system. Whilst the early colonisers of the Americas, for example, probably hunted to extinction the large ground sloth, the native horse, and other ruminants, those niches were soon filled by different creatures more suitably adapted to the punishment of new hunting technologies. In the region of Göbekli Tepe, individual animals were not removed from the natural habitat. Rather, the ecological niches themselves had been destroyed by the unintended consequences of over-farming, specifically, erosion, salinisation, and soil fertility loss.¹⁹ Hunter-gatherers may have thrown a few species out of the ecological basket; the farmers of Anatolia burned the basket, contents, and all. That is what I mean by ecocide.



Figure 2.2 The mound of Göbekli Tepe viewed from the south. Photo courtesy of the German Archaeological Institute, Klaus Schmidt.

The deforestation of fragile headwaters is an oft-told tale the world over. While Göbekli Tepe may have been the first human-induced ecocide, such a pattern of ecological mismanagement has become so widespread across history and globe that Jared Diamond could cogently argue that ecocide could be considered amongst the primary threats to any society, both past and present.²⁰ From Easter Islanders to the Greenland Norse, to the modern state of Haiti, the pattern of Göbekli Tepe would appear horribly unlearned. And that, I intend to argue, is a most poignant irony. The most famous book in the world begins with the story of an ecocide in the Taurus mountain range, at the headwaters of the Tigris and Euphrates Rivers. That well-read book is the Bible and that ecocide is commonly referred to as the banishment from the Garden of Eden.

Mapping Eden

Let us turn to the second chapter of Genesis, and the exacting biblical location of the Garden of Eden:

A river flowed out of Eden to water the garden; from there it divided into four headwaters.²¹ The name of one is Pishon. It surrounds the entire land of Havilah where there is gold. And the gold of that land is good, also there are pearls and precious stones. The name of the second is Gihon. It surrounds the land of Kush. The name of the third river is the Tigris,²² which flows to the east of Assyria. The fourth river is the Euphrates.²³

The Bible begins the story of Eden with a location on the map.²⁴ Yet, the geography is filled with lacunae. The Tigris and Euphrates emerge from springs in the Anatolian highlands,²⁵ but the location of the first two rivers mentioned, the *Pishon* and the *Gihon*, and their respective watersheds of *Havilah* and *Kush*, have been variously located in Turkey,²⁶ India,²⁷ Egypt,²⁸ Ethiopia,²⁹ and Israel.³⁰ This bewildering array of potential locations has spurred some modern scholars to conclude that Eden only describes a mythical, supernal realm. “A real locality answering to the description of Eden exists and has existed nowhere on the face of the earth,” writes Skinner decisively.³¹

Yet, there are two reasons to pull Eden down to earth. First, as Hiebert will note repeatedly throughout his scholarly review of the Yahwist’s account of Genesis, the narrator ‘J’ is consistently oriented towards a non-mythical, tangible landscape. Even the Yahwist’s God, argues Hiebert, “lives a very earthly life,” entirely reduced to activities within the “terrestrial sphere.”³² “J’s treatment of the *primaeval* era is concrete and down-to-earth, drawing its details from ordinary human experience. J is not constructing an other-worldly, ‘mythic’ realm within this narrative.”³³ Second, irregardless of the Pishon and Gihon ambiguity, the headwaters of the Tigris and Euphrates would serve as precise and well-known geo-locators for a Levantine readership. Thus, it would be as difficult to imagine the story in an entirely mythic sphere, as to pen a story about the upper reaches of the Thames River today and then have future scholars debate on whether such a place ever existed, simply because some of the Thames tributaries remained obscure to them.

Though Hiebert is mainly dismissive of a heavenly or mythic garden locale, the ambiguity of the Pishon and Gihon watersheds, coupled with the Yahwists’ “parochial perspective,”³⁴ forces him to dismiss the Anatolian highlands in favour of a more local palestinian biome for Eden. Based primarily on a passing simile in Genesis 13:10 between the Jordan Valley desert oasis of Sodom and Gomorrah with the “garden of YHVH,” Hiebert concludes that Eden was a spring-fed orchard oasis in a desert valley, perhaps even in the Jordan Valley itself.

There are a number of fundamental difficulties in mapping Eden upon an oasis per Hiebert’s suggestion. First, in order to locate the Garden of Eden in a low-lying desert valley, Hiebert is forced to reconfigure and submerge the highland headwaters described in Genesis 2:10–14 into a fantastical subterranean sea that waters the Eden oasis in some inchoate manner with the four major rivers from below. “The only explanation for the relationship between such varied watercourses as the Tigris, Euphrates, Gihon, and Pishon may lie in the ancient conception of a subterranean sea linking all fresh water sources. According to this view, a deep fresh water current ran below the surface of the earth and all the springs and rivers flowing with fresh waters were believed to have their origin in this underground stream.”³⁵ This subterranean source of all rivers creates a very forced reading of Genesis 2:10. According to Hiebert’s geographical model, a river would emerge, presumably from beneath the earth, water the desert garden, and then dip back below the surface to pop up again at

various other locations in Anatolia, Egypt, and Israel as the headwaters of four other waterways.

Furthermore, it is unclear why Hiebert does not read the alliteration to a “Garden of YHVH” in the story of Sodom and Gomorrah as simply alliteration. “Lot raised his eyes and saw that the whole region of the Jordan, all the way to Zoar, was entirely irrigated *like* the Garden of YHVH, *like* the land of Egypt ...”³⁶ Linking Sodom and Gomorrah to both Eden and Egypt may only imply that Sodom was also irrigated and lush. Or that all three ecosystems shared similar trajectories, for example, the ironic fate of collapse despite the constancy of their respective water supplies.

In light of Hiebert’s reliance on other biblical passages for edenic references, it is peculiar that he makes no mention of Ezekiel’s repeated claims that Eden is located on a mountain³⁷ as opposed to a valley.³⁸ Presumably, this is because this prophetic work is not attributed to the Yahwist³⁹ and therefore irrelevant for a true mapping of the Yahwist’s Eden. However, Hiebert has no qualms in using other passages from Ezekiel to support his own oasis reading,⁴⁰ which indicates that he will engage with non-Yahwist biblical passages when they work to his advantage. Likewise, the Yahwist and Ezekiel accounts may both be variant derivatives from a pre-biblical “epic tradition,” as Cassuto hypothesises,⁴¹ which would imply that Eden’s location upon a mountain is of ancient stock.⁴²

Furthermore, it remains unclear why Hiebert insists that the Yahwist cannot narrate a story set beyond his local milieu. Yet, Hiebert maintains that “placing Eden in a distant, foreign, terrain would be in some tension with the local, native backdrop of the pre-flood narrative as a whole.”⁴³ Perhaps, Hiebert is unaware of the intensive trading culture of the time⁴⁴ that linked communities thousands of miles apart in goods and, one can venture to assume, in traveller’s tales of the other and their landscape.

Finally, Hiebert never seriously dwells on the ruin of Eden. “With its perennial water source and bountiful orchards,” Hiebert optimistically intones, “the valley oasis was a setting in which the major difficulties of hill country agriculture were substantially diminished. For this reason, the valley oasis would provide an appropriate image for the comfortable existence in Eden that preceded the rigors of dryland farming.”⁴⁵ Because Hiebert’s edenic oasis is never destroyed,⁴⁶ there is no threat of fragility, no uncertainty or violence, no fall. Paradise is never lost. Despite Hiebert’s attempt to reclaim the ecological milieu of the biblical agrarian community, he never quite grasps that the subtext of the creation narrative is the link between farming and the collapse of Eden.

A mountain zone Eden at the headwaters of the Tigris and Euphrates catchment basin would locate the biblical garden at the epicentre of the Neolithic. And there are a handful of additional scriptural and extra-biblical indicators placing Eden in the hills above the Mesopotamian alluvial plain. For instance, the land of *Kush* may refer to the ancient province or city of Kish, located east of Babylon in the Tigris and Euphrates plain and referenced in the Sumerian King List.⁴⁷

The appellation Kush appears again within the biblical account, personified as the son of Ham, and the grandson of Noah.⁴⁸ The Bible goes on to name Havilah as one of the sons of Kush.⁴⁹ The genealogical father–son relationship could indicate close territorial proximity. Both the anthropological⁵⁰ and biblical⁵¹ records are replete with overlapping familial and locational relationality. Thus, when the Bible tells us that the Pishon River encircles a “land of Havilah” and the Gihon a “land of Kush” in Genesis 2:13 and 2:14, we may assume from the genealogical connection that we should look for relatively close, perhaps even contiguous, areas for Havilah and Kush, and by assumption, a shared source for the two rivers that feed these neighbouring lands.

If we conjecture further between genealogical and geographical proximity, we can note that another of Kush’s sons, Nimrod, was the first ruler of Babel and Akkad, amongst other surrounding city-states,⁵² which would again link Kush and Havilah to Nimrod in northern Mesopotamia. This possible mapping would hint back to a highland Eden upstream from these cities built in the alluvial valley and fed by the Tigris, Euphrates, and various tributaries.

The Taurus hills of Anatolia, where wheat was first domesticated, shares yet further links with the Garden of Eden. A minor kingdom or city called Bet Adini (a House of Eden) located on a bend of the Euphrates River, within 50 miles of Göbekli Tepe, is mentioned in Assyrian texts of 8500 BCE.⁵³ The author of 2 Kings chastises the “people of Eden who were in Tel Assar,”⁵⁴ a city located in modern-day Syria, at the foot of the Taurus range.⁵⁵

Assuming that the preliminary mapping of Eden in the Anatolian highlands is a viable theory, we can now ask why the narrator of the Genesis account chose to begin in just this place. The Tigris and Euphrates Rivers, and the uplands from which these rivers emerged, were familiar locales. Modern readers often overlook the salient fact that the Levantine audience knew of these rivers directly, perhaps even personally. Yet, the headwaters of the Tigris and Euphrates system contained no edenic garden and no forest of fruit trees. What it did contain, as it still does, is a biome of perennial grasses, sage and mastic bushes, dotted with the occasional oak or stunted pine. Those first regional listeners may have been struck by an ecological dilemma. How had Eden disappeared from that particular hillside? Why had paradise become this scrubland? The mapping of Eden is not an extraneous detail of the narrative, but an engine to drive the plotline of the story. From the beginning, the biblical author is raising ecological questions. A nearby mountain range now barren, once held the luxurious trees of Eden. From the beginning of the textual encounter, Eden is a tale of fragility, because a readership familiar with the existing terrain was implicitly aware that *Eden would not last*.

Mapping Eden in Time

In order to delve further into the transience of the Garden of Eden narrative, it is instructive at the onset to recall an important textual point. The Eden story begins in the second of two tales of creation offered by the biblical text

of Genesis. Due to the twice-told nature of creation, the reader can glean much in exploring the contrast and tension between the two juxtaposing stories. In holding up the first story against the second version, the tremulous condition of Eden emerges more completely.

The first chapter of creation offers seven⁵⁶ ordered days of world making in a precision march of progression and distinction. “There was evening and there was morning ...”⁵⁷ follows like clockwork from one day to the next. Earth and heaven, land and sea, are all apportioned divine boundaries, and each species of plant, fish, fowl, and animal are created to fill a unique niche, “according to its kind.”⁵⁸ The second chapter of Genesis offers another version of creation all together.⁵⁹ No march of progress, no structured plan, this time around there is lack and need and emptiness. Here is the beginning of this second account,

These are the unfoldings of heaven and earth in their creation, on the day the Lord God made earth and heaven. And all the field shrubs (אֵדֶם) —not yet on the earth, and all the field grasses (מִאֲדָה) —not yet sprouted, because the Lord God had not brought rain on the earth and the human (אָדָם) was absent from working (עֲמָלָה) the humus (רֶשֶׁף).⁶⁰ And a spring rose from the earth and watered the face of the humus. And the Lord God formed the human out of dust from the humus, and breathed into his nostrils the breath of life, and the human was alive. And the Lord God planted a garden ...⁶¹

These five verses from the second chapter of Genesis encapsulate the DNA of two thematic strands of the edenic narrative that I will revisit throughout this and subsequent chapters. The first trope is the linguistic dovetailing in the Hebrew between human and humus, *Adam* (אָדָם) and *adamah* (אֲדָמָה).⁶² We will have occasion to return to the consequences of Adam (the human) emerging from the *adamah* (soil/humus/ground) and the recurring verb of *avodah* (עֲמָלָה) (work/enslavement) that often connects them.⁶³ For now, however, I want to focus our attention on the second emergent strand contained within these verses. For at this stage, the Bible injects a radical departure from the ordered and fecund world making of Genesis, chapter 1, to a chapter 2 that opens with a lack of grass and bush, the space of the “not-yet.”

For the rabbinical exegetes, whose faith demanded a narrative continuity between the first and second creation accounts, the herbage state of the “not-yet” only implied that whilst the world’s flora was all created on day three,⁶⁴ the grasses and bushes waited at the cusp of the earth for the first farmers to pray for the first rains, before visibly sprouting out of the ground.⁶⁵

For more modern scholars, these pithy utterances of the not-yet could have multiple interpretations. Julius Wellhausen believed that these lines formed a vestige of a longer account on the nothingness that preceded creation, shortened by a later priestly reviser.⁶⁶ Gerhard von Rad concluded that the “not-yet” simply “sketches the original state as a desert in contrast to the sown.”⁶⁷

Whilst Theodor Gaster saw within these lines a reflection of the autumn season in the Mediterranean hills, when the earth is at its driest, the rains have not-yet fallen, and the only water sources are the mountain springs.⁶⁸

Each of these commentaries, however, is built on the assumption that the shrubs (אֲרָבִים) of the field, and the grasses (מִצְמָחִים) of the field of Genesis 2:5 connotes the entire vegetable kingdom, and thus we have a narrative that begins without any plants or trees on earth.⁶⁹ Yet, as Cassuto notes cogently, the text is explicit in which plants are lacking, implying that others, such as trees, are alive and well. “When the verse declares that these species were missing, the meaning is simply that *these* kinds were wanting, but no others.”⁷⁰ And here, Cassuto makes a startling claim. The reason that אֲרָבִים and מִצְמָחִים specifically were not yet in the field is because these two plant types only come into existence after the curse of farming and the edenic expulsion. In fact, the מִצְמָחִים of the field is the identical expression used within the divine curse itself, “you shall eat the מִצְמָחִים of the field,”⁷¹ and אֲרָבִים, continues Cassuto, is synonymous with the accursed “thorns and thistles.”⁷² “These species did not exist, or were not found in the form known to us, until after Adam’s transgressions ... Man, who was no longer able to enjoy the fruits of the garden of Eden, was compelled to *till the ground* (Gen 3:23—the same phrase as in our verse [Gen 2:5]) in order to eat bread ... Thus the term מִצְמָחִים of the field comprises wheat and barley and the other kinds of grain from which bread is made; and it is obvious that fields of wheat and barley did not exist in the world until man began to *till the ground*.”⁷³ The act of farming, that cursed occupation, casts its foreshadow of grain and briar over creation at the very onset of its fragile beginnings. The “not-yet” of farming haunts Eden, like a menacing harbinger, from its very inception.

Following Cassuto’s insight, Genesis 2:5 emerges in a new light. Let me suggest the following interpretation of this verse, coloured by this foreknowledge of Eden balanced at the threshold of farming. “Before the expulsion, no fields of thorns and thistles (אֲרָבִים) were yet predominant on the earth, for farming had not yet denuded the forest ecosystem to leave a briar scrubland in its furrow; and no fields of wheat and barley grain (מִצְמָחִים) had yet sprung up, for God had not yet brought the rain and there was no human yet enslaved⁷⁴ to the humus.” In short, what Genesis 2:5 alludes to is the not-yet time before propertied fields⁷⁵ of mono-cropped grain and barren hills of thorny bush; the time before the curse. And a time before the earliest states had marked the earth (Figure 2.3).

One particular midrash⁷⁶ may support Cassuto’s reading. “Over here [in Genesis 2:5] it states: ‘All the field shrubs-not yet on the earth, and all the field grasses-not yet sprouted.’ And yet over there [Genesis 2:9] it states: ‘And from the ground the Lord God caused every tree to grow,’ [which would contradict the idea that the herbage required human tillage to sprout]? Rav Hanina answered: ‘Over there [in Genesis 2:9 scripture is describing the flora] of the garden of Eden, and here [in Genesis 2:5 scripture is describing the flora] of the domus.’”⁷⁷ The midrashist is struck by the incongruence between the self-populating plants in



Figure 2.3 Fragment of a palace or temple wall relief depicting ripe barley. Excavated in Egypt, ca. 1353–1356 BCE. Courtesy of the Metropolitan Museum of Art, New York.

Eden, versus the domesticated grasses that require human tillage. The dilemma is simply solved. Genesis 2:5 is describing the cultivated grains that are not-yet, but will soon demand a ceaseless toil in a sedentary plot of land. Whereas the Eden of Genesis 2:9 contains only the wild varieties. Over there, natural growth; over here, human cultivars.⁷⁸ Over there, Eden; over here, the farming curse. Within this midrash, the Neolithic revolution, that dividing line between wild and domesticated crops, is synonymous with the banishment from Eden.

Both Göbekli Tepe's ancient stones and the biblical account of Eden record the stories of humanity just before the first embarkation into agriculture. Both inhabit the temporal 'not-yet' before the Neolithic's discontents. They are both tales of soil fragility, of a forest paradise lost, and of hilly headwaters above the Mesopotamian plain that is accursed because it is farmed.

Though it is only a fleeting biblical verse, the implications of Genesis 2:5, I feel, are revolutionary. Here is an indication of a life before the onset of farming and the state. Here is the scriptural intonation, whispering of a time before field crops and deforestation. Before Adam is bound to serve the soil. It is worth stressing how deeply the Bible cuts against the grain of the status quo. According to Hiebert, this creation account stands in relief against "a common ancient Near Eastern tradition, present in Sumerian and Babylonian literature alike, that the human race was created to farm for the gods and to feed them with the results of their harvests."⁷⁹ Thorkild Jacobson emphasises that the Sumerian creation myths were a "charter for the state, specifically the city state," with its strong focus on the "potential of irrigation agriculture and the dependence of the latter on strong governmental organization for its success."⁸⁰

In Genesis, human beings are created not as farmers in the city, but gatherers in the hilly forests. And the singular divine injunction offered to the first human is not to eat from the forbidden tree of wheat.

The Forbidden Tree of Wheat

There is a startling midrash told by a rabbi of the second century CE⁸¹ that attempts to fill a lacuna in the Eden account. What species of tree was the “tree of the knowledge of good and evil”⁸² from which Adam and Eve ate? “Rabbi Meir said: ‘It was a tree of wheat.’ And the rabbis questioned him: ‘But the text says a *tree*!’ Responded Rabbi Meir: ‘Once wheat grew as tall as cedar trees.’”⁸³

Even in the loose reality of midrashic parlance, the association of the forbidden fruit with wheat is a non-sequitur. And the midrash will go on to quote other rabbis with actual arboreal options such as fig and grape.⁸⁴ However, Rabbi Meir’s odd assertion appears to take its cue from the convoluted biblical episode of the forbidden fruit itself.

First, there is God’s initial warning to Adam: “But from the Tree of Knowledge of Good and Evil, do not eat, for on the day you eat of it, you will definitely die.”⁸⁵ Yet Adam, we are told in a later biblical chapter, would go on to live for hundreds of years after his banishment from the garden.⁸⁶ Second, when God curses Adam for eating the forbidden fruit, he seems to miss the mark and curse the ground, instead of the man. “Damned be the earth on your account! You will derive food from it with anguish all the days of your life. It will bring forth thorns and thistles for you and you will eat the grains of the field. With sweat upon your face you will eat bread until your return to the ground, because you were taken from the ground. You are dust and to dust you will return.”⁸⁷ Why this curse upon the ground “on your account”? And why is the curse of eating the forbidden fruit the transformation of man into a perspiring farmer? Finally, why would the land, the fecund paradise, now become an ecology of thorns and thistles?

Rabbi Meir’s radical midrashic claim of the divine taboo on wheat is the lever by which each of these odd peculiarities in our story mesh together into a fictionalised and value laden recounting of that monumental shift in history that we now entitle the Neolithic revolution. If the tree of knowledge is a “tree” of domesticated wheat, then God’s very first dictat to humanity is a plea *against* agriculture. The curse of eating the forbidden “fruit” is simply the consequence of leaving the garden of mixed subsistence strategies and adapting full-time agrarian production.

Before we delve back into the problematic elements of the Eden narrative, I would like to address a more fundamental challenge to such a reading, namely how primitive hunter-gatherers with shorter lifespans, violent and retributive tribal warfare, and near ubiquitous use of infanticide could ever be considered as the edenic ideal.

At first glance, it may seem difficult to equate a hunter-gatherer lifestyle with a paradisiacal existence.⁸⁸ Jared Diamond predicates his own pivotal

study of why Fertile Crescent society adapted agriculture around 8500 BCE, by disabusing his reader of a basic misconception regarding hunter-gathering societies as Hobbesian places, where people lived lives that were “poor, nasty, brutish and short.”⁸⁹ In a brilliant perspective shift, Diamond contends that we, as modern affluent westerners, are enamoured of farming because we rarely do it ourselves. Therefore, we enjoy the fruits of agriculture without shouldering the majority of its considerable risks. “Most peasant farmers and herders,” Diamond contends,

Who constitute the great majority of the world’s actual food producers, aren’t necessarily better off than hunter-gatherers. Time budget studies show that they may spend more rather than fewer hours per day at work than hunter-gatherers do. Archaeologists have demonstrated that the first farmers in many areas were smaller and less well nourished, suffered from more serious diseases, and died on the average at a younger age than the hunter-gatherers they replaced.⁹⁰

Clive Ponting provides the example of the Kalahari bushmen of South West Africa, to illustrate the relative ease in which hunter-gatherers can obtain food resources. The mainstay of their diet is the highly nutritious mongongo nut obtained from a drought-resistant tree. It contains five times the calories and ten times the amount of protein of an equivalent amount of cereal crops, and half a pound (about 300 nuts) has the calories of two-and-a-half pounds of cooked rice, and the protein of almost a pound of beef. “Compared with modern recommended levels of nutrition the diet of the bushman is more than adequate: calorie intake is higher, protein consumption is about a third higher and there are no signs of any deficiency diseases.”⁹¹

In addition, notes Ponting, the amount of time it takes to collect this food is surprisingly luxurious, even to a modern standard. Women (who are responsible for gathering) tend to work about 1–3 hours a day, and the men will spend on average one week of hunting for every two to three weeks of leisure activities.⁹² As one bushmen told an anthropologist, “Why should we plant when there are so many mongongo nuts in the world?”⁹³ According to Ponting, similar works to leisure ratios have been found amongst the Hadza of east Africa and the Aborigines of Australia.

James Scott likewise points out the increased risk to food security involved in planting, tending, and guarding a narrow bandwidth of fragile and vulnerable crops. “An illness—of crops, livestock, or people, a drought, excessive rains, a plague of locusts, rats or birds, could bring the whole edifice down in a blink of an eye.”⁹⁴ Farming was a risky business in comparison to the relative subsistence safety of hunters and gatherers, who could adapt and move to exploit a diversity of food sources under similarly deleterious circumstances.⁹⁵

If hunter-gathering was more leisurely, nutritious, secure, and visibly less destructive of natural resources, then why was it ever abandoned in favour

of agriculture? Or put into the context of Göbekli Tepe, why didn't the inhabitants of the place simply abandon their wheat fields, after the first season of soil erosion and tremendous effort? What pushed them to continue despite the negative impacts of harder work, for less nutrition, with an increasingly damaging effect on their highland forest?

Jared Diamond, Clive Ponting, and other archaeologists who study the Neolithic revolution have theorised that at least one of the factors that contributed to the transition from hunter-gathering to farming was the reinforcing link between a rise in plant domestication and the rise in human population density. In Diamond's words,

The adaptation of food production exemplifies what is termed an autocatalytic process—one that catalyses itself in a positive feedback cycle, going faster and faster once it has started. A gradual rise in population densities impelled people to obtain more food, by rewarding those who unconsciously took steps toward producing it. Once people began to produce food and become sedentary, they could shorten the birth spacing and produce still more people, requiring still more food.⁹⁶

In another of Diamond's works, *The Third Chimpanzee*, this idea is stated even more succinctly and violently. "Ten malnourished farmers can still outfight one healthy hunter."⁹⁷

One of the first "curses" of a society adapting agriculture would be its dire effects upon mothers. Whilst nomadic hunter-gathering women generally raise one child at four year intervals, as a woman can only carry one infant at a time on long journeys, sedentary farming mothers can and do bear children every two years.⁹⁸ Birthing more children coupled with a protein poor diet⁹⁹ would sharply increase the risks of pregnancy and childbirth. Rabbi Meir's vision of a curse derived from the domestication of wheat fashions new context into the divine rebuke of Eve. "To the woman He said: I will greatly increase the suffering of your childbirth, in pain shall you bring forth children."¹⁰⁰

Cassuto notes the Bible's phraseological link between the forbidden tree (אֵץ) and the specific labour pains (מַעַה) that haunt this woman who cultivates and shares that first wheat harvest. "The very fact that scripture does not employ the usual phrases found in connection with the suffering of childbirth, like (מַהַ) [pain], (מַסַּ) [writhing, anguish], (מַעַק) [writhing, pang], (עַק) [distress], but chose expressions derived from the root (מַעַה), proves that it was with some specific intention—for instance, to allude to the tree (אֵץ)—that these words were selected."¹⁰¹

Following Cassuto, might we conjecture that it was the tree itself, specifically that cursed act of eating from a tree of wheat, rather than the diverse foodstuff of the edenic highland forest, that is the very cause of Eve's labour pain? Scott argues that the malnutrition that haunted the late Neolithic

communities disproportionately affected women, and specifically, women during their childbearing years:

Much of the malnutrition detected in what we might call ‘agricultural women’—for women, owing to blood loss with menses, were the most severely affected—seems due to iron deficiency. Preagricultural women had a diet that supplied abundant amounts of omega-6 and omega-3 fatty acids derived from game, fish, and certain plant oils. These fatty acids are important because they facilitate the uptake of iron necessary for the formation of oxygen-carrying red blood cells. Cereal diets, by contrast, not only lack the essential fatty acids but actually inhibit the uptake of iron.¹⁰²

The increasing risk of iron-deficiency anaemia for female farmers was coupled with a sharply increased birthrate, which would compound the risks of pregnancy, childbirth, and women’s and children’s overall health.

The “tree of wheat,” Rabbi Meir’s fantastic association, also dovetails firmly with the gendered sequence of tasting the forbidden fruit, where Eve experiments first. Evan Eisenberg notes that the Neolithic, like the turn towards the “tree” of wheat, was initiated by women. “In place of the herbs and fruits of paradise, man will eat bread. As we have said, the culture of barley and wheat—first for beer and toasted seedheads, then for bread—did apparently begin in the uplands of the near east, some ten thousand years ago. That it was woman, not man, who surely began it—being the foremost gatherer, she must have been the first farmer—may be dimly recalled in the story that it was Eve who first tasted the forbidden fruit, then handed it to Adam.”¹⁰³

Archeologist Ian Morris makes a similar case for a female first-crop cultivator. He cites the tendency in ancient graves at Levantine hunter-gatherer sites at the cusp of the Neolithic, in which men’s graves contain more spears and arrowheads, whilst women’s have more seed grinding tools.¹⁰⁴ For Morris, like Eisenberg, it was women’s greater intimacy with the growth cycles of the wild cereals in the Mesopotamian foothills that fingers them as the likely culprit for the first farming act.¹⁰⁵

The Hebrew root-word for anguish (מַעַה) is used in both the curse of Eve and the curse of Adam; anguish in child-bearing linguistically linked to anguish in eating the farmed produce of the cursed earth.¹⁰⁶ The 19th-century Hebrew philologist and rabbi, Sampson Raphael Hirsch, lingers over the fullest meaning of this word in his biblical commentary on Genesis. “The root מַעַה ... leaving something against one’s will, through force, harshness or violence. So that מַעַה is the feeling that we have to give something up that we would have liked to keep or to have attained.”¹⁰⁷ Hirsch’s definition of anguish provides a possible answer to the question I posed earlier. Namely, why would the people of the Taurus highlands adapt agriculture despite a negative return on their health and time, and a clear degradation to their vital natural resources? In the Genesis narrative, we are told that Adam and Eve live out their choice in anguish, which implies, in Hirsch’s lexicon, forced

against their will. They chose the forbidden tree of wheat, either consciously or unconsciously, because population dense farmers win wars over dispersed hunter-gatherers. Adam and Eve must farm or they will be overridden by those who farm around them. The anguish of child-birth and the anguish of eating of the domesticated grain are inexorably bound to each other in a vicious cycle of increasing food surplus catalysing increasing population that forces Adam and Eve ever further from Eden.

The semi-arid farmers reading Genesis would hear the quiet ecological and societal lament of the Eden story: maternal mortality, fickle crop yields, intensive labour, and all under the shadow of a paradise lost. The ecological consequences of farming are indeed dire: the ground will be accursed for the sake of the human. Deforestation and intensive ploughing create a chain reaction of soil erosion, saline seepage, and loss of natural flora and fauna which transformed the Taurus hills into a land of thorns and thistles. Read in this context, it is not the humans who left Eden, but an Eden that literally left the humans. From beneath the feet of those very first proto-farmers, the rains washed the fertile soil downstream, reducing Eden to a shared folk memory.

The curses God metes out to Eve and Adam are not fickle or capricious, but the natural consequences of adapting farming. God's curses are not vengeful punishments, but statements of impending facts. In the shade of the cultivated "tree" of wheat, Eve's birth intervals shorten whilst her body becomes anaemic on a grain diet. With the "fruit" of wheat in his gullet, Adam must sweat ever harder in a field of growing sterility, where only thorns and thistles prosper. The fall from Eden is not a cruel punishment, but the simple story of arable topsoil washing and falling downstream when the deep roots of Eden's trees, that solid architectural matrix of the earth, are slashed, burned, and cut to make way for the shallow and unstable root-tendrils of wheat that could never hold onto Eden's dirt.

Though speculative, might we conjecture that even the curse upon the snake is itself a direct outflowing from wheat domestication and permanent settlements? Ian Morris speculates that "permanent villages changed the rules for rodents. Fragrant delicious mounds of garbage became available 24/7 ... within a few dozen generations ... rodents in effect genetically modified themselves to cohabit with humans."¹⁰⁸ As vermin proliferated in homes and grain stores, their predators did too. Thus, unlike the rare encounter with a snake in the field or forest suffered by hunter-gatherers, these early farmers must have found a profusion of snakes underfoot. The divine curse on the serpent may simply represent this bitter and new relationship between farmers and snakes. "The Lord God said to the serpent ... I will put enmity between you and the woman, and between your seed and her seed; he shall crush your head, and you shall crave his heel."¹⁰⁹ Here, again, these are not curses, but consequences. Remarking on the transformative power of domesticating grains and livestock, Scott focuses our attention on the utter change wrought by the choice to establish sedentary farmsteads, what he refers to as the "domus." "The domus was a unique and unprecedented

concentration of tilled fields, seed and grain stores, people and domestic animals, all coevolving with consequences no one could have possibly foreseen. Just as important, the domus as a module for evolution was irresistibly attractive to literally thousands of uninvited hangers-on who thrived in its little ecosystem ... not a single critter emerged from its sojourn at the late-Neolithic multispecies resettlement camp unaffected.”¹¹⁰

What of the initial divine warning: “from the day you [Adam] eat of it, you will die”?¹¹¹ While Adam does not die on the day wheat is cultivated, the soil—the *adamah*—begins to die on that day. When Adam turns the edenic forest to a field of wheat, on that very day, the ground begins its inexorable spiral of decline. Where *adamah* goes, Adam will follow. Arthur Waskow calls this entwinement between the two Hebrew words of *Adam* and *Adamah* the “most profound Jewish statement about the relationship between human beings and the earth ... in English, this connection would be obvious only if the everyday word for ‘human being’ were ‘earthling’ or perhaps if the ordinary word for ‘earth’ was ‘humus’.”¹¹² Thus, the ground receives the full brunt of Eve and Adam’s terrible choice to farm. “Damned be the earth on your account!”¹¹³ is not a misfired curse; it is an eco-lamentation for the Neolithic. Genesis is the story of the entanglement of earth and earthling. From the ground Adam is made, from the ground he will return, and from the ground will he exact a heavy price for his bread.

Babes in Eden

I have, thus far, based my argument on the cultivation of the forbidden fruit of wheat followed by a subsequent exile from the highland ecosystem of Eden, on Malthusian assumptions. That is to say, that technological innovation, in this case food cultivation, causes a demographic spike, which would put an increasing strain on natural resources. Danish agro-economist and historian Ester Boserup sums up this line of causation:

Malthusian theory considered the emergence of agriculture to be the result of the fortunate invention of food production techniques some ten millennia ago, and the increase of world population in the following period to be an adaptation to the increase in the world’s carrying capacity obtained by this technological innovation ... a radical improvement in nutrition due to the invention of agriculture was thought to have caused a shift from stagnant population to rapid growth, as agricultural populations and agricultural methods invaded more and more of the world.¹¹⁴

As noted by Boserup, as well as Phillip Smith and Cuyler Young,¹¹⁵ James Scott,¹¹⁶ and Susan Pollack,¹¹⁷ this shift from extensive to intensive food procurement precipitated the resource degradation of the Hilly Flanks¹¹⁸ of the Levant and corollary abandonment of the first agricultural settlements. “In many regions,” Boserup observes, “destruction of topsoil at the upper run of a

river serves to fertilise areas at the lower run In other words, population pressure in a mountain region, by inducing cultivation of mountain slopes and overgrazing of mountain pastures, helps to create those ‘natural conditions’ which permit the use of intensive agriculture in broad valleys at the lower run of the rivers. This type of development seems to have occurred in ancient Mesopotamia.”¹¹⁹

The early chapters of Genesis would appear to narratively trace these early Neolithic stages down from the highlands. First, Eden’s location in the mountainous zone at the headwaters of the Tigris and Euphrates river system combined with the transgressive eating from the “tree” of wheat mirrors the general locale where emmer and einkorn wheat, the first domesticated crops, originated.

Second, one of the divine curses of a sedentary, grain-based diet is the suffering of more births over shorter spans to women of poorer health than their hunter-gatherer predecessors.¹²⁰ Yuval Noah Harari describes these salient curses of the “wheat bargain.” “With the move to permanent villages and the increase in food supply, the population began to grow. Giving up the nomadic lifestyle enabled women to have a child every year. Babies were weaned at an earlier age—they could be fed on porridge and gruel ... With time the ‘wheat bargain’ became more and more burdensome. Children died in droves, and adults ate bread by the sweat of their brow.”¹²¹ Notice how inexorably Harari binds together the Neolithic with the curses upon Adam and Eve. Adam’s sweaty engagement with the crop and Eve’s hard labour and high child mortality rate are the specific outflowings of that initial “wheat bargain.”

After wheat is domesticated upon the Hilly Flanks of Mesopotamia, and the rise in birthrate drives population density upwards, the ground itself becomes cursed by continual overuse. “And to Adam He said: Because you have listened to the voice of your wife, and have eaten of the tree of which I commanded you, saying: ‘You should not eat of it!’ Damned be the earth on your account! You will derive food from it with anguish all the days of your life. It will bring forth thorns and thistles for you and you will eat the grain¹²² of the field.”¹²³ Cassuto notes that in order to comprehend the full brunt of the biblical curse of the ground, we must determine the converse method by which God blesses the earth. There are numerous passages, such as Deuteronomy 33:13–15, where a blessed land is described as well-watered and fertile. “It follows,” concludes Cassuto, “that a land cursed is when it lacks water and fertility. In the garden of Eden, which was well watered and produced abundant fruit, man ate to his satisfaction ... from now on, the earth would yield its harvest to him only with difficulty and in meagre measure.”¹²⁴ What Cassuto recognises within the divine curse is that the hard labour of farming is intrinsically bound up with a depleted natural landscape. “Only through hard work,” Cassuto delimits the concept of toil, “will you [Adam] be able to obtain and eat the fruit of the earth.”¹²⁵ The curse, thus framed, resonates so harmoniously with the advent of cultivation that Scott, in his overview of Neolithic cropping, alludes to just this biblical moment.

“Some combination of population growth, the decline in wild protein to hunt and nutritious wild flora to gather, or coercion, must have forced people, reluctantly, to work harder to extract more calories from the land they had access to. This demographic transition to drudgery has been read by many as metaphorically captured in the biblical tale of Adam and Eve being expelled from Eden to a world of toil.”¹²⁶

The overuse and subsequent depletion of natural resources is the third step of the Malthusian progression, after new food resourcing creates more people. Agricultural innovation allowed for ever increasing returns on the same area of space, which caused an increase in population density, followed by a decline of natural resources.¹²⁷ The forested Mesopotamian highlands, as the first locus of farming, suffered just such a collapse. Clive Ponting offers a detailed description of agriculture’s cursed effect in the region:

Agriculture involves clearing the natural ecosystem in order to create an artificial habitat where humans can grow the plants and stock the animals they want. The natural balances and inherent stability of the original ecosystem are thereby destroyed. Instead of a variety of plants and permanent natural ground cover, a small number of crops make only a part-time use of the space available. The soil is exposed to the wind and rain to a far greater extent than before, particularly where fields are left bare for part of the year, leading to a much higher rate of soil erosion. Forests suffered most as the requirement for wood to build houses, heat homes, and cook food rose steadily.¹²⁸

As I have stressed previously, the root-stem for Eve’s birthing pangs (מַעַה) and the toil (מַעַה) of Adam eating from the cursed ground is the same. Cassuto’s sensitive reading alerts the ear to the repetitive sounds of recurring stem words curled within the edenic drama. Thus, both Eve’s pangs and Adam’s labour are linguistically bound to the forbidden tree (מַעַה). “Apparently,” Cassuto points out, “we have here a play upon words with reference to (מַעַה) [tree]: it is with respect to מַעַה that the man and the woman sinned, and it is with (מַעַה) [pain] and מַעַהוּ[toil, suffering] that they were punished.”¹²⁹ Linguistic repetition could imply a three part chain of causation. Eating from the “tree” of wheat engenders more painful births, which creates ever more suffering farmers tilling a cursed crop, sweating ever more intensively to make up in labour input what is lost in soil fecundity. God’s curse is less punishment than a harbinger of Malthusian design.

The expulsion from Eden and the inception of agriculture align. First, the domestication of the “tree” of wheat, then Eve’s cursed birthrate, and finally, the ground accursed and worn due to the farming act. Here is a clear progression from technological innovation to population growth, leading inexorably to natural resource depletion, following the Malthusian model. But Malthus may well have been wrong.

Modern Critiques on the Malthusian Hypothesis

First, let us recall the salient points of the Malthusian theory itself. Malthus assumed that before the invention of agriculture, world population density remained stagnant precisely because human fertility consistently outgrew the supply of wild flora and fauna, leaving societies precariously balanced in a state of semi-starvation. According to this theory, agriculture created a radical improvement in nutrition as each acre of arable land could now produce a markedly higher degree of food supply, which caused a rapid population growth rate.¹³⁰ However, there are a number of interlinking critiques of this approach. First, the latest synthesis of data on the composition of the diet of gathering populations, both from prehistoric archaeological sites, and anthropological studies of the approximately 27 hunter-gatherer societies still existing today all indicate that hunter-gatherers usually had sufficient food resources. Boserup argues that whilst there is no denying that intermittent starvation may have been the lot of some prehistoric people, “there is little reason to believe that lack of food was a more frequent and widespread phenomenon among hunter-gatherers than among food-producing populations.”¹³¹ Thus, we must look to other causes, beyond simply a lack of farming acumen, for why the population rate remained relatively flat prior to the Neolithic.¹³²

The second problem with the Malthusian model by which the innovation of agriculture caused a widespread population explosion is that very many societies, both in prehistory and amongst the hunter-gatherer tribes of today, operate well below their local environment’s long-term productive capacity. This implies that societies do not automatically rush to adopt crop cultivation unless another factor catalyses them to do so. “So,” notes political scientist Francis Fukuyama, “the mere technological possibility of increased productivity and increased output, does not necessarily explain why it came about.”¹³³ If hunter-gatherer societies were really starving before the dawn of agriculture, they should have immediately embraced the new technology just as soon as possible, to avert the constant spectre of famine. James Scott notes that precisely the opposite situation prevailed. “The breathtaking four-millennia gap between the first appearance of domesticated grains and animals and the coalescing of agro-pastoral societies we have associated with early civilization commands our attention. The anomaly of such a stretch of history, when all the building blocks for a classic agrarian society are in place but fail to coalesce begs an explanation.”¹³⁴ Melinda Zeder, a prominent theorist of domestication, states the rather obvious here; clearly this 4,000 year gap, or approximately 160 generations, reveals that pre-Neolithic societies were content to utilise some cultivated crops and animals strategically and only periodically without sole reliance on them. “Stable and highly sustainable subsistence economies based on a mix of free-living, managed, and fully domesticated resources seems to have persisted for 4,000 years or more before the crystallization of agricultural economies based primarily on domestic crops and livestock in the Middle East.”¹³⁵ In a separate article, Zeder concludes that,

“neither the presence of domesticated or domesticable resources nor the diffusion of food producing technologies is sufficient to induce the adaptation of food production as a guiding principle of subsistence economy.”¹³⁶ In fact, as Scott points out, ancient societies would routinely abandon settlement crops to take up pastoralism and migratory forest foraging.¹³⁷ Such a reversion simply makes no sense within the Malthusian model, where crop cultivation leads in a one-way trajectory towards ever expanding population density, and thus, ever more intensive agricultural activities *de rigueur*.

Boserup elucidates two reasons why societies were so reluctant to fully embrace the food production revolution for so long, and why, even today, Boserup reports, agricultural advisors in sparsely populated areas of the globe regularly complain that the local population will neglect their crops in order to engage in gathering activities.¹³⁸ One reason that societies choose gathering over farming is due to the intrinsic fragility of a food procurement regime that relies on a limited scope of food sources. Writes Boserup, “People consuming a varied diet of wild foods were probably less exposed to hunger in years of bad climatic conditions than agricultural peoples heavily dependent upon the harvest of one or a few crops.”¹³⁹ Another reason, argues Boserup, why people don’t naturally adopt farming is that more intensive cultivation increases labour input exponentially. Earlier in this chapter, I cited the bushmen of the Kalahari, where women who do the gathering use only two or three days per week to provide the necessary calories to feed their extended families, and there are no seasonal hunger periods. Additionally, Boserup points out that the Kalahari, and most modern gathering societies, have been forced into the marginal areas of the globe such as deserts, tundras, and mountain forests, where poor soil, unfavourable climate, or both limit the supply of wild food. “In prehistoric times,” Boserup recalls, “most food-gathering populations lived in fertile areas with favourable climate ... It seems realistic to assume that they were able to find sufficient wild food as long as the population densities were low.”¹⁴⁰

Ester Boserup’s Theory of Population and Technological Change

In light of these serious critiques on the Malthusian model where farming technology causes the rate of population density to increase, Boserup offers an exactly opposite causative paradigm: increasing population density catalyses ever more intensive modes of agriculture. Such a population increase could have been influenced by one or more factors other than farming itself such as a climate change which boosted the local supply of wild foodstuff, or new migrants entering the region, or implementation of new cultural mores which acted to increase the population surplus such as curtailing infanticide or decreasing breastfeeding intervals.¹⁴¹

Thus, it was not farming which caused more people to be born, but more people living on the same plot of land which forced these societies to reluctantly embrace farming. Writes Boserup, “It seems that production of crops is

preferred only when a certain population density is exceeded and wild food falls into scarce supply. In other words, it indeed seems plausible that increasing population density promotes changes of food supply technology, while areas with stagnation or decline of population are much less likely to adopt this type of change.”¹⁴² Scott, quoting Boserup, points out that societies are incredibly reluctant to embrace intensive agriculture as their main source of calories until their “backs-are-to-the-wall.” Population growth combined with declining wild protein to hunt, and nutritious wild flora to gather, may have forced people to turn to farming as a last resort to extract more calories from their land, in exchange for far more drudgery.¹⁴³

Boserup’s theory, though not without its detractors,¹⁴⁴ offers a straightforward explanation on why sparsely populated societies were slow to adapt intensive agriculture, if they adapted it at all. These groups had an adequate diet and plenty of leisure time. Why dismiss such advantages for a food production regime fraught with uncertainty and drenched with more sweat upon the brow?

Likewise, Boserup’s model correlates with the seemingly haphazard spread of wheat and barley crops and techniques of production diffused from the early centres in western Asia. In the millennia after 8000 BCE, these crops appeared in northern India, China, and Europe, leapfrogging societies much closer to these early grain cores, which continued to base their subsistence on gathering or pastoralism. Malthusian models have a hard time making sense of such variation. In the Malthusian model, all pre-Neolithic societies were at the cusp of starvation. Thus, every group should unanimously race towards crop cultivation; yet again and again, we witness societies living cheek by jowl to agricultural settlements that rejected the new farming technology. Boserup theorised that it was differences in population densities, rather than proximity to the first cores of farming, which can explain this patchwork pattern of early agricultural distribution.¹⁴⁵ Only those who had hit the population wall would choose the drudgery of farming when it arrived at their doorstep.

A Midrashic Support for Boserup’s Theory

With Boserup’s thesis, our Garden of Eden narrative runs amok. For there was no population expansion, beyond the primordial pair of Adam and Eve. No children, as yet, to force that first couple to devour the forbidden fruit of domesticated wheat in the garden. What then would motivate Eve to cultivate the first grain? Must we conclude that scripture chose Malthus over Boserup? Yet, the midrashist, at least, locates the birth of Cain and Abel within the garden, prior to eating the forbidden fruit, and the subsequent expulsion from Eden. “‘And she conceived and gave birth to Cain’: Said Rabbi Eleazar ben Azariah: ‘Three wonders happened on that day [the sixth day of creation]: on that day they [Adam and Eve] were created, on that day they had relations, on that day they had offspring.’”¹⁴⁶ It is an incongruous disjointing of the accepted narrative. In the biblical account, Cain and Abel are born directly after God drives Adam out of the Garden. Only after the expulsion does the

procreative story unfold. “Now Adam knew Eve, his wife, and she conceived and bore Cain, saying: I have acquired a man with the Lord.”¹⁴⁷ Why does the midrashist defy the narrative chain of events so blatantly?

Three Theories on the Making of Midrash

Before exploring this particular question in detail, it is worthwhile to briefly outline some of the latest theories on the reasons the rabbis embarked on and developed their midrashic project, generally. According to the post-biblical scholar James Kugel, midrash grew out of the exegetical tradition forged in the Second Temple Period. Kugel argues that the rise in a written exegetical discourse emerged after the Babylonian exile, as scribes attempted to bridge the diverse and widening gaps between the ancient biblical texts and the perceptions and needs of their current readership.¹⁴⁸ Midrash, latterly, stamped this early exegetical literature with its own particular hermeneutical and literary form. For only in midrash do the authors anchor their tales and aphorisms with direct scriptural citations. Yet, what was the rabbinical purpose in creating this rich cultural hallmark? Or, as Dina Stein delineates the poles of this question: “Did the rabbis deliver their exegesis in good faith or did they consciously exploit the biblical text to express whatever preconceived ideas they may have had.”¹⁴⁹ Thus, returning briefly to the midrashic birth-story of Cain and Abel under query, did our midrashist refigure the boys’ emergence in the garden because of an impulse from within the text itself or were the rabbis reframing the timeline and appropriating the narrative arc to tell another story suited to fulfil their own goals and purposes?

Until Daniel Boyarin’s revolutionary study, *Intertextuality and the Reading of Midrash*, most modern scholars of rabbinical literature, including Isaac Heinemann, Joseph Heinemann, and Jonah Fraenkel, agreed that midrash was a type of pseudo-exegesis.¹⁵⁰ That is, the exegesis was primarily a respectable hook for the rabbinical authors to hang their own values and ideologies upon.¹⁵¹ Not so in Boyarin’s reading. Through a close analysis of the *Mekhilta de-Rabbi Ishmael*,¹⁵² Boyarin argued that the rabbis were primarily interested in making sense of a maculate¹⁵³ yet entirely numinous text. Though the midrashists could not help but be informed by their own ideological biases, their main concern in presenting midrash was to draw meaning out of the incongruous layers, gaps, inconsistencies, and poetic tangle of the scriptural canon itself.¹⁵⁴

There is a third interpretive method for analysing certain midrashic stories.¹⁵⁵ Specifically, that a scattering of midrashic traditions were neither later historical ideologies foisted on scripture per Heinemann, Heinemann and Fraenkel, nor again insights gleaned from careful reading of the primary source, per Boyarin, but instead represent the trace of folk traditions that predate the Bible. Avigdor Shinan and Yair Zakovitch suggest that certain Second Temple and rabbinical exegetical works do not emerge *sui generis*, but rather record ancient oral traditions that were simply too radical to be included in the Bible itself. These ancient narratives were kept alive only in spoken form

until their original, counter-biblical impetus was entirely forgotten and later exegetes could safely commit them to the written record as aggadic tales. As Shinan and Zakovitch note: "This enormous corpus [of post-biblical literature] preserves more than a few remnants of ancient traditions that the Bible barred from its pages but that continued nonetheless to be told and retold, transmitted orally until much later. When they were no longer deemed threatening to anyone's beliefs or ideologies, they reemerged and were recorded."¹⁵⁶

Umberto Cassuto similarly identified certain elements of rabbinical literature as a revival of pre-biblical ancient epics erased or even refuted, albeit obliquely, from the biblical account.¹⁵⁷ Cassuto argued that the biblical record is often underscored with a dialectical protest against an entire corpus of pre-existent creation myths and paradisiacal garden stories. For example, when the Bible employs such phraseology as: "And God said: Let the waters be gathered together,"¹⁵⁸ and then concludes definitively that, "It was so."¹⁵⁹ That, tucked neatly into this pithy account of an entirely passive and malleable watery body lies a subtle diatribe against the Ugaritic poetics of a sea "endowed with an autonomous divine power that fought, as it were, against the Creator of the Universe."¹⁶⁰ In Cassuto's reading, the early chapters of Genesis "voiced a kind of protest against these myths whose pagan origins were still discernible, and more particularly against the concepts of the heathens themselves."¹⁶¹ The servile passivity of the primordial sea, the simple "It was so," was a rebuke against all those who would worship such watery powers.

The midrashists, however, had no such qualms about breathing life back into the ancient myths. "Rabbi Judah said in the name of Rab: When the Holy One, blessed be He, desired to create the world, He said to the lord of the sea: 'Open thy mouth and swallow up all the waters of the world.' The latter answered: 'Sovereign of the universe, I have enough with my own!' Thereupon God instantly trod him down and slew him."¹⁶² Cassuto, foreshadowing Shinan and Zakovitch, posits that the midrashists, unlike the biblical authors, could afford to include these mythic relics preserved up to that point in oral tradition alone, because "they did not feel any misgivings about those elements in it that resembled heathen mythology, since in their day paganism was no longer a danger to Judaism."¹⁶³

In summation of the midrashic project, launched by the Second Temple exegetes and reaching cultural fruition within the Rabbinical era, we are immediately faced with, at minimum, a threefold branching. Is any particular midrashic apologue before us either a folk tale revived; or an act of nearly pure engagement with the biblical gaps; or an ideological argument foisted upon those gaps; or some entwined amalgamation of all three branches, now impossible to disentangle?

Midrashic Children in Eden

Returning to the particular midrashic tradition of babies born in Eden, I can imagine all three variants. Perhaps, the reason that the Genesis Rabba

tradition reversed the scriptural timeline was precisely so that Eve could give birth in Eden prior to the “sin” of the forbidden fruit. This would demphasise the link between the tree of knowledge and the first sexual experience, and thus undermine the concept of original sin which could play well amidst the fifth-century Christian and Jewish polemics of Palestine, the temporal and geographical locus for the Genesis Rabba compilation.¹⁶⁴

Or, perhaps, the catalyst for birthing Cain and Abel in Eden solves a narrative dilemma within the text itself. For the first divine injunction given to Adam on the sixth day of creation, namely “Be fruitful and multiply, and fill the earth and subdue it,”¹⁶⁵ is entirely premature. As Cassuto points out, if Adam and Eve had conversely “hearkened to the voice of the Creator and had been content with what He had given them, they could have eaten from the tree of life and lived forever in the garden of delight prepared for them. In that eventuality, they would have had no need, of course, to propagate their species or to spread abroad through the earth and fill it.”¹⁶⁶ Cassuto solves this discrepancy by arguing that the first chapter of Genesis is simply recounting the “gist” of creation, right through the banishment from the Garden. Therefore, God’s blessing to propagate was actually intoned much later and only in response to the expulsion from Eden and the loss of immortality, but recorded briefly—highlight-reel fashion—in this first overture. Perhaps, the midrash was equally sensitive to the narrative misplacement of the procreative blessing and offered a variant solution. Giving birth to children and eternal life need not contradict one another. Adam and Eve could indeed procreate fruitfully in the Garden per God’s blessing, and then remain there as an immortal family, rather than a couple, having all eaten from the tree of life, though the midrash, thus construed, would still not be able to explain how a family content to dwell in Eden *ad infinitum* might “fill the earth and subdue it.”

Or, perhaps again, the midrashist’s intent was to heal another chronological aberration. For how could Adam call his wife’s name Eve, as the “mother of all living,”¹⁶⁷ despite the fact that they had no children at this point in the narrative. The midrash does not attempt to read this nomenclature as anticipatory, which would strain the past tense phrasing of the verse. Nor does it prefigure Cassuto, who both elucidates the problem here, “How could Adam have known that she would be the mother of all living?” and provides a solution that, therefore, “these words have been added to the Bible” as a sort of atemporal narrative gloss.¹⁶⁸ The midrash, however, simply reworked the story to make Eve a mother before Adam’s pronouncement. She had already given birth to her two children before the name-change. Revisionist history to smooth over a contradictory account.

Finally, the midrash itself could be of ancient stock.¹⁶⁹ A folkloric tradition, in the fashion of Ester Boserup, by which a slight population increase, in this case just two small mewling babies, is precisely what catalyses Eve to become enthralled with a new method of food technology, the forbidden wheat, to feed her growing family. Two more hungry mouths require Eve, as the prototypical hunter-gatherer, to make a heart-wrenching choice. Ester

Boserup anticipates Eve's dilemma at the moment when two more are born in Eden. "Prehistoric populations had a choice between adapting population to resources by means of fertility control or adapting resources to population by means of changes in consumption patterns, migration, or technological change. The peoples who chose the first alternative and killed the population surplus ... might end up subservient to another people with increasing population or they might continue as hunter-gatherers in inaccessible areas."¹⁷⁰ Did the midrashist place Cain and Abel in the garden to give context and mitigation to Eve's extraordinary act of defiance? With two little ones in tow, Eve's decision is not one of capricious innocence lost, but a terrible and complex choice, between killing that "population surplus" of her womb or finding another way, a forbidden way, to feed her hungry children. We might even conjecture that the horrific decision imposed by more children in a garden perfectly calibrated for the needs of two, rather than four stomachs, is precisely why the narrative arc on this matter is so disjointed. As if to imply that the text itself would rather hide the new small lives emerging into Eden, because the implications of these births, and their impact on the ecology of paradise, present choices so deeply anathema to Eden—either the babies are killed or a transgression made to feed them—that the narration simply leaves the children out of Eden all together, and only takes up their birth and subsequent troubled lives when the reader is prepared for a milieu of banishment, sweaty sorrow, and death. Paradise cannot hold such tension. Babies are too disruptive of Eden. Only the unflinching gaze of the midrashist will allow these babies to stir up trouble directly in paradise.

Perhaps, as well, the midrashist was attuned to the thematically dim view of population increase in these early chapters of Genesis. For later, in the sixth chapter of Genesis, it is the multiplication of human beings on "the face of the ground"¹⁷¹ that leads directly to the denouement whereby the "Lord saw that the wickedness of man was multiplied on the earth,"¹⁷² and ultimately to the flood. Perhaps, the midrashic twist of Eve birthing her babies in the garden and only then seeking out the cursed fruit is a thematic trace of the inevitable link between population growth and destruction. A tacit implication that as population increases, beginning with just two small babies, ever more must be devoured. Or, stated historically in the framing of Boserup: "Probably the people with increasing populations sometimes hastened the decline of the others by killing the remaining ones when they had become too few to defend themselves sufficiently."¹⁷³ A multiplying of insatiable appetites against weaker parties which is precisely how the exegetical Book of Enoch frames the wickedness that engulfed pre-diluvian society. A society where women gave birth to giants of endless appetite. "And they became pregnant, and they bore great giants, whose height was three thousand ells, who consumed all the acquisitions of men. And when men could no longer sustain them, the giants turned against them and devoured mankind. And they began to sin against birds, and beasts, and reptiles, and fish, and to devour one another's flesh, and drink the blood."¹⁷⁴

Ultimately, what the midrashic retelling allows for is the inclusion of yet another alternative on what drives humanity out of Eden. A choice made by a mother seeking new fruit for her hungry family. As I shall examine in Chapter 3, that story of hunger and scarcity haunts Eve's children and lays the bitter seeds for fratricide.

Notes

- 1 Otto Silbrig and Dorothy Silbrig, *So Shall You Reap: Farming and Crops in Human Affairs* (Washington DC: Island Press, 1994), 16 and 19.
- 2 Silbrig, *Shall You Reap*, 22. "In the Middle East agriculture goes back about 10,000 years, but it would take 4,000 years before a society totally dependent on agriculture would evolve there." See also Scott, *Against the Grain*, 58.
- 3 Tom Knox, "Do these mysterious stones mark the site of the Garden of Eden?" *Mail Online*, September 5, 2009. <http://www.dailymail.co.uk/sciencetech/article-1157784/Do-mysterious-stones-mark-site-Garden-Eden.html>.
- 4 Tom Knox, "Mysterious stones."
- 5 Konstantin Pustovoytov, "Soils and Sediments at Göbekli Tepe, Southeastern Turkey: A Preliminary Report," *GeoArchaeology: An International Journal* 2, no. 7 (2006): 699–719.
- 6 Tom Knox, "Mysterious stones."
- 7 Klaus Schmidt, "Göbekli Tepe, Southeastern Turkey: A preliminary report on the 1995–1999 excavations," *Paleorient* 26, no. 26 (2000): 46. "The analysis of the animal bones revealed a rich fauna of wild species, including wild cattle, wild ass, gazelle, and wild pig, but no domesticated species have been identified. The same is the case with the botanical remains. Only wild species such as almond and pistachio as well as wild grain have been found up to this point."
- 8 Sean Thomas, "Göbekli Tepe-Paradise Regained?" *Fortean Times*, 2 March 2007, <https://www.dailysgrail.com/2008/10/7000-years-before-stonehenge-unveiling-gobekli-tepe/>
- 9 Sean Thomas, "Göbekli Tepe-Paradise Regained?"
- 10 Jared Diamond, *Guns, Germs, and Steel: A Short History of Everybody for the Last 13,000 Years* (London: Vintage, 2007), 97.
- 11 Bruce D. Smith, *The Emergence of Agriculture* (New York: Scientific American Library, 1998), 46.
- 12 Smith, *Emergence*, 70–71.
- 13 Diamond, *Guns*, 100, and table 5.1.
- 14 Peter Bellwood, *First Farmers: The Origins of Agricultural Societies* (Oxford: Blackwell Publishing, 2005), 66. "At Ain Ghazal [in modern day Jordan], increases occurred in infant mortality, and in the importance of domesticated goats and the legume species which they probably liked as fodder. Many authorities see this trajectory as recording the local collapse of a cereal-based agricultural economy due to environmental degradation and extensive deforestation, with a consequent shift toward an increasingly pastoral economy and consequent human population decline or dispersal. Zarins (1990) has suggested that a major phase of dispersal by pastoralists occurred from the PPN [Pre-Pottery Neolithic] Levant in the seventh millennium BC ... similar hints of stress are reported at other sites."
- 15 Diamond, *Collapse: How Societies Choose to Fail or Survive* (London: Penguin Books, 2005), 546. "Why, after leading the world for so long, did the Fertile Crescent decline? Much of the explanation has to do with deforestation in the low rainfall environment of the Fertile Crescent, and salinization that permanently ruined some of the world's oldest farmlands."

- 16 Konstantin Pustovoytov, "Soils and Sediments," 704. "It may be that agriculture spelled the end for Göbekli Tepe. Early agriculture was primitive and not very productive. People may have not recognized the need to rotate crops, fertilize fields, and prevent the erosion of topsoil. It's been speculated that early agriculture led to deforestation of this region, with the loss of game animals. The fields may have turned fallow eventually, and with the lack of proper management the whole area turned into a dustbowl. With no game animals left, there was no way to support a large population, and the site may have been abandoned for this reason. At any rate, Göbekli Tepe was deliberately abandoned about 8,000–7,500 BCE, soon after the development of agriculture in the region. The entire site was deliberately covered with earth before it was abandoned, which is why it is so well preserved today."
- 17 Diamond, *Collapse*, 6.
- 18 Diamond, *Collapse*, 9.
- 19 Diamond, *Collapse*, 6.
- 20 Diamond, *Collapse*, 7. "The risk of such collapses today is now a matter of increasing concern, indeed collapses have already materialized for Somalia, Rwanda, and some other Third World countries. Many people fear that ecocide has now come to overshadow nuclear war and emerging diseases as a threat to global civilization." However, Diamond is careful to point out that the collapse of historical and modern societies is never simply due to ecological causes alone, but involves a complex web of contributing factors.
- 21 Following Hiebert's translation of יִקְרָא as "headwaters," in Hiebert, *Yahwist's Landscape*, 53.
- 22 זִשְׁחֹס, see Daniel 10:4. The Tigris/Hiddekel river is known as the *Idiklat/Diklat* in Assyrian. See Hiebert, *The Yahwist's Landscape*, 53; Aryeh Kaplan, *The Living Torah: The Five Books of Moses and the Haftaret* (New York: Moznaim Press, 1981), 8 and ft. '2:14 Tigris'. Also the translation of the *JPS Tanakh* (Phil: Jewish Publication Society, 2003), 4.
- 23 Gen 2:10–14.
- 24 See W. F. Albright, "The Location of the Garden of Eden," *AJSL* 39 (1922): 17. "The author is obviously trying to locate a definitive place."
- 25 See Hiebert, *Yahwist's Landscape*, 53. Also, Albright, "Eden," 18.
- 26 Albright, "Eden," 17–18.
- 27 See Targum Onkelos on *Havila*, Gen 10:7.
- 28 See Cassuto, *Genesis I*, 115–120; also Albright, "Eden," 18–24; and Hiebert, *Yahwist's Landscape*, 53, where Pishon and Gihon are equated with the headwaters of the Nile River.
- 29 Kush is commonly translated as "Africa" or more specifically "Ethiopia." Referenced in Aryeh Kaplan, *The Living Torah*, 2:11, 10:6, & 10:29, Kaplan cites Herodotus 2:19, and Josephus, *Antiquities* 1:1:3, to buttress this locale.
- 30 See Jon Levenson, *Sinai and Zion: An Entry into the Jewish Bible* (San Francisco: Harper & Row, 1985), 127–137; and Hiebert, *Yahwist's Landscape*, 53. "Gihon is the name of the spring in the Kidron valley that was Jerusalem's major source of water, and it is hard to imagine an Israelite not making this association."
- 31 John Skinner, *Genesis, International Critical Commentary*, 2nd ed. (Edinburgh: T & T Clark, 1980), 62.
- 32 Hiebert, *Yahwist's Landscape*, 64.
- 33 Hiebert, *Yahwist's Landscape*, 59.
- 34 Hiebert, *Yahwist's Landscape*, 56; See also 52.
- 35 Hiebert, *Yahwist's Landscape*, 58.
- 36 Gen 13:10.
- 37 Ezek 28:14 & 16.

- 38 Evan Eisenberg, *The Ecology of Eden*, 90, "Today it is not common to think of Eden as a mountain. But in earlier times—from at least the sixth century B.C., when Ezekiel prophesied, to the seventeenth century A.D., when Milton wrote *Paradise Lost*—it was very common."
- 39 In current biblical scholarship, the Yahwist (J) authorship is generally linked to the period of the Divided Monarchy (922–587/6 BCE), whereas Ezekiel emerged in approximately 593–565 BCE, during the Exilic period. See Eric Meyers and John Rogerson, *The Cambridge Companion to the Bible*, ed. Bruce Chilton (Cambridge: Cambridge University Press, 1997), 52.
- 40 Hiebert, *Yahwist's Landscape*, 57. "Most explicit in this regard is a text from Ezekiel's program of restoration that evoked images of Eden in its picture of the valley's renewal (Ezek 47:1–12)."
- 41 Cassuto, *Genesis I*, 72. "My hypothesis that the Israelites had an epic tradition concerning the story of the Garden of Eden before the Torah was written, and that this tradition had already received a definite literary form in one or more epic poems, finds support in a number of Biblical verses some of which belong to our section and some to other sections or other books."
- 42 See Cassuto, *Genesis I*, 77. "Unquestionably, Ezekiel did not invent the idea of the mountain, but found it in the poetic tradition."
- 43 Hiebert, *Yahwist's Landscape*, 52.
- 44 See Matt Ridley, *The Rational Optimist* (London: Fourth Estate, 2011), 127: "One of the intriguing things about the first farming settlements is that they also seem to be trading towns. From 14,000 years ago, much valued obsidian from the Cappadocian volcanoes in Anatolia was being transported south along the upper Euphrates, through the Damascus basin and down the Jordan valley. Seashells from the Red Sea were going the other way ... it is easy to think of early farmers as sedentary, self-sufficient folk. But they were exchanging harder in this region than anywhere else, and it is a reasonable guess that one of the pressures to invent agriculture was to feed and profit from wealthy traders—to generate a surplus that could be exchanged for obsidian, shells, or more perishable goods. Trade came first."
- 45 Hiebert, *Yahwist's Landscape*, 55.
- 46 Though Hiebert does briefly conjecture that "details within the narrative may imply that the primeval garden was wiped out in the catastrophe that reduced the [Jordan] valley to a desert," he dwells no further on the subject of the fall, *Yahwist's Landscape*, 57.
- 47 Thorkild Jacobsen, trans., *The Sumerian King List* (Chicago: Chicago University Press, 1939).
- 48 Gen 10:6.
- 49 Gen 10:7.
- 50 See Scott, *Art of Not Being Governed*, 261. "The importance of 'positionality' ... is so common in the creation of ethnic boundaries that what begins as the term for location or a subsistence pattern comes to represent ethnicity."
- 51 See Cassuto, *Genesis II*, 181–182. "Without doubt the genealogical terms in our chapter [Gen 10] are employed in figurative senses. Words used in such metaphorical connotations are found in great number in the genealogical lists of the Book of Chronicles ... in these verses of Chronicles the 'father' of a given city is its founder, or the one who rebuilt it, or the eponym to whom the tribe dwelling their traces its descent; and his 'sons' are not actually sons, but are called sons only metaphorically ... The scholars who attribute to this chapter naive and primitive conceptions, as though it intended to tell us that a man named Egypt begot a man called Ludim ... fail to understand the nomenclature of the passage."

- 52 Gen 10:10.
- 53 Phillip King, *Amos, Hosea Micah: An Archeological Commentary* (Philadelphia: The Westminster Press, 1988), 51.
- 54 2 Kings 19:12.
- 55 *The Anchor Bible Dictionary*, ed. David Noel Freedman, 6 vols. (New York: Doubleday, 1992), 344–345. “Tel-Asar: An Aramean city inhabited by the people of Eden ... situated on the middle Euphrates ... the kingdom of Eden was located between Assyria and Syria-Palestine. Barsip, the capital city of the house of Eden, was conquered by Shalmanaser III. He renamed it after himself and annexed Bit Adini to Assyria in 855 B.C. Afterward Amos denounced the house of Eden (Amos 1:5). It probably rebelled against Assyria, and in the 8th century B.C. it was destroyed.”
- 56 By including the seventh day of creation (the Sabbath) under the heading of “Chapter 1” requires that I digress from the chapter allocations created by Stephen Langton in the 13th century. Though Langton’s chaptering is standardised, it would appear haphazard (or perhaps motivated by non-textual considerations) in separating between Day Six and Day Seven of the creation sequencing. The Masoretic text itself is not separated into chapters, but is spaced into paragraphs (פָּרָשׁ, sl.) of vastly varying lengths. Therefore, the first “chapter” is divided into seven coherent sub-units, one sub-unit per day, which marks the first seven days of creation. This is visually distinguished from the long single paragraph which tells the second creation story and the narrative of Eden account, which I refer to as the “Second Chapter Genesis Story.”
- 57 Gen 1:5, 1:8, 1:13, 1:19, 1:23, & 1:31.
- 58 Gen 1:12 & 1:13 (for plants), 1:21 (for sea creatures and birds), & 1:24 (for land creatures).
- 59 The prevailing view from biblical source criticism is to divide the first and second account of creation in the middle of Gen 2:4; see Marc Zvi Brettler, *How to Read the Jewish Bible* (Oxford: Oxford University Press, 2005), 34. However, I quote Gen 2:4 in full within the context of the second account. I have taken this approach for two reasons. First, in order to maintain the normative integrity of the established verses. Second, because new research in the computational linguistic analysis of the Bible attributes both creation narratives to a single author. See Moshe Koppel, Navot Akiva, Idan Dershowitz, Nachum Dershowitz, “Unsupervised Decomposition of a Document into Authorial Components,” *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics* (June 19–24, 2011): 1363. Perhaps, then, what distinguishes these two accounts is not different sources, but a deep narrative transition between the “before” and “after” of farming. As Jacob Weisdorf notes, “The adoption of agriculture in the Stone Age certainly did more, in the long run, to alter the world, than any previous human innovation,” Jacob L. Weisdorf, “From Foraging to Farming: Explaining the Neolithic Revolution,” *Journal of Economic Surveys* 19, no. 4 (2005): 563. Could the narrative have shifted so radically to mirror this very profound change to human society? As if to say that farming changes everything, even God’s name.
- 60 Following the initiative of Arthur Waskow, I have offered the rather awkward translation in this instance of *adamah* (אֲדָמָה) as “humus,” in the place of more normative translations, such as “ground” or “earth,” in order to reflect the Hebraic root link between *adam* (אָדָם) and *adamah* (אֲדָמָה). One might likewise translate this link as “earth” and “earthling.” See Arthur Waskow, *Torah of the Earth: Volume 1* (New York: Jewish Lights Publishing, 2000), vi. See also Hiebert, *Yahwist’s Landscape*, 36. “While the linguistic relationship between these two terms is regularly noted, the thorough way in which they together establish the agricultural character of human existence has not been widely recognized.”

- 61 Gen 2:4–8.
- 62 Westerman, *Genesis 1–11*, 199. Adam-Adamah “points to the basic relationship between the soil and the person which in reality characterizes agricultural life.”
- 63 See Hiebert, *Yahwist’s Landscape*, 65. “The term for cultivation in J is *abad*, the customary verb in biblical Hebrew to express servitude. It is used, for example, to express the servitude of a servant to his master (Gen 12:16) or of one people to another (Gen 27:40). It is also used for Israel’s service to God in its worship. (e.g. Exod 4:23; 7:16; 26) ... there can be no doubt that at some level the use of *abad*, ‘serve,’ for ‘cultivate’ represents the Israelite sense of dependence of human life on the land.”
- 64 Gen 1:12.
- 65 See bTalmud Chullin 60b; and Rashi commentary to Gen 2:5.
- 66 Julius Wellhausen, *Prolegomena to the History of Ancient Israel* (New York: Meridian Books, 1958), 299. See also J. Estlin Carpenter and George Harford Battersby, eds., *The Hexateuch* (London: Longmans, Green and Co., 1902), 1.97; and David Damrosch, *The Narrative Covenant* (Ithaca, NY: Cornell University Press, 1987), 121.
- 67 Gerhard von Rad, *Genesis* (Philadelphia: Westminster, 1972), 76.
- 68 Theodor H. Gaster, “Cosmogony,” in *Interpreter’s Dictionary of the Bible*, ed. Keith R. Crim (Nashville, TN: Abingdon Press, 1992), 1.704.
- 69 See Cassuto, *Genesis I*, 101. Here he proposes and rejects both the interpretations of Dillmann and Proksch, in which the אֲדָמָה and מִצְרַת imply the entirety of the vegetable world.
- 70 Cassuto, *Genesis I*, 101.
- 71 Gen 3:18.
- 72 Gen 3:18.
- 73 Cassuto, *Gen I*, 102. For similar readings of מִצְרַת as grain, see also E.A. Speiser, *Genesis, Anchor Bible* (Garden City, NY: Doubleday, 1964), 14; also Robert B. Coote and David Robert Ord, *The Bible’s First History* (Philadelphia: Fortress, 1989), 42; likewise, Hiebert, *Yahwist’s Landscape*, 37.
- 74 Here I am employing the negative implication of the word מִצְרַת as akin to bondage (as in Gen 12:16 and 27:40). In line, I believe, with Hiebert’s translation as “cultivation” and his subsequent explanation that “cultivation was an act of service to that which held absolute power over one’s survival and destiny.” *The Yahwist’s Landscape*, 66. However, the word need not necessarily connote enslavement, for מִצְרַת can also express service to God in terms of worship (as in Exod 4:23 and 7:16). Or to a more benign form of natural conservation of Eden (as in Gen 2:15). Here Cassuto translates מִצְרַת as “dressing” the garden. *Genesis I*, 122. Perhaps, the word holds both positive and negative meanings, implying that human activity can become either a service to the earth or an enslavement to the field.
- 75 I allude here to the commentary of Hirsch, *Genesis*, 51. “The words אֲדָמָה and רֶשֶׁת which occur here for the first time, show at once that we are now in the realm of mankind. Until now it has always used the general term רֶקֶן. But אֲדָמָה is that piece of ground that a man *lays claim to* for the purpose of obtaining food” [my italics].
- 76 Midrash, from the Hebrew verb “to search,” “to inquire,” “to interpret” is a type of Biblical interpretation found in the Talmuds and in the collection of exegeses edited during the Talmudic period and after. *The Cambridge Companion to the Bible*, 2nd ed., Anthony J. Salderini as revised by Amy Jill-Levine (Cambridge: Cambridge University Press, 2008), 465. A broader definition is offered by Jacob Neusner, *What is Midrash?* (Philadelphia: Fortress Press, 1987), xi, as “biblical exegesis of ancient Judaic authorities.” An even more amorphous and tautological definition of midrash is offered by Daniel Boyarin, who, indeed, refuses to

- define midrash at all, and instead will only delimit the genre broadly. “Midrash is the type of biblical interpretation which is found in the Jewish biblical commentaries which the Jews call ‘midrash.’” *Intertextuality and the Reading of Midrash* (Bloomington: Indiana University Press, 1994), viii. One senses in Boyarin’s definition a strong sentiment against all external terminological shackles, and perhaps this leads inevitably to the remark of James Kugel, “since these (earlier) studies have already not defined midrash in ample detail, there is little purpose in our not defining it again here.” James Kugel, “Two Introductions to Midrash,” in *Midrash and Literature*, eds. Geoffrey H. Hartman and Sanford Budick (New Haven: Yale University Press, 1986), 91.
- 77 Genesis Rabba, 13.1 in *Midrash Bereshit Rabba*, eds. Julius Theodor and Hanock Albeck [Heb], 3 vols. (Jerusalem: Wahrman, 1965), 113–114. I have here translated “וְהָיָה הָאָדָם כְּעֵץ הַגָּן” as “of the domus.”
- 78 Another variant edition of this midrash (Munich 97 MS) cited in Theodor and Albeck (*ibid*) further reinforces the notion that the fundamental demarcation between the within and without of Eden was the labour of crop production. “Over there in Eden [flora] grew without work or toil, here [flora] grows in domesticated fields.”
- 79 Hiebert, *The Yahwist’s Landscape*, 74. See also Eric Meyers and John Rogerson, *The Cambridge Companion to the Bible*, 57. “The whole picture contrasts sharply with other ancient Near Eastern texts. Both Sumerian and Akkadian texts assert that humans were created so that they could perform manual labor for the gods, such as building canals and cities.”
- 80 Thorkild Jacobsen, “Eridu Genesis,” 526; as quoted by Hiebert, *The Yahwist’s Landscape*, 71. Jacobson alludes to the terminology of Malinowski in the conception of myth as a state “charter.”
- 81 Evan Eisenberg, *The Ecology of Eden* (London: Picador, 1998), 95.
- 82 Genesis 2:9.
- 83 Gen R. 15.7 in *Midrash Bereshit Rabba*, eds. Theodor and Albeck, 139. See also bTalmud Sanhedrin 70b, and bTalmud Berachot 40a. The Pesiqta (20), cited in Theodor and Albeck (*ibid*), adds an intriguing addendum to the midrash (according to the hand-written edition of the Carmoli manuscript): “[the wheat ‘tree’] rattled the berries [in its sheaf] like a [shaking] cedar tree.” This midrashic addition succinctly defines the most basic quality of wheat domestication, the retention of the fully ripened seeds on the stalk until the harvest. The very fact that domesticated wheat seeds, unlike the wild variety, simply rattles in its husk when dry and ripe, but does not fall to the ground and shatter, is what allows for a determinate gathering by human hand. Domesticated wheat’s nonshattering (indehiscent) grain spikes that stay firmly attached to the stalk via a tough rachis ensure its place as the world’s most cultivated crop, as so few seeds will fall naturally to the earth before the human harvest. See Ian Morris, *Why the West Rules for Now: The Patterns of History and What They Reveal About the Future* (London: Profile, 2011), 98–99. Also, Scott, *Grain*, 74, and Solbrig and Solbrig, *So Shall You Reap*, 37. Writes Morris: “The excavated finds suggest that even around 8500 BCE fully domesticated wheat and barley were still almost unheard of. By 8000, though, about half the seeds we find in the Hilly Flanks have the tough rachis that would wait for the harvester; by 7500, virtually all do.” This thousand-year window of wheat domestication, this ever expanding rattle of hard seeds encased in their dry husk, carried more powerful consequences than the tallest cedar.
- 84 Though, in the original Biblical Hebrew, perhaps a “tree” of wheat is not quite so paradoxical as it sounds in English translation, for the Heb. designation of “tree” (עֵץ) may extend further than the English vocabulary would allow, e.g. Joshua 2:6,

- "She hid them in flax stalks [lit. *tree* (עץ)] which she had arranged on the roof." For a similarly loose designation of tree in Mishnaic Hebrew, see mShabbat 2:3, "No product from a tree may be used as a wick for the Sabbath Lamp, except flax."
- 85 Gen 2:16.
- 86 Gen 5:3–5, which places his age at 930 years.
- 87 Gen 3:17–19.
- 88 We may owe such a beneficent view of farming's origins as a progressive evolutionary step in complexity and advancement to the 18th-century Enlightenment philosophers. See B.G. Trigger, *A History of Archeological Thought* (Cambridge: Cambridge University Press, 1989). Or perhaps we owe such a positive outlook to the ancient Greek thinkers. See E. Isaac, *Geography of Domestication* (London: Prentice Hall, 1970), 3.
- 89 Jared Diamond, *Guns*, 104.
- 90 Diamond, *Guns*, 105–106. See also Jacob Weisdorf, "From Foraging to Farming: Explaining the Neolithic Revolution," *Journal of Economic Surveys* 19, Issue 4 (2005): 566. "In so-called 'affluent societies' farming was not desirable; hunters and gatherers would not embark upon time-costly methods of agriculture unless there was good reason to do so. Farming was in fact a last resort." Also, Marshall Sahlins, notably his first chapter in *Stone Age Economics* (Chicago: Aldine, 1974), where he coined the term "the original affluent society" to describe these hunter-gatherer subsistence regimes.
- 91 Clive Ponting, *A Green History of the World* (London: Sinclair-Stevenson Ltd., 1991), 20.
- 92 Ponting, *Green History*, 21.
- 93 Ponting, *ibid*.
- 94 Scott, *Against the Grain*, 112–113.
- 95 See Jared Diamond, "The Worst Mistake in the History of the Human Race," *Discover Magazine*, May 1, 1999, 64. "There are at least three sets of reasons to explain the findings that agriculture was bad for health. First, hunter-gatherers enjoyed a varied diet, while early farmers obtained most of their food from one or a few starchy crops. The farmers gained cheap calories at the cost of poor nutrition ... Second, because of dependence on a limited number of crops, farmers ran the risk of starvation if one crop failed. Finally, the mere fact that agriculture encouraged people to clump together in crowded societies, many of which then carried on trade with other crowded societies, led to the spread of parasites and infectious disease ... Besides malnutrition, starvation, and epidemic diseases, farming helped bring another curse upon humanity: deep class divisions. Hunter-gatherers have little or no stored food, and no concentrated food sources, like an orchard or a herd of cows: they live off the wild plants and animals they obtain each day. Therefore, there can be no kings, no class of social parasites who grow fat on food seized from others. Only in a farming population could a healthy, non-producing elite set itself above the disease-ridden masses. Skeletons from Greek tombs at Mycenae c. 1500 B. C. suggest that royals enjoyed a better diet than commoners, since the royal skeletons were two or three inches taller and had better teeth (on the average, one instead of six cavities or missing teeth)."
- 96 Diamond, *Guns, Germs, and Steel*, 111–112.
- 97 Diamond, *The Third Chimpanzee: The Rise and Fall of the Human Animal* (New York: HarperCollins, 1993), 190. See also Robert Wright, *NonZero: History, Evolution and Human Cooperation* (London: Abacus, 2001), 73–74. "How would war encourage agriculture? In primitive war few things come in handier than sheer manpower. And agriculture supports much larger settlements than hunting and gathering does. One of the earliest known farm towns, the ancient, excavated

village of Jericho, housed hundreds of people on around six acres. Not huge by modern urban standards, but compare it to what lies beneath: remnants of a hunter gatherer camp one-fifth as large. Imagine a battle between these two villages, and you'll see that farming was a compelling lifestyle. Whether or not early farmers thought about the military edge their lifestyle offered, war would have helped the lifestyle spread."

- 98 Diamond, *Guns, Germs, and Steel*, 190. Also Jared Diamond, *The World Until Yesterday: What Can We Learn from Traditional Societies* (London: Penguin Books, 2012), 180–181. "The ultimate evolutionary causes and the proximate physiological mechanisms responsible for those long birth intervals of nomadic hunter-gatherers have been the subject of much discussion. It appears that the ultimate reasons are two-fold. First a mother without access to cow's milk or cereal gruel, and hence likely to nurse a child until the age of three or more years, cannot produce enough milk to nurse both a newborn and a not-yet weaned child. If she tried, one of these children would be likely to starve for milk. The other reason is that only when a child is around four years old or more does it become capable of walking fast enough to keep up with its parents when they are shifting camp ... We thus have a second ultimate evolutionary factor contributing to the rapid decrease in birth interval when nomadic hunter-gatherers settle down to become farmers: most farmers live in permanent villages and don't face the challenge of having to carry all children less than four years old whenever they shift camp ... Those two ultimate causes responsible for the long birth intervals of hunter-gatherers leave open the question of the proximate physiological mechanism ensuring that there are not two children less than several years old to be cared for simultaneously. One mechanism is the resort to neglect or (less often) infanticide ... The other proximate factor is that physiological mechanisms operating in a mother nursing according to the on-demand schedule of frequent feedings characteristic of hunter-gatherer babies ... makes it less likely that a nursing mother will become pregnant even if she resumes sex while nursing."
- 99 Studies of skeletal remains between the last hunter-gatherers and first farmers in regions where agriculture was subsequently adopted indicate significantly increased nutritional stress on the early farmers. See M.N. Cohen and G. Armelagos, *Paleopathology at the Origins of Agriculture* (New Haven: Yale University Press, 1977).
- 100 Gen 3:17.
- 101 U. Cassuto, *Genesis I*, 165.
- 102 Scott, *Against the Grain*, 108–109.
- 103 Evan Eisenberg, *The Ecology of Eden*, 95.
- 104 Ian Morris, *Why the West Rules for Now: The Patterns of History and What They Reveal About the Future* (London: Profile Books, 2011), 89. "... a Western way of life distinct from other ways—grew out of the ingenuity of women living in the Hilly Flanks nearly fifteen thousand years ago."
- 105 See also Graeber and Wengrow, *Dawn of Everything*, 237. "But where evidence exists, it points to strong associations between women and plant-based knowledge as far back as one can trace such things."
- 106 Gen 3:16. "I will greatly multiply your anguish, especially of your child-bearing, in anguish you shall bring forth children ..." and Gen 3:17. "... in anguish you shall eat of it all the days of your life...."
- 107 S.R. Hirsch, *The Pentateuch: Translation and Commentary by S.R. Hirsch: Genesis*, trans. Isaac Levy (New York: The Judaica Press, 1971), 83. Hirsch offers Kings I 1:6 as a proof text of this definition. "David had never asked his son to give up (Etzvu) anything."

- 108 Ian Morris, *Why the West Rules*, 88.
- 109 Gen 3:14–15.
- 110 James Scott, *Against the Grain: A Deep History of the Earliest States* (New Haven: Yale University Press, 2017), 73.
- 111 Gen 2:17.
- 112 Arthur Waskow, *Torah of the Earth: Volume 1* (New York: Jewish Lights Publishing, 2000), vi.
- 113 Gen 3:17.
- 114 Ester Boserup, *Population and Technological Change: A Study of Long-Term Trends* (Chicago: University of Chicago Press, 1981), 31–32.
- 115 Philip E.L. Smith and Cuyler T. Young, “The Evolution of Early Agriculture and Culture in Greater Mesopotamia,” in *Population Growth: Anthropological Implications*, ed. Brian Spooner (Cambridge: MIT Press, 1972).
- 116 James C. Scott, *Against the Grain: A Deep History of the Earliest States* (New Haven: Yale University Press, 2017), 113. “Neolithic agriculture was far more fragile than hunting and gathering or even shifting cultivation, which combined mobility with a reliance on a diversity of foods.”
- 117 Susan Pollock, *Ancient Mesopotamia: The Eden that Never Was* (Cambridge: Cambridge University Press, 1999), 38–39. “In the mountain and piedmont zones surrounding the alluvial {Mesopotamian} plains, the use of trees and other woody vegetation for timber and fuel has contributed to the deforestation of huge areas; the once forested Zagros Mountains, for example, are now largely treeless ... Without the deep-rooted perennials there is little to prevent large scale erosion of soil.”
- 118 In both the geographical circumscription, as well as the capitalisation of the term Hilly Flanks, I follow Ian Morris. *Why the West Rules for Now: The Patterns of History and What They Reveal About the Future* (London: Profile Books, 2011), 86. “The clearest evidence for clumping and settling comes from what archeologists call the Hilly Flanks, an arc of rolling country curving around the Tigris, Euphrates and Jordan valleys in southwest Asia.”
- 119 Boserup, *Population and Technology*, 50.
- 120 Gen 3:16.
- 121 Yuval Noah Harari, *Sapiens: A Brief History of Humankind* (London: Vintage, 2011), 96–97.
- 122 I follow Cassuto’s translation of מֵאֵה: “That is, the bread that you will produce from grasses like wheat and barley and other kinds of cereals. This too recalls what is stated in [Genesis] ii 5: *and no grain of the field* had yet sprung up ... At that time there were cornfields yet; from now on, by dint of man’s labour, cornfields would multiply upon the earth.” Cassuto, *Genesis I*, 169.
- 123 Genesis 3:17–18.
- 124 Cassuto, *Genesis I*, 168.
- 125 Cassuto, *Ibid.*
- 126 James Scott, *Against the Grain*, 71–72.
- 127 Ester Boserup, *Population and Technological Change*, 5.
- 128 Clive Ponting, *A New Green History of the World* (London: Random House, 2007), 68–69.
- 129 Cassuto, *Genesis I*, 163.
- 130 I rely upon Ester Boserup’s definition of Malthusian theory, here and throughout later chapters of this work. See, for example, Boserup, *Population and Technological Change*, 31–32; and Boserup, *The Conditions of Agricultural Growth: The Economics of Agrarian Change Under Population Pressure* (Abingdon: Earthscan, 1965), 117. One may argue that as Boserup describes Malthus’ theoretical frame only to undermine it for her own theory, it is a bit of a jaundiced account. However,

- see also Francis Fukuyama, *Origins of Political Order: From Prehuman Times to the French Revolution* (London: Profile Books, 2012), 84, for a similar definition of Malthus, also debunked.
- 131 Boserup, *Population*, 32.
 - 132 Boserup, *Ibid.*
 - 133 Fukuyama, *Origins*, 84.
 - 134 Scott, *Against the Grain*, 58.
 - 135 Melinda Zeder, "The Origins of Agriculture in the Near East," *Current Anthropology* 52, no. S4 (2011): S230-S231.
 - 136 Melinda Zeder, "After the Revolution: Post Neolithic Subsistence in Northern Mesopotamia," *American Anthropologist* new ser. 96, no. 1 (1994): 99.
 - 137 Scott, *Against the Grain*, 64.
 - 138 Boserup, *Population*, 40.
 - 139 Boserup, *Population*, 36.
 - 140 Boserup, *Population*, 33.
 - 141 Boserup, *Population*, 38. "The intervals between births vary widely in societies that are not using deliberate birth control. The length of the period of breastfeeding, together with involuntary abortion, seems to be the most important factor influencing the time interval between births. Unless they were breastfed for a long time, infants probably had little chance of survival in groups of prehistoric hunter-gatherers, and continued breastfeeding for years may have contributed to wide spacing of births in many, or perhaps all, prehistoric populations."
 - 142 Boserup, *Technology*, 41–42.
 - 143 Scott, *Against the Grain*, 71–72. See also Boserup, *The Conditions of Agricultural Growth: The Economics of Agrarian Change Under Population Growth* (Abingdon: Earthscan, 1965), 117–118. "The complex changes which are taking place when primitive communities change over to a system of shorter fallow are more likely to raise labour costs per unit of output than to reduce them. Therefore, it seems implausible to explain upwards changes in rates of population growth as a result of this type of agrarian change. It is more sensible to regard the process of agricultural change in primitive communities as an adaptation to gradually increasing population densities, brought about by the changes in the rates of natural population growth or by immigration."
 - 144 Scott, *Against the Grain*, 72.
 - 145 Boserup, *Technology*, 41.
 - 146 Gen. R. 22.2 in *Midrash Bereshit Rabba*, eds. Theodor and Albeck, 205.
 - 147 Gen 4:1
 - 148 James L. Kugel, *The Bible as It Was* (Cambridge, MA: Harvard University Press, 1997), 17–23.
 - 149 Dina Stein, "Rabbinic Interpretation," in *Reading Genesis: Ten Methods*, ed. Ronald S. Hendel (Cambridge: Cambridge University Press, 2010), 124–125.
 - 150 I owe this terminology to Dina Stein, "Rabbinic Interpretation," 125.
 - 151 Joseph Heinemann, *Aggadah and Its Development* [Hebrew] (Jerusalem: Keter, 1974). Isaac Heinemann, *Darkhe ha-Aggadah*, 2nd ed. [Hebrew] (Jerusalem: Magnes Press, 1954). See also Joshua Levinson, *The Twice-Told Tale: A Poetics of the Exegetical Narrative in Rabbinic Midrash* [Hebrew] (Jerusalem: Magnes Press, 2005), 30–35. Also, Daniel Boyarin, *Intertextuality and the Reading of Midrash* (Bloomington: Indiana Press, 1990), 1–11. On page 3, Boyarin quotes a particularly paradigmatic passage of Joseph Heinemann: "Much of aggadic exegesis is, therefore, a kind of parable or allegory. The aggadists does not mean so much to clarify difficult passages in the biblical text as to take a stand on the burning questions of the day, to guide the people and strengthen their faith. But since they addressed themselves to a wide audience—including simple folk and

- children—they could not readily formulate the problems in an abstract way, nor could they give involved theoretical answers.” As Boyarin notes, this line of reasoning tacitly denies the “hermeneutical function” of the aggadic project.
- 152 A third-century Palestinian exegetical compilation on the book of Exodus. Stein, *Reading Genesis*, 125.
 - 153 I have borrowed the language of a scripture rife with maculations from David Weiss Halivni. See *Revelation Restored* (London: SCM Press, 1997), 2. “We must therefore begin with the premise that the literal surface of the canonical Pentateuch is marred by contradictions, lacunae, and various other maculations.”
 - 154 Boyarin, *Intertextuality*, 39. In the passage cited above, Gerald Bruns remarks that, “the Bible, *despite* its textual heterogeneity, can be read as a self-glossing book. I would like to go further than that formulation and claim that the Bible, *because* of its textual heterogeneity, allows for the multiple self-glossing readings of midrash. The heterogeneity—the multivocality of the biblical text itself, its hiatuses and gaps, creatively but not open-endedly filled in by the midrash—allows it to generate its meanings—its *original* meanings—in ever changing social and cultural situations.” [italics are Boyarin’s]; See also, Boyarin, “*Midrash in Parables*,” *AJS Review* 20, no 1 (1995): 123–138. “Midrash, like all interpretation of narrative, whether formal or informal, involves the filling in of gaps in the narrative text. The gap itself is a complex concept, which essentially means any place in the text that requires the intervention of the reader to make sense of the story. Gap filling, in accord with the theories of Menahem Perry and Meir Sternberg (or for that matter those of Hayden White), involves the application of cultural knowledge, i.e. the mobilization of narrative schemata which are in the repertoire or socio-elect of the culture in question. In this sense, the schemata are both intertextual and ideological in nature, but in neither case is this nature generally available to the consciousness of people in the culture.”
 - 155 In reality, the scholarship on midrashic patterning and purpose is much more varied and malleable than these three simplistic approaches will allow. See, for example, the work of Jacob Neusner, who considers the midrashic project as a bridge between the literal sense of scripture and the socio-historical realities of Israel occurring at the time of authorship. *What is Midrash?* (Philadelphia: Fortress Press, 1987), 3. I have taken the specific evocative metaphor for midrash as a “mediating bridge” between scripture and history, from the review of Neusner’s book by Stephen D. Freade, *The Journal of Religion* 69, No. 3, (Jul. 1989): 440. Also H.L Strack and G. Stemberger suggest that the beginnings of midrashic exegesis are already contained within the biblical text. *Introduction to the Talmud and Midrash*, trans. Markus Bockmuehl (Edinburgh: T&T Clark, 1991), 257.
 - 156 Avigdor Shinan and Yair Zakovitch, *From Gods to God: How the Bible Debunked, Suppressed or Changed Ancient Myths and Legends* (Philadelphia: Jewish Publication Society, 2012), 7.
 - 157 Cassuto, *Genesis I*, 70. “It would have been natural to suppose that, when the Torah came to instruct us concerning the creation of the world and man, it would not overlook the existing traditions, whether they emanated from the schools of the sages or from popular folklore. We should have expected Scripture to adapt a given attitude towards them, and to teach us how to extract from them the kernel of real worth and to reject the husk, how to refine and purify them in order to, on the one hand, to remove from them whatever was incompatible with its ethos, and, on the other hand, to harmonize them with its own doctrines and beliefs ... Now this is precisely what the Torah did.” Boyarin argues that both Cassuto and Benno Jacob are the most widely recognised advocates for this role of midrash as the “true representation of the intertext of the Bible.” See Boyarin, *Intertextuality*, 2.

- 158 Gen 1:9.
- 159 Gen 1:9.
- 160 Cassuto, *Genesis I*, 39.
- 161 Cassuto, *Ibid.*
- 162 bTalmud Baba Bathra 74b, as cited and translated by Cassuto, *Genesis I*, 38.
- 163 Cassuto, *Genesis I*, 39.
- 164 Dina Stein, *Rabbinic Interpretation*, 122. See also Joel Rembaum, "Medieval Jewish Criticism of the Doctrine of Original Sin," *AJS Review* 7 (April 1982): 353–382. Though Rembaum only cites medieval Jewish sources which argued against the hermeneutical integrity of original sin, it would be difficult to contend that such a fully formed polemic began in the 12th century, rather than having its formative roots in the classical period. Rashi cites our midrash (Gen. Rab. 22:2), specifically to point out that the first couple had sexual relations prior to the sin, in his commentary to Gen 4:1, s.v. לֹא יָדָע אָדָם. "And Adam knew [Eve], already before the events related above took place — before he sinned and was driven out of the Garden of Eden. So, also, the conception and birth took place before this."
- 165 Gen 1:28.
- 166 Cassuto, *Genesis I*, 162.
- 167 Genesis 3:20.
- 168 Cassuto, *Genesis I*, 170.
- 169 Regarding a pre-biblical origin to midrash, and in addition to the previous cited work of Shinan, Zakovitch and Cassuto, see also M. Stern, *A History of the Jewish People*, ed. H.H. Ben Sasson (Mass.: Harvard University Press, 1997), 360. "In the fifth, sixth and seventh centuries and somewhat later, we find independent collections of aggadah. The earliest of these that have survived are aggadic midrashim (homiletic commentaries) on Genesis, Lamentations and Leviticus from the fifth century." However, Stern notes an interesting etiological hypothesis that these aggadic narratives had been handed down orally by means of the public sermon before collection in written form. Also Emmanuel Levinas, *In the Time of the Nations*, trans. Michael B. Smith (Bloomington: Indiana University Press, 1994), 50. "An order of written meaning exists via a constant appeal to a prior tradition" (italics are mine).
- 170 Boserup, *Population and Technological Change*, 38–39.
- 171 Genesis 6:1.
- 172 Genesis 6:5.
- 173 Boserup, *Population and Technological Change*, 39.
- 174 Isaac Ephraim, "1 Enoch: A New Translation and Introduction," in *The Old Testament Pseudepigrapha*, vol. 1, ed. James Charlesworth (New York: Doubleday, 1983), 56.

3 Acquiring Man

The first man, who after enclosing a piece of ground, took it into his head to say, *this is mine*, and found people simple enough to believe him, was the real founder of civil society. How many crimes, how many wars, how many murders, how many misfortunes and horrors, would that man have saved the human species, who pulling up the stakes or filling up the ditches should have cried to his fellow: Beware of listening to this imposter; you are lost, if you forget that the fruits of the earth belong equally to us all, and the earth itself to nobody!¹

-Jean-Jacques Rousseau, *The Social Contract and Discourses*

What comes after Eden? Evan Eisenberg in his far-reaching treatise on the imagery of paradise in Western culture, *The Ecology of Eden*, observes, “The Hebrew phrase usually translated ‘cursed by the ground for thy sake’ (Gen 3:17) can, with a bit of license, be read to mean ‘cursed is the ground by thy passing over.’ As the waves of human expansion move across the earth, Eden is trampled underfoot.”²

Whilst we must be wary of drawing too heavily on the scant details of Göbekli Tepe to do more than frame the Eden narrative, the epitaph to Göbekli Tepe is itself a trampling underfoot. In approximately 8000 BCE, two millennia after the first stones of Göbekli were erected, the inhabitants of the region backfilled their magnificent structures with thousands of tons of dirt and pebbles in an engineering feat as equally daunting as the erecting of the site itself.³ As the wild game and lush ecology dwindled into oblivion, so too did these first farmers erase the images of their lost paradise.

If Göbekli Tepe freeze-frames the moment just before the onset of agriculture, 50 miles across the mountain range, archaeologists are digging at another site, Çayönü Tepesi, uncovering the ruins of an 8,000-year-old settlement just on the agrarian side of the Neolithic divide. The site of Çayönü Tepesi squats along the banks of an upper tributary of the Tigris, in the Diyarbakir province of Turkey, distinguished against the treeless inundation of brown hillocks by a low, uniform oval mound about 250 by 150 m, with a flattened top. Radiocarbon dating reasonably dates the prehistoric occupation of Çayönü between 7300 and 6500 BCE,⁴ housing between 100 to 200

inhabitants at any given time.⁵ Archaeologists have identified Çayönü as one of the earliest sites of pig domestication, and arguably sheep as well.⁶ Its stone storehouses contain evidence of some of the earliest forms of domesticated strains of wheat in the world.⁷ These storehouses also testify to another legacy of agriculture, that unique ability of the farmer to hoard large quantities of food and exchange it over time for goods and services. Whilst game needs to be consumed in a relatively short span of time and wild harvests are sporadic,⁸ staple cereal crops such as wheat, rice, and barley produce a relatively abundant and reliable food supply that stores well.⁹

When food can be effectively amassed and stored, it becomes owned. And with ownership emerges economic stratification.¹⁰ As the archaeological description of Çayönü's urban plan indicates:

To the north, this space was terminated by three large, manorial houses that had identical fronts, alignment and distance from each other. These houses stood on an elevated platform on massive foundations made from big hewn blocks and had carefully constructed stone walls, a verandah and stone stairs. In these three houses, the wealth of the society was concentrated: large blocks of crystals, stone sculptures, shells from the Mediterranean Sea and from the Red Sea, as well as imported weapons of high quality. In the Western part of the settlement, the houses were only half as large, of distinctly poorer quality, without any additional features and were not built according to a standardized plan. Only the few tools needed for daily living were found there.¹¹

"Hunter gathering societies," Jared Diamond notes in his seminal analysis of global power and historical advantage, *Guns, Germs, and Steel*, "tend to be relatively egalitarian, to lack full-time bureaucrats and hereditary chiefs, and to have small scale political organization at the level of the band or tribe. That's because all able bodied hunter gathers are obliged to devote much of their time to acquiring food. In contrast, once food can be stockpiled, a political elite can gain control of food produced by others, assert the right of taxation, escape the need to feed itself, and engage in full-time political activities."¹²

The confluence of storehouses and grand homes at Çayönü would imply that a feedback loop had quickly taken root in the sedentary agrarian society of Southwest Asia. More stored grain means more wealth and power could accumulate in fewer hands. The more power possessed by the elite, the greater the capacity to dictate the unequal distribution of grain, creating yet more power and more unequal distribution.¹³ Archeologist Katherine Wright documented a general architectural shift in food storage techniques within Levantine Neolithic houses, from open, easy-access patterns in the early Neolithic villages to a propensity towards "sheltered, private spaces in larger, more enclosed houses" in the late Neolithic.¹⁴ Ethnographic studies have bolstered the hypothesis that more intensive plough or irrigation-based

subsistence systems create higher levels of wealth inequality and inheritance regimes than societies practising low-intensity agriculture such as swidden or hand-tool cultivation.¹⁵ Such cross-cultural and historical surveys indicate that as agricultural practices intensify, material inequality grows apace.¹⁶ It was farming, it would seem, that led the Rousseauian “imposter” of our opening epigraph to claim a plot as one’s own.¹⁷

The Bible’s First Owner and Its First Taxman

As Adam and Eve leave the garden of Eden they give birth to a child. The name Eve foists on that baby is telling. He is called Cain, Hebrew for “acquisition”.¹⁸ As Eve informs the reader in the naming act, “I have acquired a man with God.”¹⁹ It is the very first time the concept of ownership is explored in the biblical account. Cain, the first biblical acquirer, is also a farmer (Gen 4:2), a murderer (Gen 4:8), and eventually, the builder of the prototypical city (Gen 4:17). In broad strokes, Cain is the embodiment of early agricultural society, that is to say, concerned with capital ownership,²⁰ possessive of land,²¹ and magnetised towards an urban core.²² I will argue that Cain’s transformation from agrarian agriculturalist to urbanist, shot through with violence, can also be read as a warning against the incursion of the state and its agents.

Before dwelling on the premeditation and aftermath of Cain’s fratricide, I want to draw the reader’s attention to the complex phrasing contained within the divine warning to Cain when he grasps that his brother’s offerings have elicited more favour than his own. Here, the verses are simply so difficult to translate, that we can offer numerous ancient and modern variants, each with uniquely robust scholarly backing.²³ “This is one of the most difficult and obscure biblical sentences,” laments Cassuto.²⁴ Its meaning, and even correct punctuation, have flummoxed biblical exegetes from the Talmudists²⁵ to Gunkel, who left a portion of the verse untranslated within his own biblical translation.²⁶

I will rely on Cassuto’s interpretation as a baseline, as it offers an intriguing allusion to nascent state authority. “The Lord said to Cain: Why are you vexed, and why has your countenance fallen? Surely, if you do well, you shall be upstanding; but if you do not do well, sin shall be a government official (מִשְׁלָטָה) at your door, its desire shall be for you, but you will be able to master it.”²⁷ Cassuto’s reading of מִשְׁלָטָה as a government official, “not necessarily one of the highest order, but one most hated by the people,” is derived from the similar title for an Akkadian functionary, *rabisu*.²⁸ Though conjectural, it would mark the first incursion of state authority into Genesis.²⁹ And what better entrance than at the doorway of the newly emergent domus of these early sedentary communities.³⁰ The hated government official, perhaps the taxman, who now knows the precise threshold where one lives.³¹ In the eyes of the state tax collector, the grain farmer was the perfect host. Their crops grew at determinate times with nearly simultaneous ripening, so the

harvest tax schedule could be accurately fixed on the calendar. And grain farmers were sedentary and perpetually rooted to their fields, and thus legible to state records. Nongrain peoples, contrastingly, such as hunters and gatherers, foragers, and swidden agriculturalists were “fiscally sterile” in state eyes, argues James Scott. “In one way or another, nongrain peoples ... embodied forms of livelihood and social organization that defeated taxation: physical mobility, dispersal, variable group and community size, diverse and invisible subsistence goods and few fixed-point resources.”³²

If Cassuto’s translation of *נִצַּח* is authentic, then the divine warning to Cain is twofold. First, that one should be as circumspect of one’s own interior anger as one is of the creeping³³ state bureaucrat who comes calling at the door. As Cassuto adds, “Sin wishes to master you and have dominion over you like the state officials who seek to impose their authority over the people.”³⁴ And second, God warns Cain that the tax collector *desires* him, wants Cain, the grain farmer, in his grasp, and knows which door to stand against and when to do so. The pristine state could not have existed unless it could effectively call at the door of the sedentary grain farmer and demand his surplus. God’s admonition to Cain, which binds together the threat of violence and the crouching taxman, is the first measure of the biblical stance on emergent statecraft. And the warning is clear: do not let it cross your threshold.

The Ellipsis in the Field

Cain’s killing is not a murder of passion; it is a murder for capital. A particular ellipsis in the biblical scene of Abel’s murder reveals this key node of the narrative by omission. A literal translation of Gen 4:8 would read as follows: “Cain said to Abel his brother [...] and when they were in the field, Cain rose up to Abel his brother and murdered him.”³⁵ The “[...]” indicates a gap in the narrative. The Hebrew text skips over just what Cain said to Abel on that fateful day. Most translations have attempted to cover over this rather glaring hole in the flow of events.³⁶ Yet, perhaps the text is silent with purpose. As though to say: and if you will ask what was said that led to this murder, the answer is that it hardly matters what was *said* between the brothers, the real impetus for this act was the salient fact that they were *in the field*. The farmer and the herder were fighting violently over the limited resources of arable land.³⁷

Rabbi Hirsch, the master Hebrew philologist, entwines Cain’s acquisitional name with the farmer’s desire to possess a particular parcel of land. “He (Cain) becomes totally absorbed in his ‘profession’ of gaining a scanty subsistence for his bodily requirements, and what we found of pride of possession, and pride of one’s accomplishments in the conception of *עֲשָׂה, דַּשׁ* is most predominantly to the fore in the occupation of agriculture. The soil, fertilized with his own sweat is something which is precious to him, it contains part of his very self, it chains him, he becomes stationary, earthbound.” In Hirsch’s reading, the impetus to

acquire, which is the very Cain-ness of the farmer, leads to an eventual subjugation of both the self and the other. "In bending his neck under the yoke of the effort to attain real property his soul too becomes bowed. He gets caught by this effort. It begets subjugation, one man enslaved by the other."³⁸ Cain has become so caught in his effort to obtain "real property" that he commits the ultimate act of subjugating another human being: murder.

Hirsch concludes that the Hebrew appellation for bread (לֶחֶם) is the root-word, and thus the etiological foundation for war itself (מִלְחָמָה).³⁹ Hirsch invokes Hobbes' imagery of the war of all against all to describe the post-edenic age of grain farming. The thick walls and numerous private storehouses of Çayönü would imply that these first Neolithic farmers would understand precisely this definition of bread and the struggle it gives rise to of "all against all."

Economic historians sterilise such vicious associations between property rights and comparative advantage within the clean lines of pure models. Yet, even within such modelling lies the hint of the ferociously violent potential for the hoarding of land and food to create clear winners, farmers, and losers, foragers. Douglass North and Robert Thomas argue that agriculture ultimately prevailed over other subsistence strategies because of the increasing exclusivity of property rights associated with farming.⁴⁰ "It is inconceivable," they postulate, "that, from the very beginning, the first farmers did not exclude outsiders from sharing the fruits of their labour."⁴¹ Exclusive property rights incentivised superior technology, efficiency, and productivity because each new fix directly rewards the owners, and offers a progressive advantage over non-owners. Cain and his offspring prosper precisely because he is willing to kill for the property under his feet.

It is worth noting that whilst the curse of Adam was directed primarily at the ground "on account of" an act of the human, the curse of Cain is in precise reversal. "And now, you [Cain] are cursed due to the ground that opened its mouth to take the blood of your brother from your hand."⁴² Now the curse is directed at the man, because of the action of the ground. A cursed ground has produced a cursed man. Whilst the fall of Eden begins with repercussions to the soil, these repercussions soon begin to infect the human community itself. Eden tells the story of how the act of agriculture damaged the natural environment. The Cain and Abel narrative follows that story, to comment on how agriculture recreated our relationships with one another.

In the aftermath of Abel's murder, God's curse of Cain extends beyond a symbolic infection passed from a cursed earth to a cursed human. Cain's home soil becomes infertile, and the farmer is forced to wander. "When you work the ground, no longer will she give her strength to you; restless and itinerant (פָּרָד) will you be on the earth."⁴³ And here, unlike the silence of Adam, Cain rails against the curse and its repercussions. "Cain said: My sin is too great to bear! Behold, today You have banished me from the face of the ground, and I am to be hidden from Your face. I will be restless and itinerant on the earth, and whoever finds me will kill me."⁴⁴ Cain reiterates the divine

curse, but concludes with a new and startling conclusion: “whoever finds me will kill me.” Why should Cain equate nomadism with certain death?

Dangerous Wandering

With the modern bureaucratic nightmare associated with obtaining a visa to live in a foreign country, and horrific tales of migrants turned away at the borders of the West, we may consider modernity to be narrow and parochial in its organisation of space. We may conjecture that an itinerant Cain would be more severely hampered by passport control or barbed wire boundaries than within the relative freedom of a non-nationalized geography. Yet, traditional peoples, especially those in dense population areas, would probably never venture further than a few miles from their home terrain throughout the course of their lifetime. And this, Jared Diamond notes, is because every encounter between two strangers was life-threatening. “Rarely or never do members of small-scale societies encounter strangers, because it is suicide to travel into an unfamiliar area to whose inhabitants you are unknown and completely unrelated.”⁴⁵ Diamond considers traditional societies to array themselves on a sliding scale towards more or less mutual exclusivity in defended territories, and thus more or less inherent hostility to the stranger. Four conditions tighten the boundaries and limit the mobility of neighbouring residents. First, a large and dense population that can spare a few of its members to patrol boundaries. Second, a productive, stable, predictable environment within which the territory owners can usually source most or all of their necessary resources, and thus need not venture beyond their own boundaries. Third, a valuable fixed resource worth defending. Finally, a relatively closed circle of group membership to maintain a level of recognised distinction from other groups.

The shift from hunting and gathering to agriculture in the Anatolian hills would reinforce the first three, if not all four, of Diamond’s fundamental conditions towards exclusive land ownership. Farming increases population density and allows for a surplus number of workers to be used in fighting and boundary protection. Farming also creates a total package of food, clothing, and shelter resources in a limited and defensible area of land. Agriculture likewise requires a large input of initial work and time and is thus consistently worth protecting to ensure that the labour involved yields a significant return.

Agricultural innovation creates the perfect storm for inelastic territories and heightened intolerance for strangers. What about Diamond’s fourth factor, a closed circle of distinctive group membership? On this point, we simply lack the requisite data to evaluate the relationships between stranger and native-dweller in the early Neolithic of the Levant. The excavation of early villages does reveal considerable evidence of a trading network in the Near East, in such objects as obsidian stone and marine shells, that extends from Anatolia to locations over thousands of miles away.⁴⁶ However, Diamond opines that though the objects themselves may have travelled for

many hundreds of miles and across numerous land boundaries, the traders did not. "Most trade in traditional small-scale societies was short range trade between neighbouring groups because intermittent warfare made it dangerous for people to make trading trips that passed through several different populations."⁴⁷

As the population shifted towards reliance on domesticated flora and fauna, group territories would gradually become smaller and more inelastic, boundaries would become more defined and concomitantly more dangerous to transverse. Thus, when Cain states that "whoever finds me will kill me," he is simply foretelling the inevitable result of leaving his home territory, labelled as a stranger in the newly emergent spatial categories of the farming age.

God's response to Cain is, in essence, an acknowledgement of the deadly threat of strangerhood in this new world of agriculture. "The Lord said to him, indeed! Whoever kills Cain will be punished seven times as much. And the Lord placed a sign for Cain so that whoever would find him would not kill him."⁴⁸ And because the text does not dwell on the details of that "sign," it is left to the midrashic imagination to fill the biblical lacuna. The rabbinical mind provides a full scope of possibilities: ranging from full-blown leprosy to a horn atop Cain's head. I am drawn to the least fanciful, and thus perhaps the most authentic, description "Rav said: '[the mark was] A dog for him.'"⁴⁹ A stranger alone was in mortal danger, but a stranger accompanied by a dog who could *signal* him at the approach of danger,⁵⁰ particularly if the story unfolds in the threshold moment in which domesticated dogs were rare,⁵¹ holds an immediate and "marked" advantage.

City of Strangers

There is a jarring postscript to this story of the wandering Cain. From the moment he leaves God's presence, Cain adamantly refuses to wander. "Cain left the presence of the Lord. He settled (יָשָׁב) in the land of Nod (נוֹד), to the east of Eden."⁵² Instead of an itinerant lifestyle, Cain dwells steadfastly in the land of Nod. Intriguingly, the very name for the land, Nod, where Cain settled and does *not* wander, is derived from the Hebrew root-stem to wander used in the Divine curse upon Cain (פָּרַד) in Gen 4:12. The 11th-century Biblical commentator, Rashi, catches this intriguing refrain of a verb turned into a proper noun, and of Cain's exilic curse transformed into his sedentary homeland. "*In the land of Nod*: [for it was] a land where all the exiles wandered there."⁵³ There are two ways of interpreting Rashi's terse exegesis on this paradoxical juxtaposition of wordplay between Cain's settled state (יָשָׁב) in a region called Wandering (נוֹד).⁵⁴ First, that Nod was the land that gathered in wandering exiles and itinerants, concentrating strangers into a settled core. In this reading, the Bible is elegantly and economically describing the radical new social order that we describe as urbanism. For the first time in human societal evolution, the bundling of various tribes and villages into a single concentrated zone will necessarily break down the barriers between strangers and coerce former enemies to dwell

together in relatively close proximity.⁵⁵ The land of Nod is the land where migrants stop migrating and dwell permanently and together. Thus, it is not surprising in the least that as soon as Cain settles in Nod he builds the Bible's first cityscape (Gen 4:17). Cain, the foreigner *par excellence*, can live unmolested in his primordial city, where everyone is inherently foreign.

Jared Diamond notes a very similar shift in the Papua New Guinea society following the 1931 date of first contact between small settlement Guineans and Europeans, which catalysed a headlong burst into the modern urban age. Recording his observation in the airport of Port Moresby, Papua New Guinea's capital city, Diamond writes: "Still another distinction of the 2006 crowd compared to the 1931 crowds was a feature that we take for granted in the modern world: most of the people crammed into that airport hall were strangers who have never seen each other before, but there was no fighting going on among them. That would have been imaginable in 1931, when encounters with strangers were rare, dangerous and likely to turn violent."⁵⁶ In a sense, Cain's biblical narrative encapsulates the last 70 years of Papua New Guinea's history, as well as the longer shift of populations towards urban hubs in the Mesopotamian plain. Rural villages with closely guarded rigid boundaries and limited resources engendered violent confrontation between strangers. But due to that very violence, and soil depletion in the higher elevations, migrant populations began to coalesce in defensible, well-irrigated, and fertile urban zones. The many faces of Cain, namely farmer, murderer, exile, and city-maker, are perhaps the unfolding drama of that societal process etched and embodied into the life of a single, paradigmatic man.

In order to articulate the second approach to the paradoxical way that Cain could enact a sedentary life in a land named the Wandering place, let us return to God's curse upon this murderous farmer. "When you work the ground, no longer will she give her strength to you; restless and itinerant shall you be on the earth."⁵⁷ Cain will presumably try to grow crops, but the soil will not cooperate. The earth is steeped in Abel's blood, yet it remains nutrient poor, forcing Cain to seek his sustenance always elsewhere. According to anthropologist Hugh Brody, the exilic plight of Cain is not a specific punishment, but is really just an expression of the natural state of the early agriculturalist. "Farmers appear to be settled, and hunters to be wanderers. Yet a look at how ways of life take shape across many generations reveal that it is the agriculturalist, with their commitment to specific farms and large numbers of children, who are forced to keep moving, resettling, colonising new lands. Hunter-gatherers, with their reliance on a single area, are profoundly settled ... Agriculture evokes the curses of Genesis."⁵⁸ Brody offers a radical paradigm shift. The domus is fragile and vulnerable because it is reliant on a narrow food web comprising just a few domesticates. Thus, it is threatened at its core by crop failure, weather fluctuations, resource depletion, warfare, and dense concentrations of threatening pathogens unknown to mobile peoples.⁵⁹ Hunter-gatherers and shifting cultivators operated within a set boundary of food subsistence zones in a regular and consistent rhythm. The early Mesopotamian farming communities, conversely,

established and abandoned their homes throughout the alluvial valley in fraught and jerky pulses of feast and famine. Writes Scott, "Over the roughly five millennia of sporadic sedentism before states, archeologists have recorded hundreds of locations that were settled, then abandoned, perhaps resettled, and then abandoned."⁶⁰

So after Cain depleted the soil and sapped its strength, he abandoned his home. Thus Cain is entrapped in a cycle of fragile sedentism followed hard by forced migration. The farmer is consistently building an unsustainable domus upon a landscape of future exile. Is this what the narrative implies in the idiosyncratic juxtaposing of Cain's settling in a wandering land? That the farmer's narrow "bandwidth" of crops and cattle is so fraught with impermanence that it becomes an act of dwelling in the wandering place of Nod. A land, if we are to reread Rashi in this light, in which "all the exiles wandered" from one collapsed grain settlement to another.

As we leave Cain upon the urban alluvium, I am struck again by the pendulation of his journey. Here is the farmer at the cusp of the proto-city-state. Could Cain have ever escaped his name, turned away the crouching demon-taxman at his door, spoke with his brother rather than kill him in the field, and uplifted the curse of the earth?⁶¹ Or, is the farmer inevitably entangled in an inexorable trajectory into hoarding, violence, and a cursed journey towards statehood?

Notes

- 1 Jean-Jacques Rousseau, *The Social Contract and The First and Second Discourses*, Susan Dunn, ed. and trans. (New Haven: Yale University Press, 2002), 113.
- 2 Eisenberg, *The Ecology of Eden*, 96–97.
- 3 Konstantin Pustovoytov, "Soils and Sediment at Göbekli Tepe, Southeastern Turkey: A Preliminary Report," *Geoarchaeology* 21 (October 2006): 712. "Filed observations suggest that the architectural constructions ... were intentionally covered by fill consisting, for the most part of limestone fragments and fine earth matrix. However, it remains unclear whether the burial by stones and earth was a single event or a more complex process. In addition, the chronology of the burial event(s) is not established. One of the last stages of the heaping activity at Göbekli Tepe took place about the beginning of the 8th Millennium BC. This date is important because it may correspond to the time of the final sealing of the PPN {pre-pottery Neolithic} buildings by fill and therefore approximate the age of the abandonment of the site."
- 4 Charles Redman, *The Rise of Civilization: From Early Farmers to Urban Society in the Ancient Near East* (San Francisco: W.H. Freeman, 1978), 153.
- 5 Redman, *The Rise of Civilization*, 154.
- 6 Redman, *The Rise of Civilization*, 162–163. "In the later subphases, the evidence was reversed: sheep and goat bones together were about thirteen times as numerous as those of aurochs and red deer. Such a shift in the principal food animals is by itself a strong indication of animal domestication by the latest subphase—and indication that is reinforced by the slightly smaller size of the sheep and goats by certain changes in their body proportions ... Pigs are abundant in all subphases and there is some evidence, although not conclusive, that they had been domesticated by the last subphase."

- 7 Redman, *The Rise of Civilization*, 163. "The people at Çayönü engaged in agriculture, if only to a limited extent, even in its earliest occupation; through time, its importance increased. Evidence of the cereals at Çayönü included domesticated einkorn and emmer wheat."
- 8 Peter J. Wilson, *The Domestication of the Human Species* (New Haven: Yale University Press, 1988), 37. "Sharing among hunter/gatherers is ubiquitous, though the pattern may differ between societies. A virtually universal rule is the sharing of the meat of large animals ... One or two individuals can neither consume nor store a large animal on their own. Given the uncertainties of hunting, those who are successful one day may fail on another."
- 9 Clive Ponting, *A New Green History of the World: The Environment and Collapse of Great Civilisations*, rev. ed. (London: Vintage, 2007), 52. "The greater amount of effort involved in farming did mean that in most circumstances more food was produced and often this was more than the individual farmer and his family could eat. It was this food surplus that was the foundation for all later social and political change."
- 10 Ponting, *Green History*, 52. "The move to growing crops in fields and herding and breeding animals opened the way to seeing land, resources and food as 'property'. The far greater amount of time and effort required by farming reinforced this trend. Perhaps at first land and food was owned by the community but rapidly individuals did so and access to land and food became unequal."
- 11 Mehmet Özdoğan, "Neolithization of Europe: A View from Anatolia. Part 1: The Problem and the Evidence of East Anatolia," *Poročilo XX* (1995): 32.
- 12 Diamond, *Guns, Germs and Steel*, 90. See also Nils-Petter Lagerlöf, "Slavery and Other Property Rights," *The Review of Economic Studies*, Volume 76, Issue 1 (January 2009): 319. "Most human societies started off in a relatively egalitarian state as low-level technology and an abundance of land provided few incentives to enclose land. Over time, many of these societies transformed first into a state of despotism and slavery, with a political elite owning both people and land."
- 13 See Diamond, *The Third Chimpanzee*, 189. Whilst we focus here on the cumulative negative societal impacts of agricultural adaptation, many historians point to the dawn of the Neolithic as the concomitant turning point in the rich and varied artistic, intellectual, and technological revolution that could only be realized by artisans, priests, engineers, craftsmen, organizers, and the like, newly freed from the immediate need to source their own sustenance, and now able to devote time and capital to these projects. "However," notes Diamond, "the whole emphasis on leisure time as a critical factor in explaining artistic differences among human societies seems to me misguided. It's not lack of time that prevents us today from surpassing the beauty of the Parthenon. While post-agricultural technological advances did make new art forms possible and art preservation easier, great paintings and sculptures on a smaller scale than that of the Cologne Cathedral were already being produced by Cro-Magnon hunter-gatherers fifteen thousand years ago. Great art was still being produced in modern times by hunter-gatherers such as Eskimos and Pacific Northwest Indians. In addition, when we count up the specialists whom society became able to support after the advent of agriculture, we should recall not only Michelangelo and Shakespeare, but also standing armies of professional killers."
- 14 K.I. Wright, "The Social Origins of Cooking and Dining in Early Villages of Western Asia," *Proceedings of the Prehistoric Society* 66: 1 (2000): 89–121. With later support from S. Kadowski, "Ground Stone Tools and Implications for Use of Space and Social Relations at 'Ain Abu Nukhayla, a PPNB Settlement in Southern Jordan," in *Domesticating Space: Construction, Community, and Cosmology in the Late Prehistoric Near East*, eds. E.B Banning, M. Chazan (Berlin: Ex Oriente, 2006), 53–64.

- 15 Jack Goody, *Production and Reproduction: A Comparative Study of the Domestic Domain* (Cambridge: Cambridge University Press, 1976), 90.
- 16 Jack Goody, *Production and Reproduction*, 97. Because of the “high productivity of the plough, the specialization that this permits, the scarcity of land which it creates or aggravates, and the differential holdings of land and capital that then become important.” Yet, cf. Graeber and Wengrow, *Dawn*, who argue that the first agricultural settlements and even the first cities do not uniformly skew towards increasing inequality, but instead represent a wide variety of social and economic structures, sometimes even altering from season to season. “Settlements inhabited by tens of thousands of people make their first appearance in human history around 6000 years ago, on almost every continent, at first in isolation. Then they multiply. One of the things that makes it so difficult to fit what we know about them into an old-fashioned evolutionary sequence, where cities, states, bureaucracies and social classes all emerge together, is just how different these cities are. It’s not just that some early cities lack class divisions, wealth monopolies, or hierarchies of administration. They exhibit such extreme variability as to imply, from the very beginning, a conscious experimentation in urban form.” Graeber and Wengrow, *Dawn*, 283.
- 17 However, it’s important to stress that even Rousseau himself never intended his description of the first land owner to be considered anything other than a model, as opposed to a realistic description, of prehistory. “Let us begin, therefore, by laying aside the facts, for they do not affect the question [of the origin of inequality]. The researches, in which we may engage on this occasion, are not to be taken as historical truths, but merely as hypothetical and conditional reasonings, fitter to illustrate the nature of things, than to true their true origins, like those systems, which our naturalists daily make of the formation of the world.” Jean-Jacques Rousseau, *Discourses*, 88.
- 18 See E.A. Spieser, *The Anchor Bible: Genesis* (New York: Doubleday, 1964), 29. Also, U. Cassuto, *Genesis I* (Jerusalem: The Magnes Press, 1961): 197–203. Cassuto offers the translation, *to acquire*, as the “usual signification of the stem שׁוּב .” Josephus, in his *Antiquities*, likewise interprets Cain’s name as a derivative of “possession,” *Ant.* 1.52–53, as cited in John Byron, *Cain and Abel in Text and Tradition: Jewish and Christian Interpretations of the First Sibling Rivalry*, (Leiden: Brill, 2011), 213. However, Cassuto prefers a rendering of the word related to fashioning, shaping, forming, or giving birth to. For other variant translations, see also the translations of Bernhard Stade, Hermann Gunkel (in the third edition of his commentary to Genesis) and Sigmund Mowinckel, who derive the name of Cain from the Semitic word for metal workers or *smiths*, Arabic *qeynun*, and Aramaic *qena’a*, as cited in John Day, “Cain and the Kenites,” in *Homeland and Exile*, eds. G. Galil, M. Geller, A. Millard, (Leiden: Brill, 2009), 335–346. However, Cassuto finds the translation of Cain’s name to mean *smith* unsatisfactory, as this is only a secondary sense of this root.
- 19 Gen. 4:2.
- 20 See Josephus’ negative portrayal of Cain. Not only did he set the world’s first land boundaries, but he also invented the first weights and measures of uniform commerce, which “changed the world into cunning craftiness,” *Ant.* 1.61, as cited by Byron, *Cain and Abel*, 213.
- 21 See Hirsch, *Genesis*, 96–97.
- 22 Scott, *Against the Grain*, 166. “The earliest states are population machines seeking to maximize their manpower base by all possible means.”
- 23 Nahum Sarna, *The JPS Torah Commentary: Genesis* (Philadelphia: The Jewish Publication Society, 1989), 33 and ft. 8. “The Hebrew text bristles with difficulties” notes Sarna. He cites the various and divergent classical exegetes, including Rashi,

Ibn Ezra, Radak, Bekhor Shor, and various Targums. See also the Septuagint's translation, which understands God's response within the context of a cultic sacrifice ("Is it not that if you rightly offer, but do not rightly divide you sin? Be quiet!"), which may be the best "contextual guess." As cited by Mark Scarlata, "Genesis" in *T and T Clark Companion to the Septuagint*, ed. James K. Aitken (London: Bloomsbury T&T Clark, 2015), 23. See also Benno Jacob, *The First Book of the Bible: Genesis*, eds. and trans. Ernest I. Jacob and Walter Jacob (New York: Ktav Publishing, 1974), 35. "This is the most obscure verse in the entire book; a satisfactory explanation of the linguistic difficulties has not yet been found." Also, Speiser, *Anchor Bible*, 32. "The oldest versions are no less perplexed than the most recent interpreters."

24 Cassuto, *Genesis I*, 208.

25 bTalmud Yoma 52a-b.

26 Herman Gunkel, *Genesis*, trans. Mark Biddle (Georgia: Mercer University Press, 1997), 44. "The text is apparently severely corrupt."

27 Gen 4:6–7. as translated in Cassuto, *Genesis I*, 205.

28 Cassuto, *Genesis I*, 210.

29 Though conjectural, Cassuto bases his translation on two strong evidential markers. First, the grammatical irregularity between the feminine קלמה and the masculine מלך implies that the word is a substantive, rather than a participle. Second, the established link between the stem מלך and the Akkadian word for a specific class of demon, the *rabisu*. And the elision between the 'demon' and 'government official' label in specific Akkadian texts. See also E.A Speiser, *The Anchor Bible: Genesis* (New York: Doubleday, 1964), 33. "Now the stem *rbš* in Hebrew signifies "to couch." A pertinent noun is otherwise unattested in this language, but is well known in Akkadian as *rābišum*, a term for "demon." These beings were depicted both as benevolent and malevolent, often lurking at the entrance of a building to protect or threaten the occupants. Phonologically, *rābišum*, both noun and participle, would be matched in Hebrew by *rōbēš*." And John H. Walton, *Zondervan Illustrated Bible Backgrounds Commentary: Genesis* (Michigan: Zondervan, 2013), 38. "An alternative is available if we access earlier Akkadian texts where the *rabišu* is not a demon but an important administrator who served a judicial function. In Ur III texts he was responsible for preliminary examination at trials. By the mid-second millennium, texts from Amarna and Ugarit showed the role of the *rabišu* respectively as local ruler and important witness of documents or at trials." See also the Chicago Assyrian Dictionary XIV, 10–13, as possible loan-word from Akkadian denoting some type of doorstep-demon, as cited by Mark Scarlata, "Genesis" in *T and T Clark Companion to the Septuagint*, ed. James K. Aitken (London: Bloomsbury T&T Clark, 2015), 22.

30 Scott, *Against the Grain*, 90. "The domestication of plants as represented ultimately by fixed-field farming, then, enmeshed us in an annual set of routines that organised our work life, our settlement patterns, our social structure, the built environment of the domus, and much of our ritual life."

31 Scott, *Against the Grain*, 130. "The fact that cereal grains grow above ground and ripen at roughly the same time makes the job of any would-be tax-man that much easier. If the army or the tax officials arrive at the right time, they can cut, thresh, and confiscate the entire harvest in one operation ... Better yet, a tax collector or enemy can simply wait until the crop has been threshed and stored and confiscate the entire contents of the granary."

32 Scott, *Against the Grain*, 135–136.

33 This is the normative meaning of the Heb. root מלך, see Gen 49:9, Exod. 23:5, and Num. 22:27.

34 Cassuto, *Genesis I*, 211.

- 35 Gen 4:8.
- 36 For example, see Aryeh Kaplan, *The Living Torah*, 17. "Cain said [something] to his brother Abel." See also *The Holy Bible: New Revised Standard Version* (Oxford: Oxford University Press, 1995), 4. Interestingly, the Samaritan Pentateuch, Septuagint, and other ancient translations of Genesis include the words spoken by Cain at this point in the story. According to the Samaritan Pentateuch, Cain here says to his brother: "Let us go to the field." According to the Septuagint: "Let us pass through into the field/plain." Both variants are cited by Mark Scarlata, "Genesis," 22–23. Perhaps these ancient translators were equally bothered by the textual ellipsis, and, like modern biblical translations, attempted to fill this unseemly gap in scripture. The rare exception to this phenomena is the Jewish Publication Society's translation, which leaves intact the same lacuna in the English as it appears in the Hebrew. *The JPS Torah Commentary: Genesis*, general ed. Nahum Sarna (Philadelphia: The Jewish Publication Society, 1989), 33. "Cain said to his brother Abel ... and when they were in the field, Cain set upon his brother Abel and killed him."
- 37 According to one midrashic tradition, this was precisely what the brothers were arguing on. "'And Cain said to Abel his brother ... (Gen 4:8)' What were they arguing about? They said: come let's divide up the world, one will take all the land and one will take all the moveable property. This one said: the ground you are standing on is mine! The other one said: what you are wearing is mine! This one said: take it off! The other one said: fly [off the ground]! Because of this '... Cain rose against his brother Abel and killed him.'" *Bereshit Rabba* 22:8, Theodor and Albeck, 213.
- 38 Hirsch, *Genesis*, 96–97.
- 39 Hirsch, *Genesis*, 86.
- 40 D.C. North and R.P. Thomas, "The First Economic Revolution," *Economic History Review* 30 (1977): 229–241.
- 41 North and Thomas, "Economic Revolution," 235.
- 42 Gen 4:11.
- 43 Gen 4:12.
- 44 Gen 4:13–14.
- 45 Diamond, *The World Until Yesterday*, 49–50.
- 46 Redman, *The Rise of Civilization*, 143.
- 47 Diamond, *The World Until Yesterday*, 61.
- 48 Gen 4:15.
- 49 *Bereshit Rabba* 22:12, Theodor and Albeck, 219. Ramban's manuscript version contains a further explanation here: "Cain was afraid of the wild animals, so God gave one of them to Cain to go before him" (Theodor and Albeck, *ibid.* ft. 3). Perhaps, this is a midrashic description of the early domestication of the wolf into the dog.
- 50 James L. Kugel, *How to Read the Bible* (New York: Free Press, 2007), 67.
- 51 Redman, *The Rise of Civilization*, 135: "There is significant evidence that dogs were among the earliest, if not the first, animals to be domesticated ... The evidence reveals that, by 7000 B.C., domestic dogs were present in several early villages in the Near East, such as Çayönü, although in small numbers."
- 52 Gen 4:16.
- 53 Commentary of Rashi to Gen 4:16, s.v. "וְאִם תִּבְרַךְ".
- 54 For a similar juxtaposition, see Gen 23:4, where Abraham describes himself to the community of Hittites as both a "stranger (אֲרֻם) and settler (דָּרֹגֶה)," and the exegetical grappling with this paradoxical description.
- 55 Redman, *The Rise of Civilization*. 266: "Between approximately 3100 to 2700 B.C., there were two major trends that were crucial to the development of urbanism. The first trend was the redistribution of population in the Warka region: the number of

rural communities reached a peak in the Late Uruk and Jemdat Nasr periods, and then declined sharply as the urban centers at Warka and elsewhere grew. It can be inferred that the growth of these centers was due less to an internal increase than to migration from scattered rural communities. The motivation to migrate was probably warfare, for large-scale warfare is documented by the construction of defensive walls around the major urban centers during the Early Dynastic period.”

56 Diamond, *The World Until Yesterday*, 4.

57 Gen 4:13.

58 Hugh Brody, *The Other Side of Eden: Hunter-Gatherers, Farmers, and the Shaping of the World* (London: Faber and Faber, 2001), 90.

59 See Scott, *Against the Grain*, 96–112.

60 Scott, *Against the Grain*, 184.

61 John Steinbeck’s *East of Eden* essentially re-envisioned the very question of Cain’s moral agency as a multigenerational novel splayed across the early 20th-century California landscape. In a pivotal scene, a study-group of Chinese elders devote themselves to a deep Hebraic analysis of Gen 4:7. “After two years we felt that we could approach your sixteen verses of the fourth chapter of Genesis. My old gentlemen felt that these words were very important too—‘Thou shalt’ and ‘Do thou.’ And this was the gold from our mining: ‘Thou mayest.’ ‘Thou mayest rule over sin.’ The old gentlemen smiled and nodded and felt the years were well spent. It brought them out of their Chinese shells too, and right now they are studying Greek.” Samuel said, “It’s a fantastic story. And I’ve tried to follow and maybe I’ve missed somewhere. Why is this word so important?” Lee’s hand shook as he filled the delicate cups. He drank his down in one gulp. “Don’t you see?” he cried. “The American Standard translation orders men to triumph over sin, and you can call sin ignorance. The King James translation makes a promise in ‘Thou shalt,’ meaning that men will surely triumph over sin. But the Hebrew word, the word *timshel*—‘Thou mayest’—that gives a choice. It might be the most important word in the world. That says the way is open. That throws it right back on a man. For if ‘Thou mayest’—it is also true that ‘Thou mayest not.’ Don’t you see?” *East of Eden* (New York: Penguin Books, 2002), 301.

4 A Tale of Two Wives and a City

The early Genesis narrative is embedded here and there with terse little snippets of evocative, yet incongruous, storytelling. And exegetes, from classical to modern, have sought to either integrate these odd discordant loose-ends into the larger tapestry of Genesis, or dismiss them as the flotsam and jetsam that comes with the tide of a multi-scribal textual project such as the Bible.¹ At the heart of this chapter is one such puzzling micro-story, enlivened with the first poetic turn of the Bible, found in the fourth chapter of Genesis. In Chapter 3, we followed Cain down into the alluvium as he established the Bible's first city and called it Enoch, after his only son.² God curses the fratricidal Cain to wander, perhaps to alert the readership that the highest ethical response to the farming life is to leave it behind altogether.³ But Cain refuses nomadism, and instead builds a city, and links its founding and its glory to his own lineage. Let us turn to the text to follow Cain's lineage:

To Enoch was born Irad, and Irad sired Mehujael, and Mehujael sired Methushael, and Methushael sired Lamech.⁴ Lamech took for himself two wives: the name of one was Adah, and the name of the other was Zillah. Adah bore Jabal who was the father of the tent-dwellers and cattle marketeers.⁵ And his brother's name was Jubal, he was the father of the lyre and pipe players. And also Zillah bore a child, Tubal-cain, sharpener⁶ of all brass and iron tools, and the sister of Tubal-cain was Naamah. Lamech said to his wives, Adah and Zillah:

Hear my voice! Wives of Lamech,
Give ear to my speech
I have slain a man for wounding me,
And a young man for my hurt.
If Cain was avenged seven-times,
Lamech will be seventy seven-times!⁷

This is a baffling text, and there is an abiding body of modern scholarship that simply sees these few verses as aberrational to the larger narrative weave of the early Genesis account.⁸ At best, these scholars conclude, we have

before us a derivative snippet,⁹ snatched piecemeal from some larger—yet unknown—earlier source material.¹⁰ Speiser transduces the inclusion of Lamech's song in the Bible to a scribal penchant towards name association. "The so-called 'song of the sword' ... owes its inclusion in the present context to the mention of Cain in the last couplet."¹¹

These interpretations may be correct, yet they are rather sterile. For they dismiss all the questions that abound both internally within these few verses and vis-à-vis their connective purpose to the overall structure of the Genesis plotlines. Other contemporary exegetes, following, in spirit, the overarching Talmudic principle that there are no extraneous stories in the Bible,¹² understand these verses as a not-so-subtle rebuke of Cain and his progeny. Thus, Lamech's "song of the sword," as the parallelistic poem has been colloquially labelled since the 19th century,¹³ is the final iteration of an inheritance of violence. Lamech is not just singing in verse to his two wives; instead he is bellowing a vicious battle dirge as a homage to Cain and the normalisation of cruelty metastasizing within Cain's lineage.¹⁴ Geula Twersky sums up this version of events in her short treatise of the sword song:

Lamech followed in the footsteps of Cain, the builder of the first city, in both deed and spirit. He fathered the inventors of fundamental societal institutions, and predicated the development of that society on a culture of violence. The *a fortiori* of the concluding verse of his song declares that if Cain was essentially able to get away with murder, then Lamech too will surely be able to do the same, and even better. His poem reflects his goals and aspirations for the future development of mankind: 'might makes right.'¹⁵

Twerski is so enamoured of just this narrative of normative belligerency that she envisions Lamech declaiming his battle song whilst brandishing a sword made by his son, Tubal-cain, despite any textual hints to such a gesture.

To some extent, Twerski makes a number of leaps of judgement about Cain and his kin. First, nowhere in the biblical text itself is Lamech's song associated with a sword or any weapon of war. Surely, whilst Tubal-cain was the father of sharpened metal tools, the term חַמֵּץ is more properly associated with farm implements,¹⁶ rather than swords. In fact, as I Samuel 13:19–20 states explicitly, honed cultivating tools, with the חַמֵּץ philological root signature, were a necessity of the farming livelihood. Theodore Hiebert specifically links Tubal-cain's metallurgy with the plough, rather than implements of war. "If the Hebrew term חַמֵּץ is translated in its familiar sense of 'plowman,' Tubal-cain's occupation may be rendered as 'the blacksmith for all who plow with bronze and iron.' He would thus represent the ancestor of village blacksmiths who forged and sharpened the metal plow points."¹⁷ Thus, we may justifiably conjecture that if Lamech was brandishing anything whilst declaiming his poem, it was likely his son's mattock, hoe, or plough.

Second, whilst nearly all modern scholars suggest that Lamech's chant is a threat of violence against his enemies,¹⁸ various classical commentaries read

Gen 4:23 as a rhetorical question.¹⁹ Instead of: “I have slain a man for wounding me . . .,” Lamech could be asking: “Have I slain a man to cause me these wounds?”²⁰ Perhaps, Lamech is not belting out a war-cry, but singing a doleful tune for his wives. This questioning Lamech is the preferred translation of Onkelos²¹ and Pseudo-Jonathan,²² as well as the textual bedrock of a midrash that focuses on an oft-overlooked element of the Lamech narrative: Lamech is the first explicit bigamist in the Bible. The very fact that his song is specifically addressed to both his wives becomes the midrashic springboard for a fantastical narrative. Before we consider the potential of just this midrashic story to reveal the fraught background of the verses before us, let us recall the methodological insights of Shinan, Zakovitch, Cassuto,²³ and Spiegel,²⁴ on post-biblical rabbinical literature brought to bear on cryptic biblical passages.

These scholars suggest that some ancient pre-biblical mythic tales were too blasphemous for whole-cloth inclusion in the Bible’s rendering, yet equally too well-known to be ignored entirely. So, the scribal redactor lobotomised these older myths until their stunted versions were fit to serve the one God and His one people in the biblical sense.²⁵ This would explain why numerous passages in the early Genesis narrative feel so incomplete, including these lines under current scrutiny. Yet, whilst the biblical scribe could cripple the myth to fit the vellum of the page, he could not ban the oral telling around the hearth.²⁶ Eventually, this fuller oral tradition drifted into various post-biblical written records. Thus, when we look to midrash, we must at least consider the possibility that we are peering through the looking glass, into a pre-biblical account, carefully committed to orality over centuries, and thus free from the censure’s pen.

The midrash is not alone in curling Lamech’s bigamy into the narrative plot. Modern exegetes have also noted that every detail a sparse text reveals about a character carries narrative heft. “Lamech is apparently the first polygamist,” notes Sarna, and adds speculatively that, “it is uncertain whether the ascription of polygamy to a descendant of Cain is meant to be a tacit condemnation of the institution.”²⁷ Other scholars are far less equivocal in asserting a negative spin to the Bible’s characterisation of the first polygamist.²⁸ Shama Gelander suggests that the poetical address to both wives is a symbolic representation of Lamech’s all encompassing pedagogy of violence, a war song impressed upon both mothers, and thus taught to the totality of the next generation. “The terrifying moral standards his [Lamech’s] words reflect are unambiguous . . . It is implied that he intended that this should be his wives’ teaching to his descendants.”²⁹ Twerski unites the dual themes of violence and implied sexual servitude which, she argues, are hinted at in the names of Lamech’s wives, to re-envision Lamech as an arch-misogynist. In this, she draws deeply from Cassuto’s entomological insights into the name Adah, meaning ornamentation with jewellery, and Zillah, implying a woman’s song, specifically for enticement.³⁰ “Lamech,” Twerski concludes, “was a man, who reduced men and women alike to nothing more than the recipients of either his vengeance and terror, or his pleasure and amusement.”³¹

Yet, all of these interpretations are predicated on Lamech's song as a menacing growl at his two wives whilst brandishing a threatening dagger. What happens when Lamech is, perhaps, just a farmer with a hoe in hand, and his song is a question: "Have *I* killed a man?" How should we make sense of the numerous curiosities of our story? Why does Lamech have two wives, and why does he sing specifically to them? Why are all his male children named versions of the same root *וָרָה* translated as "produce."³² What is the link between Lamech and the city established by his ancestor, Cain? And finally, what is Lamech's song actually about?

Bereshit Rabba sought to tie many of these odd loose threads into a single narrative weave. Here is that rabbinical oral wrestling with these knotty biblical verses:

R. Azariah said in the name of R. Yodah: The men of the generation of the flood used to act thus: each took two wives, one for procreation and the other for sexual gratification. The one used for procreation would be set aside [after giving birth] like a widow all her life. Thus her name Adah, meaning 'pregnant'. And the other one was used solely for sex. She would have to ingest a poison to keep her infertile, and she was forced to sit before her master like a prostitute. Thus, her name Zillah, meaning that she was 'in the shade' of her husband ... R. Jose son of Hanina said that Lamech summoned them for sex, and they refused saying: "Tomorrow a flood will come, are we to bear children for a curse?" Lamech answered them: "Have I slain a man to cause me these wounds?" Meaning: why should *I* be wounded with barrenness because of his [Cain's] sin [which would eventually cause the flood]. "And a child for my hurt?" Meaning: why is this hurt of not having more children, now become *my* hurt. Continued Lamech: "Cain murdered and still judgement was held back for seven generations; I, who did not murder, shouldn't my judgement [that is, no future children] wait seventy seven generations?"³³

This midrash is a gem. Its themes are idiosyncratically contemporary! Two oppressed wives, kept respectively as procreative stock and sexual object, rebel by withholding their bodies whilst their master attempts to cajole them in verse, the very first poetic display in the Bible. That these women refuse sex because they are both prescient and deeply fretful for the well-being of their children reveals that small, compelling voice of the marginal which is the midrashic gift. For a moment, however, let me draw our focus away from the gender repression, despair, and potential reconciliation (were Adah and Zillah wooed by Lamech's tune? The midrash leaves the tale unfinished) at the heart of this rabbinical dialectic, and recall that these biblical verses recounting the first act of polygamy follow hard on another aetiology, the very first city.

Modern biblicalists disagree about the moral texture of Cain's cityscaping. According to Westerman, the Yahwist writer treats the city favourably, as the first achievement of civilization.³⁴ Other modern biblical exegetes, such as

Wallis,³⁵ Bloom,³⁶ and Frick,³⁷ argue that the city is a locus of evil, founded by a murderer, Cain, and anathema to the nomadic pastoralism at the heart of the biblical milieu. Still, other scholars, namely Hiebert, conclude that Cain's city is simply a "neutral" and "cryptic" side-point, without central interest to the overall biblical account of ruralism.³⁸ Yet, like the first biblical case of bigamy, the very first city of the Bible is crucial to the overall narrative structure of Lamech and his family. Perhaps, the juxtaposed introduction of these two salient cultural innovations is indicative of a link between urbanisation and polygamy.

That the farmer Cain founded the first city in the alluvial soil of Mesopotamia is generally in keeping with the archaeological record of events. By the first half of the third millennium BCE, roughly 20 competing urban grain centres began to emerge as independent "peer polities" in the southern Mesopotamian lowlands.³⁹ City-states such as Ur, Kish, Nippur, Isin, Lagash, Eridu, and Uruk each dominated a wheat and barley producing core of 10–30 km with a fortified populace at its bureaucratic and religious centre.⁴⁰

These competing states rose and fell by the dint of their capacity to amass and concentrate people. James Scott notes that "aside from an advantageous position on trade routes and waterways or particularly clever rulers, agricultural techniques as well as the technology of warfare were both relatively static and depended largely on manpower. The state with the most people was generally the richest and usually prevailed militarily over smaller rivals."⁴¹ The research of Seth Richardson indicates that the wars fought during this era in alluvial Mesopotamia did not pit one large city-state against another, fighting over territorial expansion, but were predominantly petty wars carried out by the larger cities to "herd non-state clients into state orders by both force and persuasion."⁴² Cities domesticated people, as it domesticated plants and animals, to maximise its productive output, in order to create the necessary surplus to sustain its non-producing elites.⁴³

These early cities were hungry for manpower. And there were only two ways for the pristine city-states to increase population density. First, coaxing or coercing individuals or small groups from the periphery to enter the grain centres and become new subjects to the state. Or second, procreating new members. In fact, the two methods of peopling an urban area might coincide in the body of women. Scott argues that the slave records of the ancient states in Mesopotamia, Egypt, and China indicate a particular targeting of women for capture. "Women captives were at least as important for their reproductive services as for their labour. Given the problems of infant and maternal mortality in the early state and the need of both the patriarchal family and the state for agrarian labour, women captives were a demographic dividend."⁴⁴

Not only did the emergent city-states become "population machines"⁴⁵ forcing more women to give birth to more children, but as cities became denser and concurrently stocked with greater food and other material resources (including the humans themselves as potential war-captives), patrilocal residence

and patrilineal inheritance emerged as normative within the city ecology. Though deeply contentious, anthropologist Sarah Hrdy has long argued that the accepted norm of one 'alpha' male mating with multiple females is not written into the genetic stone of humanity.⁴⁶ In her reading of both hominid and primate societies, the conjoined patriarchal strategy of polygyny, patrilocalism,⁴⁷ and patrilinealism is simply one type amongst various and fluid methods to maximise resources and ensure infant health. "I do not doubt," writes Hrdy, "that some early hominids lived patrilocally, relied on male-biased inheritance of status and hunting grounds, and developed many of the essential ingredients of the patriarchal mindsets. But more and more I am impressed by how flexible and opportunistic many primates, including human foragers, are."⁴⁸

So what transformed a wide spectrum of breeding strategies practised by Palaeolithic foragers into male-dominate mono-behaviour? In short, according to Hrdy, agriculture and its subsequent effects, sedentism and resource hoarding:

When we turn to the human ethnographic record, patrilocal residence and polygynous marriages do indeed predominate. But this record post-dates massive population shifts caused by rapidly expanding populations of post-Neolithic agriculturalists and herders. No doubt inter-group raiding for women pre-dated the Neolithic, but defensible land and other material resources along with increasing rates of population growth would have intensified the pressure on men to remain near kin, enhancing patrilocal allegiance and the formation of fraternal interest groups.⁴⁹

Because of their upper body musculature advantage, men, more than women, are primarily suited to defend, acquire, and monopolise scarce resources. Male offspring are then the first candidates to receive resources transmitted across generations, eventually leading to patrilineal inheritance; in the anthropological parlance sons have "greater resource-holding potential" over daughters.⁵⁰ Likewise a kin group of males, over females, is an advantageous asset to control and expand resources. The emergence of fortified cities dotted across the alluvial plain of Mesopotamia seeded the ideal conditions for patriarchy: dense populations, a conglomeration of material resources worth fighting for, and neighbouring cities worth conquering. In short, the proto-city-states created the perfect storm for men to dominate women. Physical, upper body power became ever more essential, whilst the impetus to increase population upscaled the massive subjugation of women's bodies and reproductive functioning to an unparalleled new degree.⁵¹

Let us now enfold these strands of early urbanism into the biblical and midrashic landscape of Lamech, his wives and sons. The narrative conjoining of the first city and the incidental detail of the first polygamist signifies a link between farmers, cities, and a rise of patriarchal control metastasizing within those brick walls. The midrash seizes upon the biblical hint, or perhaps reveals the darker corners that were already intrinsic in the sparse text, to create a

horrorscape of sexual and reproductive male hegemony, replete with the misogynist imagery of the “sex” wife in the “shadow” of her husband looming over her drugged body, whilst the other “childbearing” wife was enslaved and pregnant. The protest of Lamech’s wives, their refusal to breed with their husband-master, is exquisitely effective because it cuts to the bone of the urban proto-states core survival mechanism, the need for an ever-expanding population. Perhaps, this is the very reason that all of Lamech’s son are named derivative forms of the term for “produce,”⁵² *Jabel*,⁵³ *Jubal*,⁵⁴ *Tubal*.⁵⁵ The core product of the Mesopotamian city-states was not grain, or goods, or herds, but people.⁵⁶

These wives, who strike with their bodies, force Lamech to sing a new tune. A tune in parallelistic verse, in which I can nearly envision him turning back and forth, from one wife to the other, to plead first to Adah, and then to Zillah: “Hear my voice! Wives of Lamech,”⁵⁷ and turning, “Give ear to my speech.”⁵⁸ According to the reading of Genesis Rabba, his cry is simple: “What have I done to deserve this? I have not killed like Cain!” And here is the subtle rebuke enfolded within the midrash’s laconic narrative. Lamech’s subjugation of his wives, the societal given of this sad state of relationality, has become so normative that Lamech simply has no idea what he has done wrong. “Have I slain a man to cause me these wounds?” Lamech, the urbanite, lives so firmly within the bounded walls of this particular brand of patriarchy that other models of male–female conjoinment are as alien as weeds in a monocropped field. “R. Azariah said in the name of R. Yodah: ‘The men of the generation of the flood used to act thus, each took two wives ...’,” not a few men, but all of that society acted thusly. And Hrdy, for one, argues that much of the established orthodoxy in her own fields of anthropology and socio-biology still continues to myopically view mating behaviour as Lamech might, that male desire for multiple partners is the universal norm. “I cannot think of any reason why the mating systems and preferences of these [pre-Neolithic] people would be less variable and less flexible than those of foragers today, or less variable than other ‘weedy’ or highly adaptable primates.”⁵⁹ The farming field with its singular mono-crop and the pastoralist herd with its domesticated mono-species would seem to have engendered a singular monoculture as well, a domestication of women’s bodies.⁶⁰

It is telling that the very next verse after Lamech’s plea to his wives departs definitively from the entire vignette. “Adam knew his wife again, and she gave birth to a son, and she called his name Seth, ‘because God has now set for me another seed, in place of Abel, who was murdered by Cain.’”⁶¹ Here is a reset narrative. Whereas Lamech “took for himself”⁶² two wives,⁶³ Adam *knows his only wife*. Though the Bible never seeks a return to the physical landscape of Eden, here there is more than a hint of a harkening backwards to a recollection of other forms of male–female relationships taking place outside the city walls. A return to a mother gifting a name to her child, not seen since Eve named Cain.⁶⁴ And a remembrance of the name of God, whose presence has not been felt since before the building of the city, when “Cain went away from the face of God....”⁶⁵

The Bible never again informs us of the wives of Lamech; yet their mid-rashic prescience is foreboding. “Tomorrow a flood will come, are we to bear children for a curse?” The curse levied at the earth for the forbidden act of farming, soaked in the fratricidal blood of Cain and Abel’s unshared field, now dwells in the angry voice of enslaved women and their children within the city walls, envisioning a coming storm.

Notes

- 1 See John Barton, *Reading the Old Testament: Method in Biblical Study* (London: Darton Longman & Todd, 1984), 24: “Before long, it became clear that the Pentateuch in particular, considered as a single text or even a corpus of five books, simply defied classification. No thinkable type of writing could possibly contain the inconsistencies, doublings back, apparently aimless changes of style, of titles for God, of narrative tone, that these books encompass. For one man to write such a text, he would have to be mentally incoherent or disturbed, or—and here source criticism really begins—he would need to be using a lot of already existing materials which, for whatever reason, he was unable to change, and setting it down in all of its inconsistency.”
- 2 See Gen 4:17. Because of the vague antecedent of the verbs in Gen 4:17, scholars have disagreed on whether the first city builder was Cain or his son Enoch. For a summary of the arguments, see Westerman, *Genesis 1–11*, 327–328. Whilst Westerman opts for Enoch as the first urban planner, here I follow the traditional opinion of Rashi in his commentary to Gen 4:17, s.v. לֵמֶךְ.
- 3 See Anne Porter, *Mobile Pastoralism and the Formation of Near Eastern Civilization: Weaving Societies Together* (Cambridge: Cambridge University Press, 2012), 323: “For early farmers, leaving the grain core was often the preferred option, and the proto-states acted accordingly to pen these farmers in. The first great wall between nomadic peoples and city-states was the 250-kilometre-long ‘wall of the land’ built around 2,000 BCE between the Tigris and Euphrates by command of the Sumerian king Sulgi. Though it was described as the wall to defend against the incursion of the Amorites, it may have had the additional purpose of keeping the southern Mesopotamia taxpaying cultivators in.” Also, James C. Scott, *Against the Grain: A Deep History of the Earliest States* (New Haven: Yale University Press, 2017), 7–8: “Still more remarkable for those interested in the state form, is the fact that the very first small, stratified, tax-collecting, walled states pop-up in the Tigris and Euphrates Valley only around 3,100 BCE, more than four millennia *after* the first crop domestication and sedentism ... It is simply assumed that weary Homo sapiens couldn’t wait to finally settle down permanently, could not wait to end hundreds of millennia of mobility and seasonal movement. Yet there is massive evidence of determined resistance by mobile peoples everywhere to permanent settlement, even under relatively favourable circumstances. Pastoralists and hunting-and-gathering populations have fought against permanent settlement, associating it, often correctly, with disease and state control.”
- 4 Following Geula Twersky, I have translated the name לֵמֶךְ as “Lamech,” eschewing a rigorous adherence to the Hebrew phonology (more accurately transliterated as “Lemekh”) for the sake of the familiar anglicised version of the name taken from earlier translations of the Bible.
- 5 See Hirsch, *Genesis*, 111. Hirsch links לֵמֶךְ to “buying and selling, earning acquiring ... could very well indicate the industrial side of cattle breeding, the substantial dealer in the market, consonant with city culture, in contrast to the דָּעַם קָלוּ, the more nomadic shepherd. Thus Joseph says the Egyptians despise all nomadic shepherds (Gen 46:34),

- and still Pharaoh had flocks, and all the Egyptians had פָּנֶס סִמְרָד and פָּנֶס סִמְרָד (Gen 47:17) which they brought to Joseph in the years of famine.”
- 6 See I Samuel 13:20. John F.A. Sawyer argues that the verb שָׁמַם describes a very specific technical process in the manufacture of weapons; the affixing of the fine edge, see John F. A. Sawyer, “Cain and Hephaestus: Possible Relics of Metalworking Traditions in Genesis 4,” *Abr-Nahrain* 24 (1986): 155-166, esp. 158. However, other scholars note the nebulous division between sharpening weapons and farm-tools in the biblical context. Geula Twersky, in particular, notes the consistent prophetic fluidity between implements of agriculture and warfare, citing Isa. 2:4, Mic. 4:3, and Joel 4:10. See Geula Twersky, “Lamech’s Song and the Cain Genealogy: An Examination of Gen 4.23–24 within its Narrative Context,” *Scandinavian Journal of the Old Testament* 31:2 (2017): 275–293. Esp 285, ft. 42.
 - 7 Gen 4:18–24.
 - 8 Brian O. Murdoch, *The Medieval Popular Bible: Expansion of Genesis in the Middle Ages* (Cambridge: DS Brewer, 2003), 71: “Neither of the Lamechs, the Cainite nor the Sethite, is really very important. Modern commentaries on Genesis do not have very much to say about either of them.”
 - 9 Speiser, *Genesis*, 36.
 - 10 Alter, *Genesis*, 20. “The narrative context of this poem is long lost, but it looks like a warrior’s triumphal song, cast as a boast to his wives.”
 - 11 Speiser, *Genesis*, 37.
 - 12 See bTalmud Sanhedrin 99b.
 - 13 See Skinner, *Genesis*, 120; Speiser, *Genesis*, 37; Hermann Gunkel, *Genesis* (Georgia: Mercer University Press, 1997), 52; Murdoch, *Expansion of Genesis in the Middle Ages*, 81; J. G. Vos, *Genesis* (Pittsburgh: Crown & Covenant Publications, 2006), 105; S. R. Driver, *The Book of Genesis* (London: Kessinger, 1926), 70.
 - 14 See Terence E. Fretheim, *The New Interpreter’s Bible* (Nashville: Abingdon Press, 1994), 375: “The song shows how violence has been intensified through the generations. This is an important point, because it shows how the sword song is part of the build-up to the moral decay that is the proximate cause of the flood. Sin progresses as civilization progresses.” See also, K. A. Mathews, *Genesis: 1–11* (Nashville: B&H Publishing Group, 1996), 289. “Lamech’s gloating over a reputation more ruthless than infamous Cain’s shows the disparagement of human life among Cain’s seed that was fostered by his murder of Abel.” And Simpson, *The Book of Genesis- Exegesis*, 522.
 - 15 G. Twersky, “Lamech’s Song and the Cain Genealogy: An Examination of Gen 4.23–24 within its Narrative Context,” *Scandinavian Journal of the Old Testament* 31, no. 2 (2017): 285.
 - 16 *BDB*, 360. Related to Phoenician, Arabic, and Farsi for *plough*.
 - 17 Hiebert, *Yahwist’s Landscape*, 44.
 - 18 Though, to be sure, many classical and mediaeval commentators also followed this reading. See Ibn Ezra, commentary to Gen 4.23. Also, *Midrash Tanhuma*, Gen 11, Warsaw edition; *Midrash Agada*, Gen 4:3, ed. Buber; Cuthbert A. Simpson also reads this poem as a war-cry in *The Book of Genesis: Exegesis*, (Nashville: Abingdon Press, 1951). The LXX, Vulgate and Peshitta also reflect this belligerent tradition in their rendering of the text, as cited in Albert Pietersma and Benjamin G. Wright, *The New English Translation of the Septuagint* (Oxford: Oxford University Press, 2007), 8; and R. Coyne, *The Holy Bible Translated from the Latin Vulgate* (Dublin: Coyne, 1847), 5.
 - 19 These variant translations pivot on the exact function of the word נָּ in Gen 4:23. Cassuto offers four variant readings: (1) that it indicates a question; (2) that it introduces *oratio recta*, akin to opening quotation marks; (3) that it is an interjection, like the word הִנֵּה [lo, behold!]; (4) that it gives a reason for the preceding

- statement. The midrash in question prefers the first reading, while Cassuto prefers the last explanation. Cassuto, *Genesis I*, 240.
- 20 See *Bereshit Rabba*, 23.1, Theodor and Albeck, 221.
- 21 Onkelos casts the poem as a question, essentially by translating קָ as the interrogative וָ. As cited in Israel Drazin and Stanley Wagner, *Onkelos on the Torah: Understanding the Bible Text: Genesis* (Jerusalem: Gefen, 2006), 27. Drazin and Wagner reliably date the Targum Onkelos to just after 400 CE (xix).
- 22 See E.G. Clarke, *Targum Pseudo-Jonathan of the Pentateuch: Text and Concordance* [Hebrew] (New Jersey: Ktav Publishing House, 1984), 6. Also *Targum Pseudo-Jonathan: Genesis*, trans. and ed. Michael Maher (Edinburgh: T and T Clark, 1992), 35. "I have not killed a man so that I should be killed for him: neither have I wounded a young man so that my offspring should be exterminated because of him." Modern scholarship generally dates the compilation of the Pseudo-Jonathan or *Targum Yerushalmi* between the 7th to the 10th century (11–12).
- 23 Cassuto, *Genesis I*, 70.
- 24 Shalom Spiegel, *The Last Trial: On the Legends and Lore of the Command to Abraham to Offer Isaac as a Sacrifice*, trans. Judah Goldin (Vermont: Jewish Lights Publishing, 1993), 57. "It may be that here [in the Mekhilta de-Rabbi Ishmael] are the last and faint echoes of some primitive pagan or prebiblical version of the Akedah story. Possibly too, 'the blood of Isaac's Akedah' may be a fossil expression from the world of idolatry, some tiny stone fragment from pagan ruins sunk into the edifice of the talmudic aggadah—and therefore we would have here, as occasionally also elsewhere in the Midrashim of early generations, some leftovers of belief before the beliefs of Israel."
- 25 Avigdor Shinan and Yair Zakovitch, *From Gods to God: How the Bible Debunked, Suppressed or Changed Ancient Myths and Legends*, 7.
- 26 Regarding the resilience of Israelite oral tradition to retain genuine memories, see, for example, Eric M. Meyers and John Rogerson, "The World of the Hebrew Bible," in *The Cambridge Companion to the Bible*, ed. Bruce Chilton (Cambridge: Cambridge University Press, 1997), 47. "In all probability written and oral versions of stories of the ancestors existed side by side, exerting profound influences on one another ... oral traditions probably do preserve genuine memories, such as that the original homeland of the Israelites' ancestors was not Canaan but northern Mesopotamia, and that the ancestors were originally city dwellers, who became pastoralists without fixed abode."
- 27 Nahum M. Sarna, *JPS Genesis*, 45.
- 28 See Harper, "The Fratricide: The Cainite Civilization. Genesis IV," *Quarterly calendar* 3.4 (1894): 264–274; also, Jacob, *Genesis*, 38; Vos, *Genesis*, 98.
- 29 Shamai Gelernder, *The Good Creator: Literature and Theology in Genesis 1–11* (Atlanta: Scholars Press, 1997), 58.
- 30 Cassuto, *Genesis I*, 233.
- 31 Twerski, "Lamech's Song," 293.
- 32 Hirsch, *Genesis*, 111. Commentary to 4:18–22. "All three names are formed from the root הִרָשׁ means to bring, to bring home, to bring in, hence הִרָשׁ produce."
- 33 *Bereshit Rabba*, 23.1–4, Theodor and Albeck, 221–223.
- 34 Westerman, *Genesis 1–11: A Continental Commentary*, trans. John Scullion (Minneapolis: Augsburg, 1984), 327–328.
- 35 G. Wallis, "Die Stadt," 133–48, cited in Hiebert, *Yahwist's Landscape*, 183, ft. 38.
- 36 Harold Bloom, *The Book of J* (New York: Grove Weidenfeld, 1990), 188.
- 37 Frick, *The City*, 205–207, cited in Hiebert, *Yahwist's Landscape*, 183, ft. 38.
- 38 Hiebert, *Yahwist's Landscape*, 42.
- 39 Scott, *Grain*, 120.

- 40 J.N. Postgate, "A Sumerian City: Town and Country in the 3rd Millennium B.C.," *Scienza dell'Antichità Storia Archeologia* 6–7 (1996): 409–435.
- 41 Scott, *Grain*, 154.
- 42 Seth Richardson, "Early Mesopotamia: The Presumptive State," *Past and Present* 215 (2012): 20.
- 43 Piotr Steinkeller and Michael Hudson, in *Labour in the Ancient World*, eds. Piotr Steinkeller and Michael Hudson (Dresden: LISLET Verlag, 2015), 21; as cited in Scott, *Grain*, 151.
- 44 Scott, *Grain*, 168.
- 45 Ogden Goelet, "Problems of Authority, Compulsion and Compensation in Ancient Egyptian Labor Practices," in *Labour in the Ancient World*, eds. Piotr Steinkeller and Michael Hudson (Dresden: LISLET Verlag, 2015), 80.
- 46 For a brief glimpse of the critique levelled against Hrdy's research that polyandrous potential existed within the landscape of the pre-neolithic hominid breeding strategy spectrum, cf. D. Symons, "Another Woman That Never Existed," *The Quarterly Review of Biology* 57, no. 3 (1983): 297–300, esp. 299. "Why should a female be better off with ... three males, each of which invests one-third unit, than with one male which invests one unit?" Also, cf. R. Wrangham and D. Peterson, *Demonic Males: Apes and the Origins of Human Violence* (Boston, MA: Houghton Mifflin, 1996), 125, where the authors argue that patrilocal behaviour is "universal" for primates.
- 47 Approximately 80% of patrilocal societies are polygynous, as cited in S.R. Witkowski and W.T. Divale, "Kin Groups, Residence and Descent," in the *Encyclopedia of Cultural Anthropology*, vol. 2, eds. D. Levinson and M. Ember. (New York, Henry Holt, 1996), 673–680.
- 48 Sarah Blaffer Hrdy, "The Optimal Number of Fathers: Evolution, Demography, and History in the Shaping of the Female Mate Preferences," *Evolutionary Perspectives on Human Reproductive Behavior* 907, Issue 1 (April 2007): 89.
- 49 Hrdy, "Optimal Number of Fathers," 89.
- 50 Hrdy and D.S. Judge, "Darwin and the Puzzle of Primogeniture: An Essay on Biases in Parental Investment after Death," *Human Nature* 4 (1993): 1–46.
- 51 See Boserup, *Agricultural Growth*, 74. "Many wives in polygamic tribes are domestic and agricultural slaves in disguise."
- 52 See Cassuto, *Genesis I*, 233. Also, Richard Hess, *Studies in the Personal Names of Genesis 1–11* (Winona Lake: Eisenbrauns, 1993), 128. However, various other interpretations for the names of Lamech's sons abound. For example, a triad of mythical entities who bestowed upon mankind the gifts of music and the tools of agriculture and warfare, cf. William R. Harper, Ernest DeWitt Burton and Shailer Mathews, eds., *The Biblical World* (Michigan: University Press of Chicago, 1894), 273; or that the root of all three names, *שמע* comes from the term for a stream, or 'bringing forth,' cf. Victor P. Hamilton, *The New International Commentary on the Old Testament: the Book of Genesis*, ed. Robert L. Hubbard Jr. (Grand Rapids: Eerdmans Publishing, 1990), 239.
- 53 Hirsch, *Genesis*, 111. Jabel is the personal active form, thus "the one who brings something in, who creates things of value, produces."
- 54 Hirsch, *Genesis*, 111. Jubal is the passive, the opposite of Jabel, he "produces nothing, things are brought to him."
- 55 Hirsch, *Genesis*, 112. Tubal implies the abstract conception of production, "He created the means for all progress and art, he creates *creating*, he creates the tools and instruments for industry and art."
- 56 See Scott, *Art*, 64. "In its crudest version, the formula goes something like this: Political and military supremacy requires superior access to concentrated manpower close at hand. Concentrated manpower, in turn, is feasible only in a setting of compact, sedentary agriculture."

57 Gen. 4:23.

58 Gen. 4:23.

59 Hrdy, "Optimal Number of Fathers," 89.

60 I owe the arresting, yet speculative, idea that early domestication of crops and animals paved the way to the domestication of human beings, to James Scott. "What if we were to examine slavery, agrarian war captives, helots, and the like as state projects to domesticate a class of human servitors—by force—much as our Neolithic ancestors had domesticated sheep and cattle?" (180). And in particular, the focus of this state domestication project was the bodies of women. "The domestication of non-slave women in the early grain state ... a combination of property in land, the patriarchal family, the division of labour within the domus, and the state's overriding interest in maximizing its population has the effect of domesticating women's reproduction in general." Scott, *Against the Grain*, 181.

61 Gen 4:25.

62 As Twersky points out, the phraseology used to describe Lamech's "taking for himself" is unusual, and thus evocative. "The bible employs an unusual phrase here for taking a wife. Whereas the term usually found for marriage in the bible is *קָבַץ*, here we read *קָבַץ*, 'and he took for himself.'" Twersky, "Lamech's Song," 288.

63 Gen 4:19.

64 Gen 4:1.

65 Gen 4:16.

5 Noah the Ploughman

The next three chapters will examine the biblical flood story, from the birth of Noah, to the survivors atop their mountain fastness, to the cusp of the Babylon urban enterprise. I have arrayed the chapters chronologically. This chapter speaks of the pre-flood milieu and the particular events that lead inexorably towards global genocide. Chapter 6 looks at the direct aftermath of the flood and the particular covenant between God and Noah that occurs then. Finally, we will examine the tragic last scene of Noah and his children, and discuss why Noah's curse upon his progeny will lead the biblical storyline back down from the mountains and into the valley, and the clutches of the state.

The arc of the flood narrative shares striking parallels with the Garden of Eden vignette.¹ As I will argue later in this chapter, by relying upon the insights of Nahum Sarna, Noah can be viewed as Adam 2.0. The flood becomes a reset button on creation, first returning the earth to primordial waters and then beginning anew with Noah in the place of Adam. If these parallels hold true, then we can also explore similarities between the fall of Adam and a fall for Noah and his descendants. Assuming that this flood narrative is primarily a mimetic description of reality, rather than a fantastical myth, we can then ask how the pristine post-flood landscape recreates the world of Eden. I will argue that both are tales from the mountains mainly beyond state control, and thus conducive to a particular type of human flourishing blessed by the divine. In both instances, this non-state existence is ephemeral. As Adam and Eve were banished from their mountain home and their child, Cain, builds a city, so too do Noah's descendants suffer a fall into the state hegemony of the alluvial building enterprise of Babel.

Mapping the trajectory of Noah's escape from the state, his short-lived sojourn in non-state altitude, and his tragic slippage back into the alluvium will be the mainstay of the three chapters before us. However, first we must plot the course of evil that tips the world towards annihilation. I will argue that this tipping point is the plough which Noah invents.

The story of Noah and the flood is a re-interpretive telling of older accounts from Mesopotamia, which have been preserved in three distinct cuneiform incarnations. These are generally classified as *the Sumerian flood story*, components of the *Atrahasis Epic*, and the *Gilgamesh Epic*, respectively.²

The earliest of these tablets, written in Sumerian and dated to approximately 1600 BCE, was discovered in the Sumerian city of Nippur, in what is now southern Iraq.³ A certain level of geological investigation has been devoted both in the past and currently to various theories of a “true” mega-flood that swept through the region. Such a flood would lay a deep and lasting impression in the minds of storytellers thereafter, and would be the catalysing progenitor of all the variant narratives before us. Indeed, various culprits have been fingered for this mega-flood, from a rogue tidal-wave, melting glaciers,⁴ earthquakes, or some combination therein. A more cautious approach is taken by Irving Finkel, curator of Mesopotamian cultural relics at the British Museum:

Psychologically it is not surprising that a flood myth should be deeply embedded in the Mesopotamian psyche, for it derived from and reflected the very landscape in which they found themselves. Their dependence on the Tigris and Euphrates waters was absolute and inescapable ... The flood ... enshrined remote memory of a real disaster or disasters. Probably some version of the story had been told for millennia.⁵

The clear external origins of the flood narrative⁶ imply a certain challenge to the general scope of this book. The flood epic does not fit neatly into any known stage of the agricultural revolution of the Levant. Its narrative cuts incongruously into the biblical genealogical lists that would otherwise bridge a seamless transition from the first city of Cain to the empire of Babel. If not for the insertion of the flood story, one could trace a clear progression in the Genesis account mirroring the onset of agriculture in the fertile crescent. The tale begins with the advent of domesticated wheat in the edenic Taurus mountains and lingers with the establishment of small sedentary villages, struggling desperately for the right to control that suddenly precious resource, arable land, along the fault lines between emergent farmers of Cain and shepherds of Abel. Then, the lens moves inexorably towards the conglomeration of those villages into urban hubs under the direction of Cain and his descendants, and from independent city-states to the rise and fall of the first empires, which we will encounter in the pithy tale of the Tower of Babel. Where does the flood narrative fit within that historical sweep?

As I have mentioned, in one sense the story of the flood is a reset button on the creation narrative in general and the origins of farming in particular. In terms of the former, Nahum Sarna highlights a number of “artful echoes”⁷ of the creation account that reappear in the flood episode. Sarna lists over a dozen parallelisms, including the classification of animal life in Gen 6:20 and 7:14 which corresponds to that in Gen 1:11–12, 21, 24, and 25. Likewise both Adam and Noah begat three sons, one of whom turns out to be degenerate (Cain and Ham, respectively), and the role of the wind in sweeping back the flood waters recalls the wind from God in Gen 1:2.⁸ In fact, as Sarna notes, the flood narrative does not simply mirror the creation story; first, it reverses

the very infrastructure of creation and then re-establishes Noah as the new Adam. "The Deluge itself," writes Sarna, "is brought about by the release and virtual reuniting of the two halves of the primordial waters that had been separated in the beginning (7:11: cf. 1:1, 6–7)."⁹ Out of the voidless waters, Noah emerges from the ark and is blessed by God in Gen 9:7 in a nearly identical manner to the divine blessing imparted to Adam in Eden (Gen 1:28).

If the flood narrative is creation redone, then the character of Noah becomes a reflection, foil, and palliative to Adam's farming curse. Indeed, Noah's life can be understood as a long struggle to overcome the fraught post-edenic relationship with an accursed earth. A struggle that is encoded into the essence of Noah at birth and imprinted into his name. "When Lamech¹⁰ had lived 182 years, he begot a son. He named him Noah, saying, 'This one will provide us relief from our work and from the toil of our hands from the ground that God has cursed.'"¹¹ If we were ever in any doubt from the Garden of Eden account itself that the divine curse given to Adam is a fervent scream against the horrors of the farming lifestyle, this passage reiterates the link between agriculture and its labour intensive costs. Lamech's charge to his newborn son to remove the "toil" (עָמָל) of working the earth is the exact word used in the Divine curse to Adam.¹² Noah is tasked at birth with somehow relieving the cursed effects of agriculture.

However, Lamech does not ask his young son to renounce farming altogether. Once the feedback loop of crop cultivation to increased yields of food per unit of territory to exponential population growth was set in motion, there was no return ticket.¹³ Thus, what Lamech demands from Noah is not an end to farming, but a particular relief from its discontents. In fact, Noah's entire life is bookended by his attempts to ease the pain of agriculture.¹⁴ As we will encounter two chapters hence, by the end Noah will plant a vineyard to soothe the farmer's toil with wine, and ease the personal labour costs by the enslavement of his own family.¹⁵ But here, in the flush of youth, according to the midrashists, Noah turns to a technological fix.

We read in the midrashic collection *Tanchuma*, "What was 'our work and the toil of our hands' [Gen 5:29] before the birth of Noah? One would plant seeds yet harvest weeds of thorns and thistles. After the birth of Noah, the world reverted to its intended state, one would plant wheat and harvest wheat, plant barley and harvest barley. Not only that, but until the birth of Noah, farming was done by hand, as it is written: 'the toil of our hands.' Then Noah fashioned for them the plough, the scythe, the spade, and all tools of labour."¹⁶ This midrash slides neatly into the biblical text. Whereas Adam's curse was to generally "toil" against the earth, Noah's embryonic mission is to assuage the "toil of our hands." Thus, he invents the implements of farming that introduce degrees of separation between the hands and the cultivated earth. Now the plough, and the concomitant beasts of burden to pull it, will do the heavy labour previously performed manually by the hands. In modern terminology, Noah inhabits that historical dividing mark between "shifting

cultivation,” the labour intensive method of farming with hand-held tools like the hoe and the digging stick, and ‘plough cultivation’ which is more capital intensive.

The concomitant transformation from a grain field thick with invasive weeds of thorns and thistles into a field where the sown seeds produce a true crop is likewise consistent with this historical moment. When the plough is used there is less need for weeding. The ploughshare’s hardened wood or bronze-tipped point, pulled by an ox or other large animal, could penetrate deeper into the soil than the hoe or other hand-held devices, thus uprooting weeds before the seeds were cast.¹⁷ Unlike a field husbanded with hoes and digging sticks and rife with weeds from embedded roots, in a well-ploughed field, when you planted wheat, a monoculture of wheat emerged.

But the technological achievements of Noah and his plough are insufficient bulwarks against the encroaching spectre of human evil that will soon bring the flood. The Bible’s poetic interweaving of linguistic irony takes the key words of hope used by Lamech holding his newborn boy, “relief” (רַחֲמֵי) from the “pangs” (אֲדָמָה) of work, and twists their meaning radically in the mouth of God planning a global genocide. “The Lord regretted (נִחַם) having made the human on earth and was grieved (אֲדָמָה) to the heart.”¹⁸ The Hebrew root for Lamech’s relief and God’s regret are one and the same (רַחֲמֵי). And the Hebrew root (אֲדָמָה) for both the pangs of Eve giving birth and the toil of Adam’s farming hands, that Lamech tasked his son to heal, is now embedded as the rooted word for the very grief (אֲדָמָה) of God. Sarna¹⁹ and Robert Alter²⁰ both note the looping verbal irony of these Hebrew roots uttered in hope by Lamech divinely reshaped into sinister connotations.

Through the lens of the midrash, the plough is the fulcrum object here. And the Bible’s dual perspective on the same root words, where a human vision of comfort is likewise God’s very regret, allows us to visualise, nearly simultaneously, how the plough can be a solution to a human sorrow, whilst also opening a Pandora’s box of new evils. Noah endeavours to fix the toil of work with mechanical and capital heavy solutions; the plough brings easy relief. Yet, via the objective gaze of God, the relief mechanism itself becomes the deepest source of grief and the catalyst for the flood. What is so problematic about the plough to cause such divine horror? How does the object of agricultural ease elide into the very source of divine regret?

The Failing and Falling Sons of God

In order to understand how the plough, and by implication, technological solutions to the ills of farming more generally, achieved this maligned biblical aura, we will need to carefully examine a few short yet baffling verses that intrude, seemingly haphazardly, between the birth of Noah and the main flood narrative. I will quote the strange passage in full, complying with the Masoretic paragraphical bookends, rather than the mediaeval Christian chaptering:

Noah had lived 500 years, Noah begat Shem, Ham, and Japheth. And it happened as humankind began to multiply over the earth and daughters were born to them, that the sons of God saw that the daughters of man were comely, and they took wives from whomever they chose. And the Lord said, 'My breath shall not abide in the human forever, for he is but flesh. Let his days be a hundred and twenty years.' It was then and later too that the *Nephilim*²¹ appeared on the earth, when they came to the daughters of man to bear their offspring, they are the strong ones of yore, the men of renown.²²

What are we to make of this odd vignette? "The account given in these few verses," observes Sarna, "is surely the strangest of all the Genesis narratives. It is so full of difficulties as to defy certainty of interpretation. The perplexities arise from the theme of the story, from its apparent intrusiveness within the larger narrative, from its extreme terseness, and from some of its vocabulary and syntax."²³ Most modern scholars, therefore, consign these difficult lines to a vestigial status. Alter²⁴ and Sarna²⁵ follow the proposal of E.A. Speiser in his assertion that this tale is a fragmented allusion to a fuller and well-known story sourced to a Hurrian myth from the middle of the second-millennium BCE.²⁶

Yet, why is this story included in the account of Noah and the flood? And why force this parenthetical narrative between the birth of Noah and the rest of the flood story? Let us turn to the midrashists. Their allegorical insights, argues Cassuto, preserve certain "folk-memories"²⁷ from a pre-scriptural tradition²⁸ that may have prefigured the sparse text. "The details of the Deluge with which the Talmudic and Midrashic literature embellish the Torah account, some of them correspond to what is narrated in the Mesopotamian inscriptions and possibly belong to the ancient Israelite store of tradition."²⁹ If this midrashic homily is of ancient stock, it preserves a very early account of powerful men abusing their power, just after the innovation of the plough.

This particular midrash begins by disabusing the reader of any notion of a divine origin story for these all-too-human elite. "'*The sons of God* (בְּנֵי הָאֱלֹהִים)' [Gen 6:2]: R. Shimon b. Yochai called them the sons of judges and nobles,³⁰ furthermore he cursed all who called them the sons of God ... R. Azariah said in the name of R. Levi: 'When the priests use stolen gods, by what can one swear or to what can one sacrifice?' Now why are they called the 'sons of God'? R. Hanina and Resh Lekish said: 'Because they lived a long time without trouble or suffering.'³¹ At the onset of this midrashic discourse, the rabbinical authorities set to work demystifying the context of the narrative.³² What we took to be semi-divine beings copulating with the "daughters of man" are only the offspring of the powerful, the entitled young playboys "taking wives from whomever they choose." And because they are the sons of the judges and the wealthy, there are no limits to their desire, and no troubles to offset their whims. R. Azariah's terse statement underscores this point. To what moral code can one possibly appeal to,

when even the priests, the arbitrators of that moral code, are themselves thieves whose very gods are acquired through violence.

The midrash continues to decode the narrative in this harsh light. “*when they saw that the daughters of man were comely* (עָמֵם)’: The word used is עָמֵם, written defectively [without a ך]. When a girl was made comely for her husband, the chief [of these young nobles] entered and had intercourse with her first. ‘*From whomever they choose*’: even married women, even males, even animals.”³³ The cruel custom—known in later times as *ius primae noctis*, the right to the first night—underscores the complete inertia of ethics in a society where the powerful can utterly subjugate the powerless. In the words of the modern biblical scholar Avivah Gottlieb Zornberg, this act of rape “exemplifies the institutionalization of barbarity. If we are to speak of passion here, it is the arrogant passion of the self, for whom no Other exists, in her own subjectivity, her own readiness to open doors or leave them closed. The ruler simply *enters* the intimate domains of others and expresses his mastery.”³⁴ Likewise, Nehama Leibowitz writes in her modern commentary: “The stronger enslave the weaker, the classic example of exploitation being the subjugation of the daughters of man.”³⁵

Yet, despite the adamant midrashic denial of the mythical implications of this narrative, the biblical labels of “sons of God” and “nephilim” carry contextual indicators of the supernal, beyond mere status as tribal big-men. After all, there are perfectly good Hebrew nouns for judges, chiefs, and rulers. Why employ descriptions particularly evocative of bridging the distance between heaven and earth? One explanation may be that the author of this vignette was simply recording the names that these big-men called themselves. As historian Ian Morris observes, in the arc of rolling country curving around the Tigris, Euphrates, and Jordan valley in Southwest Asia, the region he and others entitle the Hilly Flanks, the rise of settlements and domesticated grains and animals coincides with an upsurge of cultic ritual infrastructure. At ‘Ain Ghazal in modern Israel two pits were found containing life-size statues and busts made from bundles of reeds coated with plaster. Some had twin-heads and giant eyes. In approximately 8000 BCE in Çayönü Tepesi in southeast Turkey, the inhabitants built what its excavators labelled a “House of the Dead,” with 66 human and animal skulls and 400 skeletons stashed behind an altar stained with both human and animal blood. Though Morris is careful to point out that his claims are speculative, he theorises on a causal link between an emergent class of holymen and the increasing hierarchy uncovered in the uneven distribution of material goods and house sizes of these early farming settlements. “The statues and the House of the Dead seem to imply the emergence of religious specialists who somehow persuaded everyone that they had privileged access to the supernatural. Perhaps they could fall into trances or fits, perhaps they could just describe their visions better. Whatever the reason, priests may have been the first people to enjoy institutionalised authority. Here, perhaps, we see the beginnings of entrenched hierarchy.”³⁶

Returning to our scriptural conjecturing, the very fact that these big-men are labelled as “children of gods” may be a reflection of the specific cultural justification offered by these men and others in their sway. They were not simply powerful chieftains, but heirs to the divine. And it was this propaganda of godly status which made them deserving of power and women. Political theorist Francis Fukuyama offers a similar potential answer to the vexing question of why states formed in the first place. For why would any free tribal group willingly delegate authority to a single individual and that individual’s kin group on a permanent basis. “It seems extremely likely,” argues Fukuyama, “that religious ideas were critical to early state formation, since they could effectively legitimate the transition to hierarchy and loss of freedom enjoyed by tribal societies ... a charismatic leader asserts authority not because he is elected by his fellow tribesmen for leadership ability but because he is believed to be a designee of God.”³⁷ Perhaps, here we can detect the tongue-in-cheek subtlety of the biblical label Nephilim. The double entendre of a name that implies not only their lofty origin story but also a biblical jeer upon these men who had now “fallen flat” from their heights. Those who had legitimised their rule as “children of gods”³⁸ were found out as simply “flesh” in the radical and subversive gaze of God: “And the Lord said, ‘My breath shall not abide in the human forever, for he is but flesh.’”³⁹

How the Plough Carved the Furrow for the Sons of God

Why now? Why does the Torah obliquely conjure up the violence of powerful men against women at this precise moment in the Genesis narrative? I believe the culprit was the plough that Noah built. For it was the spread of plough technology, argues Ester Boserup, which best explains gender inequality variations from one society to another. So too, the plough invented by Noah prefigures the story of gender inequality within our text.

In 1970, the Danish agro-economist, Ester Boserup published a treatise entitled *Women’s Role in Economic Development*. She began with a basic question: why are there such wide variations in gender (in)equality from one society to another. First, she created a causative correlation between gender equality and female labour force participation. Those cultures where women actively participate in employment outside of the home reflect more positive attitudes of gender equality than those cultures in which tasks are specialised according to gender norms and women are enclosed more firmly within the domestic sphere. Thus, her next question: what determining factor led some societies to create norms and beliefs that delineated a largely domestic role for women, whilst other cultures did not. Her answer, which later studies have supported, is that the main determinant of gender norms in a society is whether that particular culture used the plough to cultivate its major crops.⁴⁰

As we have noted before, there are two main types of traditional agricultural cultivation. Shifting cultivation using hand-held tools and plough-based crop farming. Shifting cultivation is labour intensive and puts a premium on the

maximum number of people in the field constantly tending the crops and weeding out of the alien species. Women are as likely as men to participate in farm-work in shifting cultivation fields. Plough cultivation, conversely, is gender-biased. “Unlike the hoe or digging stick, the plough requires significant upper body strength, grip strength, and burst of power, which are needed to pull the plough or control the animal that pulls it. Because of these requirements, when plough agriculture is practised, men have an advantage in farming relative to women.”⁴¹ Also reinforcing this gender-bias in raw strength is the fact that ploughed fields reduce the need for weeding, a task usually performed by women and children.⁴² Finally, shifting cultivation is a slow, methodical task that allows for interruptions and contains few inherent risks. It is thus conducive to child-care responsibilities, a role universally assigned to women. Plough cultivation relies on large animals, heavier parts, and—when parts and animals are finally hitched together—large segments of devoted time, factors which removed children and their caregivers from the fields.⁴³

The result, according to Boserup, is that “societies that traditionally practised plough agriculture—rather than shifting cultivation—developed a specialisation of production along gender lines. Men tended to work outside of the home in the fields, while women specialised in activities within the home ... This division of labour then generated norms about the appropriate role of women in society.”⁴⁴ More recent studies have buttressed Boserup’s model, mainly by cross-cultural studies that reveal plough usage as the main determinant in plotting variant attitudes of gender inequality in ethnic language groups, rather than the converse causal relationship, where those societies with historical gender inequality indicators for other reasons would be more likely to adopt the plough.⁴⁵

As the first wheat farmers of the Karacadag Hills⁴⁶ moved down from the Piedmont into the Tigris and Euphrates alluvial valley, the soil and landscape favoured deeper tillage across wider tracts of land. Plough cultivation became the dominant technology, easily out-producing hand-held methods. As the plough dominated, so too did men.⁴⁷ Women’s labour participation was reduced to ever smaller spheres of private, domestic activities. These gender boundaries were not just separate, but also unequal, as the political and economic decision-making took place in the public arena, by those who wielded the plough and the food production it engendered (Figure 5.1).

Now, let us examine the narrative timeline of Noah before the flood. According to the midrash, Noah alleviates the “toil of our hands”⁴⁸ with the innovation of the plough. At first, this new technology relieves the curse upon the ground. Weeds are reduced and animal labour replaces the back-breaking drudgery of the hoe and digging stick. But the invention of the plough had unintended consequences. Men’s upper body musculature and their new-found, plough-induced status as the “strong-ones”⁴⁹ increase the likelihood of unequal gender norms. At precisely this moment in the text, in a world dominated by men and their ploughs, women now become the victims of a “rapacious egotism ... which sweeps away all other worlds, all other selves”⁵⁰



Figure 5.1 Painted wood model of a man ploughing a field; item of funerary equipment excavated in Egypt, ca. 1981–1885 BCE. Courtesy of the Metropolitan Museum of Art, New York.

in the chilling lexicon of Aviva Zornberg. The sons of the lordly and powerful take women for wives “from whomever they choose.” The act of ploughing and the act of rape share similar tropes, a penetrating with aggressive strength and a ploughing under of every “other” whether it is the wild foliage that gives way to the ploughshare or a woman’s desperate cry.

Now, the final stanza of that strange account reverberates with other meanings. “It was then and later too that the *Nephilim* appeared on the earth, when they came to the daughters of man to bear their offspring.”⁵¹ As Sarna notes, for the first time in the Bible, the act of cohabitation is not called “knowing” the other (יָדָע), but “coming upon them” (צִיב), a physical deed without intimacy.⁵² The perpetrators of this mass rape are now labelled as the *Nephilim*, literally “the fallen ones.”⁵³ The etymology of the label is uncertain, and its usage is limited to just one other biblical narrative.⁵⁴ When the spies return to deliver their report to Moses and the people on surveying the land of Canaan, they cry out in their despair that they have seen *Nephilim* and “we looked like grasshoppers to ourselves, and so we must have looked to them.”⁵⁵ The *Nephilim* are fallen, not only because they are degraded, but because they degrade others.⁵⁶ After an encounter with the *Nephilim*, one’s

own projection of self is fractured. One becomes an insect in one's own eyes. Perhaps, the Genesis account uses this term to similar effect. The power of the *Nephilim*, their status as נִפְלִימִים, strong-men, is precisely manifest in their ability to dominate others so completely that their particular vision of reality, no matter how violent or hideous, becomes the only acceptable model. Even the genetic code of the women these *Nephilim* impregnate cannot alter this totalitarian force. As the midrashist notes pithily: "these *Nephilim* would come upon them [the daughters of man] to such an extent that these women would give birth to monstrous giants just like themselves."⁵⁷

With midrashic prodding, this final verse, and verdict, on the sons of power reveals yet another dimension of the agricultural revolution. "*It was then and later too*: Judah son of Rabbi commented: 'The latter generations would not learn from the earlier ones.'"⁵⁸ As the narrative segues into the story of the flood, it is this inability to learn from the past that will become the biblical hallmark of failure, a stubborn "hardening of the heart" and "stiffening of the neck" in which aberration hardens into habit and habit becomes normative. Perhaps, there is also a way of reading this phrase, "the latter generations would not learn ..." as a rabbinical commentary bearing upon a fundamental question debated by scholars of the Neolithic. If farming was so detrimental to the local ecology, if it resulted in more work for less nutritious food, weaker teeth and bones, permanent settlements that resulted in ever-growing security concerns, and the sedentary inability to effectively shift camp when the next drought or enemy approached, why did farming win out so spectacularly against hunting and gathering? Why didn't the latter generations abandon farming and return to a mixed-subsistence regime upon hearing a tale of woe from their parents and grandparents?

The historian Yuval Noah Harari provides two possible and overlapping answers. "Partly because it took generations for the small changes to accumulate and transform society and, by then nobody remembered that they have ever lived differently. And partly because population growth burned humanity's boats. If the adoption of ploughing increased a village's population from 100 to 110, which ten people would have volunteered to starve so that the others could go back to the good old times? There was no going back. The trap snapped shut."⁵⁹ I would like to offer a thick reading⁶⁰ of this precise moment in the biblical narrative as warped on the midrashic loom. For the short vignette of Gen 5:32–6:4 hints at the formation of the earliest states and their centripetal forcefield of one-way entrapment. Here are all the hallmarks of the proto-state: an expanding population core ("as humankind began to multiply over the earth") with centralised control by a select group of men ("they took wives from whomever they choose") with power over the reproductive capacity of women ("who bore them children"), midrashically ploughing the Mesopotamian alluvial soil with an influx of grain.⁶¹ As the population of state subjects multiplied, the grain production would increase and the control of resources would become ever more centralised. The elite non-producers had less and less incentive to recall alternatives to intensive agriculture,

despite its many discontents. These strong, power-hungry men felt no compunction to learn from the encroaching sorrow of the women they impregnated, and turned no ear towards the genealogy of creeping destruction that expanded in farming's wake.⁶²

Seen in this light, this terse narrative snippet on the powerful and the fallen fits snugly into the biblical description of early statecraft that animates these first chapters of Genesis. Namely, that after the first farmers moved down from the edenic mountains to build cities in the alluvial valleys, they began to focus on producing and holding a population nucleus to cultivate a surplus of grain. We first encountered this drive to build a "population machine"⁶³ with Lamech and his two wives. Now, with the spread of plough-based agriculture, the alluvial soil can produce even more grain per acre, simultaneously spurring an even denser population core, and locking-in gender inequality into this increasingly stratified society. It is precisely at this pivotal moment in state-making that the Genesis account introduces the notion of divine regret, which will soon lead directly to the flood.⁶⁴ As if to say, that the grasp of the alluvial grain state was so pervasive, that only a divine deluge could wrench free just a single family from its clutches.

Though it can be nothing more than speculative, the curse of wheat bears striking parallels to more modern resource curses, in particular the negative effects of oil and gas booms in developing states. As in plough-centric agriculture, oil industrialization marginalises women from the labour force. As economist Michael Ross argues, oil reduces the demand for female labour by shrinking the number of jobs in the export-oriented factories.⁶⁵ Though the causes and effects in this process are complex, in a very basic model, the influx of oil causes the exchange rate to jump, which in turn causes the manufacturing sectors to lose their overseas markets. Because factories are more likely to hire women than men, a decreasing overseas market keeps women at home. This, Ross suggests, is why oil-rich developing countries have some of the lowest rates of women in the workforce in relation to overall wealth. Women who are stranded at home have less political influence,⁶⁶ less access to education,⁶⁷ less knowledge on women's health and contraception,⁶⁸ and decreased influence over family decisions about prenatal and child health.⁶⁹

One cannot directly transduce backwards from modern oil booms into the nebulous early history of farming. Yet, perhaps we can draw a few parallels. As women lost their role in the fields, they lost their power and their voice in the decision-making components of society, much as modern-day women in oil-rich countries remain underrepresented in government bodies. Perhaps, these gossamer-threaded analogies between the ancient curse of wheat and the modern one of oil can bear the weight of making an ancient text relevant. That the uneasy lessons the Bible learns from the cultivation of wheat and the covenantal response of Israel may have some small teachings to guide our modern response to the societal and environmental repercussions of the petro-revolution.

But before we enter into this long narrative of hope, the Bible demands an unflinching exploration of other potential options that will salve the afflictions of the agricultural revolution. The invention of the plough has failed to stem the curse of farming. The bold technological promise of relief from toil has led directly to a degenerate patriarchal hierarchy forcing women to breed a denser population. What Noah envisioned as an easy technological fix through God's own looking-glass becomes nothing more than the root of regret over the entire human enterprise. That which eases the human material condition brings so much sorrow to the rest of God's creation. The alluvial lowlands of ploughed fields must be drowned, and Noah must flee to the safety of the high mountains and its hardscrabble, nearly un-ploughable terrain, in order to enter into the first covenant with God.

Notes

- 1 See Cassuto, *Genesis II*, 4. "For example, God's blessing to Noah and his sons after the Flood, which repeats the *ipsissima verba* of the benison bestowed on Adam and comes as the fulfilment of that ancient covenant." See also, Nahum Sarna, *The JPS Torah Commentary: Genesis* (Philadelphia: The Jewish Publication Society, 1989), 49.
- 2 Irving Finkel, *The Ark Before Noah: Decoding the Story of the Flood* (London: Hodder & Stoughton, 2014), 87. Westerman conducts an extensive scholarly review of the similarities and differences between the Biblical account and the Mesopotamian narrative, Claus Westerman, *Genesis 1–11: A Continental Commentary*, trans. John J. Scullion (Minneapolis: Fortress Press, 1994), 398–406.
- 3 Finkel, *The Ark Before Noah*, 90.
- 4 Perhaps, the most widely cited and archaeologically rooted of these theories holds that a "perfect storm" of torrential rain, seasonal cyclones, and the early melting of the snows in the mountains of Anatolia caused the Euphrates and the Tigris rivers to burst their banks and inundate the flat alluvial valley for thousands of miles. Evidence of a catastrophic deluge from the ancient settlement of Shuruppak, where the flood survivor of the Gilgamesh epic, Utnapishtim, lived, coupled with fluvial deposits from excavations at Kish, both date a widespread flood to approximately 2900 BCE. Sarna, *The JPS Torah Commentary: Genesis* (Philadelphia: The Jewish Publication Society, 1989), 48.
- 5 Finkel, *The Ark Before Noah*, 87.
- 6 However, the Mesopotamian flood accounts might not themselves be the original source for the biblical version, as Nahum Sarna notes in his commentary to Genesis: "Yet it cannot be claimed that any version presently known is the direct source of the biblical narrative, for the latter has points of contact with each version while it also contains items independent of them all ... it is clear that our biblical account constitutes an independent Israelite version that is nevertheless closely related to the Mesopotamian traditions." Sarna, *The JPS Torah Commentary: Genesis*, 48–49.
- 7 Sarna, *Genesis*, 49.
- 8 Sarna, *Genesis*, 49.
- 9 Sarna, *Genesis*, 49.
- 10 We can also note that the story of Noah is not only a reset on creation but also in another sense the end stage of a parallel familial storyline running alongside the story of Cain and his family. As Cassuto points out, "The names of the founding fathers of the world in our chapter [Gen 5], beginning with Enosh, bear a remarkable resemblance to the names that appear in the family-tree of the sons of

Cain. Enosh corresponds to Adam, Cain's father (the two names have the same signification: man); Kenan is the equivalent of Cain; the series Mahalel-Jered-Enoch parallels, in reverse order, the series Enoch-Irad-Mehujel; Methuselah is the counterpart of Methushael; thereafter Lamech appears in both pedigrees." *Genesis I*, 266. In Cassuto's reading, the parallelisms are essentially morally comparative. Cain and his line are debased by materialism and thus doomed. However, Seth and his offspring are "heirs of the knowledge of the Lord," which eventually will lead Noah and his family to be favoured with survival. *Genesis I*, 267.

11 Gen 5:28.

12 Gen 3:17.

13 For example, see Yuval Noah Harari, *Sapiens: A Brief History of Humankind* (London: Vintage, 2014), 94: "Around 13,000 B.C.E. when people fed themselves by gathering wild plants and hunting wild animals, the area around the oasis of Jericho, in Palestine, could support at most one roaming band of about a hundred relatively healthy and well-nourished people. Around 8500 B.C.E. when wild plants gave way to wheat fields, the oasis supported a large but cramped village of 1,000 people, who suffered far more disease and malnourishment ... This is the essence of the agricultural revolution: the ability to keep more people alive under worse conditions." See also, Ian Morris, *Why the Rest Rules*, 100. "Having more babies in fact became a boon, providing extra labor, and recent skeletal studies suggest that the average woman in an early farming village, staying in one place with stores of food, gave birth to seven or eight babies (of whom maybe four would survive to their first birthday and perhaps three to reproductive age) as compared to the mere five or six live births of her roving ancestresses. The more food people grew, the more babies they could feed; although, of course, the more babies they fed, the more food they had to grow."

14 See Hiebert, *Yahwist's Landscape*, 45. "Noah was born to turn back this curse and to moderate the arduous character of agricultural labor in the primeval age. This is the very assignment that Noah accomplishes upon his emergence from the ark after the flood." Where Hiebert and I part ways is whether Noah did effectively accomplish this relief. In my mind, it remains difficult to see Noah as successful, as a key node of his vineyard farming leads directly to the curse of slavery.

15 Gen 8:20–28.

16 *Midrash Tanhuma: Mantua Manuscript 1563* from the *The First Editions of the Midrashim* series [Hebrew] (Jerusalem: Makor Publishing, 1986), 4. Also *Tanhuma-Yelammedunu*, Gen. 11, trans. Berman, 34.

17 Andrew D. Foster and Mark R. Rosenzweig, "Technical Change and Human-Capital Returns and Investments: Evidence from the Green Revolution," *American Economic Review* 86, no. 4 (1996): 931–953.

18 Gen 6:6.

19 Sarna, *Genesis*, 44. "The father {Lamech} looked to 'relief' (*yenahamenu*) from 'toil' (*'itsavon*), but instead came God's decision to wipe out civilization. God 'regretted' (*va-yinnaḥem – nihamti*) and was 'saddened' (*va-yit'atsev*)."

20 Robert Alter, *The Five Books of Moses: A Translation with Commentary* (New York: W.W. Norton & Co., 2004), 40. "The same verbal root, '-ts-b, is reflected in Eve's pangs, Adam's pain, and the 'pain of our hands' work."

21 Lit. "fallen ones," see Skinner, *Genesis*, 146, alluding to a "fall" of angels from heaven, or to a "fall" of the world through their action. Skinner also states that a connection with לֵבָד "abortive birth" "is not improbable" (citing F. Schwally, ZATW. 18, 144ff.). Also Cassuto, *Genesis I*, 298. Cassuto cites Ezekiel 32:20 where reference is made to the לֵבָד [analogous to our passage] in *Sheol*. "The correct explanation appears to be that they were called thus because, according to the story related of them they all fell by the sword and descended to *Sheol* abode of the

dead.” The only other usage of the term *nephilim* is Num 13:33, where they are described as “giants.” Westerman concludes that, “There is every reason to think that Nephilim in 4a refers to mythical semi-divine beings, the fruit of the marriages of the gods with humans, who are connected with overstepping of the bound presumed in the divine judgement of v. 3.” Westerman, *Genesis 1–11*, 378. See also P. Humbert, who translates *nephilim* as the “deposed,” in Humbert, *Démésure et Chute dans l’ancien Testament* (Montpellier: FS. W. Vischer; 1960) as cited by Westerman, *Genesis 1–11*, 378. Also, Franz Delitz’s translation as “attackers,” as cited by Westerman, *Genesis 1–11*, 378.

22 Gen 5:32–6:4.

23 Sarna, *Genesis*, 45.

24 Alter, *The Five Books of Moses*, 38.

25 Sarna, *Genesis*, 45.

26 E.A. Speiser, *The Anchor Bible: Genesis* (New York: Doubleday and Company, 1964), 46. See also Westerman, *Genesis 1–11*, 369. “One can take as a certain that it [Gen 6:1–4] did not originate in Israel. It is very likely a Canaanite myth because the *בְּנֵי הַמַּגִּל* are known from Canaanite mythology where they are found often and clearly play an important role.”

27 Cassuto, *Genesis II*, 29.

28 Cassuto, *Genesis II*, 24. “When the earliest ancestors of Israel dwelt in the Ur of the Chaldees, they were undoubtedly well versed in the local Sumerian culture ... the progenitors of Israel preserved the ancient memories, which were revised in accordance with their religious outlook.”

29 Cassuto, *Genesis II*, 26.

30 Nehama Leibowitz, *Studies in Bereshit (Genesis): In the Context of Ancient and Modern Jewish Bible Commentary*, trans. Aryeh Newman (Jerusalem: Alpha Press, 1985), 55. “This is the interpretation followed by most of the [traditional] commentators, basing themselves on the use of the word *בְּנֵי הַמַּגִּל* in Exodus 22:8, ‘And the householder shall draw near onto the judges.’”

31 *Ber. Rab.* 26.5, Theodor and Albeck, 247–248.

32 Harold Kushner intones that, following the lead of the midrashists, the “traditional commentators (Onkelos, Rashi, Hirsch) strive to avoid the mythological implications of this account. They understand the ‘divine beings’ to be the noble descendants of Seth, intermarrying with the descendants of Cain who offered physical attractiveness but no moral standards. Whatever the ancient roots of this story, no Jewish commentator accepts the notion of a sexual union between divine beings and morals, giving rise to a semi-divine race.” Harold Kushner, *Etz Hayim: Torah and Commentary* (Philadelphia: The Jewish Publication Society, 2001), 33–34. See also the translation of Harold Fisch, *The Holy Scriptures* (Jerusalem: Koren, 1997), 6. For an opposing view, cf. Cassuto, *Genesis I*, 291. “All the expositions, both ancient and modern, that regards the *sons of God* as a distinct group of human beings are unsatisfactory ... that wherever the term *בְּנֵי הַמַּגִּל* or *בְּנֵי הַמַּלְאָכִים* occur, the reference is definitely to angels (Psalm 19:1; Job 1:6, 2:1, 38:7; so too Deut. 32:8, according to the reading of the Septuagint).”

33 *Ber. Rab.* 26.5, Theodor and Albeck, 248.

34 Avivah Gottlieb Zornberg, *Genesis: The Beginnings of Desire* (Philadelphia: The Jewish Publication Society, 1995), 52.

35 Leibowitz, *Studies in Bereshit*, 55.

36 Ian Morris, *Why the West Rules-For Now* (London: Profile Books, 2011), 103.

37 Francis Fukuyama, *The Origins of Political Order: From Prehuman Times to the French Revolution* (London: Profile Books, 2012), 86–87.

38 Gen 6:2.

39 Gen 6:3.

- 40 Ester Boserup, *Women's Role in Economic Development* (London: George Allen and Unwin Ltd., 1970).
- 41 George P. Murdock and Caterina Provost, "Factors in the Division of Labor by Sex: A Cross-Cultural Analysis," *Ethnology* 12, no. 2 (April 1973): 205.
- 42 Andrew Foster and Mark R. Rosenzweig, "Technical Change and Human-Capital Returns and Investments: Evidence from the Green Revolution," *American Economic Review* 86, no. 4 (September 1996): 941.
- 43 Alberto Alesina, Paola Giuliano, Nathan Nunn, "On the Origins of Gender Roles: Women and the Plough," *Quarterly Journal of Economics* 128, no. 2 (May 2013): 469–530.
- 44 Boserup, *Women's Role*, 13–14. See also Alesina, et al. "On the Origins of Gender Roles," 472. "Societies characterized by plough agriculture, and the resulting gender-based division of labor, developed a cultural belief that the natural place for women is within the home."
- 45 Alesina, et al., "On the Origins of Gender Roles," 477.
- 46 Manfred Heun, et al., "Site of Einkorn Wheat Domestication Identified by DNA Fingerprints," *Science* 278, 5341 (1997): 1.312–1.314.
- 47 Alesina, et al., "On the Origins of Gender Roles," 482. "Boserup's hypothesis suggests that in societies that engaged in plough agriculture, cultural beliefs about gender inequality were relatively beneficial. Therefore, these norms may have evolved in plough-agriculture societies but not hoe-agriculture societies. Because of the persistent nature of cultural beliefs, norms of gender inequality may persist even after the economy moves out of agriculture or industrializes, affecting the participation of women in activities performed outside the home, such as market employment, entrepreneurship, or participation in politics."
- 48 Gen 5:28.
- 49 Gen 6:4.
- 50 Zornberg, *Genesis: The Beginnings of Desire*, 52.
- 51 Gen 6:4.
- 52 Sarna, *Genesis*, 46. "Significantly, the verb *y-d'* is not used, as in 4:1,17, and 25, but a coarser term, as befits the circumstances." For an opposing view however, see Cassuto, *Genesis I*, 294. "Simply the usual expression for legal marriage. The passage contains not a single word, either here or in the Lord's speech in v. 3, alluding to rape or adultery or to any act against the Lord's will." Also H.A. Brongers, *Vetus Testamentum* 13 (1963): 269ff, as cited in Westerman, *Genesis I–II*, 372.
- 53 See Targ. Jon. to Gen 6:4.
- 54 Sarna, *Genesis*, 46. "The etymology of Nephilim is uncertain. The obvious association with *n-f-l* yields the rendering 'fallen ones,' that is, fallen angels."
- 55 Num 13:33.
- 56 This would appear to be the reading offered by Rashi on Gen 6:4, s.v. רָא, יָרָא. "Because they were fallen and they caused the world to fall."
- 57 *Ber. Rab.* 26.7, Theodor and Albeck, 253.
- 58 *Ber. Rab.* 26.7, Theodor and Albeck, 254.
- 59 Harari, *Sapiens*, 98.
- 60 The notion of a thick reading emerges from the landmark essay on US Constitutional Law, "*Nomos and Narrative*," by Yale law professor Robert Cover. In particular, Cover's section on Biblical law entitled "The Thickness of Legal Meaning." Robert M. Cover, "The Supreme Court, 1982 Term—Forward: *Nomos and Narrative*," *Harvard Law Review* 97, no. 4 (1983). Barry Scott Wimpfheimer then applies this thick reading to the legal and aggadic matrix of the Talmud in *Narrating the Law: A Poetics of Talmudic Legal Stories* (Philadelphia: University of Pennsylvania Press, 2011).

- 61 In the next chapter, we will explore the proto-states' primary goal to attract and hold manpower. However, here it is worth outlining these earliest states' fundamental focus on population density, as described by James Scott. "The key to successful statecraft was typically the ability to attract and hold a substantial, production population within a reasonable radius of the court. Given the relative sparseness of the population and the ease of physical flight, the control of arable land was pointless unless there was a population to work it ... A growing productive population settled in the ambit of the monarch's capital was a more reliable indicator of a kingdom's power than its physical extent." James Scott, *Seeing Like a State* (New Haven: Yale University Press, 1998), 185.
- 62 See Rashi's commentary to Gen 6:4, s.v. "לֹא יָבִישׁ הָאָרֶץ" which likewise frames the moral failing of these "strong-ones" as a stubborn inability to learn from past collapse. "And later too: Although they [the generation of the flood] had witnessed the destruction during the generation of Enosh, when the ocean had wiped out a third of humanity, the flood generation did not humble themselves to take a lesson from them."
- 63 See Scott, *Against the Grain*, 151.
- 64 Cassuto notes a number of artful parallels between the story of the sons of God and the divine decree to enact the flood. For example, the linguistic link between Gen 6:1 ("Men began to multiply [וַיִּבְרְאוּ] on the face of the ground) and Gen 6:5 ("the wickedness of man was great" [וַיִּרְבּוּ]). "The verbal parallelism indicates a correspondence of theme; in this case, too, there is an antithesis between God's and man's deeds. God blessed mankind that they should be fertile and fill the earth (Gen 1:28), and He implemented His promise: men began to multiply on the face of the ground. Man, however, was an ingrate; truly, he filled the earth, but he did so with violence (Gen 6:11 & 13)." Cassuto, *Genesis I*, 302. That the events of 6:1–4 reveal the first causes for God's decision to bring the flood upon humankind, see also Westerman, *Genesis 1–11*, 370. "Even if one does not agree with E.G. Kraeling, who concludes from this that 6:1–4 and the flood narrative belong together, nevertheless one cannot say that this agreement is mere chance ... we find that 6:1–4 and 11:1–9 are in agreement in that the expansion of humankind, which is in fact a consequence of the creation blessing, begins to take on a negative side."
- 65 Michael L. Ross, *The Oil Curse: How Petroleum Wealth Shapes the Development of Nations* (Princeton: Princeton University Press, 2012), 112–113.
- 66 Nancy Burns, Kay Schlozman, and Sidney Verba, *The Private Roots of Public Action: Gender, Equality, and Political Participation* (Cambridge: Harvard University Press, 2001).
- 67 Ross, *Curse*, 112.
- 68 Ross, *Curse*, 112.
- 69 Duncan Thomas, Dante Contreras, and Elizabeth Frankenberg, "Distribution of Power within the Household and Child Health," Unpublished paper, University of California at Los Angeles (2002), as cited by Ross, *Curse*, 113.

6 The Flooded State

The arc of the biblical flood narrative so closely parallels other, perhaps earlier, Mesopotamian flood epics, that I remain somewhat wary of including it within the design of this book, devoted to reading Genesis as an outsider's subversive retelling of the encroaching hegemony of the Neolithic. Perhaps, it was simply the stylistic convention of Southwest Asian etiological accounts to include a flood hero. A genre setting device, much as we begin a fairy tale with "once upon a time . . ." And certainly the broad strokes of a heroic figure informed by divine fiat to build a boat of refuge that will allow him and a smattering of animals and family members to survive whilst the rest of the world is drowned into oblivion is an oft-repeated tale amongst Israel's neighbours. From the earliest material evidence of the flood tradition, the fragmented Sumerian text of Ziusudra from the 17th century BCE,¹ to the legend of Atrahasis from roughly 1600 BCE,² to the Epic of Gilgamesh, which took shape in the Old Babylonian period (2017–1595 BCE),³ to an early third century BCE retelling by the Babylonian priest Berossus,⁴ to the Greco-Roman version recorded in Ovid's *Metamorphoses* (43–17 CE),⁵ the flood story was remarkably adaptive and resilient throughout the ancient Near East. Perhaps, as Umberto Cassuto posits boldly, the Hebrew Bible had no choice but to include such a ubiquitous folk account. "The Torah could not very well pass over in silence the ancient poetic tradition regarding the Flood, which was already widely current among the Israelites. It was necessary to accept it for the sake of the continuity of the narrative of the history of man upon earth."⁶

Or, perhaps, the inclusion of such a familiar narrative device in the Bible is a subversion of the accepted genre, what William Ryan and Walter Pittman call the Bible's "radical departure" from Mesopotamian flood tradition.⁷ In all other Mesopotamian deluge epics, the gods are configured in pantheonic conflict amongst themselves. There are certain gods of destruction and separate gods of salvation. And the dialectic between them is the core drama of the epic. For example, in the Atrahasis story, the god "Enlil did an evil deed on the peoples," whilst "Enki opened his mouth and addressed his slave."⁸ Yet the Bible insists, against all the extant contemporaneous flood myths, that the same god who saves the lone refugee is the very god who conceived of the genocidal deluge in the first place. "In the polytheistic Mesopotamian

tradition,” remarks the natural historian Daniel Hillel, “the event was the whim of the capricious gods. The biblical account typically introduces the moral dimension. The Flood was the divine punishment for the corruption of humanity.”⁹ This clarion call of ethical monotheism is an early marker between the “us” and “them” of the Biblical perspective, notes David Pleins, “the theological line in the sand drawn between ancient Israel and the surrounding cultures.”¹⁰

Comparing the biblical flood story to other Mesopotamian accounts is a rich source of covalence and friction in ancient Near Eastern studies. Yet, there is a separate analysis that emerges wholly from within the biblical narrative arc. Why include the flood story precisely now? Why bookend this well-trod tale, at least to ancient Levantine ears, between the subjugation of the “daughters of man” at the onset of chapter 6 and the megalithic tower project in the valley of Shinar chronicled in chapter 11?

One answer as to why the deluge comes at this narrative juncture might be that flood stories arrive on the scene only after a society has gone beyond a critical human mass. Floods in Mesopotamian narratives are the result of an overpopulation problem. In the Myth of Atrahasis, the gods drown humanity because they increase and make too much noise. “600 years, less than 600, passed; And the country was as noisy as a bellowing bull. The gods grew restless at their racket, Enlil had to listen to their noise. He addressed the great gods: ‘The noise of mankind has become too much, I am losing sleep over their racket!’”¹¹ Note the similar nod to a population problem at the onset of the sixth chapter of Genesis: “And it came to pass when men began to *increase* on the face of the ground, and daughters were born to them ... and God saw that the evil of Man had *increased* on the earth.”¹² I have here juxtaposed verse 1 with verse 5, following in the wake of Cassuto,¹³ because of the parallel stem-word (ṭṣ) for *increase* that links the growth of population density with the onset of a new and odious degree of evil upon the earth. As human beings multiply, evil grows apace. And the adjunct of evil, violence, now fills the earth to the brim. “The earth was corrupt before God, and the earth was filled with violence,”¹⁴ and two verses later, “God said to Noah: the end of all flesh has come before me, because the earth is filled with violence before me”¹⁵ The pre-flood world is a metastasizing of people and evil, a claustrophobic world filled with acts of violence. So, whether we are speaking of noise pollution, as in Atrahasis, or moral pollution, as in the Bible, the basic catalyst for a catastrophic flood is a population explosion.

At this point, we must engage with some general historical perspectives on the reason for increased population density in the ancient world. We often assume, like Malthus, that population increase is brought about by technological innovation, such as the agricultural revolution, or the invention of the plough, which allows for more food production per acreage. Thus, as the carrying capacity for any given piece of land increases, people have more food surplus. And more food leads parents to have more babies, who provide more

intensive labour, and thus create more surplus food on that set parcel of land, in an ongoing one-directional feedback loop.

However, as I have detailed earlier, there are two interlinked problems with this Malthusian model of population increase across a given area of landmass. First, the Malthusian assumption that societies will always push the limits of food production to the maximum capacity just as long as they have the tools and know-how to do so does not mesh with real-world observations. A number of hunter-gatherer and swidden-based societies around the globe have both knowledge of and access to intensive agricultural techniques, and yet still don't produce the amount of food surplus of which they are technically capable.¹⁶ Groups such as the Bemba of Zimbabwe, who routinely work in their long-fallow fields at a rate of just three to five hours a day, and the Zande in Southern Sudan, whose agricultural tasks in the high season do not take up more than one-fifth to one-fourth of their daily activities, led Ester Boserup to surmise that these groups opt for leisure time over food surplus because—devoid of any external compulsion—most societies will stop working when they have achieved subsistence levels.¹⁷ Francis Fukuyama sums up this perspective well. "Some anthropologists have suggested that in certain hunter-gatherer societies, increases in food supply are met with decreasing amounts of work, because their members value leisure over work. Inhabitants of agricultural societies may be richer on average, but they also have to work much harder, and the trade-off may not seem appealing."¹⁸ James Scott suggests that the very concept of intensive farming as the intrinsically more efficient system is flawed. For whilst fixed-field grain farming is certainly more productive per unit of land than shifting cultivation or foraging, it is, however, less productive per unit of labour. Therefore, argues Scott, "where land is comparatively plentiful and labour scarce... ... shifting cultivation was more labour saving per unit of output, and hence more efficient."¹⁹

The second fault with using a simple Malthusian model to understand the increasing population density of the Mesopotamian alluvium is that there was plenty of room to expand. Thus, even if these early farmers did produce a large surplus of crops and birthed more children, those children could simply move to other arable plots in the alluvium which would disperse population rather than increase density in any given area.²⁰ One archeologist, Seth Richardson, estimates that the Mesopotamian alluvium of 3000 BCE held roughly 15 ha of arable land per every nuclear family, more than enough for general sustainability with low levels of population densification.²¹

So, if we witness very many societies both past and present choosing leisure over labour, and if land was plentiful, then just what catalysed population density rise in the early Levant? In order to understand why population density increased so quickly in Mesopotamia and other proto-grain cores globally, we must return to Boserup's theory that groups do not normatively relinquish their many free hours to become year-round, full-time intensive grain cultivators unless *compelled* to do so.²² We generally label that external force of compulsion the "state."²³

Before delving into a further understanding of just why the state would compel its subjects to live and labour in more concentrated populated cores, we must recall, ever so fleetingly, the basic demarkers of a state versus a tribal society. Let us use the rough and ready definition provided by Fukuyama in his treatise on state formation, *The Origins of Political Order*. Fukuyama delineates five major signifiers that distinguish a state society from a tribal one. First, states possess a centralised source of sovereign authority such as a king, a president, or a prime minister. Second, that source of authority is backed by a monopoly of the legitimate means of coercion, in the form of an army or a police force. Third, membership in the state is based on territory rather than kin relationships. Fourth, states are far more stratified and unequal than tribal societies, with a greater proportion of coerced members, such as slaves, serfs, and corvée labourers, than tribal societies. Fifth, and finally, lists Fukuyama, “states are legitimated by much more elaborate forms of religious belief, with a separate priestly class as its guardian.”²⁴

Whilst these descriptive points offer a general blueprint of the state, there is an implicit bias towards statehood contained within this definition. Fukuyama presents the state not in terms of what it looks like per se, but rather how much more muscular, hierarchical, and complex it is in relation to less “evolved” social groupings such as the tribe or band. Here Fukuyama follows in the long line of social theorists, stretching back to Giovanni Batista Vico at the start of the 18th century, who argue that over the millennia of hominid organisation, societies have progressed from small bands to larger tribes and chiefdoms to eventually reach the apex of complexity, scale, and energy consumption, in the form of the state.²⁵ What this would imply is that the remaining pockets of modern bands and tribes marginalised beyond the boundary of the state represent a more primitive, ‘throwback’ model of social organising that had failed, for some reason or another, to make the transitional leap to statehood. James Scott reminds us of the problematics of such one-directional social evolutionary theories as implying both a need for those left behind to become the object of development efforts to integrate them into the cultural and economic life of the state, and the concomitant view that fragmentation of any state represents a regression.²⁶

In fact, what Scott urges is a reconceptualisation of those pockets of non-state peoples, not as some ancestral throwback of a bygone social taxonomy, but as the “dark-twin”²⁷ of the state itself. These unincorporated peoples existing just beyond the control of the state are bound up in a symbiotic relationship with the state itself. Such non-state zones are a catchment area for the dissidents and refugees who have consciously rejected the state’s authority. And the line between the two zones is porous. Scott writes, “Nor did movement in one direction or the other preclude subsequent moves. Depending on the circumstances, groups have disengaged themselves from a state and then, later, sought to affiliate themselves (or been seized by!) the same or another state. A century or two later, they might again be found

outside that state's grasp, perhaps because they had moved away or perhaps because the state in question had itself collapsed."²⁸

The simple ability for state subjects to flee the state's reach into these non-state boundary zones of symbiosis and opposition, when taxes, conscription, forced labour, disease, or environmental conditions became too dire, was the main check on state power. Scott argues persuasively that nearly every state exudes former subjects outward into complementary "shatter zones,"²⁹ and the more onerously the state exerts pressure, the greater the population haemorrhaging of its core. Scott notes the variety of ethnic groups, including the Lahu, Hani, Akhi, Miao, and Hmong people who fled to the hills of mainland Southeast Asia in response to successive waves of Han expansion, from its earliest dynastic heaving southward to the Yangzi (202 BCE–220 CE) right through the Ming, the Qing, and its successors, the Republic, and the modern-day People's Republic.³⁰ These refugees eschewed wet-rice farming to take up foraging and shifting cultivation in order to aid mobility and evade state control and taxation.³¹

States consistently generate non-states at their margins. In the Maghreb, Ernest Gellner records how the Berbers of the High Atlas have consciously chosen to reject the authority of the lowland Arab townships to occupy the hilly "land of dissidence (*bled-es-siba*)," in direct opposition to those who dwell in the "land of the government (*bled el-makhzen*)."³² In the new world, this process was mirrored during the Spanish colonisation. The anthropologist Pierre Clastres argues that the hunting and gathering societies of South America, far from being fossilised remnants of left-behind pre-Neolithic tribes, had previously lived as sedentary state-subjects practising fixed-field agriculture. These "tribes" abandoned their fields, and dissolved into small foraging bands, probably as a mechanism to evade capture and European-borne diseases.³³

For the premodern state, the ideal subjects were densely packed cultivators of fixed fields of grain who could generate a large surplus for appropriation by rulers, soldiers, priests, and artisans. Georges Condominas described this perfected state space as *emboiment* ("bundling") of a state-controlled population.³⁴ For only within such a clearly defined and confined zone were subjects *legible*, by dint of their sedentary food supply, *taxable*, in that a field of non-shattering husked grain ripens all at once and above ground, unlike tuber crops, so that it is easily assessed by the passing tax collector, and *conscriptable*, by living close to the state core.³⁵ Michael Mann's description of the "social cageing"³⁶ employed by the archaic states might ring closer to the real, for it reminds us again of Boserup's theory that it is compulsion from external pressures that led societies to produce more food than they required, by forcing a population into "thickly-settled clumps" of demographic, and thus production, density.³⁷ Describing the formation of the earliest Mesopotamian states, Scott writes: "Such a surplus does not exist until the embryonic states creates it. Better put, until the state extracts and appropriates this surplus, any dormant activity that might exist is 'consumed' in leisure and cultural elaboration."³⁸

In order to create and control this surplus, the state was obligated to concentrate its subjects within a limited and “frictionless” geographic spread. As we explored in the thumbnail itinerary way back in our first chapter, for the premodern state, a flat expanse of arable farmland and a densely packed population of farmers concentrated around the state core were absolutely essential to its existence. If the farmers were too far removed from the non-producing elites, artisans, and specialists at the core, the effort of transporting bulk goods would outweigh their value. As an ancient Han proverb states: “Do not make a grain sale over one thousand *li*”—415 kilometers.³⁹ And so too, if the friction of terrain increases because of marshes, swamps, and especially hilly areas, though they might be quite close to the court centre, these zones were generally excluded from the state’s control.⁴⁰ Their elevation and topography made them essentially state-repellent. The terrain was too rugged for easy assessment, the transport too challenging for simple appropriation back to the core, and the population too dispersed to tax effectively. The early states only retained political control across relatively compact horizontal spaces.⁴¹ Beyond these flatlands stood mountains, marshes, and forests, intrinsically resistant to state administration.

The interplay between river valleys and mountains, between state space and non-state space, is more than a bifurcation of political realms. For the anthropologist Robert Carneiro, the diverse terrain of a wide riverine floor-bed set between rugged mountains was the key ingredient in the formation of the pristine states of Mesopotamia. States require dense populations producing a surplus crop and available for military conscription.⁴² In un-circumscribed regions, subjects could simply flee beyond the state’s reach rather than hand over their grain and their bodies into state control. But in regions bounded by mountains, deserts, oceans, or a combination of these geographic obstacles, fleeing en masse was substantially more challenging, and the proto-state could coerce its subjects more extensively without fear of mass exodus.⁴³ And when the geographic barriers were insufficient, the ancient grain states might erect walls to hold its population within concentrated space. Owen Lattimore argues that the Great wall(s) main function was not to keep the barbarians out, but to keep a tax-shy Chinese workforce from “going over to the barbarians.”⁴⁴ Similarly, Anne Porter conjectures that the frontier forts of Ur III (2,112-2,004 BCE) were an early attempt at rooting down formerly mobile pastoralists into sedentary state cultivators.⁴⁵

Yet, these early grain states were brittle and transient projects. “Interregna were more common than ‘regna’,” observes Scott of the Mesopotamian alluvium statelets, “and long episodes of collapse and disintegration were commonplace.”⁴⁶ The relatively stable kingdom of Ur III lasted a mere hundred years. And the Mesopotamian historian Robert Adams notes that such longevity was the exception to the rule of ancient statecraft.⁴⁷ Victor Lieberman details how such early attempts at statecraft had a definitively “convulsive quality,”⁴⁸ and Scott describes how the early irrigated rice “padi” states of Southeast Asia were intrinsically “self-liquidating”:

The heartland or core region of the padi state is the most legible and accessible concentration of grain and manpower. Other things being equal, it is this population from which it is easiest and most efficient to extract the resources necessary to sustain the state and the elite. The fiscal temptation was to press heaviest on this core population and, as a result, it was likely to be the most beleaguered... This population operated like something of a homeostatic device for the state as a whole: the greater the pressure exerted on it, the more likely it would simply flee out of range.⁴⁹

As long as the coercive power of the state remained intact, it could counterbalance the drain of fleeing subjects by incorporating previously non-state peoples, mainly via slave raids, or increasing its birthrate by intensifying agricultural production, or a combination of the two.⁵⁰ The proto-state was not just a fragile political body, but epidemiologically and ecologically vulnerable as well. These early states were crowded settlement camps, with animals and people transmitting germs and parasites in an infectious reservoir of population on a scale ten times greater than the small towns they displaced.⁵¹ Along with increased risk of plague, the earliest states were environmentally unstable as they tended to rely on just a handful of cultivated crops and animals to produce the bulk of their food. When the narrow strains of domesticated grain harvest failed en masse due to climatic changes, increasing salinisation, and over-extended soil usage, or—and perhaps concomitantly—the domesticated herds, often descended from a single genetic ancestry, all contracted the same disease, the effects to the state would be catastrophic. Unlike foragers, hunters, pastoralists, and swidden farmers with a multiplicity of available foods sourced from diverse ecological zones, these immobile state subjects had no alternative food system to adequately rely on when their narrow spectrum of crops and animals collapsed.

Though disease and famine might spell the end for the state core, scholars too often misinterpret the disintegration of these early states as the onset of a “dark age” for its former subjects, argues Scott, Adams, and Lattimore. This may be due to the reliance on written state records and large state-building projects as the reliable measures of a population’s overall health. Certainly, such a keen cry as the Sumerian poem *Lamentation over Ur*: “Hunger filled the city like water, its people are as if surrounded by water, they gasp for breath, its king breathes heavily in his palace, all alone, its people dropped their weapon”⁵² indicates that the collapse was devastating for the rulers of Ur and its army. However, it is less clear that the end of Ur III marked a definitive decline for the average state subject. “The ‘collapse’,” argues Scott, “of an ancient state centre is implicitly, but often falsely associated with a number of human tragedies, such as a high death toll. To be sure, an invasion, a war, or an epidemic may cause large-scale fatalities, but it is just as common for abandonment of a state centre to entail little if any loss of life. Such cases are better considered a redistribution of population.”⁵³ Scott urges modern scholarship to embrace a more flexible definition of an “amphibious

peasantry”⁵⁴ that could move fluidly between geographic and political niches, either farmers or foragers, sedentary agrarians within large, complex and hierarchical ruling systems, or small acephalous bands of mobile swiddeners, as a flexible coping mechanism vis-à-vis the waxing or waning reach of the state. Likewise, Adams argues that in the early state formation epoch of the Tigris and Euphrates alluvial plane, “the connection between nomads and sedentaries was a two-way street, with individuals and groups moving back and forth along this continuum as a response to environmental and social pressure.”⁵⁵

I have all too briefly considered the ancient grain states oscillating existence, in order to set the scene for the biblical flood telling. Let us reconsider our earlier query. Why is the flood story narratively bookended between the population expansion instigated by the “sons of the powerful” on the one hand, and the building of “a tower and a city” in the Mesopotamian alluvium on the other? Perhaps, amidst this steady metastasizing of statecraft on the valley floor, the flood can be read as a decisive and divine ‘no’ against the otherwise inexorable grasp of the state. The deluge pushes Noah out of the flooded state zone and into the non-state mountains. The flood becomes, then, the biblical prelude to the unfolding refugee experience of a people who are continuously eluding, entrapped by, and exiting state appropriation.

Thus far in the biblical account we have met the first farming family in the Hilly Flanks of the Levant. We have followed their descendants in a violent land-grab, and the erection of the first city, presumably on the alluvial plane along a watercourse, where the nutrient-rich soil, year-round access to irrigation, and frictionless transport along a river and flat ground are ideal for cityscaping. Next we listen, as the first bigamist consolidates his power with an expanding family, and watch as Noah invents the first plough to increase crop yield in that flat expansive valley floor, inadvertently furrowing patriarchy in its wake. We shudder as powerful men coerce women to produce more children, and have just come to realise that this population expansion is intrinsically linked to a suffocating and violent process of coercion to cage a densely sedentary labour force.⁵⁶ Now the divine voice calls for an escape for a single segmented family.⁵⁷ A flood that will wipe clean the alluvial grain core, and push Noah, floating within an ark of diverse animals, into the mountains, stateless. In fact, I will argue, the post-diluvian Noah is a model of precisely the kind of social structuring, agrarian practice, and food subsistence traits that are “state-repellent,” to invoke Scott’s evocative phrase.⁵⁸ These include inhabiting a rugged terrain with a high degree of inaccessibility as opposed to the state favoured, flat and frictionless alluvial plane. In place of concentrated grain crops that ripen simultaneously, and facilitated easy appropriation by the tax collector, the stateless agrarian would favour shifting, diverse crops with variant harvests, that could be cultivated in small rocky plots of soil with little management. In place of a dense sedentary population, the inhabitants of this non-state zone would live in fluid mobile groups, capable of fissioning into small family units, to elude capture and

enslavement. When Noah and his family step out of the ark in the Anatolian highlands, the world they recreate is not the crowded cityscape left behind, but an artfully elusive arena designed in dialectical dichotomy to the density of violence which pervaded the valley floor.

Noah: The Homeless, Mountain-Man

Noah appears to represent an odd phenomenon in the biblical milieu. A farmer, literally “a man of the earth,”⁵⁹ who lives in a tent.⁶⁰ Whereas a nomadic pastoralist, who must source ever new grazing pasture for his herd, such as Yubal (Genesis 4:20), would favour a tent and its rapid ease of pitching and dismantling, Noah is slowly cultivating a grape vine, which would imply a more permanent dwelling. As I will elucidate further in Chapter 7, the vine is so slow to maturation, that it was primarily planted for the subsequent generations. And thus, a productive vineyard represents such a tangible inheritance that Navoth would rather scorn King Ah'av than part with his ancestral vineyard.⁶¹ However, Noah does not plant in the state centre, which is where Navot's vineyard was unfortunately situated,⁶² but—we presume—on the hillsides of Ararat, in rocky terrain, where a vine could work its roots effectively through rough soil⁶³; but the grain and the plough, the accoutrements of the state, could not take root. For Noah and his family, the vine was simply one amongst various subsistence mechanisms all driven towards non-accessibility and evasion. Why else remain in the hills planting grape vines, when the rich alluvial soil below beckoned for grain? Perhaps because the vine stands in some opposition to wheat or barley agriculture. And thus, whereas wheat was a state-promoting crop,⁶⁴ vines were a state-evading one. Where wheat ripens within a short fixed window, grape vines have variant and long harvest cycles, making them hard to assess and appropriate. Where wheat, especially of an irrigated field, requires intensive manual labour to plough, fertilise, sow, weed, water, and harvest, the vine, once planted on a hilly rain-fed slope, requires little daily care, so a shifting agriculturist can visit his vine plots without building a house there, and can slip away quickly before he or his crop become ensnared by state forces. “A hill farmer,” notes Richard O'Connor, specifically describing the small Tai statelets of the mountain zones, but applicable to non-state farmers everywhere, “had no corvée obligations.”⁶⁵ But it was not simply vines which endowed Noah's food selection as a politically motivated escape strategy against state power, the post-diluvian covenant with the Divine also demanded that Noah embrace foraging and seasonal crop variety, namely a subsistence difficult to confiscate or tax.

The State-Repellent Subsistence Covenant

Noah's emergence from the ark in the highlands is met with a series of divine promises and covenantal blessings that demand two new modes of acquiring

foodstuff, one clearly elucidated, the other only elliptically referenced. "Every moving thing that lives," God informs Noah, "shall be food for you; as the green grasses, I now give you everything."⁶⁶ Here is the straight warrant for Noah to embrace a truly omnivorous diet. Now, every living creature is a potential meal. From the talmudic exegetes⁶⁷ to Cassuto,⁶⁸ this verse is nominally considered as a divine loosening of the original injunction to Adam to eat only the grasses. "Behold," God tells Adam, "I have given you every plant yielding seed which is upon the face of all the earth, and every tree, with seed in its fruit; you shall have them for food."⁶⁹ In contradistinction to Noah's blessing to eat "everything," God offers Adam only plants and fruit. Yet, as Cassuto concedes in his commentary on this ostensibly vegetarian command to Adam, "It is true that the eating of flesh is *not* forbidden here."⁷⁰ And further we know that Adam wore animal skins,⁷¹ and that Abel and Noah each sacrificed animals,⁷² which all implies that animals in the Bible were killed, if not eaten, before God's post-flood omnivorous blessing. Perhaps, the command to Noah to eat of everything that creeps on the earth and grows from the ground is not a new carnivorous concession to a bloodthirsty post-diluvian world,⁷³ but a reminder to a once sedentary grain farmer to embrace the mobile foraging strategy of the hillsides, a strategy that extended beyond a simple food procurement regime suited to the local environment, but can be read as a God-given mechanism to marry the ideal non-state terrain of rugged hills with subsistence techniques that were inherently so multitudinous as to be illegible, and thus inappropriable, by the state. In order to catch and eat such "creeping" critters, Noah must himself remain on the move, slipping ever beyond the reach of authority.

Yet, Noah is not just told by God to forage in the hills, but, I will tentatively propose, he is likewise commanded to transform his family into shifting agriculturalists, using various cultivars on a staggered harvesting routine. Let us return again to the human-divine interplay at the dawn of the post-diluvium. "The Lord smelled the pleasing fragrance, and the Lord said in His heart, never again will I curse the ground because of humanity... While the earth remains, seedtime and harvest, cold and heat, summer and winter, day and night, shall not cease."⁷⁴ Many modern biblical scholars consider this verse to reference God's reversion to normative seasonal and diurnal patterns after disruption. Cassuto conjectures that God is recreating a sense of order after certain dichotomies of day and night, hot and cold, were "blurred or altogether obliterated during the year of the Deluge."⁷⁵ J. Martin argues that this verse obliquely references a devastating drought and famine that effectively shut down the seasons before the flood, as in the Epic of Atrahasis.⁷⁶ God then flicked the post-diluvian restart switch whilst declaiming these verses. Ellen Davis, meanwhile, argues that the flood washed away the human-initiated "ruination" of the earth, which led God to restore a steady cycle of seedtime and harvest that had been previously gummed-up by moral turpitude.⁷⁷ But each of these interpretive gestures assumes that the seasonal cycle had been broken in the past either by God or humanity, despite the fact that the text

makes no allusion to such a breakdown. Rather, the foregrounded interest of this verse is to elucidate the seasonal shifts with unique specificity, and assure Noah that they will remain intact, as though, the verse implies, Noah had never known of the seasons, nor the differentiation between day and night, until this very moment of leaving the ark! It is Noah, rather than the earth itself, who seems strangely unacquainted with nuanced periodicity.

What makes a great deal of sense if Noah and his family are escaping a mono-cultural, irrigated grain core in the Tigris and Euphrates alluvium. The state leans towards legibility and standardisation in order to control, coerce, and appropriate its subjects and their products. State planting and harvesting were carried out at uniform junctures that could be manipulated to maximise a single focused harvest period across large swaths of fields, making tax collection expedient.⁷⁸ Such legible harvests on a massive scale required a coordinated irrigation regime, rather than the whims of seasonal precipitation.⁷⁹ In such a system, seasons became deemphasised and the farmers would be dulled to the nuances of six seasonal variations⁸⁰; in the same way they were dumb to the variety of animals that one could eat in the hills. Seasons, like wild animals, play little role in the day-to-day activities of a sedentary state subject. By far, the most ubiquitous artefacts of the ancient city of Umma in Mesopotamia are the bevelled rim bowls that each held exactly two litres of barley and may have served as the daily food ration for the lowest classes of labourers.⁸¹ In such a regimented landscape, variety of seasons and crops was probably anathema. Perhaps, to push this reading a bit further, if these state subjects were not free farmers, but *corvée* labourers, indentured servants, or outright slaves, then the distinction between day and night—between labour and leisure—would blur; slaves cannot sing, as the psalmist does: “Man goes out to his work, and to his labour *until* evening,”⁸² for a slave has no evening time for his own use. Day and night are equally times of servitude. Slaves are not attuned to the diurnal rhythm of the day, but rather to the whims of the master and the task.⁸³

The homogeneity of crop monoculture is conjured up in a remarkable midrashic aphorism. “Rabbi Yochanan said: ‘Why did they [the generation of the flood] revolt? It used to be that they would sow for a single year, and yield grain for 40 years. God said to them: Is this how you act!? While the earth remains, seedtime and harvest, cold and heat, summer and winter, day and night, shall not cease.’”⁸⁴ It is certainly reasonable to read Rabbi Yochanan within a supernatural context linked to a moral message, namely that a magic super-abundance of crops leads to indolence, and indolence leads to sin. And further, the midrashists could dovetail these superabundant harvests with Noah’s invention of the plough.

Yet, there are other, less fantastical ways to situate this midrash. Ironically, in order to frame the midrashic commentary as mimetic realism, we would do well to recall the hypothesis of Cassuto that certain midrashic narratives, and specifically those related to the flood genre, can be traced

back to ancient Mesopotamian myths.⁸⁵ Relying on strong linguistic parallels between cuneiform tablets and midrashic folios, Cassuto asserts that,

Details of the deluge with which the Talmudic and Midrashic literature embellishes the Torah account are in part homiletical interpretations of biblical texts or later legends, but some of them correspond to what is narrated in the Mesopotamian inscriptions ... even when the old [Mesopotamian] epic poetry was forgotten, its subject matter was preserved in the folk-memory in a manner that made it possible for it to be reflected in the Talmudic and Midrashic literature.⁸⁶

It is quite conceivable, therefore, to read Rabbi Yochanan's 40 years of a singular harvest as a reflection of the 40 years of farming and canal-digging drudgery suffered by the lower-class Igigi gods in the Atrahasis epic. In the Sumerian account, the 40 years of monotonous, back-breaking labour unleashes a revolt, and the gods burn their farming tools in protest. "Great indeed was the drudgery of the gods, the forced labor was heavy, the misery too much ... They counted years of drudgery... ... and forty years, too much!... ... forced labor they bore night and day. They were complaining, denouncing, muttering down in the ditch."⁸⁷ Rabbi Yochanan's pithy commentary pulls at the threads of the Atrahasis account. The binary of day and night is blurred with weary labour, the 40 years of singular harvest is no blessing, but a curse of unvarying work, 40 years of tending and irrigating the mono-cultural field. In both midrash and epic, one senses the underlying homogeneity of state life. "Valley [kingdoms] flatten," Scott notes of symmetrical ancient Southeastern polities, "A padi-planting society also shaped a common material culture—diet, cuisine, farm implements, plow animals, household architecture, and so on."⁸⁸ And flat, homogenous societies concretise cross-generational inequality with nearly inflexible, and often state-enforced, control and inheritance of property.⁸⁹ Thus, after 40 flat years, which is to say a biblical generation,⁹⁰ of labour-intensive irrigation and harvesting of a singular crop, both midrash and epic poem speak of rebellion.

In the ancient Mesopotamian account, the ruling gods create the first humans as replacement slaves in the fields and ditches. This paradigm the cruel logic and centripetal force⁹¹ of archaic statecraft. When its subjects revolted or fled, the state would recuperate its losses via new slave raids, purchases of slaves from slave takers, and forced resettlement of whole communities near the grain core.⁹² In effect, it was as if the gods themselves were always willing to birth new creatures to "bear the yoke,"⁹³ as the Atrahasis epic recounts, after the last lot had burned their tools and escaped, whereas, in the midrashic reinterpretation, the divine response is to institute seasons which seems an odd amelioration indeed. Yet, what is seasonal variation if not the opportunity to cultivate a variety of crops in shifting climes, with heterogeneous harvests in multiple terrains and altitudes. Rashi, in his mediaeval exegesis on our verse, hints to such a diverse subsistence strategy. "Spring"⁹⁴ is for planting the quick

ripening crops of barley and pulses... whilst early summer is for the ingathering of the figs and drying them in the fields.”⁹⁵ Theodore Hiebert deftly aligns these Noahide verses on various cropping times with two extrabiblical calendars, the Solomonic agricultural calendar of Gezer and the Rule of the Community text of Qumran, to reveal the intricate subtlety and variety of ancient Levantine highland harvest events. “The planting of grain after the arrival of the first good rains in the late fall (November and December), is the first event in the agricultural year... It is followed by the grain harvest, the first harvest in the highlands that occurs in the spring... The harvest of fruit follows in the summer and early fall... The cuttings of June and July likely refers to the grape harvest, the summer fruit of August to the fig harvest in particular ... and the ingathering of September and October to the olive harvest.”⁹⁶ After the mono-cropped wheat and beer-barley culture is submerged, Noah learns, seemingly anew, of the nuanced tempo of the seasons, of a short spring season for an early lightning crop of legumes and green barley, and the onset of summer when the fig tree with its indeterminate, staggered harvest is gleaned.

God, it would seem, is awakening Noah to a particular type of farming deeply resistant to state appropriation on two fronts. First, Noah will farm the high hills where certain crops can “escape” the state.⁹⁷ As the friction of terrain increases, state seizure becomes increasingly less efficient. The difficulty in capturing the crop in the first place, and then transporting it back to the state centre, lessens its worth to those at the core of state-making. As well as rugged geographical friction, hill farming represents a patchwork of small plots that are hard to trace, and ripen at different times dependent on crop type, as well as slight variations in altitude, rain-shadow, and soil structure and chemistry. “To practice hill farming” concludes Scott, “is to choose a social and political life outside the framework of the state... for the state or a raiding party, it represents an agricultural surplus and population that is difficult to assess, let alone seize.”⁹⁸ Second, to grow diverse crops at various points in the year encourages mobility of movement and flexibility in social structure. Malcolm Yapps designates such hill swiddeners as “jellyfish tribes,” for their intrinsic plasticity and ability to disaggregate in confrontation with the state.⁹⁹ A family that can quickly harvest lentils and barley before disappearing back into the high hills only to return on an intermittent basis to pick and dry figs would be difficult to capture. Noah’s social disaggregation into a minimal family unit coupled with this shifting subsistence strategy created elusive escapees from a drowned state core. Ernest Gellner describes a parallel mechanism amongst the Berbers to segment under threat into small and scattered bands, with the aphorism “Divide that ye not be ruled.”¹⁰⁰

In fact, the ancient state records abound with boastful rulers confidently proclaiming the total annihilation of one non-state group or another, only for the self-same group to return (and usually be destroyed) again at some later point in the historical record. These flexible and mobile people could dissolve into atomised segments and thus evade the ruler’s heavy fist and premature

death warrant. For example, the minor village of Pi-naratim, “mouth of the canals,” lay 34 miles upstream from the state metropolis of Larsa on the Tigris. Samu-el, King of Larsa, claimed to have destroyed it utterly in 1886 BCE. Yet, just 47 years later the little village required “seizing,” this time by the Larsan king, Sin-iqisam, in 1839 BCE. The recalcitrant Pi-naratim required a further reconquest in 1808 BCE.¹⁰¹ But perhaps the braggadocio of the Egyptian king Merneptah inscribed on a stele in 1207 BCE, claiming to have wiped out a hill people named “Israel,” takes pride of place in ancient state mis-forecasting.¹⁰² “Ashkelon has been overcome/ Gezer has been captured; Yano’am is made non-existent/Israel is laid waste and his seed is not.” For jellyfish tribes are, by design, evasively structured into numerous break-away sub-segments, and thus difficult to seize or extinguish.

If we but recall that Noah, akin to nearly all swidden cultivators, was enjoined to supplement his diverse crop strategy with a variegated hunting diet, it seems clear that the divine covenant with Noah is, at least in part, fashioned with multiple and interlocking systems to continuously evade state control. In the midrashic parlance of Rabbi Yochanan, the divine response to the grain state is seasonal, and thus subsistence complexity. Scott describes the upland peoples of Southeast Asia in similar terms. “By pursuing such a broad portfolio of subsistence strategies they spread their risks, they ensure themselves a diverse and nutritious diet, and they present a nearly intractable hieroglyphic to any state that might want to corral them.”¹⁰³

In effect, Noah the ploughman has been flooded from his monoculture field of grain to become a hill-swiddener and forager. God witnesses the violent population herding techniques intrinsic to the Mesopotamian alluvium and its city-states,¹⁰⁴ and designates Noah as the state elusive survivor. The divine post-diluvian gift, the Noahide covenant, is the wherewithal to live and thrive in artful and purposeful avoidance of state hegemony. Perhaps, this is alluded to when God states, just prior to describing the dizzying seasonal harvest calendar, that “never again will I curse the ground because of humanity (אדם).”¹⁰⁵ Hiebert alerts us to the paradigm shift of God’s pronouncement. “Until this point, arable land lies under the divine curse (Gen 3:17, 4:11, 5:29); but once the curse is removed after the flood, arable land is never again so described in the epic.”¹⁰⁶ A farming system that constantly shifts with the hillside seasons cannot be accursed, in as much as it cannot be easily taxed or taken. And a blessed subsistence strategy that utilises every variant of flora and fauna, specifically in the mountain ranges, curtails the curse of Adam because it hallows the migration of peoples away from the state alluvium. Not all ground is cursed, just mono-cropped state ground. And when a covenant is enacted to develop other means of food production, the curse can cease¹⁰⁷ (or at least be held in obeisance, as we shall explore in Chapter 7).

It is impossible from such a long historical distance to clarify whether Noah and his family fled an actual flood in the Tigris and Euphrates riverine valley, or whether the flood event serves as a politically and narratively expedient device

to effectively remove Noah from state space into a state-evasive, shatter zone¹⁰⁸ in the hills. Perhaps, it is sufficient that floods, as in the Gilgamesh epic, force a sedentary character to become mobile and essentially hard to locate. The urbanite king Gilgamesh must journey far to find the Noah-esque hero, Utnapishtim, in the marginal and “far-away” marsh zone near the “Mouth of the Rivers” that feeds the Persian Gulf.¹⁰⁹

Yet, despite the fluidity of Noah’s flight, the intrinsic plasticity of a tent abode, a diverse and moveable feast of animals and shifting cultivars, a mountainous terrain of friction, and a segmented, adaptable social unit of a single family, in short, despite strategies aplenty to keep Noah and his children at a geographic, agro-ecological, and political remove from the state’s centripetal force, the Bible will draw Noah’s descendants back to the valley floor as urban builders. The Book of Jubilees¹¹⁰ foregrounds the Bible’s implied crisis of descent from the Ararat highlands into the valley of Shinar, and all the concomitant evil of the city-state, with just a few terse words in the mouth of the minor biblical character, Peleg (Genesis 10:25). “He [Peleg] said: Behold, the sons of man have become evil with perverse counsel so that they are building a city and a tower for themselves in the land of Shinar. For they departed from the land of Ararat toward the east into Shinar.”¹¹¹ In Chapter 7, I will explore the moment when Noah’s children descend and revert stateward, the very moment in which one of their number is cursed with the perversity of enslavement.

Notes

- 1 J.B. Pritchard, *Ancient Near Eastern Texts* (Princeton, NJ: Princeton University Press, 1969), 42–44. However, some scholars assert that the narrative flood genre dates from at least ca. 2500 BCE in ancient Sumer. Cf. Eric M. Meyers and John Rogerson, “The World of the Hebrew Bible,” in *The Cambridge Companion to the Bible*, ed. Bruce Chilton (Cambridge: Cambridge University Press, 1997), 55. See Y.S. Chen, *The Primeval Flood Catastrophe: Origins and Early Development in Mesopotamian Traditions* (Oxford: Oxford University Press, 2013) for a comprehensive survey of relevant cuneiform texts and scholarly debate on an earlier (Early Dynastic III, c. 2600–2350) or later (Old Babylonian, c. 2000–1600) provenance of the Mesopotamian flood myth. Chen argues for the later date for the flood tradition and describes all earlier flood attestations as figurative allusions to human and divine acts of power.
- 2 S. Dalley, *Myths from Mesopotamia: Creation, the Flood, Gilgamesh, and Others* (Oxford: Oxford University Press, 1989), 9–35.
- 3 A. George, *The Epic of Gilgamesh: The Babylonian Epic Poem and Other Texts in Akkadian and Sumerian* (London: Penguin, 1999).
- 4 A. Parrot, *The Flood and Noah’s Ark* (New York: Philosophical Library, 1955), 37–38. See also, G.P. Verbrugghe and J.M. Wickersham, *Berosos and Manetho, Introduced and Translated: Nature Traditions in Ancient Mesopotamia and Egypt* (Ann Arbor: University of Michigan Press, 1996), 50.
- 5 J. David Pleins, *When the Great Abyss Opened: Classic and Contemporary Readings of Noah’s Flood* (Oxford: Oxford University Press, 2009), 109.
- 6 U. Cassuto, *A Commentary on the Book of Genesis: Part II, from Noah to Abraham (Gen VI 9–XI 32)*, trans. Israel Abrahams (Jerusalem: The Magnes Press, Hebrew University, 1964), 27.

- 7 W. Ryan and W. Pitman, *Noah's Flood: The New Scientific Discoveries About the Event That Changed History* (New York: Simon and Schuster, 1998), 246–247.
- 8 W.G. Lambert and A.R. Millard, *Atra-hasis: The Babylonian Story of the Flood* (Oxford: Clarendon Press, 1969), 87, 89.
- 9 Hillel, *The Natural History of the Bible*, 51.
- 10 Pleins, *When the Great Abyss Opened*, 121.
- 11 Dalley, *Myths from Mesopotamia*, 31.
- 12 Gen 6:1 & 5.
- 13 Cassuto, *Genesis II*, 302. “The verbal parallelism indicates a correspondence of theme.”
- 14 Gen. 6:11.
- 15 Gen 6:13.
- 16 See Fukuyama, *The Origins of Political Order*, 84.
- 17 Boserup, *The Conditions of Agricultural Growth*, 51–54.
- 18 Fukuyama, *Origins*, 84.
- 19 James C. Scott, *The Art of Not Being Governed: An Anarchist View of Upland Southeast Asia* (New Haven: Yale University Press, 2009), 192.
- 20 The ability for state subjects to physically flee state space into non-state zones constrained pre-colonial regimes the world over. As Scott notes, “Much of the classical literature on statecraft is preoccupied with the techniques of attracting and holding a population in an environment where they can flee to the nearby frontier or settle under the wing of another nearby prince.” Scott, *Seeing Like a State*, 185.
- 21 Seth Richardson, “Early Mesopotamia: The Presumptive State,” *Past and Present* 215 (2012): 4. “In very broad terms, lower Mesopotamia, an area of roughly 110,000 square kilometres, probably possessed a human population of around half a million people in 3000 BC, with enough land to give every nuclear family more than 15 ha — more than enough for general sustainability.”
- 22 Boserup, *The Conditions of Agricultural Growth*, 54.
- 23 See Giles Deleuze and Felix Guattari, *A Thousand Plateaus* (London: Bloomsbury, 2017), 521, in which the authors label the State the “Apparatus of Capture” (Chapter 13), referring to the state’s insatiable desire to acquire and hold (1) land (for rent), (2) work/labour (for profit), and (3) money (for taxes). Paraphrasing Marx, Deleuze and Guattari argue that the means of capture is intrinsically with violence, and the state is essentially a “machine of enslavement.”
- 24 Fukuyama, *Origins*, 80–81.
- 25 Fukuyama, *Origins*, 52. “There is an obvious analogy between Darwin’s principle of natural selection and competitive human evolution. This recognition led to a revival of theorizing at mid-century by anthropologists such as Leslie White, Julian Steward, Elman Service, Morton Fried, and Marshall Sahlins, who argued that there was a clear progression over time in complexity, scale, and energy use of societies.”
- 26 Scott, *The Art of Not Being Governed*, 128.
- 27 Scott, *The Art of Not Being Governed*, 99. Scott also calls these “tribal peoples” the “pole of comparison, and at the same time, the antidote,” to the state form.
- 28 Scott, *The Art of Not Being Governed*, 27.
- 29 Scott, *The Art of Not Being Governed*, 24.
- 30 Scott, *The Art of Not Being Governed*, 129.
- 31 This pattern is paralleled amongst the *remontados* in the Philippines. This was the label given to Tagalog subjects who would flee “back up to the hills” to escape the harsh labour regimes imposed by Roman Catholic Friars in the early years of Spain’s dominion in the Philippines. See Scott, *Seeing Like a State*, 186.

- 32 Ernest Gellner, *Saints of the Atlas* (London: Weidenfeld and Nicolson, 1969), 14, 31.
- 33 Pierre Clastres, *Society Against the State: Essays in Political Anthropology*, trans. Robert Hurley (New York: Zone, 1987), 79–99.
- 34 George Condominas, *From Lawa to Mon, from Saa' to Thai: Historical and Anthropological Aspects of Southeast Asian Social Spaces*, trans. Stephanie Anderson et al. (Canberra: Australian National University, 1990), as cited by Scott, *The Art of Not Being Governed*, 65.
- 35 Scott, *The Art of Not Being Governed*, 65. See also, Scott, *Against the Grain*, 129 “The key to the nexus between grains and states lies, I believe, in the fact that only the cereal grains can serve as a basis for taxation: visible, divisible, assessable, storable, transportable, and ‘rationable.’ Other crops—legumes, tubers, and starch plants—have some of these desirable state adapted qualities, but none has all of these advantages.”
- 36 Michael Mann, *The Sources of Social Power* (Cambridge: Cambridge University Press, 1986), 56.
- 37 Robert Elson, “International Commerce, the State, and Society: Economic and Social Change,” in the *Cambridge History of Southeast Asia*, vol. 2, *The Nineteenth and Twentieth Centuries*, ed. Nicholas Tarling (Cambridge: Cambridge University Press, 1992), 131.
- 38 Scott, *Against the Grain*, 151.
- 39 As quoted by Scott, *Art of Not Being Governed*, 41.
- 40 Scott, *Art of Not Being Governed*, 54.
- 41 F.K. Lehman estimates that the precolonial state’s maximum size could not have been more than 160 kilometers in diameter. F.K. Lehman, “Burma: Kayah Society as a Function of the Shan-Burma-Karen Context,” in *Contemporary Change in Traditional Society*, as cited by Scott, *Art of Not Being Governed*, 44.
- 42 See Scott, *Seeing Like a State*, 186. “The classical state envisaged a concentrated population, within easy range, producing a steady supply of easily transportable, storable grain and tribute and providing a surplus of manpower for security, war, and public works.”
- 43 Robert L. Carneiro, “A Theory of the Origins of the State,” *Science* 169 (1970): 733–738. See also Fukuyama, *Origins*, 86.
- 44 Owen Lattimore, “The Frontier in History,” 473.
- 45 Anne Porter, *Mobile Pastoralism and the Formation of Near Eastern Civilization: Weaving Together Societies* (Cambridge: Cambridge University Press, 2012), 347.
- 46 Scott, *Against the Grain*, 122.
- 47 Robert Adams, “Strategies of Maximization, Stability, and Resilience in Mesopotamian Society, Settlement and Agriculture,” *Proceedings of the American Philosophical Society* 122, no. 5 (1978): 329–335.
- 48 Victor Lieberman, *Strange Parallels: Southeast Asia in a Global Context, C. 800–1830, Vol I: Integration on the Mainland* (Ann Arbor: Univ. of Michigan Press, 2003), 61.
- 49 Scott, *Art of Not Being Governed*, 95.
- 50 See Scott, *Against the Grain*, 29. “Evidence for the extensive use of unfree labor—war captives, indentured servitude, temple slavery slave markets, forced resettlement in labour colonies, convict labour, and communal slavery (for example, Sparta’s helots)—is overwhelming . . . The attention to ‘husbanding’ the subject population, including women, as a form of wealth, like livestock, in which fertility and high rates of reproduction were encouraged, is apparent.”
- 51 Scott, *Against the Grain*, 191.
- 52 Quoted in Ian Morris, *Why the West Rules-For Now* (London: Profile Books, 2011), 194.

- 53 Scott, *Against the Grain*, 210.
- 54 Scott, *Against the Grain*, 212.
- 55 Adams, "Strategies of Maximization, Stability and Resilience," 334.
- 56 See Scott, *The Art of Not Being Governed*, 24. "The concentration of people and production in a single location required some form of unfree labour when population was sparse."
- 57 Scott, *Art*, 24. "The effect of all state-making projects of this kind was to create a shatter zone or flight zone to which those wishing to evade or to escape bondage fled."
- 58 Scott, *Art*, 178.
- 59 Gen 9:20. This translation follows the more literal reading offered by Ibn Ezra and Nachmonides in their respective commentaries on the verse, but differs from that of Rashi, and latterly, Cassuto (*Genesis II*, 158-9), who translate the text more figuratively as "master of the earth." In Chapter 7, I will translate this nomenclature as "dirt-man," in keeping, I will argue, with scripture's increasingly negative portrayal of Noah, culminating in his angry curse of enslavement upon his own grandchild.
- 60 Gen 9:21.
- 61 I Kings 21:3.
- 62 I Kings 21:2. "And Ah'av spoke to Navot, saying, Give me your vineyard, that I may have it for a vegetable garden, because it is near my house."
- 63 Emily Bell, "Why Does Bad Soil Make for Great Wine," *Vinepair* (November 2015), <https://vinepair.com/wine-blog/why-does-bad-soil-make-for-great-wine/>. "There are a few reasons bad soil yields higher quality grapes. In poor soil, roots have to work harder, 'ramifying,' or branching off, to gather nutrients. Not only does this increase the surface area of root-to-soil—meaning more nutrients ultimately get to the grape—but it also helps regulate water absorption... ... Whatever the quality of the soil, there's one unifying 'must' for all good vineyards: drainage. Rockier soils actually have better drainage—as do slopes."
- 64 See Scott, *Against the Grain*, 117. "In Weberian terms, we are dealing here with something like 'elective affinity' rather than cause and effect. Thus it was possible and not uncommon at the time to have a sedentary farming population on alluvial soils practicing irrigation without any state. But there was no such thing as a state that did not rest on an alluvial, grain-farming population."
- 65 Richard O'Connor, "A Regional Explanation of the Tai Muang as a City-State," in *A Comparative Study of Thirty City-States*, ed. Mogens Herman Hansen (Copenhagen: Royal Danish Academy of Sciences and Letters, 2000), 434.
- 66 Gen 9:3.
- 67 bTalmud Sanhedrin 59b.
- 68 Cassuto, *Genesis I*, 58–59; and *Genesis II*, 126.
- 69 Gen 1:29.
- 70 Cassuto, *Genesis I*, 59. Italics are mine.
- 71 Gen 2:2.
- 72 Gen 4:4 and Gen 8:20.
- 73 See Cassuto, *Genesis I*, 58.
- 74 Gen 8:21–22.
- 75 Cassuto, *Genesis II*, 121.
- 76 J. Martin, *Journal of Biblical Literature*, Vol. 14 (1926): 129–133, as cited by Cassuto, *Genesis II*, 122.
- 77 Ellen F. Davis, *Scripture, Culture, and Agriculture* (Cambridge: Cambridge University Press, 2009), 32.
- 78 See Scott, *Against the Grain*, 133. Also, *the Art of Not Being Governed*, 73. "A pre-modern ruler in mainland Southeast Asia would have been less interested in what

- today would be called the gross domestic product (GDP) of his kingdom, than what might be called its 'state-accessible product' (SAP)."
- 79 See Scott, *Against the Grain*, 143–144. "Some of the Sumerian polities, especially Ur III, look like command-and-control economies, heavily centralised (on paper—or rather, on tablet), militarized, and regimented... One tablet recorded 840 rations of barley, meted out, in all probability in the (mass produced?) beveled bowls holding two liters of barley. Other rations mention beer, groats, and flour. Labour gangs, whether of war captives, slaves, or corvée laborers, seem ubiquitous."
 - 80 See Scott, *The Art of Not Being Governed*, 252. "Strong reliance on a single crop... comes to dominate the work routines and social organisation of much of the population. Each household planted, transplanted, weeded and harvested, the same crop at virtually the same time, and in much the same way. Coordination of water use required a certain level of institutionalized cooperation and dispute settlement." For a deeper perspective on how agriculture monocropping created materialistic and ritualistic uniformity in the early valley states, see also Richard O'Connor, "Agricultural Change and Ethnic Succession in the Southeast Asian States: A Case for Regional Anthropology," *Journal of Asian Studies* 54 (1995): 968–996.
 - 81 Scott, *Against the Grain*, 132. See also Hans Nissen, *The Early History of the Ancient Near East, 9000-2000 B.C.* (Chicago: University of Chicago Press, 1988), 84–85. However, arguments about the purpose of these bowls continue in Mesopotamian archeology, and variant proposals include presentation bowls for votive offerings, cf. Thomas Beale, "Bevelled Rim Bowls and Their Implications for Change and Economic Organization in the Later 4th Millennium B.C." *Journal of Near Eastern Studies* 37 (1978): 289–313; or containers for salt-making, cf. Giorgio Buccellati, "Salt at the Dawn of History: The case of the bevelled-rim bowls," *Resurrecting the Past*, eds. Paola Matthiae, Maurits van Loon, and Harvey Weiss (Leiden: Nederlands Instituut voor het Nabije Oosten, 1990), 17–40; or bread-baking, cf. A. Millard, "The Bevelled-rim Bowls: Their Purpose and Significance," *Iraq* 50 (1988): 49–57.
 - 82 Psalm 104:23. For an intricate reading of the link between Psalm 104 and the measure of the workday, see Emmanuel Levinas, "Judaism and Revolution," in *Nine Talmudic Readings*, trans. Annette Aronowicz (Bloomington: Indiana University Press, 1990), 103. "I am even inclined to believe that there are not many other ways to love God than to establish these working hours correctly, no way that is more urgent. A psalm is after all not such a bad text on which to found justice for the toiling man."
 - 83 Perhaps, this is the midrashic implication behind the power of the "mighty men" of Gen 6:4 to "rebel against light" (citing Job 24:13). That is to say, their power of coercion was so pervasive that the transition from day into night became of little consequence; these powerful men did not need the light to see and to control. This would indeed be a rebellion, insofar as the dark of night was normally a time of cessation and rest. See *Tanchuma*, Bereshit 12, Mantua Edition, c. 1563, 4.
 - 84 *Tanchuma*, Bereshit 12, Mantua Edition, c. 1563, 4.
 - 85 Cassuto is certainly not alone in asserting that certain midrashic elements may have long-lived folk roots that precede the Bible itself. In Shalom Spiegel's classic scholarly treatise on classic and medieval commentary to the *Akedah*, *The Last Trial*, he references a particular midrashic tradition found in the Mekilta de-Rebbi Ishmael, which states that God will eschatologically save Jerusalem in the merit of the "blood of Isaac's *akedah*," which presumes, in contradistinction to the normative Biblical reading, that Isaac was slaughtered by his father, Abraham. Spiegel then adds, "The blood of Isaac's *Akedah*' is quite possibly a

survival of ancient legends which once upon a time celebrated and memorialized saving miracles in the month of Nisan, and brought together the Akedah theme with the paschal sacrifice, possibly even with the archaic sacrifice of the first born. It may be that here are the last and faint echoes of some primitive pagan or prebiblical version of the Akedah story... therefore we would have here, as occasionally also elsewhere in the Midrashim of early generations, some leftovers of belief before the beliefs of Israel.” *The Last Trial: On the Legends and Lore of the Command to Abraham to Offer Isaac as a Sacrifice*, trans. Judah Goldin (Vermont: Jewish Lights Publishing, 1993), 57.

86 Cassuto, *Genesis II*, 26 and 29.

87 W.G. Lambert and A.R. Millard, *Atra-hasis: The Babylonian Story of the Flood*, 89.

88 Scott, *The Art of Not Being Governed*, 252.

89 See Scott, *The Art of Not Being Governed*, 253.

90 See Exodus 14:29-33.

91 Hjørleifur Jonsson, *Shifting Social Landscapes: Mien (Yao) Upland Communities and Histories in State-client Settings* (New York: Cornell University Press, 1996), 218. Jonsson compares the centripetal forces of valley systems to the centrifugal forces of hill systems. As Scott notes (*The Art of Not Being Governed*, 251), this results in a “striking uniformity” of valley kingdoms vis-à-vis the diversity of hill societies, in terms of religion, language, and ethnicity.

92 See Scott, *Against the Grain*, 153–154. “Warfare in the Mesopotamian alluvium beginning in the late Uruk Period (3,500-3,100 BCE) and for the next two millennia was likewise not about the conquest of territory but rather about the assembling of populations at the state’s grain core.” see also Seth Richardson, “Early Mesopotamia: The Presumptive State,” *Past and Present* 215 (2012): 9. “The local skirmishes of the Old Babylonian period are thus misunderstood when they are viewed as border conflicts, mere by-products of competition between territorial states. Rather, they should be seen as a recrudescence of an epochal competition for the allegiance of rural and non-state constituencies which were increasing in number during this time.”

93 W.G. Lambert and A.R. Millard, *Atra-hasis: The Babylonian Story of the Flood*, 90.

94 I am quoting the Rashi who states that **וְהָיָה בְּשֵׁשֶׁת יָמִים** encapsulates the second half of the month of *Shevat*, *Adar*, and the first half of *Nissan*; this follows the Rabbinical interpretation that this passage refers to an annual seasonal cycle in six parts which would render the verse: “late-autumn and late-spring, winter and late-summer, early summer and early spring ...” **וְהָיָה בְּשֵׁשֶׁת יָמִים** **וְהָיָה בְּשֵׁשֶׁת יָמִים** (see Tosefta Ta’aanith I:7, bTalmud Bava Metzia 106b, Bereshish Rabba 34:11, and others). The discord in listing the seasons out of order in this way (as well as the clearly non-season “day and night” tacked on the end of the verse) leads me towards Cassuto’s translations, where we have four diplotic pairings, “Seedtime and harvest, cold and heat, summer and winter, day and night.” Neither understanding, I believe, would discourage the overall point that God’s first covenant embraces shifting patterns of weather, ecology, and agriculture that seem previously to have been forgotten, or simply—in the way we urbanites know of stars, but rarely encounter them—to which Noah and his family were habitually oblivious before the flood.

95 Rashi, commentary to Gen 8:22, s.v. **וְהָיָה בְּשֵׁשֶׁת יָמִים**.

96 Hiebert, *Yahwist’s Landscape*, 45–47, with particular reference to Table 2.1 on page 46.

97 “Escape crops” are those flora that may be well adapted to environmental niches that are difficult to map and control: high rugged mountains, swamps, deltas, mangrove groves, and so on. They may also be fast growing, of staggered maturity, and easily hidden. Scott, *The Art of Not Being Governed*, 199.

- 98 Scott, *The Art of Not Being Governed*, 193-4.
- 99 Cited by Scott, *The Art of Not Being Governed*, 210.
- 100 Gellner, *Saints*, 43.
- 101 Richardson, "Early Mesopotamia," 20. "The record of Larsa's efforts to control Pi-naratum puts the village in a larger category of unpacified and non-aligned non-state actors, not merely border towns that were traded between major states."
- 102 As quoted in Kyle McCarter, "The Patriarchal Age," in *Ancient Israel*, ed. Hershel Shanks (Washington, DC: Biblical Archeology Society, 1999), 17. On early Israel as a loose confederation of hill peoples, see *ibid.*, 18. "The strong insistence in the patriarchal narratives that the ancestors of Israel were *not* Canaanites... probably derives from the early conflict between hill-dwelling people from whom Israel emerged and the population of the cities of the plains and valleys controlled by Egypt in the Late Bronze Age." That pre-dynastic Israel inhabited non-state zones, see Marvin L. Chaney, *Peasants, Prophets, and Political Economy: The Hebrew Bible and Social Analysis* (Oregon: Cascade Books, 2017), 3. "All would agree that Late Bronze Canaan comprised a miscellany of agrarian city-states, each with its own petty kinglets, but under the nominal suzerainty of Egypt. The power and control of the local dynasts were centered in walled cities which clustered near the steady water supplies of the piedmont springline and the rich, alluvial soils of the plains. Connecting these concentrations of populations with each other and the world beyond were overland routes, also favoured by the relatively level topography. While constituting only a small minority of the population, the ruling elite were able to dominate the other inhabitants of these plains because they alone could field chariots armed with composite bows. Conversely, premonarchic Israel's poor and mostly unwalled towns and villages were concentrated where the control of the Canaanite kings had never been strong—in the rugged terrain and scrub woods of the hill country. Although this territory was less desirable economically because of its steep, brushy hillsides, thin soils, and relative lack of perennial water sources, it effectively neutralized the tactical advantage of the chariots and composite bows with which the ruling classes held sway." That premonarchic Israel may have originally fled from state-controlled regions into the non-state hillsides, see George E. Mendenhall, "The Hebrew Conquest of Palestine," in *The Biblical Archeologist Reader*, eds. Edward Campbell and David Noel Freedman (Garden City, NY: Doubleday, 1970), 114. Mendenhall speaks of some groups which "migrated to the fringe areas from more populous regions."
- 103 Scott, *The Art of Not Being Governed*, 195.
- 104 I draw the notion of the ancient Mesopotamian city-states "herding" together an urban populace from Seth Richardson's "Early Mesopotamia," 9. Richardson conjectures that these early states were mainly concerned with increased clientele rather than land holdings, and thus adopted a "strategy to herd non-state clients into state orders by both force and persuasion." Though Richardson himself does not directly make the link between the herding of domesticated animals and the herding of people, he does reference an ancient Sumerian text (c.1860–1620 BC) which does so explicitly, "the [non-state] people 'like a scattered herd of cattle'." 18. See Scott, *Against the Grain*, 180–182, where he makes some preliminary claims regarding state slavery as the first large-scale *domestication* of people. Also Scott, *Seeing Like a State*, 184. "Seen from the center, the royal court or the seat of the state, this process has often been described as a 'civilizing process.' I prefer to see it as an attempt at domestication, a kind of social gardening devised to make the countryside, its products and its inhabitants more readily identifiable and accessible to the center."

- 105 Gen 8:21.
- 106 Hiebert, *Yahwist's Landscape*, 47–48.
- 107 See Hiebert, *Yahwist's Landscape*, 45. “In explicit reference to Genesis 3:17, God decides ‘not to curse arable land again’—or ‘not to continue to curse arable land’ (both translations are possible)—on account of *adam* (here used with its generic meaning ‘the human race’).”
- 108 “Shatter zones,” according to Scott, aptly describes the non-state regions of flight from state-making. See *The Art of Not Being Governed*, 24.
- 109 Kovacs, *The Epic of Gilgamesh*, 103. See Scott, *Seeing Like a State*, 187. “Thus marshes and swamps (one thinks of the now beleaguered Marsh Arabs on the Iraqi-Iranian border), ever shifting deltas and estuaries... have all served as ‘nonstate space’.”
- 110 A literary work of the Second Temple period, particularly in the last two centuries BCE and the first century CE. The Book of the Jubilees retells the biblical history from creation to the Giving of the Torah at Mount Sinai. See Dina Stein, “Rabbinic Interpretation,” *Reading Genesis: Ten Methods*, ed. Ronald Hendel (Cambridge: Cambridge University Press, 2010), 120.
- 111 James Kugel, “Jubilees,” in *Outside the Bible: Ancient Jewish Writings Related to Scripture*, eds. Louis Feldman, James Kugel, and Lawrence Schiffman (Lincoln: Nebraska University Press, 2013), 331–332.

7 A Slave to Your Brothers

In this third and final flood chapter, we close the narrative of Noah with a tragic family tale. We first encountered Noah, pre-deluge, as a “righteous man, blameless in his time.”¹ Yet, by Genesis chapter 9, Noah is reframed as a naked drunk, cursing his grandson into slavery. I will argue that this personal family horrorshow can also be read as a precisely choreographed account on what causes non-state peoples to revert into state subjugation. I will suggest that the account of Noah tending his vineyard, drinking until he blacks-out, waking and realising that his youngest son has done something unspeakable to him in his stupor, and then cursing his own grandson with enslavement is a pithy unit that answers a basic narrative question in the trajectory of the Genesis story. Namely, how did the God-blessed, high altitude, state evasive family of Noah become the God-cursed, low altitude, city-state makers of Babel? What made Noah’s family, like Adam’s beforehand, slip down from the mountains and into the clutches of state space? Perhaps, this parting story of Noah offers a few possible clues.²

Let us turn to the primary text under discussion. “Noah, a man of the soil (אִם מֵאֲדָמָה), was the first to plant a vineyard. And he drank of the wine and became drunk and exposed himself within his tent. And Ham, the father of Canaan, saw his father’s nakedness and told his two brothers outside.”³ Noah’s introduction of viticulture into the agricultural tableau and his subsequent inebriation then spirals even further downwards with a nearly hysterical curse. “Noah awoke from his wine and he knew what his youngest son had done to him. And he said: ‘Cursed by Canaan, a slave of slaves shall he be to his brothers.’”⁴

This is an incongruous story. The obvious question here is why Noah curses his innocent grandson, Canaan, Ham’s fourth child,⁵ instead of Ham, the actual tattle-tale. Many modern biblical scholars suggest that Canaan alone receives the full force of the curse, because this passage is essentially anti-Canaanite propaganda. “The story seems clearly designed to discredit the Canaanites,” asserts Frederick Bassett, “and justify the Israelite and Philistine hegemony over them.”⁶ In this reading, the entire tale revolves around the curse of Canaan, either as an Israelite justification for the subjugation of the indigenous people,⁷ or to stigmatise pseudo-sexual practices associated with the

lascivious “nakedness” of the Canaanites,⁸ or some combination of the two.⁹ However, as Speiser laments, “the passage thus supplies more questions than answers.”¹⁰ Indeed, difficulties abound here. Why does the narrative begin with the onset of viticulture, if its fixation is really about sexual impropriety, or anti-Canaanite polemics? And further, what is so problematic about the act of Ham simply encountering his naked father, to which Noah takes such great offence?

I don’t believe we can simply follow the reading of Speiser and assume that these difficulties only emerge because this particular story is “merely a splinter from a more substantial narrative.”¹¹ For a strong textual cohesion bookends the beginning and end of Noah’s life into a formative whole. In this last chapter of the flood narrative, Noah is introduced literally as a dirt-man (אִם מֵאֲדָרָם).¹² Back in Chapter 5 in the tale of Noah’s birth and naming, his father brands a life-task into Noah’s name: “This one will console us for the pain of our hand’s work from the dirt (מֵאֲדָרָם) which the Lord cursed.”¹³ Notice the ironic contradistinctions at the inception and conclusion of Noah’s life. Instead of providing salvation *from* the dirt, Noah becomes a being *of* the dirt; instead of hopeful fathers calling on their children to heal the dirt’s curse of toil, Noah becomes a creature of the dirt cursing his own children with greater toil. Perhaps, the dialectic force that drives this concluding narrative is not just a negative framing of Canaan, but, as we have encountered within both the Adam and the Cain narratives of previous chapters, a particular relationship with the soil (אֲדָרָם) that is manifestly harmful to societal relationships. For Noah, that particular engagement with the earth begins with planting grapes and ends with a vicious curse of slavery upon his grandchild. Perhaps, to understand why Canaan is cursed instead of Ham, and to delve further into Ham’s own terrible act against his father, we should start by tracing the underlying dialectic between these two disparate events of viticulture and bondage.

Violent Inheritance

We can recall that Sarna draws on the numerous “artful echoes”¹⁴ by which Noah becomes the literary refiguring of Adam. Thus, just as Adam is the first farmer of wheat, Noah is the first cultivator of grapes. In the midrashic tradition, that role is expanded to include not only grapes, but the other primary orchard taxa of the Levant as well: figs and olives. Genesis Rabba reveals: “From where did Noah obtain [these grape vines]? Answered Rav Abba bar Kahana: ‘Noah brought them on the ark, along with young shoots of figs and planting roots of olives.’”¹⁵ Another midrashic work states even more succinctly, “Noah introduced tree-crop planting.”¹⁶ Adam is the Bible’s first agriculturalist, and Noah its first arboriculturalist.

This staggered depiction of crop domestication mirrors the archaeological record. The cultivation of annual cereals and legumes marked the onset of the Neolithic revolution and the advent of agriculture. But some 5,000 years later at the conclusion of Chalcolithic period and the cusp of the Early Bronze

Age, a “second wave of agricultural innovation”¹⁷ emerged in the Levant with the domestication and extensive use of orchard crops, primarily grapes, olives, and figs.¹⁸

One of the fundamental societal shifts that occur as the result of increased reliance on orchard crops is an increase in sedentism. The cultivation of annual crops allows for, and may even favour, opportunistic seasonal agriculture by non-sedentary populations,¹⁹ whereas orchards and vineyards, in contrast, inherently require a higher degree of permanence and long-term investment.²⁰ Most vines and fruit trees require three to eight years before bearing fruit, and may not become fully productive until 10–20 years after planting.²¹ As John Boardman observes, “A farmer plants... his olives for his old age or his children.”²²

Long-term investment in a particular tract of orchard perennials spurred increasingly complex societal strategies around land tenure, ownership, and inheritance.²³ Tree crops exacerbated the contentious undertones of land privatisation, as successive generations had a great deal at stake in maintaining their ownership of sizable investments in tree cultivation expended by previous generations. As vines and orchards become a “focal point for the new economy” of the Early Bronze Age,²⁴ inheritance was now worth fighting over. No longer could one send one’s offspring away to cultivate a virgin field of arable soil. Now fathers left their children a fixed number of productive trees and no more.

The narrative of Noah’s vine planting placed within the context of emergent Early Bronze Age conflicts around arboriculture and land ownership may help to explain the violence that pervades the aggadic telling of Ham’s superficially benign act of encountering his naked and intoxicated father. Here is the Talmudic account,

Noah awoke from his wine, and knew what his youngest son had done to him (Genesis 9:24). [With respect to the last verse] Rab and Samuel argue: one maintains that he castrated his father, whilst the other says that he sexually abused his father. He who maintains that Ham castrated Noah, [reasons thus:] Since Noah cursed Ham’s fourth son [Canaan], Ham must have injured Noah with respect to [procreating] a fourth son... But according to the interpretation that Ham sexually abused his father, why did Noah curse Ham’s fourth son; Noah should have cursed Ham directly?—We must conclude that according to this view Ham both castrated and abused his father.²⁵

The Talmudists are wrestling with an elliptical text. How could Noah, upon awakening, “know what his youngest son had done to him”²⁶ if Ham had only gazed upon his naked and somnolent form? There must be some tangible and obvious crime wreaked upon the body of Noah and traceable to Ham.²⁷ The rabbinical association between the phraseology of “seeing nakedness” and a sexual, usually incestual, activity is straightforward.²⁸ As we read explicitly in

Leviticus 20:17: "If a man marries his sister, the daughter of either his father or his mother, so that he sees her nakedness and she sees his nakedness"²⁹ When the biblical text speaks of "seeing nakedness," the rabbinical ear is attuned to sexual impropriety. Thus, the concept of Ham raping his father makes textual sense. Noah awakes violated, and knows—we know not how—that Ham is to blame. Yet, what are we to make of Ham's cruel act of castration? The act to which both Talmudic opinions are in agreement.

Robert Alter, echoing the classical commentary of Philo of Babylos,³⁰ glimpses here a "Zeus-Chronos story in which the son castrates the father, or alternatively, penetrates him sexually."³¹ According to Ulf Oldenburg, Noah's castration is an Israelite iteration of the Canaanite pantheonic epic where the son, Ba'al-Hadad, castrates his father, El, to establish the younger god's right to the throne.³² Yet, if this tale is simply the stuttered retelling of a neighbourhood myth of usurping children, then why does Noah awaken to curse his *grandson*, rather than his son?

On this point, Bassett argues that Canaan is specifically cursed, instead of Ham, because the deed itself was not castration or abuse of Noah. Instead, it was Ham's incestuous relationship with his own mother, Noah's wife, carried out on the night of Noah's stupor, and resulting in the conception of the accursed child, Canaan. Bassett leans heavily on the particular idiom employed by the biblical narrative, "Ham, the father of Canaan, saw the nakedness of his father."³³ Here, Bassett employs Leviticus 18 as a definitive lexicon to create a general pattern that "To see a man's nakedness means to have sexual relations with his wife."³⁴ To some degree this reading, latterly buttressed by John Sietze Bergsma and Scott Walker Hahn in their nuanced retelling of the narrative,³⁵ solves many of the literary inconsistencies. Noah's curse of Canaan is not a capricious outburst, but particularly directed at the bastard fruit of Ham's incest, the child Canaan. And inebriation is a plot device, used to somewhat similar effect in a parallel child-parent incestual union, again occurring after a family survives a catastrophic natural disaster, where Lot's daughters ply him with wine before sleeping with him to produce yet further enemies of Israel, Moab and Ammon.³⁶ However, the wine parallel with the Lot affair in Genesis 19:30-38 is not entirely precise. There the daughters use wine to dumb Lot to the sexual act itself, whereas here Noah's drunkenness keeps Noah dumb to the incest between his son and wife. Nonetheless, wine clearly plays an integral part in both cases when illicit sexual activity is underway.

However, the maternal incest interpretation creates its own textual lacuna. In fact, an extended nine-month gap between verses 24 and 25! "Noah awoke from his wine and he knew what his youngest son had done to him. And he said: 'Cursed by Canaan, a slave of slaves shall he be to his brothers.'"³⁷ Were Canaan indeed the product of the affair between Noah's wife and Ham that occurred on that drunken night, then this second verse, the curse of Canaan, would require a holding of the breath, mid-narrative, for a nine-month time-lag between Noah's sober awakening and the actual birth of the accursed

progeny and the realisation of the curse.³⁸ And further, why curse Canaan with slavery, precisely? Further still, if the Lot story is our intertextual touchstone, then the Bible shows few qualms in lewdly detailing the incestuous origins of its tribal enemies; why does the Bible not mention incest explicitly in the curse of Canaan itself? Why the focus solely on the bondage of Canaan, rather than his ignominious conception?

Let us attempt a more ecologically minded integration of the story within its grisly Talmudic framing. Ham castrates his father, just after Noah cultivates a vineyard, because tree crops invariably engender scarcity and competition amongst inheritors. There are simply not enough vines to sustain a fourth child, and Ham must guard his inheritance at all costs. Just as the monocultural predominance of the wheat of the forbidden “tree” destroyed the bio-diverse paradise of Eden, so too do the vines and orchards of the arboricultural revolution drive a new and violent competition amongst Noah’s children. Another midrash certainly hints at just such an interpretation: “The Holy One, blessed be He, meant to issue four sons from Noah who would inherit the four winds of the earth. Ham said: ‘I will castrate my father so that he will not produce a fourth son, in order that [this fourth son] will not share the world with us.’”³⁹

Now, the curse upon Ham’s own fourth child,⁴⁰ Canaan, makes for a narrative *gestalt*. “Cursed is Canaan, he shall be a slave’s slave to his brothers!”⁴¹ Noah’s curse is the concretisation of Ham’s fear of competitive inheritance, foreseen for Ham’s own kin. Four children cannot be sustained by the orchards and vineyards of the previous generation. Four children cannot “share a world” of precious vines. There will not be enough land-wealth, and the fourth child, landless, will become a slave to the more entitled older siblings. The midrash explicitly acknowledges that it is this very “fourth-ness” of too many children that fuels the violence of Ham and the subsequent curse of slavery. “Noah said to Ham: ‘[when you castrated me] you negated my chance for a fourth child! By your life, I will curse your own fourth child!’ And thus Noah said: ‘Cursed is Canaan, a slave of slaves.’”⁴²

When the planting of the grape vine is taken as integral, the story uncurls with a dark logic. Noah’s tree crops are worth inheriting and thus worth winnowing the potential pool of inheritors. So Ham castrates his father before Noah can conceive of a fourth competitor. Noah wakes and curses the fourth and extraneous child of Ham with servitude, reinforcing the narrowing limits of land inheritance. Canaan is enslaved by his brothers because there is simply no property wealth left over for him.

A State of Slavery

The curse of slavery is the slip back down from the mountains of Ararat and into the alluvial core of Babel. For it was slavery, on a mass scale, which was the life-blood of the early grain states. Inevitably, the state’s relentless drive to acquire ever more subjects ensnared non-state peoples at the margins of the

kingdom. In this first pronouncement of human subjugation in the Bible, we can tentatively encounter allusions to the growing magnitude of slavery manifesting within the proto-states of Mesopotamia. For whilst slavery was surely not invented by the early grain cities along the Tigris and Euphrates,⁴³ the sheer scale of enslaved human beings that were required to uphold the infrastructure of the urban grain complex was simply unprecedented. James Scott, in a conservative estimate, stresses that at least 20 percent of the city of Uruk's population in the year 3000 BCE were imprisoned women and juvenile textile workers sequestered in gulag-style workshops.⁴⁴ That fully one-fifth of Uruk was enslaved in the textile workforce does not account for prisoners of war or slaves in various other sectors of the Uruk economy. Historian Moses Finley estimated that slaves comprised a clear majority, and perhaps as much as two-thirds, of Athenian society, and he projects similar figures onto earlier pre-Greek grain states. "The pre-Greek world—the world of the Sumerians, Babylonians, Egyptians and Assyrians, was, in a very profound sense, a world without free men, in the sense in which the west has come to understand the concept."⁴⁵

The reason that the elites of the ancient world required such coercive pressure to motivate a proletariat to work was due to the underlying fact that people are disinclined to produce a surplus once they have met their basic needs. "A peasantry," notes Scott, "will not automatically produce a surplus that elites might appropriate, but must be compelled to produce it. Under the demographic conditions of early state formation, when the means of traditional production were still plentiful and not monopolized, only through one form or another of unfree, coerced labour—corvée labour, forced delivery of grain products, debt bondage, serfdom, communal bondage and tribute and various forms of slavery—was a surplus brought into being."⁴⁶ As Ester Boserup sums up succinctly, "Where population is sparse and fertile land abundant and uncontrolled, a social hierarchy can be maintained only by direct, personal control over members of the lower class. In such communities therefore, both subjugated peoples and individual captives of war are kept in personal bondage."⁴⁷

Could there be a more suitable landscape to engender slavery than the post-diluvian setting? Sparse population, plenty of arable land to cultivate, and the patriarch Noah keeping his children close, as though they were all still entrapped on the ark. Perhaps, Noah's desire to bind his family in a state of intimate servility is already anticipated in the incongruous hiatus of nearly two months between the time Noah opens the ark to spy the dry earth and the time he and his family actually leave the ark. "And it was in the 601st year, *in the first of the month* the waters were dried up from the earth and Noah removed the ark's covering and he looked, and behold, the face of the ground was dry. And *in the second month, on the 27th day of the month* when the earth was dry; God spoke to Noah, saying: 'Get out from the ark, you and your wife and your sons and your sons' wives with you.'"⁴⁸ Noah opens the ark to observe the habitable land before him, and then sits for weeks and months

within the confines of his boat, allowing no one to venture forth, until the divine call intervenes.

As Scott notes, the agrarian family unit was a microcosm of the coercive forces at work in the polity of the early grain states “Holding onto labour—physical and reproductive—of the women in the family as well as the labour of the children is central to its success, especially the success of its CEO, the patriarch.”⁴⁹ In effect, the father was the representative of the elite class, and the women and children were enslaved to create the surplus necessary to uphold his lifestyle. This reading of the father as family slavemaster resonates with a particular midrash found in *Genesis Rabba*, where the patriarch Noah reveals his motivation for cursing Canaan. “You [Ham] prevented me [Noah] from producing a youngest son who would serve me; consequently that man [Canaan] will be his brother’s slave.”⁵⁰ Noah yearned for a life of leisure. Ease was imprinted into the DNA of his very name⁵¹ and fated by his father’s desire: “He named him Noah, saying, ‘this one will bring us relief from our work, and the anguish of our hands from the dirt that God has cursed.’”⁵² So Noah planted a vineyard, an entirely extraneous crop, and envisioned a future where his children cum slaves would feed and clothe him. His son, Ham, rebels, chafing against his father’s yoke. Finding his father indolent, Ham cuts away his penis, shouting out that no more children would be forced into this servitude.

Noah responds with a threefold curse upon his grandchild, Canaan. “He said: ‘Canaan is cursed, a slave’s slave he will become to his brothers!’ He then said, “Blessed be the Lord, the God of Shem, Canaan will be his slave. May God enlarge Japheth, and he will dwell in Shem’s tents, Canaan will be his slave.”⁵³ Tentatively, let us read Noah’s curse as we earlier read the divine curse on Adam after tasting from the forbidden wheat “tree.” Curses are not a punishment here, but an inevitable harbinger of what will come. Accordingly, Noah intuitively discerns his children’s future where all great achievements, each fraternal enlargement and every blessing, will be founded on the enslavement of others. For the very surplus gained by some brothers, invariably meant forcing others to produce that surplus.⁵⁴ Expansion and growth are predicated on this curse against the child, Canaan, “a slave’s slave to his brothers.”⁵⁵ Beyond the normative understanding of this word repetition as “extremely enslaved,”⁵⁶ exegetes offer two other variant readings. That Noah curses Canaan with endemic, multi-generational patterns of slavery: a slave *from* slaves.⁵⁷ Or, that Canaan will be enslaved by others who are themselves enslaved, a slave *to* slaves.⁵⁸

If Canaan’s enslavement implies a vicious cross-generational cycle, then Noah’s curse foreshadows the historical record,⁵⁹ as interpreted by Ester Boserup:

The slave raids of old times had the effect of increasing the contrast in population densities and levels of civilizations as between neighbouring tribes and peoples. The numerically weak were frequently decimated by the

slave raiders and those left behind could easily find subsistence by means of long-fallow systems with the result that they were never able to break the vicious circle of sparse population, long-fallow agriculture, and low levels of civilization. They continued, sometimes for millennia, to serve as reservoirs of potential slave labour for their more prolific neighbours. The numerically strong become still stronger.⁶⁰

If Boserup is generally correct that increasing population density drives technological innovation and “pristine” state formation,⁶¹ rather than the reverse Malthusian causality in which technological advances, such as the agricultural revolution, drive population growth and statehood, we are still left with a basic question.⁶² What causes population density to sharply increase amongst a particular tribe or band in the first place? For Boserup, one answer is slavery. The import of slaves would hasten the sort of economic development that results from dense and permanent settlements, whilst allowing the raiders themselves to concentrate on the art of war to enslave yet other people and grow larger still.⁶³ Meanwhile, the surrounding peoples would be continuously decimated by the slave raids, their population dwindling, their labour at a premium; they could never turn to the kind of agricultural techniques that required a vast workforce. And thus, these vulnerable people were entrapped in a cycle of slavery, raid after raid, from the proto-state population machines growing ever more hungry at their threshold.

Noah’s curse hints towards Boserup’s thesis. Indeed for Yafeth to expand and for Shem to be blessed, in the particular biblical sense of blessing which implies material benefit in the form of land, children, or goods,⁶⁴ requires a continuous influx of slaves upon slaves. In order for the early grain states to make up for the leakage of its subjects fleeing the core, it would be necessary to raid the less populous marginal zones again and again.⁶⁵ Canaan’s curse is *ad nauseum*, irrevocable, and entirely foundational for growth in the agricultural age. Canaan is not just bound into servitude; rather, his entire lineage will be forced into a multi-generational trap of enslavement: a slave who fathers slaves, who fathers slaves, without end. The blessing of surplus that flows to the elite, that constant “enlargement,” is predicated on Canaan’s constant bondage.

Alternatively, “a slave *of* slaves” could imply that the post-diluvian civilisation of intensive cultivation, and thus very permanent settlements, diminishes leisure time for all classes.⁶⁶ In effect, agriculture forces everyone to become slaves to the crop. For it takes a massive input of labour to clear the land, sow, tend, irrigate, harvest, and store a grain crop, when more nutritional, reliable, and varied wild sources are readily available with a fraction of the labour expenditure.⁶⁷ Both the archaeological record of pre-Neolithic bands of hunter-foragers and still existing modern examples in the marginal corners of the globe reveal a surprising reluctance to adapt intensive agriculture. Recent experiments using only stone sickles modelled on pre-Neolithic types to harvest the large stands of the wild ancestors of current

domesticated crops that still grow in Southwest Asia have shown that yields of 800 kilos per hectare, equivalent to the rate of one year's harvest in a mediaeval English farm, could be obtained in just three weeks of gathering.⁶⁸ Scott offers the following explanation for humanity's refusal to adopt intensive farming for millennia:

Planting techniques were long known and occasionally used; wild plants were routinely gathered and their seeds stored; all the tools for grain processing were at hand, and even a captive animal or two might be held in reserve. Nevertheless, planting and livestock rearing as *dominant* subsistence practices were avoided for as long as possible because of the work they required. And most of the work arose from the need to defend a simplified, artificial landscape from the resurgence of nature excluded from it; other plants (weeds), birds, grazing animals, rodents, insects and the rust and fungal infections that threatened a monocropped field.⁶⁹

Perhaps, this is the underlying hermeneutic of Noah's curse. "A slave of slaves shall you be to your brothers!" All my children, cries Noah, are effectively coerced to farm by the patriarchal head of the domus,⁷⁰ whereas you, Canaan, the youngest of the grandchildren, with the least of all inheritance, will be a slave to enslaved men.⁷¹

The biblical logic is frightening in its fatalism. Once Noah cultivates a lifestyle of surplus wealth, embodied in the grape vine and inebriation, the irrevocable twisting of the curse on the ground into the curse of slavery is irreversible. The farmed earth has so deeply infected humanity that for the first time in the Bible, it is a human, rather than God, who issues a curse, perhaps to imply that accursed bondage was now socially normative, rather than some transcendent device. Slavery is simply what is required to build the pristine grain state. Again quoting Boserup: "The toil of the slaves made it possible for a leisured class with a high urban civilisation to emerge."⁷² Imagined more graphically by Fukuyama, "People in early societies would not produce a surplus on their own until compelled to do so by rulers who could hold a whip hand over them."⁷³ Slavery, in all of its various incarnations, was the prerequisite of early state-making. So too, Noah's curse of bondage ultimately drives Noah's children out of their hilly non-state enclave, to descend into the plains of Shinar to make bricks for the imperial city of Babel.

If Noah is the narrative embodiment of the orchard crop revolution and its concomitant effects on land tenure, then the constituent components of this parting story of Noah can be regarded within the broader context of the biblical unease against the encroaching agricultural hegemony of the Levant. After the displacement of a bio-diverse edenic forest with a wheat monoculture, the land becomes accursed, "The soil will be cursed because of you."⁷⁴ After Cain kills Abel in a land dispute, the curse metastasizes outward from the earth into the fraternal fabric of human relations, "Now you

are cursed from the soil that opened its mouth to take your brother's blood from your hand."⁷⁵ Now, after the cultivation of the first grape vines, the curse infects across generations. "Cursed is Canaan," shouts Noah, his grandfather, "he shall be a slave's slave to his brothers!"⁷⁶ Noah, we recall, has become the dirt-man (אִם מְאֵדָרֵם).⁷⁷ In a sense, he takes the cancer of farming that was dwelling in the dirt and injects it into the blood and bone of human society. It is relatively straightforward to escape from a thorny or tired earth accursed, but when Canaan is cursed with slavery, he cannot run from such a damning label which latches to his very self. The slave cannot easily shake off the cursed earth from her bent back.

The word "curse" (אָשַׁשׁ) is the linguistic trace that binds these Genesis narratives. A cursed ground yields a perilous crop, worth hoarding and fighting over. That curse rises out of the ground to curse the hoarder. Now, that curse mutates into power over the bodies and labour of other human beings. Some of Noah's children will own land, whilst others will be slaves to landowners. An ecological crisis evolves into a societal one. Everett Fox tersely links the vineyard of Noah with the destruction of Eden, "The soil, which evidently has not entirely shaken off its primeval curse,⁷⁸ proves once again to be a source of trouble."⁷⁹

A Drunken Stumble Down to Babel

Finally, in our re-reading of this story of drunkenness and nakedness as a homiletic memory of the onset of orchard cultivation, we have not yet contextualised Noah's inebriated state. Why use grapes qua wine as the paradigmatic tree crop? Why not make Noah an olive or fig tree grower? Why is Noah's drunkenness so essential to this tale? The midrashic collection *Tanchuma* mixes up the order of events in our story. Now cursing and slavery stand between the cultivation of the vineyard and the inebriated aftermath: "Noah... ... introduced four things. Noah introduced [tree-crop] planting, as it is written: *And he planted a vineyard* (Genesis 9:20); Cursing, when he said: *Cursed be Canaan* (Genesis 9:25); slavery when he said: *He shall be a slave onto his brothers* (Genesis 9:25); and drunkenness when *he drank of the wine, and was drunken* (Genesis 9:21)."⁸⁰

Though we run the risk of reading too much into a singular midrash, it is indeed odd that the introduction of drunkenness occurs as the final phenomenon, rather than its rightful place in the story after the planting of the vine and before the curse. It would appear from this midrash that inebriation follows on the heels of slavery, rather than preceding it. Perhaps, and this is very speculative, we are touching upon a quality of drinking which anthropologist Mary Douglas describes as constructing an "ideal world." "There is also a sense in which drinks perform the... ... task of ritual. They make an intelligible, bearable world which is much more how an ideal world should be than the painful chaos threatening all the time."⁸¹ The midrashic misplacement would be intentional. As production and trade of olive-oil, wine,

date-honey, and similar arboreal products increased, inherited orchard land-wealth would become increasingly centralised under elite control. Agrarians who once cultivated small communal plots on the hilly terrain were removed from their land to make way for lucrative tree crops, and forced—perhaps cursing—into corvée labour and outright slavery. Slaves seek out the salve of drink to cope with their slavery. Thus, as in the midrashic sequence, tree domestication engenders a cursed state of slavery, and the horror of slavery is numbed with alcohol.

In a cruel feedback loop, the more the oppressed drink, the more power they forfeit. An economy awash in alcohol, notes Douglas, solidifies wealth: “Alcohol production and distribution as an industry is particularly conducive to monopoly. It provides a uniform product, its manufacture offers large economies of scale, it has low carrying costs and is highly in demand.”⁸²

Thus, the Bible’s liminal story at the conclusion of the Noah narrative contains the elements of a monopolising market, ever increasing economies of scale, and an emergent servant workforce, precisely the components from which to construct the Bible’s first totalitarian regime: the Tower of Babel.

Notes

- 1 Gen 6:9.
- 2 My claim that this story is a uniquely biblical linking narrative between the flood and the tower accounts may be somewhat buttressed by Sarna’s assertion that Gen 9:21–27 has no known counterpart in Mesopotamian traditions. See Sarna, *Genesis*, 63.
- 3 Gen 9:21–22.
- 4 Gen 9:24–25.
- 5 See Gen 10:6. Some modern biblical critics suggest that the story belongs to a lost stratum of tradition which identified Noah’s sons as Shem, Japheth, and Canaan, directly. See Gerhard von Rad, *Genesis, A Commentary*, trans. John H. Marks (Philadelphia: SCM Press, 1961), 131–132. However, J. Hoftizer argues persuasively that the current biblical genealogy that places Canaan as Noah’s grandson is probably the correct one. See “Some Remarks to the Tale of Noah’s Drunkenness,” in *Studies on the Book of Genesis*, eds. B. Gemser et. al. (Leiden: Oudtestamentische: XII, 1958), 22–25.
- 6 Frederick W. Bassett, “Noah’s Nakedness and the Curse of Canaan, a Case of Incest?,” *Vetus Testamentum* 21, Fasc. 2 (April 1971): 232.
- 7 Walter Brueggemann, *Genesis: Interpretation-A Bible Commentary for Teaching and Preaching* (Atlanta: John Knox Press, 1982), 90. “As a result of the narrative’s political interest in Canaan, a rival of Israel is imposed on a symmetrical family genealogy originally unrelated to Canaan. Ham is the third son of Noah in the genealogies (9:18; 10:6–20). And he is the key actor in this narrative. But the poetic course of verses 25–27 ignores Ham and centres on Canaan. It is clear that the means of a family narrative is used to characterize political realities at some point in Israel’s history.”
- 8 E.A. Speiser, *Genesis*, 62.
- 9 Alter, *Five Books*, 52–53.
- 10 Speiser, *Genesis*, 62.
- 11 Speiser, *Genesis*, 62.
- 12 Gen 9:20.

- 13 Gen 5:29.
- 14 See Alter, *The Art of Biblical Narrative*, 12; also, Judy Klitsner, *Subversive Sequels in the Bible*, xvii-xviii, particularly footnote no. 7. Here, Klitsner attempts to unite modern "intertextual methodology" with classical and mediaeval rabbinical exegetical hermeneutical principles, for example: R. Eliezer son of R. Yossi Ha-Gelilli said: "A text is not fully expounded in its place, but is fully expounded in another" (Principle 17 of the 32 Principles).
- 15 *Gen. Rab.* 36:2, Theodor and Albeck, vol. 1, 337.
- 16 *Tanchuma*, Noah 14, Mantua Ed., c. 1563, 10. See also Samuel A. Berman, *Midrash Tanchuma-Yelammedenu: An English Translation of the Genesis and Exodus from the Printed Version of the Tanchuma-Yelammedenu* (New Jersey: KTAV Publishing House, 1996), 69.
- 17 Patricia Fall, Steven Falconer, and Lee Lines, "Agricultural Intensification and the Secondary Products Revolution along the Jordan Rift," *Human Ecology* 30, no. 4 (Dec 2002): 451.
- 18 Fall, et al. "Agricultural Intensification," 451: "Olive, grape, and fig products became the three most important crops grown in the dry farming regions of the Southern Levant and traded throughout the Mediterranean basin."
- 19 O.S. LaBianca, *Sedentarization and Nomadization: Food System Cycles at Hesban and Vicinity in Transjordan* (Michigan: Andrews University Press, 1990), 43-44. Also, D. Zohary and M. Hopf, *Domestication of Plants in the Old World* (Oxford: Clarendon Press, 1988).
- 20 Fall, et al., "Agricultural Intensification," 454.
- 21 L.E. Stager, "The First Fruits of Civilization," *Palestine in the Bronze and Copper Ages*, *Institute of Archaeology Occasional Papers* 11 (1985): 172-188.
- 22 J. Boardman, Kathleen M. Kenyon, E. J. Moynahan, and J. D. Evans, "The Olive in the Mediterranean: Its Culture and Use [and Discussion]," *Philosophical Transactions of the Royal Society of London*. Series B, Biological Sciences 275, no. 936 (1976): 188.
- 23 Fall, et al., "Agricultural Intensification," 446.
- 24 Alexander H. Joffe, "Alcohol and Social Complexity in Ancient Western Asia," *Current Anthropology* 39, no. 3 (June 1998): 301.
- 25 *b. Sanh.* 70a.
- 26 Gen 9:24.
- 27 Though this is clearly the view of the midrashists, Cassuto, however, argues that the very act of Ham telling his two older brothers about their naked father was a sufficiently unchaste deed for Noah to "know what his youngest son had done to him" with derisive speech, alone. *Genesis II*, 152. Yet, Cassuto's reading feels forced, as nowhere in the text does it mention that Noah overheard what Ham had said to his brothers.
- 28 Cassuto argues, however, that Ham did nothing more than gaze upon his nude father. As we can learn from the later verse (Gen 9:23) in which Shem and Japheth, in contradistinction to Ham, "did not see their father's nakedness," from which, notes Cassuto, "we may infer, conversely, that Ham's sin consisted in seeing only." see Cassuto, *Genesis II*, 151.
- 29 According to Alter, this is one of the implied undertones of Joseph's rebuke to his brother's when they come down to Egypt to buy grain, and Joseph—in the guise of Tzapnath Paaneach, Viceroy of Egypt—accuses them of spying on "the nakedness of the land." Writes Alter: "All the other biblical occurrences of the common idiom 'to see the nakedness of,' or 'to uncover the nakedness of' are explicitly sexual, usually referring to incest... and perhaps Joseph feels a kind of incestuous violence in what the brothers have done to him and through him to his father." Alter, *The Art of Biblical Narrative*, 164.

- 30 Philo of Byblos interprets this encounter as a reflection of a Phoenician tradition about the god Kronos, founder of the city Babylos, who castrates his father, Uranus, with Uranus' own knife. Cited in Shinan and Zakovitch, *From Gods to God*, 135.
- 31 Alter, *Art of Biblical Narrative*, 40. For further mythological examples of the castration and usurpation motif, see also Robert Graves and Raphael Patai, *Hebrew Myths: The Book of Genesis* (Garden City, 1964), 120.
- 32 Ulf Oldenburg, *The Conflict Between El and Ba'al in the Canaanite Religion* (Leiden: E.J. Brill, 1969).
- 33 Gen 9:22a.
- 34 Bassett, "Noah's Nakedness," 235.
- 35 Bergsma and Hahn interweave a careful reading of such key words as אִמָּם (Gen 9:21), which they interpret as "her tent" (ie Noah's wife's tent), with strong intertextual links to the parallel story of Lot's daughters sleeping with their drunken father to produce other enemy progenitors of Israel (i.e. Moab and Ammon) to support this interpretation of maternal incest. John Sietze Bergsma and Scott Walker Hahn, "Noah's Nakedness and the Curse on Canaan (Genesis 9:20-27)," *Journal of Biblical Literature* 124, no. 1 (Spring 2005): 25–40.
- 36 For a thorough review of the similarities between these two passages, see Bergsma and Hahn: "Noah's Nakedness," 31, and Susan Niditch, *Chaos to Cosmos: Studies in Biblical Patterns of Creation* (Atlanta: Scholars Press, 1985), 53–55.
- 37 Gen 9:24-25.
- 38 In fact, Bergsma and Hahn also address the awkward missing nine months in the maternal incest interpretation, insisting that, "the narrator may have simply compressed the chronology at this point, as he does elsewhere." Bergsma and Hahn, "Nakedness," 39.
- 39 *Midrash Ha-Gadol (Haggadol)* 9:25, ed. Mordechia Margolies [HEB] (Jerusalem: Mossad Harav Kook, 1947), 190. See also Rashi to Gen 9:25, s.v. אִשְׁשׁ בֶּשֶׁ. "Cursed be Canaan: [Noah said]: 'You [Ham] have brought it about that I cannot beget a fourth son to serve me; cursed, therefore, be your fourth son to serve under the descendants of my elder children, who will now require slaves.' What did Ham envision that motivated him to castrate his own father? He said to his brothers: 'Adam had two sons and one killed the other in order that he might possess the whole world: our father already has three sons, and he wishes to have a fourth!'" Rashi's commentary is generally based on *Gen. Rab.* 36:4, Theodor and Albeck, 339.
- 40 See Gen 10:6.
- 41 Gen 9:25.
- 42 *Midrash Ha-Gadol*, ed. Margolies, 190.
- 43 For just one salient example, one can cite the extensive research into pre-Columbian Latin American slavery, See Fernando Santos-Granaros, *Vital Enemies: Slavery, Predation, and the Amerindian Political Economy of Life* (Austin: University of Texas Press, 2009). However, other scholars argue that it was rarely practised amongst hunter-gatherers to any great degree, cf. Nils-Petter Lagerlöf, "Slavery and Other Property Rights," *The Review of Economic Studies* 76, Issue 1 (January 2009): 321.
- 44 Scott, *Against the Grain*, 159. "Various estimates put the Uruk population at around forty thousand to forty-five thousand in the year 3,000 BCE. Nine thousand textile workers alone would represent at least 20 percent of Uruk's inhabitants, not counting the other prisoners of war and slaves in other sectors of the economy." See also, Guillermo Algaze, "The End of Prehistory and the Uruk Period," in *The Sumerian World*, ed. Harriet Crawford (London: Routledge, 2013), 81.
- 45 Michael Finlay, "Was Greek Civilization Based on Slave Labour?," *Historia: Zeitschrift für alte geschichte* 8, no. 2 (1959): 164.

- 46 Scott, *Against the Grain*, 152. Scott bases his understanding on a substantial body of scholarship, notably Marshall Sahlins, *Stone Age Economics* (Chicago: Aldine, 1974); A.V. Chayanov, *The Theory of Peasant Economy*, eds. Daniel Thorner, Basile Kerblay, and R.E.F. Smith (Homewood, Ill.: Richard D. Irwin, 1966); and Ester Boserup, *The Conditions of Agricultural Growth*.
- 47 Ester Boserup, *The Conditions of Agricultural Growth: The Economics of Agrarian Change Under Population Pressure* (Chicago: Aldine, 1965), 73.
- 48 Gen 8:13-16.
- 49 Scott, *Against the Grain*, 270, footnote 5.
- 50 *Gen. Rab.* 36:7, eds. Theodor and Albeck, 341.
- 51 רַחֵם meaning *rest* in Hebrew. Also related to the word בָּקַל meaning “to bring relief” or “comfort.” Cited in Aryeh Kaplan, *The Living Torah* (Jerusalem: Moznaim Publishing Co., 1981), 25.
- 52 Gen 5:29.
- 53 Gen 9:25-27.
- 54 According to the enlightenment social theorist, Jean-Jacques Rousseau, this zero-sum relationship between slaver and enslaved is the inevitable end-game of property ownership. “Riches, before the invention of signs to represent them, could scarcely consist in anything but lands and cattle, the only real goods which men can possess. So, when estates increased so much in number and in extent as to take in whole countries and touch each other, it became impossible for one man to aggrandise himself but at the expense of some other ... And from that began to arise, according to their different characters, domination and slavery, or violence and rapine. The rich on their side scarcely began to take the pleasure of commanding, when they preferred it to every other; and making use of their old slaves to acquire new ones, they no longer thought of anything but subduing and enslaving their neighbours; like those ravenous wolves, who having once tasted human flesh, despise every other food, and thereafter want only men to devour.” See “The Second Discourse,” in *The Social Contract and the First and Second Discourses*, ed. and trans. Susan Dunn (New Haven: Yale University Press, 2002), 123.
- 55 Gen 9:25.
- 56 See Levi Ben Gerson (Ralbag) commentary to Gen 9:25, in *Commentary of Ralbag on the Torah: Book of Genesis*, [HEB], ed. Y. Levi (Jerusalem: Mossad Harav Kook, 1992), 94. “This phrase implies intensive and extreme enslavement. Similarly to the phraseology of ‘Holy of Holies’ which implies a degree of holiness above that of the merely ‘holy.’”
- 57 Potentially, the interpretation favoured by Cassuto, *Genesis II*, 166.
- 58 This is the interpretation offered by Hirsch, *Genesis*, 188. “נֶחֱדָל נֹחַ not the lowest slave, but just as אֱלֹהֵי מֶלֶךְ does not mean the highest god, but the God of the gods, so also נֶחֱדָל נֹחַ, the others too are נֶחֱדָל, but Canaan will be נֶחֱדָל.”
- 59 That the Bible is speaking in general terms of bondage, rather than about a particular individual or family, see Cassuto, *Genesis II*, 166. “The name Canaan does not denote here an individual but the Canaanite people, and his brothers are the other nations in general.”
- 60 Boserup, *Agricultural Growth*, 74.
- 61 See Francis Fukuyama, *The Origins of Political Order*, 84, who succinctly explains Boserup’s argument: “The dense populations around river systems in Egypt, Mesopotamia, and China spawned intensive systems of agriculture involving large scale irrigation, new higher-yielding crops and other tools. Population density promotes state formation by permitting specialization and a division of labour between elites and non-elite groups. Low density band or tribal-level societies can mitigate conflict simply by moving away from one another, hiving off segments when they find they can’t coexist. Dense populations in newly created urban

- centres do not have this option. Scarcity of land or access to certain key public resources are much more likely to trigger conflicts, which then might require more centralized forms of political authority to control.”
- 62 Some of the problematics of Boserup’s theory are explored by Francis Fukuyama, *The Origins of Political Order*, 84.
- 63 Boserup, *Agricultural Growth*, 74–75.
- 64 Cassuto, “Blessings,” in *Encyclopaedia Biblica* [Hebrew] (Jerusalem: Bialik Institute, 1954), 357.
- 65 Scott, *Against the Grain*, 153. “Most archaic states sought to replace their losses by various means, including wars to capture slaves, purchases of slaves from slaves-takers, and forced resettlement of whole communities near the grain core.”
- 66 See Weisdorf, “The Neolithic Revolution,” 565–566. “Evidence started to appear which suggested that early agriculture had cost farmers more trouble than it saved. Studies of present-day primitive societies indicated that farming was in fact back breaking, time consuming, and labour intensive.” Also, Boserup, *Agricultural Growth*, 43; and R.B. Lee and I. Devore, *Man the Hunter* (Chicago: Aldine, 1968).
- 67 See Clive Ponting, *A New Green History of the World*, 40–41.
- 68 See Ponting, *A New Green History of the World*, 41; and Scott, *Against the Grain*, 11.
- 69 Scott, *Against the Grain*, 96.
- 70 In fact, notes E. Boserup, when actual slaves are unavailable, relatively wealthy households and kingroups will purchase children in some form of adaptation to increase labour output. *Conditions of Agricultural Growth*, 74, ft. 1.
- 71 Rousseau similarly argues that inequality grew so rapidly after the adaptation of farming and property rights because human beings are willing to become slaves themselves just as long as they can enslave others. “Besides, citizens only allow themselves to be oppressed in proportion as hurried on by blind ambition, and looking rather below than above them, they come to love authority more than independence. When they submit to fetters, it is only to be the better able to fetter others in their turn.” “The Second Discourse,” 133–134.
- 72 Boserup, *Conditions of Agricultural Growth*, 75. See also, Scott, *Against the Grain*, 93.
- 73 Fukuyama, *Origins*, 84.
- 74 Gen 3:17.
- 75 Gen 4:11.
- 76 Gen 9:25.
- 77 Gen 9:20.
- 78 And it feels, almost, that even the Almighty cannot shake off this curse entirely! A curse remains in the mouth of Noah, the dirt-man, despite the divine post-diluvian pronouncement of removal. “God said in his heart: No longer will I curse the dirt because of Adam” (Gen 8:21).
- 79 Everett Fox, *In the Beginning: A New English Rendition of the Book of Genesis* (New York: Schocken Books, 1983), 37.
- 80 *Tanchuma*, Noah 14, Mantua edition, c. 1563, 6b. See also Samuel A. Berman, *Midrash Tanhuma-Yelammedenu*, 69.
- 81 Mary Douglas, *Constructive Drinking: Perspectives on Drink from Anthropology* (Cambridge: Cambridge University Press, 1987), 11.
- 82 Mary Douglas, *Constructive Drinking*, 12–13.

8 King of the Four Corners of the Universe

The transition from independent villages, such as Çayönü Tepesi, to larger city-states vying for power throughout the fertile Tigris–Euphrates catchment basin, to the first full-scale multinational empire has, according to climate journalist Elizabeth Kolbert, “come down to us in a form somewhere between history and myth.”¹ Forty-three hundred years ago, Sargon of Akkad, a “cup-bearer to the ruler of Kish,”² itself one of ancient Babylonians most powerful proto-city-states, usurped the kingly rulership and subdued the other fractious statelets between the Tigris and Euphrates. Sargon’s empire eventually extended from his capital city of Akkad, believed to lie south of Baghdad, to the Habur plains of northeastern Syria, a prized grain production site of the age.³ Sargon’s own scribes described him as “king of the world”; one of his later descendants would enlarge his own title to “king of the four corners of the universe.”

Akkadian rule was bureaucratic and centralised. The Akkadian administration levied taxes and tributes, introduced standardised weights and measures throughout the empire,⁴ and imposed a uniform dating system, under which each year was assigned the name of a major event that had recently occurred, for instance, “the year that Sargon destroyed the city of Mari.”⁵ During the empire’s hegemonic dominance of the Habur plains in the fourth and fifth millennia BCE, the population was displaced from small- to medium-sized villages and concentrated into a few strategically located cities with massive wall fortifications.⁶ These cities housed both the workers and extensive storehouses of barley and wheat harvested in irrigated cultivation techniques on the large land tracts of the area. The vast majority of these grains were transported back to the hub of the empire, whilst the farmers themselves received a daily ration of food in standardised one litre bevelled rim bowls found at all imperialised Akkadian sites in the region.⁷ One extensively excavated city in the Habur plains, Tell Leilan, reveals significant structural remains of the centralised economy and authority of the Akkadian imperial system. At the centre of the city complex lies the “Unfinished Building,” identified by lead archaeologist Harvey Weiss as the “administrative and revenue fortress” of the Akkadian regional authority. This structure was designed with imposing 1.86 metre wide walls built of basalt

blocks which had been transported some 40 kilometres, specifically for this building project.⁸

Undergirding the mega-construction and centralised regime of Akkadian imperialism was a highly developed and uniform writing system. Akkadian was not the first written script that is attributed to the Sumerian system of cuneiform from roughly 800 years earlier.⁹ Under the reign of Sargon, however, the Akkadian cuneiform system of impressing images on soft clay with a reed stylus became standardised across the length of the emergent empire.¹⁰ The complex system of Akkadian writing, consisting of a weave of logograms, phonetic signs, and non-pronounced determinatives which clarified ambiguities between identical images, contained over 600 separate signs, and thus was the purview of a few highly trained scribal specialists. Describing writing's earliest inception amongst the elite power base of ancient Sumer, Jered Diamond argues that it was the very complexity of the writing system which was sought by the elite. "The *intended* restricted uses of early writing provided a positive disincentive for devising less ambiguous writing systems. The kings and priests of ancient Sumer wanted writing to be used by professional scribes to record numbers of sheep owed in taxes, not by the masses to write poetry and hatch plots. As the anthropologist Claude Levi-Strauss put it, ancient writing's main function was 'to facilitate the enslavement of other human beings'"¹¹ (Figure 8.1).

Within the fortress-like administrative store-houses dispersed throughout the Akkadian empire, dwelt a class of scribes "nourished by stored food surpluses grown by food producing peasants,"¹² recording the tributes of intensive grain cultivation and herd animals in an inaccessible script. These tributes from the empire's periphery were transported to soldiers, administrators, and priests at the heart of the kingdom. According to one cuneiform tablet entry, "daily five thousand four hundred men ate at his [Sargon's] presence"¹³ at the capital of Akkad (Figure 8.2).

Then, around 2200 BCE, the Akkadian empire suddenly collapsed. "Who was king, who was not king?" asked the Sumerian King List,¹⁴ describing this precipitous decline. The lamentation of the "Curse of Akkad" written within a century of the empire's fall offers a detailed description of the end:

For the first time since cities were built and found.
The great agricultural tracts produced no grain,
The inundated tracts produced no fish,
The irrigated orchards produced neither syrup nor wine,
The gathered cloud did not rain, the *masguram* did not grow... ...
He who slept on the roof died on the roof,
He who slept in the house, had no burial,
People were flailing at themselves from hunger.¹⁵

When Harvey Weiss and his team began to excavate the Akkadian city of Tell Leilan, they encountered a ubiquitous layer of soil roughly three feet

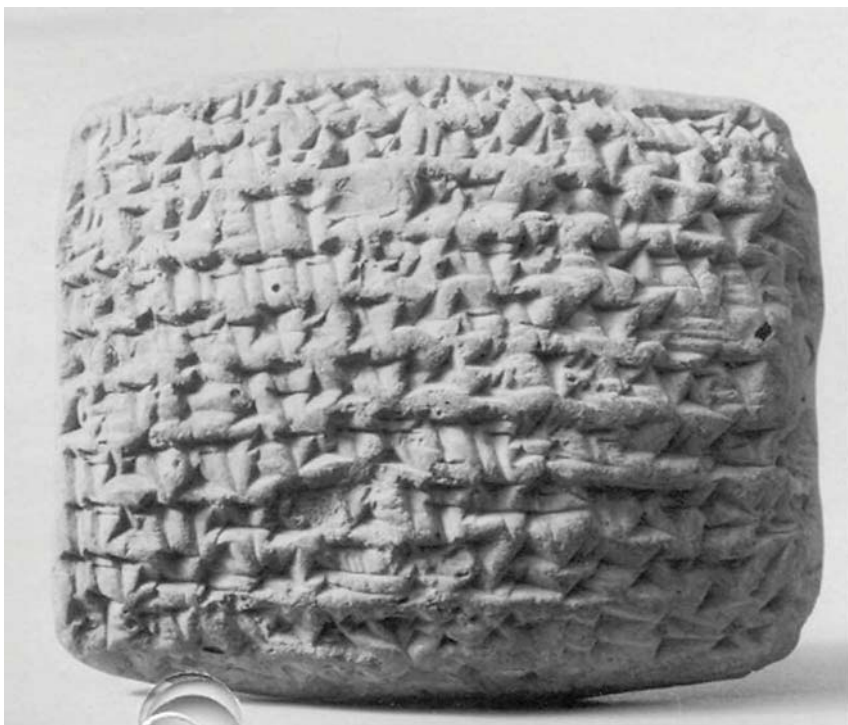


Figure 8.1 Cuneiform tablet of a slave sale from 554 BCE, excavated at Babylon in Central Iraq. Courtesy of the Metropolitan Museum of Art, New York.

deep that corresponded to the years 2200–1900 BCE. This layer of dirt contained no sign of human habitation and later analysis revealed that even the city’s earthworms had died out.¹⁶

Weiss and others have hypothesised that the specific abandonment of Tell Leilan and the general end of the Akkadian empire were the analogous products of an “abrupt climate change” of severe and prolonged drought throughout the Levant.¹⁷ This theory was further buttressed by paleoclimatic indicators from sediment cores taken from the Gulf of Oman, which revealed sharply increased levels of drought-induced dolomite dust during this specific period.

This mega-drought occurred so precipitously that buildings throughout the empire, such as the ‘Unfinished Building’ at Tell Leilan, were abandoned in mid-construction.¹⁸ “Nothing allows you to go beyond the third or fourth year of a drought,” Weiss remarked in a 2004 interview, “and by the fifth or sixth year you’re probably gone.”¹⁹ The settlement of Tell Leilan and other regional city centres in northern Syria were abandoned. And it was not until nearly 300 years later, with the return of stable climate conditions to the Levant, that new construction fully resumed in the form of thousands of small village sites in the area.²⁰

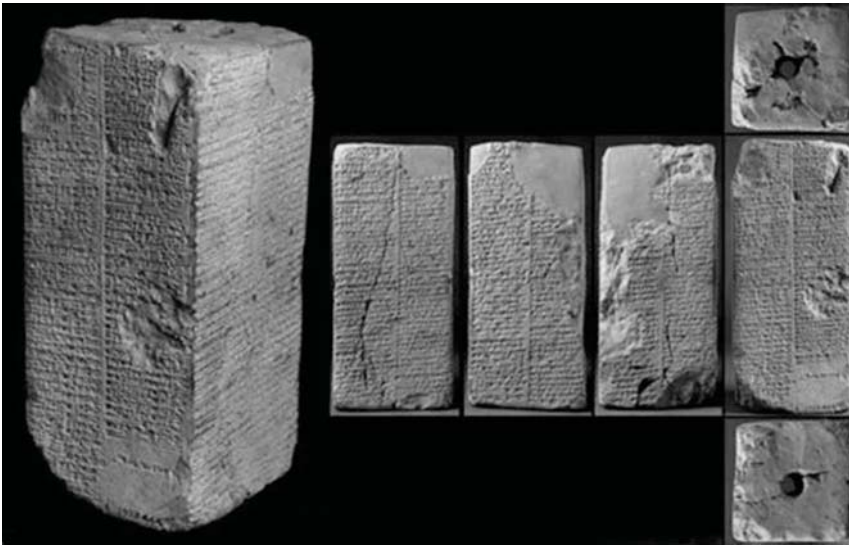


Figure 8.2 Sumerian King List clay-pillar, with separate images showing all four sides of cuneiform text and the pillar's top and bottom, c. 1800 BCE. Excavated at Larsa, approx 20 miles from Uruk, in Southern Iraq. Image courtesy of the Ashmolean Museum, Oxford.

Thwarting State Centropy

The history of the rise and fall of Akkad, as laid bare in the stones of Tell Leilan, conjures up strong echoes of the familiarity with the biblical narrative of the Tower of Babel. Yet, as I have stressed in the earlier parallelisms of ancient sites and biblical happenings, the scripture need only echo, or point obliquely, to an actual event, which was itself shrouded in a veil of mythological retelling, in order to ring with an authentic resonance for the listening audience. The biblical narrative need not directly shout out that the Tower of Babel was an Akkadian fortress in order for those early listeners to feel that here was a genuine and familiar hermeneutical insight into the ephemeral power of empire, be it Akkadian, the Third Dynasty of Ur, Babylonian, or otherwise.

Nor, I think, should the “tongue-in-cheek” satirical artistry of this narrative gem entirely diminish its adherence to mimetic realism. Indeed, these verses can be considered an outright parody of the Babylonian urban project, as Cassuto alerts us to in his close analysis. Whilst the inhabitants of Shinar blueprint for a grand tower “with its head in the heavens,”²¹ God must “come down to see the city and the tower,” in a mocking reminder of its puny dimensions vis-à-vis the divine gaze. Whilst the cityscapers intended to make a “name”²² for themselves via the muscular dimensions of their singular city and tower, “lest we be scattered abroad over the face of the whole earth,”²³ God’s ironic response is framed in antithesis. “So the Lord scattered them

abroad from there over the face of all the earth,”²⁴ and made the precise act of scattering and disintegration the very name of that ground zero of urban folly. “Therefore its name was called Babel (𒅗𒄿), because there the Lord confused (𒅗𒄿) the language of the whole earth, and from there God scattered them abroad over the face of the whole earth.”²⁵ “Behold,” narrates Cassuto, channelling the voice of the biblical satirist confronting Babylonian empire, “how fitting is the name that you have given to your city! It is true that in *your* language it expresses glory and pride, but in *our* idiom it sounds as though it connotes confusion—the confusion of tongues heard therein, which caused its destruction and the dispersion of its inhabitants in every direction.”²⁶ This is no lamentation over the collapse of empire, but a mockingly subversive satire on the limits of power and the state.²⁷

Though the tone is satirical, its themes trace a familiar realpolitik. The Bible, again, pulls the reader fleetingly within the grasp of the state. Here is the coercive “social caging”²⁸ of the alluvial grain state, what Scott somewhat ladenly entitles the Mesopotamian “resettlement camps.”²⁹ Here in the Babylon story is the homogeneous valley flattening of languages and labour into a single speech and a singular task of cityscaping. Not only do the engineers build a city to hold its subjects, but a watchtower as well, shadowing its populace on the flat valley floor.³⁰ One of the earliest megabuilds of the Mesopotamian alluvium was the 10-metre-high stepped platform constructed in the city of Susa during the Late Ubaid period (approximately 4500–4000 BCE). This monumental construction combined a temple, a grain storehouse, and a probable residence of an important family. Archeologist Susan Pollock argues that the imposing presence of this structure juxtaposed against the flattened horizon was a tangible message of authoritative control:

The association of a monument, easily visible from afar because of its elevation above a very flat landscape, with elements of the sacred world (a temple), an important family, and the products of human labour (the construction itself, as well as the goods stored in the storehouse) was a deliberate expression of the rights of some people over the labour and products of others. The spatial proximity of a house of god to the products of human labour helps to make this assertion of hierarchy palatable.³¹

The Tower of Babel story runs deeply against the grain of state texts. The vignette underscores the positive potentiality of dispersive population. Cuneiform state tablets, conversely, held to a “grand narrative”³² in which primitive, non-state people could enter the sphere of civilisation, but “civilised” urbanites would never willingly or rationally return to primitivism. Reviewing the scribal works of the Third Dynasty of Ur, a mere 98 year empiric effort (2112–2004 BCE) that flashed briefly across the Mesopotamian alluvium, Robert Adams notes the myopic quality of state texts to dismiss the dialectical “two-way street”³³ between state subjects and non-state peoples.

For the scribal documentarians of Ur, however, all roads, goods, and progress led radially into a centralised core:

The rationale of the whole social order was to sustain civilization by assuring the necessary flow of agricultural and other commodities along the incoming arteries. Retention of wealth in the countryside, let alone a reverse flow, contradicted this 'natural' order. Hence rurally based, tribal units with that as their objectives were not readily conceptualised as component parts of the same system. More often they were regarded as ethnically distinguishable 'barbarians' invaders, or else were both distrusted and dismissed as an ignorant, inferior rabble who fled from proper urban discipline into dissidence and banditry.³⁴

Like the regime of Ur III, the Babel builders could scarcely conceive of any political identity, of a recognisable *name*, beyond the walled city. "They said, Come, Let us build a city for ourselves, and a tower with its head in the heavens, and let us make a name for ourselves, lest we become scattered across the face of the whole earth."³⁵ In the purview of the urban elites, the building of the city-state and the retention of a whole identity was the same activity. Beyond the state's centripetal pull, all was an amorphous, nameless, and scattered barbarism.

However, argues Scott, the magnetic inward lure of the state is only half the story. "We now know, this view to be, on the historical evidence, fundamentally wrong,"³⁶ Wrong, mainly because the state was constantly disgorging its subjects into complementary and surrounding non-state zones of hills, marshes, deserts, forests, and the like, in a process that the anthropologist Pierre Clastres labelled "secondary primitivism."³⁷ Examining the post-Conquest New World, Clastres noticed a hitherto overlooked phenomena. Many of the primitive hunting-and-gathering "tribes" of South America were not the "Palaeolithic survivors, living in a veritable state of nature,"³⁸ as earlier anthropological surveys had readily classified them. Rather, these peoples were former grain farmers who had dispersed into the forests in order to avoid state subordination and infectious pathogens. One paradigmatic case is the Siriono of Eastern Bolivia. When the anthropologist Allan Holmberg described the Siriono in his classic 1950 survey, he viewed their inability to make clothing or even fire, their rude shelters, and their lack of domesticated animals or a numbering system, as definitive indicators of a fossilised tribe, left behind in the social evolutionary process by more civilised societies such as the Aztec, Maya, and Inca.³⁹ However, Clastres conclusively proved that the Siriono had been living in sedentary villages and growing crops until roughly 1920, when first diseases (smallpox and influenza) followed by raids by numerically superior slavers caused the survivors to abandon their fixed fields and divide into nomadic subsistence bands that could evade density-dependent diseases⁴⁰ and capture.⁴¹ Adams describes similar patterns in ancient Mesopotamia of semi-nomadic "adaptive strategies" which rose to the fore during the "troughs between the peaks of dynastic consolidation."⁴²

Such numerous and deliberate choices to abandon sedentarism, and flee beyond state control run counter to a “hegemonic”⁴³ civilisational narrative to use Scott’s phrasing. “The narrative derives its hegemonic status not only from its affinity with social Darwinism but from the fact that it maps nicely on the stories most states and civilizations tell about themselves. The schema assumes movement in a single direction towards concentrated populations and intensive agricultural production; no backsliding is envisioned; each step is irreversible progress.”⁴⁴ Adams laments the lack of testimony that could correct the record on the “resilient strategies”⁴⁵ of non-state peoples living beyond the pale of ancient Ur, Babylon, and the like. However, I will argue that one such record of backsliding away from the state, not simply as an alternative political strategy, but as a divinely ordained one, does indeed exist within the ancient canon of texts. That text is the Tower of Babel account.

For if one widens the frame, as I will attempt later in this chapter, to the verses on either side of the Babel story, one recognises an emergent pattern. A diversity of peoples and languages bookend both sides of the totalitarian cityscape tale. I will argue that this is a narrative mirror of the migratory flux of peoples into and out of the state core. And it is a mirror as well of the real geographic patterns of these early states. A corporate centre with a singular culture and language, surrounded by a crazed-quilt of heterogeneous absorption zones of foragers, swiddeners, pastoralists, and others fleeing state control.⁴⁶

The irony of the Babel story is that the diversity at the gates, the very diversity the city builders wanted expunged from their midst, would instead become the milieu for the real story traced from Abraham to Ezra. Seth Richardson astutely captures the cognitive dissonance of the ancient Mesopotamian state. On the one hand, we know from numerous Old Babylonian letters and administrative texts that urbanites travelled outside their city boundaries with a good deal of trepidation. “As you have heard,” writes one beleaguered administrator, “the country is in confusion and the enemy has settled in the countryside.” “We could not stay overnight in the *dimtu*-fortress area because of the enemy,” another letter laments; while a third writer asks, “have you not heard that an enemy is camping in the land?”⁴⁷ Yet, no court tablets, kingly steles, or palace wall engravings ever hint at such discord. Richardson points out that “it is striking to recall that these letters were drafted in the dusty neighbourhoods of cities where, just down the street, the kings in their palaces confidently issued boastful statements of total control over their lands, extending even to the limits of the four quarters of the earth.”⁴⁸

Remarkably, the Babel story takes all the ideatic hubris of the palace perspective, and twists it and twists it until the story is turned inside out, and the narrative is pulled down the dusty road. Thus, the utopian vision of the urbanites “not to be dispersed over the face of all the earth” of Gen 11:4 becomes inverted by the divine override “to disperse them over the face of all the earth” by verse 8. In the hills beyond the alluvium plain, God had ordered Noah and his sons to “fill the earth,”⁴⁹ yet the urbanites of Babel sought to

build and hold the population in one place, rather than scatter and fill the whole. Cassuto conjectures that the tower itself was a “sign-post of their point of assembly,”⁵⁰ a monument to consolidation. Yet, by the end, the city is the very source of the multiplicity of cacophonous tongues and the springboard for diversified peoples. “Therefore its name is called Babel, because from there the Lord confounded the language of all the earth, and from there the Lord dispersed them on the face of the whole earth.”⁵¹

In the biblical narrative, unlike every other cuneiform tablet, wall relief, royal stele, or engraved statue ever uncovered in ancient Mesopotamia, the dispersal and fleeing of subjects and the collapse of empire are lauded. The centralised building project is given the short-shrift of just 11 tangential and marginal verses, while the nomadic peoples and non-state pastoralists will become the true *dramatis personae* of Genesis.

A Proto-State Island in Genesis

James Scott writes: “Looked at from outer space in 2,500 BCE, the very earliest states in Mesopotamia, Egypt, and the Indus Valley would have been scarcely visible ... States, being largely agrarian phenomena, would, with the exception of some intermontane valleys, have looked like small alluvial archipelagos, located on the floodplains of the handful of major rivers.”⁵² The Tower of Babylon is the narrative equivalent of Scott’s imagined space perspective. Bookended on both sides by elaborate lists of peoples and languages, it is a small island of homogeneous proto-state formation, held briefly against the noise of diversity. Because it is so pithy, it offers a very pure and concentrated insight into the Bible’s analysis of early state-making. Like Darwin’s Galapagos, isolated islands can frame models that would be difficult to spot across wider swaths. Thus, a close reading of the micro-patterns and linguistic idiosyncrasies within the encapsulated Babel tale may yield insights on why Genesis cursed the state so thoroughly.

“And it came to pass, that all the land was of one language (לשון אחת, literally ‘lip’⁵³) and of uniform words (דברים אחדים).”⁵⁴ Both ancient and modern exegetes have questioned the repetition of the nearly identical “one language” and “uniform words,” and offered various interpretations.⁵⁵ Hirsch conjectures that the term “one language” implies the “phonetic sameness of speech.”⁵⁶ He points out that there is yet another Hebrew word for language, **לשון**, which does not appear in the account of Babel, though it was used in a similar, if not identical, context just two verses earlier in Gen 10:31. “These are the sons of Shem, according to their families, according to their languages (לשונם) in their lands, according to their nations.”⁵⁷ In this earlier verse, language is the indicator of a multiplicity of families and tribes that descended from Noah in the post-flood setting. **לשון** literally translates as tongue, and thus here the nations are speaking in many tongues.⁵⁸ Perhaps, the transition from a multiple taxonomy of tongues (לשונם) to the singular **לשון** of lip-speech at the threshold of the tower narrative hints to a new type of speech that was barely

lip-deep, and did not extend down to the tongues of the people. The city of Babel spoke a singular “lip” language, not various “tongue” languages, implying, perhaps, a language imposed on the external lips, yet foreign to one’s own “mother tongue.” A hegemony of speech, granted only “lip-service” by those assembled at the city core.

“Uniform words,” continues Hirsch, introduce another element entire, a “sameness of the formation of words and sentences brought about by mental agreement in the way that things and their relations are looked at.”⁵⁹ For Hirsch, whilst *מִצָּה* relates to the biological process of speech itself, *רִאשִׁים* implies a common system of communication understood between companions. Going slightly further than Hirsch, we note that *רִאשִׁים*, from the stem-root *ראש*, has the alternative meaning of “object” or “thing.”⁶⁰ This would imply that the uniform convention of understanding which stands as the prerequisite to the building of the tower was a tangible system of communication via *thing-ness*, language objectified. That is to say, a writing system. Perhaps, here is the biblical nod to the introduction of writing into state design. Before the first large-scale coordinated building project can begin, a particular uniformity of speech and writ must already be realised.

But the Babel narrative is not simply the story of community cohesion. According to the 19th-century Lithuanian biblical exegete, Naftali Tzvi Yehuda Berlin, known by his acronym Netziv, the tower project is the tale *par excellence* of coercive hegemony. “It is undoubtedly illogical to assume that there would be but one city in the entire world,” Netziv notes, “Rather, they thought that all cities would be connected and subsidiary to that one city in which the tower was to be built. And the purpose of the tower was to look out upon the distance over all their dwellings to ensure that none would split off into another land.”⁶¹ The Netziv’s reading of enforced uniformity in Babel goes beyond population control, for the textual indicators reveal a level of ideological manipulation as well. “This was certainly related to the *רִאשִׁים קלרים*, the ‘one speech’ among them. They feared that since not all human thoughts are identical, if some would leave they might adopt different thoughts. And so they saw to it that no one left their enclave.”⁶²

The creation of any city in the Mesopotamian alluvium implies a wall,⁶³ a wall not just to keep invaders out but also to keep the urban population within.⁶⁴ Now, Netziv’s commentary keeps pace with early state-making. The builders of Babel envisioned a walled city to keep the population enclosed within a single modality of thought and speech, with a tower to ensure that runaways were spotted and recaptured across the flat plain.

Arnold Ehrlich, the 19th-century master of biblical grammar,⁶⁵ translated *רִאשִׁים קלרים* as the formation of a limited vocabulary⁶⁶ in Babel. This would seem an odd reversion against the consistent rhythm of branching language trees spreading intricately after the flood. In the tenth chapter of Genesis, we listen to a tripartite story of linguistic diversity. “From these spread the coastline peoples, each one with his own language for its families in its nations.”⁶⁷ And again, “These are the sons of Ham, according to their families,

by their languages, by their lands and nations.”⁶⁸ And finally, “These are the descendants of Shem, according to their families, by their languages, by their lands and nations.”⁶⁹ Yet in Babel, language is confined and limited into uniform singularity. To what purpose? Perhaps, a prosopographic case study of the Ur III scribal enterprise may provide some clues. Approximately 150 years after the fall of the Akkadian regime, in the newly emergent empire of Ur III, King Shulgi (2094-2047 BCE) instituted scribal schools teaching strictly in Sumerian, an archaic language that was only taught within these state schools and solely used for administrative purposes in the Third Dynasty.⁷⁰ Piotr Michalowski questioned the purpose of teaching native speakers of various Semitic languages and dialects a clearly fossilised language no longer in common use at the time. “The answer, I think, is quite clear,” he concludes, “The school was an ideological mold of minds, the place where future members of the bureaucracy were socialized, where they received a common stock of ideas and attitudes which bound them together as a class, and in many ways separated them from their original backgrounds.”⁷¹

Our various interpretations of “the whole earth was of one language and of uniform words”⁷² sync with Michalowski’s rendering of Third Dynasty Sumerian as both a wedging device to elevate and maintain an elite administrative core and a mechanism of control. In the short-lived Ur III regime, Sumerian was a lip language imposed by institutional education across the whole empire, rather than the dialectical tongue languages of birth. Likewise, the Sumerian scribal academies of Ur III align with Ehrlich’s definition of a scant vocabulary in Babel juxtaposed against the multiple languages newly emergent in the post-diluvium. A fossilised orthography such as Sumerian, used precisely and solely to record state events and appropriations, would be paradigmatic of such a limited language type. A pure language of a totalitarian state that could mobilise a population to build monuments and enumerate the specific materials for that build, but lacked local idiom and spoken texture.

Not only did the pedagogy of the scribal schools work to separate the bureaucratic functionaries from everyone else, but the recording act itself was a gesture of coercion. Adams emphasises the detachment and indifference of the cuneiform records, and by implication the entire Ur III polity, to engage or even take note of the hazards, needs, and personnel requirements of its labourers in the fields and pastures. “This neglect seems to extend to the roots of the operations and effectiveness of the entire agricultural—hence also economic—system. The very large quantities of cuneiform records create an initial impression of broad and painstaking supervision. In fact, however, they focus only on a handful of control points chosen to meet a given set of state goals for resource expropriation.”⁷³ Any state that employs a language that lacks the linguistic breadth to convey the day-to-day needs of its populace, whilst simultaneously demonstrative in its own expropriative demands is very fragile indeed. James Scott specifically blames the fall of Ur III on its blunt extractive grain policy that was “all thumbs and no fingers for fine-tuning.”⁷⁴ “Core officials,” he writes, “could and did require more grain to be

planted and fallows to be shortened in order to maximise immediate returns at the cost of long-term productivity... ... when Ur was threatened, it appears, by Amorite incursion, the defending generals pressed so hard on Ur's cultivators for grain that they either resisted or fled."⁷⁵

Similarly, the scene is set in Babel with the רֵאשִׁים קְלָרִים. These "uniform words" imply a singular bureaucratic corporate language quite literally intent on creating a state monument with its "head in heaven,"⁷⁶ and thus, far removed from the earthy details of seasonal shifts and multiple tongues. The midrash intuitively that Babel was essentially so extractive in its core outlook that bricks took precedence over life. "(The labourers) who took up the bricks went up on the eastern (ascent), and those who descended went down on the western (descent). If a man fell and died they paid no heed to him, but if a brick fell they sat down and wept, and said: Woe is us! when will another one come in its stead?"⁷⁷

The Babel building project mirrors some of the piercing fracture lines of the earliest states. The urban plan is predicated on a superficial "lip" lexicon, a bureaucratic veneer of communication confined to bricks and mortar,⁷⁸ seemingly dismissive of the depth of mother tongues. The tower's lofty position is indicative of the constant threat of population leakage. That is why there is a watchtower in Babel to scope out fleeing subjects across the valley of Shinar. And Babel is intrinsically appropriative. Here, bricks are more valuable than the people who carry them. Babel is a project of extraction and internal fragility. In short, the scene is set for that state to collapse.⁷⁹

Let us proceed to the second verse of chapter 11: "And it came to pass, as they journeyed from the east that they found a plain in the land of Shinar, and they settled there."⁸⁰ The repetition of the opening phrase, "And it came to pass," in the first two verses of the Tower of Babel narrative is revealing. Jonathan Sacks notes the Talmudic rule of thumb brought to bear whenever this passive phraseology opens any biblical narrative. "Wherever it says 'and it came to pass' it is a sign of impending tragedy (bTalmud Megillah 10b)." Adds Sacks, "Passivity allows bad things to happen."⁸¹

So the adaptation of a singular "lip" language and the migration to the valley of Shinar are both described as enforced practices upon a passive populace. The biblical voice is not the mouthpiece of some powerful tyrant, hubristically informing his audience: "I moved the people to Shinar" or "I forced them to speak one language." Rather, here is the voice of those who are disempowered, and are doing the migrating, the speaking, and the building in coerced uniformity. This is not to imply that the Bible is not familiar, even in these decisive first chapters, with just these types of powerful rulers. Already in chapter 10, we learn of the birth of Nimrod who "started to be a mighty man in the land."⁸² And it is this self-same Nimrod who unites "Babel and Erech and Akad and Calneh in the land of Shinar."⁸³ Thus an early multinational confederacy is united under the rule of Nimrod in the very land of Shinar in which the tower will take shape.

However, the project of building a “city and a tower” is abruptly halted by divine intervention. “The Lord came down to look at the city and the tower that humanity had built, and the Lord said: ‘If, as one people with one language, this is how they have begun to act, then nothing that they may propose to do will be out of their reach. Come, let Us go down, so that their language will dry up (בָּאֵדָה), so that the one may not understand the speech of his friend.’”⁸⁴ Though the Hebrew word בָּאֵדָה is more commonly translated as “confusion,” I have followed the linguistic expertise of Hirsch, who links this Hebrew root to the concept of “withering and drying up.”⁸⁵ Though subtle, it hints to a divine dry-spell which causes disintegration in the Tower of Babel project. Could it be the faintest of sign-posts to the mega-drought that brought down the Akkadian empire?⁸⁶

A State of Rupture

Our narrative now reaches its climactic conclusion: “The Lord scattered them from there over the surface of the whole earth, and they ceased to build the city. Therefore, He called its name Babel, for there the Lord mixed the language of all the earth, and from there the Lord scattered them abroad over the face of the whole earth. These are the descendants of Shem...”⁸⁷ I have taken the unorthodox step of continuing the Biblical narrative of the Tower of Babel beyond the conclusive dispersion of verse 9 to incorporate the lineage list of Noah’s progeny at the start of verse 10. This genealogical survey had previously commenced in chapter 10, “These are the sons of Shem, according to their families, according to their languages, in their lands, according to their nations.”⁸⁸ Now, in chapter 11, after the collapse of Babel, the Bible simply picks up that recounting from nearly the exact point of caesuration. The entire account of the Tower of Babel can be viewed as an external implant literally forced into the natural progression of the biblical recounting of bifurcating descendants and devolving nation-states. In fact, the injection of the tower project demarcates such a clean tear at the fabric of the narrative, that the verses in the tenth chapter of Genesis that precede the tower would perfectly dovetail into Genesis 11:10 without missing a beat. Here is the reading then, if we excised the nine verses describing Babel:

These are the sons of Shem, according to their families, according to their languages, in their lands, according to their nations. These are the families of the sons of Noah according to their descendants, and in their nations, and from these, the nations separated in the land, after the flood⁸⁹... ... [the Tower of Babel story]. These are the descendants of Shem. Shem was 100 years old and begat Arpaxad, two years after the flood. After the birth of Arpaxad, Shem lived 500 years and begat sons and daughters.⁹⁰

Seen through this wider lens, the Babel project is a jarring incursion into the narrative sweep of Genesis. It ruptures the text, for a brief spell, from its

unfolding of diversifying family units with multiple languages of the tongue, towards a project that runs roughshod over complexity. Many tongues become one lip language, many families are coalesced into one labour unit, forced to build one city and one tower in one location.

Perhaps, in the Bible's laconic style, even the overarching desire of the tower builders, "Let us make for ourselves a name (שם),"⁹¹ is a textual invocation, a *leitwort*,⁹² of word repetition. The pattern would begin with Noah's son, Shem (שם), whose name means "name." Shem creates a multitude of descendants, and thus a plurality of names, "These are the descendants of Shem, according to their families and languages, by their lands and nations."⁹³ The tower builders, conversely, seek to transform the cacophony into a singular name, their own. "Let us build for ourselves a city and a tower whose top shall reach the heavens, and make for ourselves a name, lest we be scattered over the face of all the earth."⁹⁴ By the end of the tale the "name" *leitwort* will return twice more. First, as divine mockery, "He named it Babel, because this was the place where the Lord confused the speech of all the land"⁹⁵ The state constructors had envisioned for themselves a hegemonic single name. And indeed they do achieve a notorious label; only it is a derisively satiric one. Babel is the place of dissolution, of nameless and withered abandonment. And with the collapse of Babel, the biblical text can return to this other and profusive reading of naming. Thus, we re-encounter Shem and his lineage, "These are the generations of Shem."⁹⁶ But what precisely is the problem with this singularity on the alluvial floodplain of Mesopotamia? Why does God despise the work of the state-makers of Babel?

The Imperial Problem

Political scholar Daniel Gordis claims that the Bible is arguing for a particular type of nation building, namely a wide diffusion of states and peoples. The tower project's fleeting derailment into empire and the divine intervention to re-assert this political meta-direction are indicative of the underlying biblical blessing of ethnically diverse nation-states. He begins his analysis with the hinted role of Nimrod as the "first imperialist"⁹⁷ of the Shinar valley:

That the story of the Tower of Babel takes place in the region of Nimrod's rule may serve to explain the mysterious homogeneity that besets humankind just as it was beginning to divide itself into nations. For what is imperialism if not the forceful binding of different peoples under a common regime? 'The same language and the same words' at the opening of Genesis 11 may well be the consequence of an imperialist attempt to artificially hold humanity together.⁹⁸

In Gordis' careful reading, the genealogical lists that bookend the tower narrative reveal an intrinsic divine directive for humanity, which was momentarily thwarted by the rule of empire. "Humankind, the Bible has already stated,

ought to separate into distinct nations, each with its own land and language. Dispersion, in this sense, is part of a divine plan ... Mankind had taken the first steps towards this ideal after the flood, but its progress was effectively stunted. The Tower builders of Babel ... sought to sustain this uniformity. That was their sin, and that, ultimately, is why they had to be stopped.”⁹⁹

For Gordis, an intellectual disciple of John Stuart Mill and Isaiah Berlin, the real sin of the Tower of Babel is not hubris or cityscaping, but the derailment of “positive liberty” (to use Berlin’s phraseology)¹⁰⁰ in which different national entities require a large degree of freedom in order to establish their own distinct settings in the search for the good life. Nimrod and his minions eschewed that vision of unique nation-states, and the Bible tells of the rise and fall of that imperial tower as a moral-political warning. God, as it were, simply despises empires and loves political freedom.

Modern biblical exegete Judy Klitsner sees in the very ordering of events in Babel a similar alienation from self-determined goals, albeit on the granular plan of the individual rather than the nation-state. In her reading, the fact that the inhabitants of Shinar are first commanded to make bricks, “Come, let us make bricks and burn them hard” (Gen 11:3), and are only later informed to make those bricks into a tower, “Come, let us build a city” (Gen 11:4), implies a labour force who must work, assembly-line style, on menial tasks divorced from outcomes.¹⁰¹ She writes: “Initially, the builders receive only a very partial description of the overall plan. The minute details are not accompanied by an explanation of the larger purpose of their exertions. In this way, individuals are kept at a distance from their goals, from each other, and from any personal satisfaction they might derive from their efforts.”¹⁰²

Absent, however, from both Gordis’ and Klitsner’s analyses is a reflection of the historical and ecological record in this drama of empire. The Tower of Babel may indeed be a treatise on the monumental struggle for positive liberty in the face of tyranny and imperialism, and a small cry of individualism against the alienation of the working class; yet it might also tell a story of a particular Levantine community living under a Mesopotamian grain-state in a time of catastrophic climate change.

Blessed Collapse

What emerges from an eco-mimetic reading of the first 11 chapters of Genesis is a wary stance towards grain farming, in general, and cityscaping, in particular. How uniquely aberrant for an ancient text to eschew the glorification of building-works. How subversive to describe empire, in all its Ozymandias-esque glory, as lasting just nine verses. An unsustainable intrusion into the real story of unfolding diversity, quickly rectified by divine fiat. Compare our text with the Gilgamesh Epic, in which the founding king Gilgamesh is praised specifically for erecting the city walls.¹⁰³ Outside those walls exists the non-state habitat of Enkidu, the wild barbarian, who “knew nothing about eating bread for food,”¹⁰⁴ or how to “bend the knee,”¹⁰⁵ who must sleep with

a city woman and enter into its ramparts to become fully civilised.¹⁰⁶ This is city-state propaganda of the highest order. A tale of the feral barbarian evolving into the tamed state subject, safely enclosed within civilised walls. Or the Babylonian Creation story *Enuma Elish*, which recounts, in part, the glorified construction of the temple-tower of Marduk in Babylon, called E-sag-ila, “House of the Raised Head.” “For one whole year they molded bricks. When the second year arrived. They raised high the head of E-sag-ila equalling Apsu [the freshwater deep]. Having built a stage tower as high as Apsu [that is, as high as the waters were deep].”¹⁰⁷

Consider now the Tower of Babel story pulling in precisely the opposite direction. “God said, Behold! They are now a single people, all having one language and this is the first thing they do! Now nothing they plan to do will be unobtainable to them.”¹⁰⁸ In Hirsch’s attunement, the Hebrew for “unobtainable” (שָׁרָא) stems from the root word to fortify or strengthen a city wall. And thus, the divine lament is that “nothing will be secure”¹⁰⁹ from these city makers, no wall high enough to repel them. Or perhaps, to invert Hirsch’s reading, they threaten to create a wall that cannot be broken from within. Indeed, the walls of the first archaic states, suggests Owen Lattimore,¹¹⁰ Ann Porter,¹¹¹ and James Scott, may well have been erected “to contain mobile populations fleeing state control,”¹¹² just as much as they stood to keep out the barbarians. This reading of walls to keep unwilling citizens within dovetails with the conclusion of the Babel narrative. “God scattered them from there over the surface of all the earth, and they stopped building the city.”¹¹³ As soon as the population is dispersed, they never mount a return to Babel or attempt a rebuild. Simply put, the sedentary inhabitants of Babel had no great desire to remain sedentary.

This is the sublime about-face of the Tower of Babel story, the invitation to see through the eyes of the marginalised and disenfranchised so that the collapse of the agrarian state complex of Babel is a God-ordained progression rather than a traumatic reversion to a darker-age. That Babel is the place to leave, not lament. That fracturing into small kin-groups and eluding state-authority is in fact a positive counter to the darker mechanisms of statecraft. Seth Richardson reminds us that because archaeologists and historians focus on monumental ancient architecture, bas-reliefs, and scribal (read state-sponsored¹¹⁴) clay records, we have a tendency to see the early empires of Ur III, Akkad, and Babylon as the real achievements of ancient Mesopotamian society from 2500 to 1600 BCE, when in fact, argues Richardson, these incipient empires were a 300 year aberration against the backdrop of over seven centuries of “division and decentralisation.”¹¹⁵ Norman Yoffee and George Cowgill likewise note the typically fragile and short-lived potential of the agrarian city-state complexes of the ancient world.¹¹⁶ And Scott compares these earliest states to a dying patient, with acute and chronic morbidity.¹¹⁷

The Bible reduces these empires, with all their architectural awe, even further to a mere nine verses of annoyance before the return of diverse, localised

governance. Only recently are certain historians willing to adopt the Bible's positive attitude towards a collapsing urban centre. Writes James Scott,

What might collapse mean?... Higher order elites disappear; monumental building activity ceases, use of literacy for administrative and religious purposes is likely to evaporate; larger-scale trade and redistribution is sharply reduced; and specialist craft production for elite consumption and trade is diminished or absent... It is just as essential to emphasize what such events do *not* mean. They do not necessarily mean a decline in regional population. They do not necessarily mean a decline in human health, well-being, or nutrition, and... may represent an improvement.¹¹⁸

Scattered populations have clear advantages over and against a heavily populated grain core. They engage in less warfare, have a lower concentration of infectious diseases, retain more variability in subsistence diet—and thus have increased resilience to famine and drought, denude less concentrated areas of forest for firewood, and decrease soil exhaustion and salinisation.¹¹⁹ The divine decree of population dispersal from Babel is not a collapse but the antidote to the suffocation of humans, herds, mono-crops, pests, and pathogens. God scorns the state core and pushes the biblical narrative ever into the diversity at the margins.

Surveying the Genesis Landscape, Thus Far

The first 11 chapters of Genesis are in altitudinal pendulation between the mountains and the valley, and the concomitant entrapment and disgorging of the biblical personae in and out of the state zone. The story begins with Adam and Eve entangled with the first crop of wheat planted in the fragile headwaters of Eden. After Eden suffers the earth's primordial ecocide, the first farming family must struggle onto death over the most precious new capital, an arable field. The winner of that fraternal struggle exhausts the cursed soil, and descends into the valley to build the first city. Cain's urbanised descendants will now hoard wives to create ever denser population core zones. The tale next alludes to the first plough, and the concomitant gender inequality that follows in its furrow. Now the dense core zone is polluted with violence, so God floods state space. A single refugee family flees into the elusive hills, to learn anew the diverse practice of a gathering swidden food procurement regime that could free this small unit from the clutches of the grain states. Yet, the patriarch of that elusive unit plants a coveted vineyard seeding strife into the next generation. This limited inheritance engenders the curse of slavery for the extraneous descendants. And this enslavement is the catalyst which leads the storyline back into the alluvium, under the shadow of a totalitarian monumental city-state. But God is so exquisitely anti-state that the inward pull of peoples and goods that signifies the intrinsic function of statecraft is divinely inverted. The people are dispersed outward into small,

mobile, and diverse communities. The God of Genesis is allergic to the early states of Mesopotamia. And the story that accompanies that God is a radically subversive one.

At the beginning of this book, I posed the question: Why does the Bible curse farming before an audience of farmers? One could trace the answer by following that curse (עָרַף) like a spreading virus. The curse begins in the ground that nourished the first domesticated wheat,¹²⁰ infects the farmer who kills his brother in the field to become “more cursed than the ground,”¹²¹ and metastasizes into a father cursing his descendants into slavery.¹²² Curses prefigure migration out of non-state space and into the state core. Out of Eden and down from the hills of Ararat, towards cities and towers of the plains. The Bible curses the grain, because the grain is the foundation stone of the state, and thus Genesis 1–11 is essentially a narrative curse upon the evolving state, laced with divine attempts to shift the human gaze beyond its clutches. There is the injunction to Adam to refrain from the “tree” of wheat, which Eve and Adam ignore. Then the entreaty to Cain to keep the crouching tax collector away from one’s door, which Cain ignores. Then the call to Noah to flee the valley and head for the hills, there to be blessed with flora and fauna diversity, which Noah ignores for his wine. Until God acts without warning and simply scatters the city populace, twisting their monument to statehood into the paradigm of state collapse.

This is the landscape from which Abraham and all the genealogies of Israel will emerge. Another palliative response to the cursed state. It is fitting to conclude this Babel chapter with the midrashic tale of the outsider who passed it by with a curse:

Abram, son of Terah, passed by (עָבַר)¹²³ and saw them building the city and he cursed them in the name of his God, and said: “Swallow it up Lord, and divide their tongues” (Psalms 35.9). But they rejected his words like a stone thrown away on the ground. Yet, every chosen and good stone upholds the corner of the house. Thus scripture writes: “The stone which the builders rejected has become the head cornerstone’ (Psalms 118.22).”¹²⁴

The midrash understands that it is not the city and the state on which to build our story, but upon the shoulders of those that pass it by.¹²⁵

Notes

- 1 Elizabeth Kolbert, *Field Notes from a Catastrophe: A Frontline Report on Climate Change* (London: Bloomsbury, 2007), 93.
- 2 Kolbert, *Catastrophe*, 93.
- 3 Kolbert, *Catastrophe*, 93.
- 4 The *gur*, which was equal to approximately 300 litres. Kolbert, *Catastrophe*, 93.
- 5 Kolbert, *Catastrophe*, 94.
- 6 Harvey Weiss, Francesca deLillis, Dominique deMoulins, Jesper Eidem, Thomas Guilderson, Ulla Kasten, Torben Larsen, Lucia Mori, Lauren Ristvet, Elena

- Rova, and Wilma Wetterstrom, "Revising the Contours of History at Tell Leilan," *Annales archéologiques arabes syriennes*, Cinquantenaire (2002): 61–62. "Research at Tell Leilan has defined several attributes of the Akkadian imperialisation of the Habur plains... labour concentration by means of the city's massive circumvallation and displacement of populations at secondary centers."
- 7 Weiss, et al. "Tell Leilan," 61. See also Susan Pollack, *Ancient Mesopotamia*, 95. "Their interpretation as ration vessels, although not without problems, has been favoured by many scholars in recent years." However, as is often the case with artefacts of such antiquity, their true purpose remains unclear and often influenced by the lens of the theorist. For an entirely different view of the bowls' purpose, see specifically the work of Czech Assyriologist Petr Charvát. Charvát argues that these bowls were not containers for measured grain but were primarily used as set moulds for bread or sweetmeat eaten by everyone and at every location to publicly reinforce symbolic equality across the community. Petr Charvát, *Ancient Mesopotamia: Humankind's Long Journey into Civilization* (Prague: Oriental Institute, 1993).
 - 8 Weiss, et al., "Tell Leilan," 62.
 - 9 Diamond, *Guns, Germs, and Steel*, 218.
 - 10 Barry Powell, *Writing: Theory and History of the Technology of Civilization* (Chicester: Wiley-Blackwell, 2009), 81.
 - 11 Diamond, *Guns, Germs, and Steel*, 235.
 - 12 Diamond, *Guns, Germs, and Steel*, 236.
 - 13 Kolbert, *Catastrophe*, 94.
 - 14 Kolbert, *Catastrophe*, 95.
 - 15 Kolbert, *Catastrophe*, 94.
 - 16 Kolbert, *Catastrophe*, 97.
 - 17 Harvey Weiss, et al., "Tell Leilan."
 - 18 Kolbert, *Catastrophe*, 97.
 - 19 Kolbert, *Catastrophe*, 120.
 - 20 Weiss, et al., "Tell Leilan."
 - 21 Gen 11:4.
 - 22 Gen 11:4.
 - 23 Gen 11:4.
 - 24 Gen 11:8.
 - 25 Gen 11:9.
 - 26 Cassuto, *Genesis II*, 230.
 - 27 Theologian Jonathan Sacks also attunes us to the moral component of the ironic literary techniques employed in this short tale. "Not only is the story of Babel historically precise, it is also shot through with literary devices: inversions, word plays, ironies and puns. One of the most masterly is that the two key words, *l-v-n* [כִּבְנוּ], 'brick,' and *n-v-l* [וַחֲבָלָה], 'confuse,' are precise inversions of one another. As so often in the Torah, literary technique is closely related to the moral or spiritual message being conveyed. In this case the word play draws attention to the phenomenon of inversion itself ... The builders wanted to concentrate humanity in one place: 'Let us build a city... and not be scattered over the face of the whole earth' (11:4). The result was that they were dispersed: 'from there the Lord scattered them over the face of the whole earth' (11:8)." Jonathan Sacks, *Covenant and Conversation, Genesis: the Book of Beginnings* (Jerusalem: Maggid Press, 2009), 50.
 - 28 Michael Mann, *The Sources of Social Power*, 54.
 - 29 Scott, *Against the Grain*, 153.
 - 30 See Benstein, *The Way into Judaism and the Environment*, 65: "Since antiquity, towers have been built for command and control ... People in towers can control others' movements, gather information, survey and supervise."

- 31 Susan Pollock, *Ancient Mesopotamia: The Eden that Never Was* (Cambridge: Cambridge University Press, 1999), 176–177.
- 32 See Scott, *Against the Grain*, 230–233.
- 33 Robert McC. Adams, “Strategies of Maximization, Stability, and Resilience in Mesopotamian Society, Settlement and Agriculture,” *Proceedings of the American Philosophical Society* 122, no. 5 (Oct. 1978): 329–335, esp. 334.
- 34 Robert McC. Adams, “Strategies of Maximization,” 230. On the compelling theory that early or proto-Israelite society was also comprised, at least in part, of peoples labelled “social bandits” (‘Apiru) by the state scribes of Canaan in 14th century BCE correspondence between vassal rulers and their Egyptian overlords, see Dever, *Who Were the Early Israelites and Where Did They Come From?*, 181: “It should therefore be no surprise that I will now advocate explicitly a model for the Iron I hill country colonists—my ‘proto-Israelites’—that accounts for a variety of groups, all of them dissidents of one sort or another.”
- 35 Gen 11:4.
- 36 Robert McC. Adams, “Strategies of Maximization,” 231.
- 37 Pierre Clastres, *Society Against the State: Essays in Political Anthropology*, trans. Robert Hurley (New York, Zone, 1987), 79–99.
- 38 A.R. Holmberg, *Nomads of the Longbow: The Siriono of Eastern Bolivia* (New York: Natural History, 1950). See also Scott, *The Art of Not Being Governed*, 189.
- 39 Holmberg, *Nomads*.
- 40 In the parlance of contemporary public health, such diseases are termed “acute community infections.” As referenced in Scott, *Against the Grain*, 102.
- 41 Clastres, *Society Against the State*, 82.
- 42 Adams, “Strategies of Maximization,” 333.
- 43 Scott, *The Art of Not Being Governed*, 187.
- 44 Scott, *Art*, 187.
- 45 Adams, “Strategies of Maximization, Stability and Resilience,” 334.
- 46 Early statecraft was primarily concerned with the diminishment, or at least neutralisation, of these nearby non-state zones and their threatening aura of subversion. See Scott, *Seeing Like a State*, 187. “These stateless zones have always played a potentially subversive role, both symbolically and practically. From the vantage point of the court, such spaces and their inhabitants were the exemplars of rudeness, disorder, and barbarity against which the civility, order, and sophistication of the center could be gauged. Such spaces, it goes without saying, have served as refuges for fleeing peasants, rebels, bandits, and the pretenders who have often threatened kingdoms.”
- 47 Seth Richardson, “Early Mesopotamia: The Presumptive State,” *Past and Present* 215 (2012): 14.
- 48 Richardson, “Early Mesopotamia,” 14.
- 49 Gen 9:1.
- 50 Cassuto, *Genesis II*, 243.
- 51 Gen 11:9.
- 52 Scott, *Against the Grain*, 219–220.
- 53 See BDB, 973.
- 54 Gen 11:1.
- 55 See Rashi, commentary to Gen 11:1, s.v. פִּרְאִשִּׁים קִלְרִים “They came with a plan with a singular purpose.” Also Ibn Ezra, commentary to Gen 11:1, s.v. פִּרְאִשִּׁים קִלְרִים “Everyone was wise enough to understand what everyone else meant.” Also, Dillman, as cited by U. Cassuto, *Genesis II*, 239. However, Cassuto concludes that each of these attempts to distinguish two separate meanings in “one language” and “uniform words” is forced. “The correct explanation is that we have here a simple parallelism, the same thought reiterated in different terms.” (239)

- 56 Hirsch, *Genesis*, 204.
- 57 Gen 10:31.
- 58 See BDB, 546.
- 59 Hirsch, *Genesis*, 204.
- 60 See BDB, 183.
- 61 *Ha-amek Davar*, 11:4, s.v. *ve-rosho ba-shamayim*. As cited and translated by Judy Klitsner, *Subversive Stories in the Bible* (New Millford, CT.: Maggid Press, 2011), 40.
- 62 *Ha-amek Davar*, 11:4, s.v. *pen nafutz al panei kol ha-aretz*, as cited and translated by Klitsner, *Subversive Stories*, 40.
- 63 See Scott, *Against the Grain*, 137. "Most towns in the Mesopotamian alluvium were, by the middle of the third millennium BCE, walled." See also Porter, *Mobile Pastoralism*, 324, and Susan Pollock, *Ancient Mesopotamia: The Eden that Never Was* (Cambridge: Cambridge University Press, 1999), 47.
- 64 Scott, *Against the Grain*, 138.
- 65 Such an accolade was bestowed upon Ehrlich by Robert Gordis. "Some Hitherto Unrecognized Meanings of the Verb Shub" *Journal of Biblical Literature* 52, no. 2/3 (Jun.–Sep. 1933): 153. "The starting point for our inquiry is the acute observation of that master of Biblical Hebrew, Arnold B. Ehrlich."
- 66 As quoted by Cassuto, *Genesis II*, 239. Cassuto rejects Ehrlich's and other translations as overly "forced," and offers his own translation referring to the "single language" of the Babylonians. However, Cassuto's translation does not offer a fully satisfactory reading of what distinguishes "one speech" from a "single language" nor why the Bible offers both phrases in the same verse. See also Rashi to Genesis 11:1 and Ibn Ezra to Genesis 11:1 for further and disparate commentaries on this elusive phraseology.
- 67 Gen 10:5.
- 68 Gen 10:20.
- 69 Gen 10:31.
- 70 See Robert McC. Adams, "Shepherds at Umma in the Third Dynasty of Ur: Interlocutors with a World Beyond the Scribal Field of Ordered Vision," *Journal of the Economic and Social History of the Orient* 49, no. 2 (2006): 133–169. "The language of the administration (or at least its records) was Sumerian, taught only in scribal schools instituted by King Shulgi," 146.
- 71 Piotr Michalowski, "Charisma and Control: On Continuity and Change in Early Mesopotamian Bureaucratic Systems," *The Organisation of Power: Aspects of Bureaucracy in the Ancient Near East: Studies in Ancient Oriental Civilisations* 46 (1987): 55–68.
- 72 Gen 11:1.
- 73 Adams, "Shepherds at Umma," 165. See also Pollock, *Mesopotamia*, referring to the earlier Uruk period (4000 BCE–3100 BCE), 112. "Overall, the available evidence suggests that there was more administrative concern with the distribution of goods than with their production. Texts record goods issues and received; other administrative artifacts used in a bureaucracy concerned with control over closure, shipping, receipt, storage, and disbursement of goods; differential distribution of animal parts and animals of different ages imply some degree of control over the distribution of meat."
- 74 Scott, *Against the Grain*, 206–207. Scott quotes this metaphor in the name of his academic colleague, Ed Lindblom.
- 75 Scott, *Grain*, 207–208.
- 76 Gen 11:4.
- 77 *Pirque Rebbe Eliezer*, 24. [Heb], *editio princeps*, Constantinople, 1514. folio 16b. Digitized Copy, Hebrew Union College, Klau Library, in the Pirque Rabbi Eliezer

- Manuscript Database. <https://dornsife.usc.edu/assets/sites/1167/imgs/pre0116b.jpg>. Rabbi David Luria (HARDA"l) notes in his commentary to this midrash that the concern for bricks over and against human beings is a precursor (; װ) to the bondage under the Egyptian state regime (Jerusalem: Varsa ed., 1969), 57a.
- 78 Gen 11:3. "They said: 'Come, let us make bricks and burn them hard;' and they had bricks for stone. and bitumen as mortar."
- 79 Scott notes that the archaic states agrarian core was also its "Achilles' heel." For this was the zone that was likely to be squeezed first and hardest in any crisis. "Two facts," argues Scott, "made this a very dangerous gamble—one that could bring down states. First, for an agrarian kingdom always liable to the vagaries of rainfall, weather, pest, and human and crop diseases the annual yield, even in this most reliable of agrarian ecologies was extremely variable ... A second factor to recall in the case of pristine states was the quite rudimentary knowledge the state had of the actual acreage planted, the likely and the actual yields, district by district, for wheat and barley." *Against the Grain*, 206.
- 80 Gen 11:2.
- 81 Jonathan Sacks, *Covenant and Conversation, Numbers: The Wilderness Years* (Jerusalem: Maggid Press, 2017), 206.
- 82 Gen 10:8b.
- 83 Gen 10:10b.
- 84 Gen 10:5-7.
- 85 Hirsch, *Genesis*, 212.
- 86 Sandra Postel, *Pillar of Sand: Can the Irrigation Miracle Last?* (New York: Norton & Co., 1999), 18. "One overarching message is that a thriving, integrated, resourceful society can collapse abruptly from a cascade of ecological events. The irrigated agriculture that enabled this [Akkadian] empire to expand well beyond anything previously known also became a source of vulnerability under the strains of population pressures and water shortages induced by abrupt climactic change."
- 87 Gen 11: 8–10a.
- 88 Gen 10:31.
- 89 Gen 10:31-32.
- 90 Gen 11:10-11.
- 91 Gen 11:5.
- 92 *Leitwortstil* (Literally: "Leading word-style) is a term coined by Martin Buber and Franz Rosenzweig, in the explanatory prefaces to their German translation of the Hebrew Bible. "A *Leitwort*," writes Buber, "is a word or a word-root that reoccurs significantly in a text, in a continuum of texts, or in a configuration of texts: by following these repetitions, one is able to decipher or grasp a meaning of the text, or at any rate, the meaning will be revealed more strikingly." Cited and translated by Robert Alter in *The Art of Biblical Narrative* (New York: Basic Books, Inc. 1981), 93.
- 93 Gen 10:31.
- 94 Gen 11:4.
- 95 Gen 11:9.
- 96 Gen 11:10.
- 97 Daniel Gordis, "The Tower of Babel and the Birth of Nationhood," *Azure: Ideas for the Jewish Nation* 40 (Spring 2010): 25.
- 98 Gordis, "The Tower of Babel," 25.
- 99 Gordis, "The Tower of Babel," 25.
- 100 Isaiah Berlin, "Two Concepts of Liberty," in *Four Essays on Liberty* (Oxford: Oxford University Press, 1969), 118–172.
- 101 It is worth noting that whilst Klitsner finds the narrative ordering of building events "unusual," the Tower of Babel phraseology might simply be mirroring the

- norm of other near eastern etiological narratives such as the *Enuma Elish* Babylonian creation epic. In *Enuma Elish* we read: For one whole year they molded bricks/When the second year arrived/They raised high the head of the [tower] Esagila." *Enuma Elish*, tablet 6, lines 60–61.
- 102 Klitsner, *Subversive Sequels in the Bible*, 41.
- 103 Scott, *Against the Grain*, 138.
- 104 Maureen Kovacs, *The Epic of Gilgamesh*, 16, ft. 2, citing the parallel account in the Old Babylonian version.
- 105 As cited in Scott, *Grain*, 62.
- 106 See Anne Porter's reading of successive versions of the Epic of Gilgamesh in which the description of Enkidu shifts dramatically, and in parallel with the growth of cities. In the earliest versions of the Epic, Enkidu is merely a pastoralist, indicative of a relatively benign relationship between farmers and herders. In versions from a millennium later, Enkidu has become a subhuman barbarian from the wild who requires taming. Here, argues Porter, the ideology of the emerging grain state has imagined the mistrusted and dangerous outsider, lurking beyond the protective walls. Ann Porter, *Mobile Pastoralism and the Formation of the Near Eastern Civilization: Weaving Together Societies* (Cambridge: Cambridge University Press, 1999), 351–393.
- 107 *Enuma Elish*, tablet 6, lines 60–62, as translated by Avigdor Shinan and Yair Zakovitch, *From Gods to God*, 66.
- 108 Gen 11:6.
- 109 Hirsch, *Genesis*, 210.
- 110 Owen Lattimore. "The Frontier in History," in *Studies in Frontier History: Collected Papers, 1928–1958* (Oxford: Oxford University Press, 1962), 470. "The Great Wall(s) of China were built quite as much to keep Chinese taxpaying cultivators inside, as to keep the barbarians outside."
- 111 Ann Porter, *Mobile Pastoralism*, 324.
- 112 Scott, *Against the Grain*, 138.
- 113 Gen 11:8.
- 114 Writes James Scott on the difficulty of documenting the interregna periods of ancient Mesopotamia, "The self-documenting court center that offered convenient one-stop shopping for historians and archaeologists is replaced by a fragmentary, dispersed, and largely undocumented 'dark age.'" *Against the Grain*, 214.
- 115 Seth Richardson, "Early Mesopotamia," 16.
- 116 Norman Yoffee and George Cowgill, *The Collapse of Ancient States and Civilizations* (Tucson: University of Arizona Press, 2006), 30. See also, Norman Yoffee, "Collapse in Ancient Mesopotamia: What Happened, What Didn't?," in *Questioning Collapse: Human Resilience, Ecological Vulnerability, and the Aftermath of Empire*, eds. Patricia A. McAnany and Norman Yoffee (Cambridge: Cambridge University Press, 2012), 181. "If one 'rule' of political stability/instability can be risked, it is that the more centralized the government, the larger the bureaucracy, and the larger the army in a state, the less stable is the government and the more drastic and comprehensive is the fall of the state."
- 117 Scott, *Against the Grain*, 190.
- 118 Scott, *Against the Grain*, 186.
- 119 Scott, *Against the Grain*, 212 and 215.
- 120 See Gen 3:17.
- 121 Gen 4:11.
- 122 See Gen 9:25.
- 123 I believe the midrashic word choice here is purposefully used to imply that Abram is distinctly removed from the enterprise of Babel and the centripetal force of the state. He passes by what all the others are enraptured by. Thus, he becomes "the

one who passes over” *par excellence*, as he is designated in Gen 14:13 as קאטש הַחַשׁוּי. Gen. Rab. 42:8 offers three separate etymologies for the epithet הַחַשׁוּי. It may indicate his geographic origins: Abram hails from across, חַשׁ, the River Euphrates. It may be a genealogical appellation: Abram is a descendent of *Eber*. Or, it may indicate a key character trait: “Rav Yehuda said: All the world was on one side (הַאֶחָד) and he (Abram) was on the other.” I follow Judy Klistner, *Subversive Sequels*, 47, ft. 26, in leaning towards this third definition. “This [final] interpretation is supported by the use of the term *Ivri* throughout the Bible to denote a group distinct from their surroundings.” For a fuller review of the various definitions of חַשׁוּי, see S. Engelmey, “Ivri: Naming Ourselves,” *Judaism* 54, no. 1 (2005): 13–26.

124 *Pirke Rebbe Eliezer*, 24. [Heb], *editio princeps*, Constantinople, 1514. folio 16b.

125 An allusion to Gen 17:22 as elucidated by Rashi s.v. וַיֵּלֶךְ קִיטְשׁ. Here, Rashi is struck by the odd scriptural description, “וַיֵּלֶךְ קִיטְשׁ מֵעַל אַבְרָהָם” literally translated as, “God arose from on top of Abraham.” Rashi comments that “on top of Abraham” implies that “the righteous are the chariot of the Omnipresent.” Thus, God *is in motion*, carried on the backs of Abraham and his children.

Conclusion

Dear reader, are you convinced by this treatise? Is Adam and Eve's farming curse really a warning against the crouching power of the nascent state? Are these first 11 chapters of Genesis a warrant to wrench free from the centripetal pull of the emergent state core on the Mesopotamian alluvium, and cross over into a non-state mode of farming subsistence in the hills of Palestine? I harbour some doubts on that score. For there are numerous ambiguities buried within my argument. For example, much of this book weaves midrashic details into the very warp and woof of the biblical narrative. Consider how essential these threads are to the fundamental pattern of my claims. The forbidden tree of "wheat," Cain's domesticated dog, the banal misogyny of Lamech, the plough invented by Noah, the rapacious chiefs who call themselves children of god, Ham's castration of his father, and the bricks valued more than people in their fall from the tower heights are each key fulcrum moments missing from scripture itself. Yet, I have suggested the distinct possibility that these midrashim are mimetic reflections of life lived in Mesopotamia from approximately 10,000 to 1500 BCE.

How can we be so sure? Perhaps, these midrashists were not faithful transmitters of an earlier age, but moulders of their own tales, and the Bible simply a handy hook on which to peg new-spun narratives. After all, at the time of the redaction of *Genesis Rabba* in the fifth century,¹ the Tigris and Euphrates river basin was undergoing increased salt build-up in the topsoil.² Ground salinisation reached such damaging levels that just a few centuries later, 15,000 slaves were forced to work in the southern region peeling off the upper sterile layer of soil, to uncover more fertile layers below. These slaves repeatedly revolted, culminating in a 14-year uprising from 869 to 883 CE. This revolt, and the famines that occurred along with it, took more than a million lives.³ Maybe the midrashists were not preservers of the past, but keen observers of the salt-drenched wheat fields of their own present. And the forbidden tree of wheat was simply an external imposition into the Genesis account, shaped by the midrashists' jaundiced perspective on state-sponsored alluvial grain farming in the early stages of its Levantine decline.

The midrashists fill-in the unsaid moments of the Genesis narrative. But are these additional details the caulk of mimetic gleanings from the Neolithic,

or their particular yearnings for a story that responds to their own political and ecological crisis? And, if the latter, we have another, and far more subtler answer on why a community of farmers would choose to sacralise a text that curses the act of farming? The Bible does not curse farming, only the later rabbinical community does that. After all, when stripped of all midrashic garmenting, is the divine interdiction to eat bread via “the sweat of the brow”⁴ so distinctly different from the Atrahasis account of disgruntled gods in the fields?⁵ Grain farming is admittedly labour intensive and sweaty, but God never tells Adam to put down his farm implements altogether. Perhaps, the real invective against the grain and the anti-state tenor of these 11 chapters are the layered gloss of perceptive and subversive midrashists from a later epoch, reading their own future ills backwards into their founding texts? When we encounter the midrash on Noah’s invention of the plough just before the flood account, should we not retain a bit of scepticism? Don’t the midrashists know far less about the innovations of past millennia, and far more about their own current predicament with eroded, plough-worn fields? Plough-tilled land always displaces topsoil, which leads to erosion and siltation of the canal networks, and silted canals flood alluvial lowlands.⁶ Thus, intensive ploughing causes flooding for every age. Is the story of Noah’s plough an ancient memory or simply a bit of midrashic deduction? If Noah had a flood, then he must have started with a plough.

These questions shake my thesis. Yet, I hope that it is a beating akin to the thrashing of the olive tree at harvest, which allows the fruit to fall into the waiting cloth below. As individual accounts, each midrash may be read as a product of its own redactive time and place, but here we are far less concerned with the isolated exegesis of each midrash, and far more concerned with how a series of midrashic stories enlivens the very sweep of these first 11 chapters. For even stripped clean of all midrashic accoutrementation, there are two deeply rooted patterns that run right through these variegated stories to articulate an orientation vis-à-vis the state apparatus in its formative mode.

Patterns of Topography

These 11 etiological chapters of Genesis are on the move. They begin in the mountainous headwaters of Eden in a highland forest milieu.⁷ Yet, just one generation and two chapters later, the biblical landscape has shifted to the domus of field and flock.⁸ From here, the narrative travels down with Cain into the alluvial plain to build the first city.⁹ This first movement from non-state hills onto flat state space is the overture of Genesis, a tempo that will repeat again and again from Noah right through to Abraham’s and Jacob’s descents to Egypt. For the magnetic force of the state is fierce. Consider the flood hero Noah. Fleeing to the mountains of Ararat creates only a short reprieve for his family. Inexorably, his offspring are led in bondage back down from the hilly elevations to build another city in the valley of Shinar.¹⁰ If one were to trace the elevation flux of these first 11 chapters, it would

resemble a roller coaster track. From the heights of Eden to the lowland alluvial city of Enoch, back up to Ararat, then down again to the trough of Babel. People moving in and out of the state zone of control. And with each shifting space comes either a blessing or a curse. A blessing for those dwelling securely in non-state modalities, and a curse when a society begins to slip and fall downward into the state mode. This pendulation between blessings and curses is the second holistic pattern of these 11 chapters of Genesis.

The Pattern of Blessing and Curse

As soon as humanity blinks into consciousness, God blesses the couple, then places them in the hills of Eden.¹¹ Adam and Eve are divinely promised a diet of seeds, nuts, and fruit.¹² God enjoins them to spread out over the earth and seek dominion over the heterogeneous fauna species.¹³ Here, the fundamental strategies of non-state living are delineated and sanctified. Eden's hilly elevation ensures that it is a zone of friction, antithetical to traditional state formation with its reliance on smooth transport of people and goods. The mountain soil and topography can never contain the large-scale fields of assessable and expropriable grain on which the state depends. The blessed diversity of foodstuff implies a non-sedentary society, elusively gathering the wild harvests of the highlands. The invocation to fill the entire land is a further consecrated prod towards mobility and against the dense packing into a population core so favoured by statemakers. Yet, non-state space is tremulous. God's curse on Adam, "with sweat on your brow you will eat bread till you return to the ground,"¹⁴ encapsulates the first slip away from the hills and the variegated forest diet and mobile social units towards the totality of grain-based sedentary farming and all the implications of the Neolithic.

As the first curse leads to farming, the second engenders valley urbanism. God's curse on the fratricidal Cain, "And now you are cursed from the ground,"¹⁵ describes the farmer's dilemma of over-used, nutrient poor soil. "When you work the ground she will no longer give her strength to you."¹⁶ So Cain must migrate from the hills, where the soil is washing away, into the alluvial flood plain that has captured the run-off. There he will build the first proto-city and thus the beginning of the pristine state. In these early chapters, to be accursed is synonymous with a predilection towards state-making.

And blessing finds those who dwell beyond state reach. Like the call to Noah in the mountains of Ararat. "The Lord blessed Noah and his sons and said to them: 'be fruitful and multiply and fill the earth.'"¹⁷ Again, the invitation to spread out, to elusively multiply beyond the force of state centralization. But that force is too compelling. It tugs at the base desires of Noah and ensnares him in his drunken stupor. Where once God's curse preceded a pull stateward, now the curse-as-state-harbinger is placed in Noah's mouth. "[Noah] said: 'Cursed by Canaan, a slave's slave he will be to his brothers.'"¹⁸ That is enough, it would seem, for the narration to consolidate in Babel, again forced into the city and the alluvial plain.

The interlocking pulse of blessing to curse to blessing to curse is a prelude. We are now primed for God's blessing to Abraham, as he turns towards the hills of Palestine. We are primed to follow that blessing from Isaac to Jacob and his children. We are primed to hold fast to that blessing as Israel slips in and out of state space.

Perhaps, the midrashists magnified that pulse, amplifying its tempo and making it their own. Yet, that incessant call to the margins and the marginal, that curse upon the grain and the grain-fat state, beats its rhythm right through Genesis. Can you not hear it?

Notes

- 1 See H. L. Strack and Günter Stemberger, *Introduction to the Talmud and Midrash*, trans., Markus Bokmühl (Minneapolis: Augsburg Fortress, 2009), 160; Also Ofra Meir, *The Darshanic Story in Genesis Rabba* [Hebrew] (Tel Aviv, Hakibbutz Hameuchad, 1987), 70–72.
- 2 Sandra Postel, *Pillar of Sand: Can the Irrigation Miracle Last?* (New York: Norton, 1999), 26.
- 3 Peter Christensen, *The Decline of Iranshahr* (Copenhagen, Denmark: Museum Tusculanum Press, University of Copenhagen, 1993).
- 4 Gen 3:19.
- 5 W.G. Lambert and A.R. Millard, *Atra-hasis: The Babylonian Story of the Flood*, 89. "Great indeed was the drudgery of the gods, the forced labor was heavy, the misery too much."
- 6 Rattan Lal, "History of Erosion Research," in *Encyclopedia of Water Science*, eds. B.A. Stewart and Terry Howell (New York: Marcel Dekker, 2003), 225.
- 7 Gen 2:10.
- 8 Gen 4:2.
- 9 Gen 4:17.
- 10 Gen 11:2.
- 11 Gen 3:8.
- 12 Gen 1:29. "The Lord said: 'Behold, I have given to you every herbage seeding seed that are on the face of all the earth, and every tree that has fruit seeding seed, they will be yours as food.'"
- 13 Gen 1:28.
- 14 Gen 3:19.
- 15 Gen 4:11.
- 16 Gen 4:12.
- 17 Gen 9:1.
- 18 Gen 9:25.

Bibliography

- Adams, Robert. "Strategies of Maximization, Stability, and Resilience in Mesopotamian Society, Settlement and Agriculture." *Proceedings of the American Philosophical Society* 122, no. 5 (1978): 329–335.
- Adams, Robert. "Shepherds at Umma in the Third Dynasty of Ur: Interlocutors with a World Beyond the Scribal Field of Ordered Vision." *Journal of the Economic and Social History of the Orient* 49, no. 2 (2006): 133–169.
- Aharoni, Yohanan, and Ruth Amiran. "A New Scheme for the Subdivision of the Iron Age in Palestine." *Israel Exploration Journal* 8 (1958): 171–184.
- Albright, W. F. "The Location of the Garden of Eden." *American Journal of Semitic Languages and Literatures* 39 (1922): 15–31.
- Algaze, Guillermo. "The End of Prehistory and the Uruk Period." In *The Sumerian World*, edited by Harriet Crawford, 68–94. London: Routledge, 2013.
- Alesina, Alberto, Paola Giuliano, and Nathan Nunn. "On the Origins of Gender Roles: Women and the Plough." *Quarterly Journal of Economics* 128, no. 2 (May 2013): 469–530.
- Alt, Albrecht. *Essays on Old Testament History and Religion*. Oxford: Oxford University Press, 1966.
- Alter, Robert. *The Art of Biblical Narrative*. New York: Basic Books, 1981.
- Alter, Robert. *The Five Books of Moses: A Translation with Commentary*. New York: W.W. Norton & Co., 2004.
- Alter, Robert. *The World of Biblical Literature*. London: SPCK Publishing, 1992.
- Assmann, Jan. "Myth as Historia Divina and Historia Sacra." In *Scriptural Exegesis: The Shapes of Culture and the Religious Imagination*, edited by Deborah Green and Laura Lieber, 3–18. Oxford: Oxford University Press, 2009.
- Auerbach, Erich. *Mimesis: The Representation of Reality in Western Literature*. Translated by Willard R. Trask, Fifteenth anniversary printing. Princeton: Princeton University Press, 2003.
- Bar-Ilan, Meir. "Illiteracy in the Land of Israel in the First Centuries C.E." In *Essays in Social Scientific Study of Judaism and Jewish Society*, vol. 2, edited by Simcha Fishbane et al., 46–61. New York: Ktav, 1992.
- Bar-Yosef, Ofer. "Forward." In *The Neolithic Revolution in the Near East: Transforming the Human Landscape*, edited by Alan Simmons, ix–xiii. Arizona: University of Arizona Press, 2007.
- Barton, John. *Reading the Old Testament: Method in Biblical Study*. London: Darton Longman & Todd, 1984.

- Bassett, Frederick W. "Noah's Nakedness and the Curse of Canaan, a Case of Incest?" *Vetus Testamentum* 21, Fasc. 2 (April 1971): 232–237.
- Bellwood, Peter. *First Farmers: The Origins of Agricultural Societies*. Oxford: Blackwell Publishing, 2005.
- Benstein, Jeremy. *The Way into Judaism and the Environment*. Woodstock, VT: Jewish Lights Publishing, 2006.
- Berlin, Isaiah. *Four Essays on Liberty*. Oxford: Oxford University Press, 1969.
- Berman, Samuel A. *Midrash Tanhuma-Yelammedenu: An English Translation of the Genesis and Exodus from the Printed Version of the Tanhuma-Yelammedenu*. New Jersey: KTAV Publishing House, 1996.
- Bergsma, John Sietze, and Scott Walker Hahn. "Noah's Nakedness and the Curse on Canaan (Genesis 9:20–27)." *Journal of Biblical Literature* 124, no. 1 (Spring 2005): 25–40.
- Blidstein, Gerald. "Man and Nature in the Sabbatical Year." *Tradition* 8 (1966): 48–55.
- Bloom, Harold. *The Book of J*. New York: Grove Weidenfeld, 1990.
- Boardman, J., Kathleen M. Kenyon, E. J. Moynahan, and J. D. Evans. "The Olive in the Mediterranean: Its Culture and Use [and Discussion]." *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences* 275, no. 936 (1976): 187–196.
- Borowski, Oded. *Agriculture in Iron Age Israel: The Evidence from Archaeology and the Bible*. Winona Lake, Indiana: Eisenbrauns, 1987.
- Boserup, Ester. *Population and Technological Change: A Study of Long-Term Trends*. Chicago: University of Chicago Press, 1981.
- Boserup, Ester. *The Conditions of Agricultural Growth: The Economics of Agrarian Change Under Population Pressure*. Abingdon: Earthscan, 1965.
- Boserup, Ester. *Woman's Role in Economic Development*. London: George Allen and Unwin Ltd., 1970.
- Boyarin, Daniel. *Intertextuality and the Reading of Midrash*. Bloomington: Indiana University Press, 1994.
- Boyarin, Daniel. "Midrash in Parables." *AJS Review* 20, no. 1 (1995): 123–138.
- Brettler, Marc Zvi. *How to Read the Jewish Bible*. Oxford: Oxford University Press, 2005.
- Bright, John. *A History of Israel*. Philadelphia: Westminster, 1959.
- Brody, Hugh. *The Other Side of Eden: Hunter-Gatherers, Farmers, and the Shaping of the World*. London: Faber and Faber, 2001.
- Brueggemann, Walter. *Genesis: Interpretation: A Bible Commentary for Teaching and Preaching*. Atlanta: John Knox Press, 1982.
- Brueggemann, Walter. *The Land: Place as Gift, Promise, and Challenge in Biblical Faith*, 2nd ed. Minneapolis: Fortress Press, 2002.
- Burns, Nancy, Kay Schlozman, and Sidney Verba. *The Private Roots of Public Action: Gender, Equality, and Political Participation*. Cambridge: Harvard University Press, 2001.
- Byron, John. *Cain and Abel in Text and Tradition: Jewish and Christian Interpretations of the First Sibling Rivalry*. Leiden: Brill, 2011.
- Callaway, Joseph. "Excavating Ai (et-Tell): 1964–72." *Biblical Archeology* 39 (1976): 18–30.
- Campbell, Edward. "Two Amarna Notes: The Shechem City-States and Amarna Administrative Terminology." In *Magnolia Dei: The Mighty Acts of God: Essays on the Bible and Archeology in Memory of G. Ernest Wright*, edited by Frank M. Cross et al., 37–45. New York: Doubleday, 1976.

- Cassuto, Umberto. *A Commentary on the Book of Genesis: Parts I and II*. Translated by Israel Abrahams. Jerusalem: The Magnes Press, Hebrew University, 1961.
- Carneiro, Robert L. "A Theory of the Origins of the State." *Science* 169 (1970): 733–738.
- Chaney, Marvin. *Peasants, Prophets and Political Economy: The Hebrew Bible and Social Analysis*. Eugene, OR: Cascade Books, 2017.
- Charvát, Petr. *Ancient Mesopotamia: Humankind's Long Journey into Civilization*. Prague: Oriental Institute, 1993.
- Christensen, Peter. *The Decline of Iranshahr*. Copenhagen, Denmark: Museum Tusculanum Press, University of Copenhagen, 1993.
- Clarke, E.G. *Targum Pseudo-Jonathan of the Pentateuch: Text and Concordance* [Hebrew]. New Jersey: Ktav Publishing House, 1984.
- Clastres, Pierre. *Society Against the State: Essays in Political Anthropology*. Translated by Robert Hurley. New York: Zone, 1987.
- Clements, R.E. *Abraham and David: Genesis XV and Its Meaning for Israelite Tradition*. London: SCM Press, 1967.
- Cohen, M.N. and G. Armelagos. *Paleopathology at the Origins of Agriculture*. New Haven: Yale University Press, 1977.
- Collins, Desmond. *The Human Revolution: From Ape to Artist*. Oxford: Phaidon Press, 1976.
- Condominas, George. *From Lawa to Mon, from Saa' to Thai: Historical and Anthropological Aspects of Southeast Asian Social Spaces*. Translated by Stephanie Anderson et al. Canberra: Australian National University, 1990.
- Coote, Robert B. and David Robert Ord. *The Bible's First History*. Philadelphia: Fortress, 1989.
- Cover, Robert M. "The Supreme Court, 1982 Term—Foreword: Nomos and Narrative." *Faculty Scholarship Series Paper* 2705 (1983): 4–68. http://digitalcommons.law.yale.edu/fss_papers/2705.
- Coyne, R. *The Holy Bible Translated from the Latin Vulgate*. Dublin: Coyne, 1847.
- Damrosch, David. *The Narrative Covenant: Transformations of Genre in the Growth of Biblical Literature*. New York: Harper and Row, 1987.
- Dar, Shimon. *Landscape and Pattern: An Archaeological Survey of Samaria 800 B.C.E.–636 C.E.* Oxford: B.A.R. 1986.
- Davis, Ellen F. *Scripture, Culture and Agriculture: An Agrarian Reading of the Bible*. New York: Cambridge University Press, 2009.
- Dalley, S. *Myths from Mesopotamia: Creation, the Flood, Gilgamesh, and Others*. Oxford: Oxford University Press, 1989.
- Day, John. "Cain and the Kenites." In *Homeland and Exile*, edited by Gershom Galil, Markham Geller, and Alan Millard, 335–346. Leiden: Brill, 2009.
- Deleuze, Giles, and Felix Guattari. *A Thousand Plateaus*. London: Bloomsbury, 2017.
- Demsky, Aaron. "Writing in Ancient Israel: The Biblical Period." In *Mikra: Text, Translation, Reading and Interpretation of the Hebrew Bible in Ancient Judaism and Early Christianity*, edited by Martin-Jan Mulder, 2–20. Philadelphia: Fortress, 1988.
- Dever, William G. *Who Were the Early Israelites and Where Did They Come From?* Michigan: Erdmans, 2003.
- Diamond, Jared. *Collapse: How Societies Choose to Fail or Survive*. London: Penguin Books, 2005.
- Diamond, Jared. *Guns, Germs, and Steel: A Short History of Everybody for the Last 13,000 Years*. London: Vintage, 2007.

- Diamond, Jared. *The Third Chimpanzee: The Rise and Fall of the Human Animal*. New York: HarperCollins, 1993.
- Diamond, Jared. *The World Until Yesterday: What Can We Learn From Traditional Societies*. London: Penguin Books, 2012.
- Douglas, Mary. *Constructive Drinking: Perspectives on Drink from Anthropology*. Cambridge: Cambridge University Press, 1987.
- Drazin, Israel, and Stanley Wagner. *Onkelos on the Torah: Understanding the Bible Text: Genesis*. Jerusalem: Gefen, 2006.
- Driver, S. R. *The Book of Genesis*. London: Kessinger, 1926.
- Ellis, Peter. *The Yahwist, The Bible's First Theologian*. Collegeville, MN: The Liturgical Press, 1968.
- Elson, Robert. "International Commerce, the State, and Society: Economic and Social Change." In *the Cambridge History of Southeast Asia, The Nineteenth and Twentieth Centuries*, vol. 2, edited by Nicholas Tarling, 131–196. Cambridge: Cambridge University Press, 1992.
- Elvin, Mark. *The Retreat of the Elephants: An Environmental History of China*. New Haven: Yale University, 2004.
- Eisenberg, Evan. *The Ecology of Eden: Humans, Nature and Human Nature*. London: Picador, 1998.
- Eissfeldt, Otto. *The Old Testament: An Introduction*. Translated by Peter R. Ackroyd. New York: Harper & Row, 1965.
- Fall, Patricia, Steven Falconer, and Lee Lines. "Agricultural Intensification and the Secondary Products Revolution along the Jordan Rift." *Human Ecology* 30, no. 4 (Dec. 2002): 445–482.
- Finkel, Irving. *The Ark Before Noah: Decoding the Story of the Flood*. London: Hodder & Stoughton, 2014.
- Finlay, Michael. "Was Greek Civilization Based on Slave Labour?" *Historia: Zeitschrift für alte geschichte* 8, no. 2 (1959): 145–164.
- Fisch, Harold. *The Holy Scriptures*. Jerusalem: Koren, 1997.
- Fohrer, George. *Introduction to the Old Testament*. Translated by David Green. Nashville: Abingdon, 1968.
- Ford, David. *The Future of Christian Theology*. Chichester: Wiley and Blackwell, 2011.
- Foster, Andrew D., and Mark R. Rosenzweig. "Technical Change and Human-Capital Returns and Investments: Evidence from the Green Revolution." *American Economic Review* 86, no. 4 (1996): 931–953.
- Fox, Everett. *In the Beginning: A New English Rendition of the Book of Genesis*. New York: Schocken Books, 1983.
- Fukuyama, Francis. *The Origins of Political Order: From Prehuman Times to the French Revolution*. London: Profile Books, 2011.
- Frei, Hans W. *The Eclipse of Biblical Narrative: A Study in Eighteenth and Nineteenth Century Hermeneutics*. New Haven: Yale University Press, 1974.
- Gaster, Theodor H. "Cosmogony." In *Interpreter's Dictionary of the Bible*, edited by Keith R. Crim. Nashville, TN: Abingdon Press, 1992.
- Gelander, Shamaï. *The Good Creator: Literature and Theology in Genesis 1-11*. Atlanta: Scholars Press, 1997.
- Gelb, Ignace Jay. "Prisoners of War in Early Mesopotamia." *Journal of Near Eastern Studies* 32, no. 12 (1973): 70–98.
- Gellner, Ernest. *Saints of the Atlas*. London: Weidenfeld and Nicolson, 1969.

- George, Andrew. *The Epic of Gilgamesh: The Babylonian Epic Poem and Other Texts in Akkadian and Sumerian*. London: Penguin, 1999.
- Goelet, Ogden. "Problems of Authority, Compulsion and Compensation in Ancient Egyptian Labor Practices." In *Labour in the Ancient World*, edited by Piotr Steinkeller and Michael Hudson, 523–582. Dresden: LISLET Verlag, 2015.
- Goody, Jack. *Production and Reproduction: A Comparative Study of the Domestic Domain*. Cambridge: Cambridge University Press, 1976.
- Gordis, Daniel. "The Tower of Babel and the Birth of Nationhood." *Azure: Ideas for the Jewish Nation* 40 (Spring 2010): 19–36.
- Gottwald, Norman K. *The Tribes of Yahweh: A Sociology of the Religion of Liberated Israel, 1250–1050 B.C.E.* Maryknoll, NY: Orbis, 1979.
- Greenberg, Moshe. "Crimes and Punishments." In *The Interpreter's Dictionary of the Bible*, edited by George A. Buttrick, 733–744. Nashville: Abingdon, 1962.
- Gunkel, Herman. *Genesis*. Translated by Mark Biddle. Georgia: Mercer University Press, 1997.
- Gunkel, Herman. *The Legends of Genesis*. Translated by William Herbert Carruth. London: Open Court Publishing, 1901.
- Halivni, David Weiss. *Revelation Restored*. London: SCM Press, 1997.
- Harari, Yuval Noah. *Sapiens: A Brief History of Humankind*. London: Vintage, 2011.
- Hartman, David. *Individual and Communal Perspectives on Shabbat*. Jerusalem: The Shalom Hartman Institute for Advanced Judaic Studies, 1981.
- Hauser, Alan J. "Israel's Conquest of Palestine: A Peasants' Rebellion?" *Journal for the Study of the Old Testament* 7 (1978): 2–19.
- Heinemann, Isaac. *Darkhe ha-Aggadah*, 2nd ed. [Hebrew]. Jerusalem: Magnes Press, 1954.
- Heinemann, Joseph. *Aggadah and Its Development* [Hebrew]. Jerusalem: Keter, 1974.
- Hess, Richard. *Studies in the Personal Names of Genesis 1–11*. Winona Lake: Eisenbrauns, 1993.
- Hiebert, Theodore. *The Yawhist's Landscape: Nature and Religion in Early Israel*. New York: Oxford University Press, 1984.
- Hillel, Daniel. *The Natural History of the Bible: An Environmental Exploration of the Hebrew Scriptures*. New York: Columbia University Press, 2006.
- Hillers, Delbert R. *Covenant: The History of a Biblical Idea*. Baltimore: John Hopkins University Press, 1969.
- Hirsch, Samson. *The Pentateuch: Genesis*. Translated by Isaac Levy. New York: The Judaica Press, 1971.
- Holmberg, A.R. *Nomads of the Longbow: The Siriono of Eastern Bolivia*. New York: Natural History, 1950.
- Hrdy, Sarah Blaffer. "The Optimal Number of Fathers: Evolution, Demography, and History in the Shaping of the Female Mate Preferences." *Evolutionary Perspectives on Human Reproductive Behavior* 907, no. 1 (April 2007): 75–96.
- Hrdy, Sarah Blaffer, and D.S. Judge. "Darwin and the Puzzle of Primogeniture: An Essay on Biases in Parental Investment after Death." *Human Nature* 4 (1993): 1–46.
- Jacob, Benno. *The First Book of the Bible: Genesis*. edited and translated by Ernest I. Jacob and Walter Jacob. New York: Ktav Publishing, 1974.
- Jacobsen, Thorkild, trans. *The Sumerian King List*. Chicago: Chicago University Press, 1939.
- Joerstad, Mari. *The Hebrew Bible and Environmental Ethics: Humans, Non-Humans, and the Living Landscape*. Cambridge: Cambridge University Press, 2019.

- Joffe, Alexander H. "Alcohol and Social Complexity in Ancient Western Asia." *Current Anthropology* 39, no. 3 (June 1998): 297–322.
- Jonsson, Hjörleifur. *Mein Relations: Mountain People and State Control in Thailand*. Ithaca: Cornell University Press, 2005.
- Kadowski, Seiji "Ground Stone Tools and Implications for Use of Space and Social Relations at 'Ain Abu Nukhayla, a PPNB Settlement in Southern Jordan." In *Domesticating Space: Construction, Community, and Cosmology in the Late Prehistoric Near East*, edited by Edward Bruce Banning, and Michael Chazan, 53–64. Berlin: Ex Oriente, 2006.
- Kaplan, Aryeh. *The Living Torah: The Five Books of Moses and the Haftarat*. New York: Moznaim Press, 1981.
- Kermode, Frank. "The Plain Sense of Things." In *Midrash and Literature*, edited by Geoffrey Hartman and Sanford Budick, 174–192. New Haven: Yale University Press, 1986.
- Khazanov, A. M. *Nomads and the Outside World*. Cambridge: Cambridge University, 1984.
- Khodarkasky, Michael. *Where Two Worlds Met: The Russian State and the Kalmyk Nomads, 1600–1771*. Ithaca: Cornell University Press, 1992.
- King, Phillip. *Amos, Hosea Micah: An Archeological Commentary*. Philadelphia: The Westminster Press, 1988.
- Klein, Jacob. "Enki and Ninmah." In *The Context of Scripture: Canonical Compositions from the Biblical World*, edited by William W. Hallo, 509–521. Leiden: Brill, 1997.
- Klitsner, Judy. *Subversive Stories in the Bible*. New Millford, CT.: Maggid Press, 2011.
- Köhler, Ludwig. *Hebrew Man*. Translated by Peter Aykroyd. Nashville: Abingdon, 1956.
- Kolbert, Elizabeth. *Field Notes from a Catastrophe: A Frontline Report on Climate Change*. London: Bloomsbury, 2007.
- Koppel, Moshe, Navot Akiva, Idan Dershowitz, and Nachum Dershowitz. "Unsupervised Decomposition of a Document into Authorial Components." *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics* (June 19–24, 2011): Portland, Oregon: Association for Computational Linguistics. 1356–1364.
- Kovacs, Maureen Gallery, trans. *The Epic of Gilgamesh*. Stanford, CA: Stanford University Press, 1989.
- Kugel, James L. *The Bible As It Was*. Cambridge, MA: Harvard University Press, 1997.
- Kugel, James L. *How to Read the Bible*. New York: Free Press, 2007.
- Kugel, James L. "Jubilees." In *Outside the Bible: Ancient Jewish Writings Related to Scripture*, edited by Louis Feldman, James Kugel, and Lawrence Schiffman, 272–465. Lincoln: Nebraska University Press, 2013.
- Kushner, Harold. *Etz Hayim: Torah and Commentary*. Philadelphia: The Jewish Publication Society, 2001.
- LaBianca, O.S. *Sedentarization and Nomadization: Food System Cycles at Hesban and Vicinity in Transjordan*. Michigan: Andrews University Press, 1990.
- Lagerlöf, Nils-Petter. "Slavery and Other Property Rights." *The Review of Economic Studies* 76, no. 1 (January 2009): 319–342.
- Lambert, W.G., and A.R. Millard. *Atra-hasis: The Babylonian Story of the Flood*. Oxford: Clarendon Press, 1969.
- Lapp, Paul. "The Conquest of Palestine in the Light of Archeology." *Concordia Theological Monthly* 38 (1967): 283–300.

- Lattimore, Owen. *Studies in Frontier History: Collected Papers, 1928–1958*. Oxford: Oxford University Press, 1962.
- Leick, Gwendolyn. *Mesopotamia: The Invention of the City*. London: Penguin Books, 2002.
- Leach, Edward. *The Political Systems of Highland Burma: A Study of Kachin Social Structure*. Cambridge: Harvard University Press, 1954.
- Leiberman, Pinchas Yehuda, ed. *Exegesis of the Ramban on the Torah: Vol. 1: Genesis, with Explanation of the Tov Yerushalayim* [Hebrew]. Jerusalem: Tuviah, 1997.
- Lieberman, Victor. *Strange Parallels: Southeast Asia in a Global Context, c. 800–1830, Vol. I: Integration on the Mainland*. Ann Arbor: University of Michigan Press, 2003.
- Leibowitz, Nehama. *Studies in Bereshit (Genesis): In the Context of Ancient and Modern Jewish Bible Commentary*. Translated by Aryeh Newman. Jerusalem: Alpha Press, 1985.
- Levenson, Jon. *Sinai and Zion: An Entry into the Jewish Bible*. San Francisco: Harper & Row, 1985.
- Levinson, Joshua. *The Twice-Told Tale: A Poetics of the Exegetical Narrative in Rabbinic Midrash* [Hebrew]. Jerusalem: Magnes Press, 2005.
- Levin, Yigal. “Nimrod the Mighty, King of Kish, King of Sumer and Akkad.” *Vetus Testamentum* 52, no. 3 (2002): 350–366.
- MacDonald, Nathan. *What Did the Ancient Israelites Eat? Diet in Biblical Times*. Michigan: Eardmans Publishing, 2008.
- Mann, Michael. *The Sources of Social Power*. Cambridge: Cambridge University, 1986.
- Matchan, Erin L., David Phillips, Fred Jourdan, and Korien Oosting. “Early human occupation of southeastern Australia: New insights from ⁴⁰Ar/³⁹Ar dating of young volcanoes.” *Geology* (2020): 23–47. 10.1130/G47166.1.
- McCarter, Kyle. “The Patriarchal Age.” In *Ancient Israel*, edited by Hershel Shanks, 1–31. Washington DC: Biblical Archeology Society, 1999.
- Mendenhall, George. “Ancient Oriental and Biblical Law.” In *The Biblical Archeologist Reader*, vol. 3, edited by Edward F. Campbell and David Noel Freedman, 3–24. Garden City: Doubleday, 1970.
- Mendenhall, George. “The Hebrew Conquest of Palestine.” In *The Biblical Archeologist Reader*, vol. 3, edited by Edward F. Campbell and David Noel Freedman, 100–120. Garden City: Doubleday, 1970.
- Mendenhall, George. “The Hebrew Conquest of Palestine.” *Biblical Archeologist* 25 (1962): 66–87.
- Michalowski, Piotr. “Charisma and Control: On Continuity and Change in Early Mesopotamian Bureaucratic Systems.” *The Organisation of Power: Aspects of Bureaucracy in the Ancient Near East: Studies in Ancient Oriental Civilisations* 46 (1987): 55–68.
- Millard, Alan R. “An Assessment of the Evidence of Writing in Ancient Israel.” In *Biblical Archaeology Today: Proceedings of the International Congress, Jerusalem, April 1984*, edited by Janet Amitai, 301–312. Jerusalem: Israel Exploration Society, 1985.
- Miller, J. Maxwell. “The Israelite Occupation of Canaan.” In *Israelite and Judean History*, edited by John H. Hayes and J. Maxwell Miller, 252–262. Philadelphia: Westminster, 1977.
- Murdoch, Brian O. *The Medieval Popular Bible: Expansion of Genesis in the Middle Ages*. Cambridge: DS Brewer, 2003.
- Morris, Ian. *Why the West Rules for Now: The Patterns of History and What They Reveal About the Future*. London: Profile Books, 2011.

- Murdock, George P., and Caterina Provost. "Factors in the Division of Labor by Sex: A Cross-Cultural Analysis." *Ethnology* 12, no. 2 (April 1973): 203–225.
- Naveh, Joseph. *Early History of the Alphabet: An Introduction to West Semitic Epigraphy and Paleography*. Jerusalem: Magnes, 1982.
- Neusner, Jacob. *The Tosefta: An Introduction*. Atlanta: Scholars Press, 1992.
- Neusner, Jacob. *What is Midrash?* Philadelphia: Fortress Press, 1987.
- Nielson, Eduard. *Oral Tradition*. London: SCM Press, 1961.
- Niditch, Susan. *Chaos to Cosmos: Studies in Biblical Patterns of Creation*. Atlanta: Scholars Press, 1985.
- Nissen, Hans J. *The Early History of the Ancient Near East, 9000–2000 BC*. Chicago: University of Chicago Press, 1993.
- North, D.C. and R.P. Thomas. "The First Economic Revolution." *Economic History Review* 30 (1977): 229–241.
- Northcott, Michael S. *A Political Theology of Climate Change*. London: Society for the Promotion of Christian Knowledge Press, 2014.
- Noth, Martin. *The Deuteronomistic History*. Sheffield: JSOT Press, 1981.
- O'Connor, Richard. "Agricultural Change and Ethnic Succession in the Southeast Asian States: A Case for Regional Anthropology." *Journal of Asian Studies* 54 (1995): 968–996.
- O'Connor, Richard. "A Regional Explanation of the Tai Muang as a City-State." In *A Comparative Study of Thirty City-States*, edited by Mogens Herman Hansen, 431–443. Copenhagen: Royal Danish Academy of Sciences and Letters, 2000.
- Oded, B. "The Table of Nations (Genesis 10)—a Socio-Cultural Approach." *Zeitschrift Für Die Alttestamentliche Wissenschaft* 98 (1986): 17–21.
- Oelschlaeger, Max. *The Idea of Wilderness: From Prehistory to the Age of Ecology*. New Haven: Yale University Press, 1991.
- Oldenburg, Ulf. *The Conflict Between El and Ba'al in the Canaanite Religion*. Leiden: E.J. Brill, 1969.
- Özdoğan, Mehmet. "Neolithization of Europe: A View from Anatolia. Part 1: The Problem and the Evidence of East Anatolia." *Poročilo* XX (1995): 25–61.
- Parrot, A. *The Flood and Noah's Ark*. New York: Philosophical Library, 1955.
- Pietersma, Albert, and Benjamin G. Wright. *The New English Translation of the Septuagint*. Oxford: Oxford University Press, 2007.
- Pleins, J. David. *When the Great Abyss Opened: Classic and Contemporary Readings of Noah's Flood*. Oxford: Oxford University Press, 2009.
- Pollack, Susan. *Ancient Mesopotamia*. Cambridge: Cambridge University Press, 1999.
- Ponting, Clive. *A Green History of the World*. London: Sinclair-Stevenson Ltd., 1991.
- Porter, Anne. *Mobile Pastoralism and the Formation of Near Eastern Civilization: Weaving Societies Together*. Cambridge: Cambridge University Press, 2012.
- Postel, Sandra. *Pillar of Sand: Can the Irrigation Miracle Last?* New York: Norton & Co., 1999.
- Powell, Barry. *Writing: Theory and History of the Technology of Civilization*. Chichester: Wiley-Blackwell, 2009.
- Postgate, J.N. "A Sumerian City: Town and Country in the 3rd Millennium B.C." *Scienza dell'Antichità Storia Archeologia* 6–7 (1996): 409–435.
- Pritchard, J.B. *Ancient Near Eastern Texts*. Princeton, NJ: Princeton University Press, 1969.
- Pustovoytov, Konstantin. "Soils and Sediments at Gobekli Tepe, Southeastern Turkey: A Preliminary Report." *GeoArchaeology* 21 (2006): 699–719.

- Quinn, Daniel. *Ishmael*. New York: Bantam, 1992.
- Redman, Charles. *The Rise of Civilization: From Early Farmers to Urban Society in the Ancient Near East*. San Francisco: W.H. Freeman, 1978.
- Reid, Anthony. *Southeast Asia in the Age of Commerce, 1450–1680: The Lands Below the Winds*, Vol. 1. New Haven: Yale University, 1988.
- Richardson, Seth. “Early Mesopotamia: The Presumptive State.” *Past and Present* 215 (2012): 3–48.
- Ridley, Matt. *The Rational Optimist*. London: Fourth Estate, 2011.
- Ross, Michael L. *The Oil Curse: How Petroleum Wealth Shapes the Development of Nations*. Princeton: Princeton University Press, 2012.
- Ryan, W., and W. Pitman. *Noah's Flood: The New Scientific Discoveries About the Event that That Changed History*. New York: Simon and Schuster, 1998.
- Sacks, Jonathan. *Covenant and Conversation, Genesis: The Book of Beginnings*. Jerusalem: Maggid Press, 2009.
- Sacks, Jonathan. *Covenant and Conversation, Numbers: The Wilderness Years*. Jerusalem: Maggid Press, 2017.
- Sahlins, Marshall. *Stone Age Economics*. Chicago: Aldine, 1974.
- Sapir-Hen, Lidar, A. Sasson, A. Kleiman, and I. Finkelstein. “Social Stratification in the Late Bronze and Early Iron Ages.” *Oxford Journal of Archaeology* 35 (2016): 47–73.
- Sarna, Nahum M. *The JPS Torah Commentary: Genesis*. Philadelphia: The Jewish Publication Society, 1989.
- Sawyer, John F. A. “Cain and Hephaestus: Possible Relics of Metalworking Traditions in Genesis 4.” *Abr-Nahrain* 24 (1986): 155–166.
- Sasson, Aharon. “The Pastoral Component in the Economy of Hill Country Sites in the Intermediate Bronze and Iron Ages: Archaeo-Ethnographic Case Studies.” *Tel Aviv* 25, no. 1 (1998): 3–51.
- Scarlata, Mark. “Genesis.” In *T and T Clark Companion to the Septuagint*, edited by James K. Aitken, 13–28. London: Bloomsbury T&T Clark, 2015.
- Schneidau, Herbert. *Sacred Discontent*. Berkeley: University of California Press, 1977.
- Schmidt, Klaus. “Gobekli Tepe, Southeastern Turkey: A Preliminary Report on the 1995–1999 excavations.” *Paleorient* 26, no. 26 (2000): 45–54.
- Scott, James. *Against the Grain: A Deep History of the Earliest State*. New Haven: Yale University Press, 2017.
- Scott, James. *The Art of Not Being Governed: An Anarchist History of Upland Southeast Asia*. New Haven: Yale University Press, 2009.
- Scott, James. *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed*. New Haven: Yale University Press, 1998.
- Shanks, Hershel. *Ancient Israel: From Abraham to the Roman Destruction of the Temple*. Washington DC: Biblical Archeology Society, 1999.
- Shinan, Avigdor, and Yair Zakovitch. *From Gods to God: How the Bible Debunked, Suppressed or Changed Ancient Myths and Legends*. Philadelphia: Jewish Publication Society, 2012.
- Simmons, Alan H. *The Neolithic Revolution in the Near East: Transforming the Human Landscape*. Arizona: University of Arizona Press, 2007.
- Skinner, John. *Genesis, International Critical Commentary*, 2nd ed. Edinburgh: T & T Clark, 1980.
- Smith, Bruce D. *The Emergence of Agriculture*. New York: Scientific American Library, 1998.

- Smith, Philip E.L., and Cuyler T. Young. "The Evolution of Early Agriculture and Culture in Greater Mesopotamia." In *Population Growth: Anthropological Implications*, edited by Brian Spooner, 1–59. Cambridge: MIT Press, 1972.
- Soggin, Alberto. *Introduction to the Old Testament*. Translated by John Bowden. Philadelphia: Westminster, 1976.
- Solbrig, Otto T., and Dorothy J. Solbrig. *So Shall You Reap: Farming and Crops in Human Affairs*. Washington DC: Island Press, 1994.
- Speiser, E.A. "The Biblical Idea of History in Its Common Near Eastern Setting." *Israel Exploration Journal* 7, no. 4 (1957): 200–230. <http://www.jstor.org/stable/27924711>.
- Speiser, E.A. *The Anchor Bible: Genesis*. New York: Doubleday, 1964.
- Spiegel, Shalom. *The Last Trial: On the Legends and Lore of the Command to Abraham to Offer Isaac as a Sacrifice*. Translated by Judah Goldin. Vermont: Jewish Lights Publishing, 1993.
- Stager, Lawrence E. "Agriculture." In *The Interpreter's Dictionary of the Bible Supplementary Volume*, edited by Keith Crim, 11–13. Nashville: Abingdon, 1976.
- Stager, Lawrence E. "The First Fruits of Civilization." *Palestine in the Bronze and Copper Ages, Institute of Archaeology Occasional Papers* 11 (1985): 172–188.
- Stein, Dina. "Rabbinic Interpretation." In *Reading Genesis: Ten Methods*, edited by Ronald S. Hendel, 119–135. Cambridge: Cambridge University Press, 2010.
- Stern, Menahem "The Jews in the Land of Israel." In *A History of the Jewish People*, edited by Haim Hillel Ben Sasson, 70–335. Mass.: Harvard University Press, 1997.
- Strack, H.L. and G. Stemberger. *Introduction to the Talmud and Midrash*. Translated by Markus Bockmuehl. Edinburgh: T&T Clark, 1991.
- Symons, D. "Another Woman That Never Existed." *The Quarterly Review of Biology* 57, no. 3 (1983): 297–300.
- Theodor, Julius, and Hanock Albeck, eds. *Midrash Bereshit Rabba* [Heb] 3 vols. Jerusalem: Wahrmann, 1965.
- Tucker, Gene M. "The Rahab Saga (Joshua 2): Some Form-Critical and Traditio-Historical Observations." In *The Use of the Old Testament in the New and Other Essays: Studies in Honour of William Franklin Stinespring*, edited by James M. Efird, 71–86. Durham: Duke University Press, 1972.
- Twersky, Geula. "Lamech's Song and the Cain Genealogy: An Examination of Gen 4.23–24 within its Narrative Context." *Scandinavian Journal of the Old Testament* 31, no. 2 (2017): 275–293.
- Van Der Toorn, Kerel. *Scribal Culture and the Making of the Hebrew Bible*. Cambridge, MA: Harvard University Press, 2007.
- Vanstiphout, H.L.J. "The Disputation Between Ewe and Wheat." In *The Context of Scripture: Canonical Compositions from the Biblical World*, vol. 1, edited by William W. Hallo, 570–578. Leiden: Brill, 1997.
- Van Wolfswinkel, Martine. "Intercropping of Annual Foodcrops." *Agromisa-Agrobrief* 4 (July 2003): 4–18.
- Verbrugghe, G.P., and J.M. Wickersham. *Berosos and Manetho, Introduced and Translated: Nature Traditions in Ancient Mesopotamia and Egypt*. Ann Arbor: University of Michigan Press, 1996.
- von Rad, Gerhard. *Genesis*. Philadelphia: Westminster, 1972.
- Walton, John H. *Zondervan Illustrated Bible Backgrounds Commentary: Genesis*. Michigan: Zondervan, 2013.
- Waskow, Arthur. *Torah of the Earth*, Vol. 1. New York: Jewish Lights Publishing, 2000.

- Weippert, Manfred. *The Settlement of the Israelite Tribes in Palestine: A Critical Survey of Recent Scholarly Debate*. Translated by James D. Martin. London: SCM, 1971.
- Weisdorf, Jacob L. "From Foraging to Farming: Explaining the Neolithic Revolution." *Journal of Economic Surveys* 19, no. 4 (2005): 561–586.
- Weiss, Harvey, Francesca deLillis, Dominique deMoulins, Jesper Eidem, Thomas Guilderson, Ulla Kasten, Torben Larsen, Lucia Mori, Lauren Ristvet, Elena Rova, and Wilma Wetterstrom. "Revising the Contours of History at Tell Leilan." *Annales archéologiques arabes syriennes*, Cinquantenaire (2002): 59–74.
- Wellhausen, Julius. *Prolegomena to the History of Ancient Israel*. New York: Meridian Books, 1958.
- Westerman, Claus. *Genesis 1-11: A Continental Commentary*. Translated by John J. Scullion. Minneapolis: Fortress Press, 1984.
- Wilson, Peter J. *The Domestication of the Human Species*. New Haven: Yale University Press, 1988.
- Wimpfheimer, Barry Scott. *Narrating the Law: A Poetics of Talmudic Legal Stories*. Philadelphia: University of Pennsylvania Press, 2011.
- Wirzba, Norman. *The Paradise of God: Renewing Religion in an Ecological Age*. New York: Oxford University Press, 2003.
- Witkowski, Stanley R., and William Divale. "Kin Groups, Residence and Descent." In *the Encyclopedia of Cultural Anthropology*, edited by David Levinson and Melvin Ember, 673–680. New York: Henry Holt, 1996.
- Wright, Chris. *Old Testament Ethics for the People of God*. Illinois: Intervarsity Press, 2004.
- Wright, Ernest G. *Biblical Archeology*. Philadelphia: Westminster, 1962.
- Wright, K.I. "The Social Origins of Cooking and Dining in Early Villages of Western Asia." *Proceedings of the Prehistoric Society* 66, no. 1 (2000): 89–121.
- Wright, Robert. *NonZero: History, Evolution and Human Cooperation*. London: Abacus, 2001.
- Yoffee, Norman, and George Cowgill. *The Collapse of Ancient States and Civilizations*. Tucson: University of Arizona Press, 2006.
- Yoffee, Norman, and George Cowgill. "Collapse in Ancient Mesopotamia: What Happened, What Didn't?" In *Questioning Collapse: Human Resilience, Ecological Vulnerability, and the Aftermath of Empire*, edited by Patricia A. McAnany and Norman Yoffee, 176–206. Cambridge: Cambridge University Press, 2012.
- Young, Ian M. "Israelite Literacy: Interpreting the Evidence." *VT* 48 (1998): 239–253.
- Zeder, Melinda. "After the Revolution: Post Neolithic Subsistence in Northern Mesopotamia." *American Anthropologist* new ser. 96, no. 1 (1994): 97–126.
- Zeder, Melinda. "The Origins of Agriculture in the Near East." *Current Anthropology* 52, no. S4 (2011): S221–S235.
- Zohary, D., and M. Hopf. *Domestication of Plants in the Old World*. Oxford: Clarendon Press, 1988.
- Zoloth, Laurie. "Justice in the Margins of the Land: Jewish Responses to the Challenges of Biotechnology." In *The Routledge Companion to Religion and Science*, edited by James Haag, Greg Peterson, and Micheal Spezio, 476–484. New York, NY: Routledge, 2011.
- Zornberg, Avivah Gottlieb. *Genesis: The Beginnings of Desire*. Philadelphia: The Jewish Publication Society, 1995.

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